

**The data file AMMONIAEX  
(based on AMUJEL format):  
Additional Atomic and Molecular Data for EIRENE for  
AMMONIA**

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### 3 Appendix

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# Introduction

## Record:

- April, 17: First version
- July 3, 17: Adding cross sections
- July 31, 17: Database update. Thanks to Ms S. TAMODARIN and Mr S. KETHESWARAN for their work.

## to be done:

- To be done

Additional atomic data fits, read by EIRENE Specific reaction for AMMONIAC  
Format as HYDHEL DATEN [2] or METHANE DATEN [1] or AMJUEL DATEN  
SPECIFIC BIBLIOGRAPHY IS AVIABLE

## End of preface:

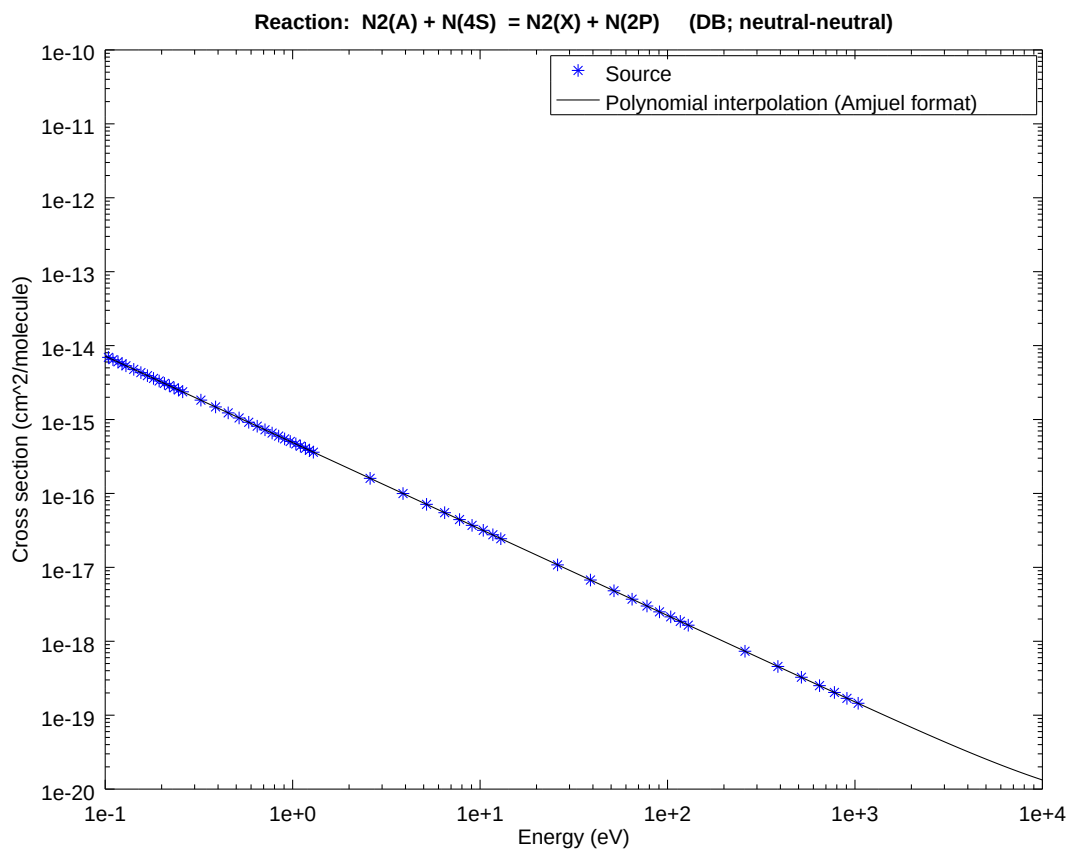
This next string is searched by EIRENE in subroutine SLREAC to initialize search for a particular set of fit coefficients. From here on, a character string ‘**H.n**’, n an integer, must only appear in the section title, but not in the text. Likewise: identifiers p0, a0, b0, ..., h0, k0 are used in SLREAC and must not appear in the text elsewhere, from here on.

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.      ##BEGIN DATA HERE##  
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# 1 H.1 : Fits for cross section

## 1.1 Reaction DB N2(A) + N(4S) = N2(X) + N(2P)

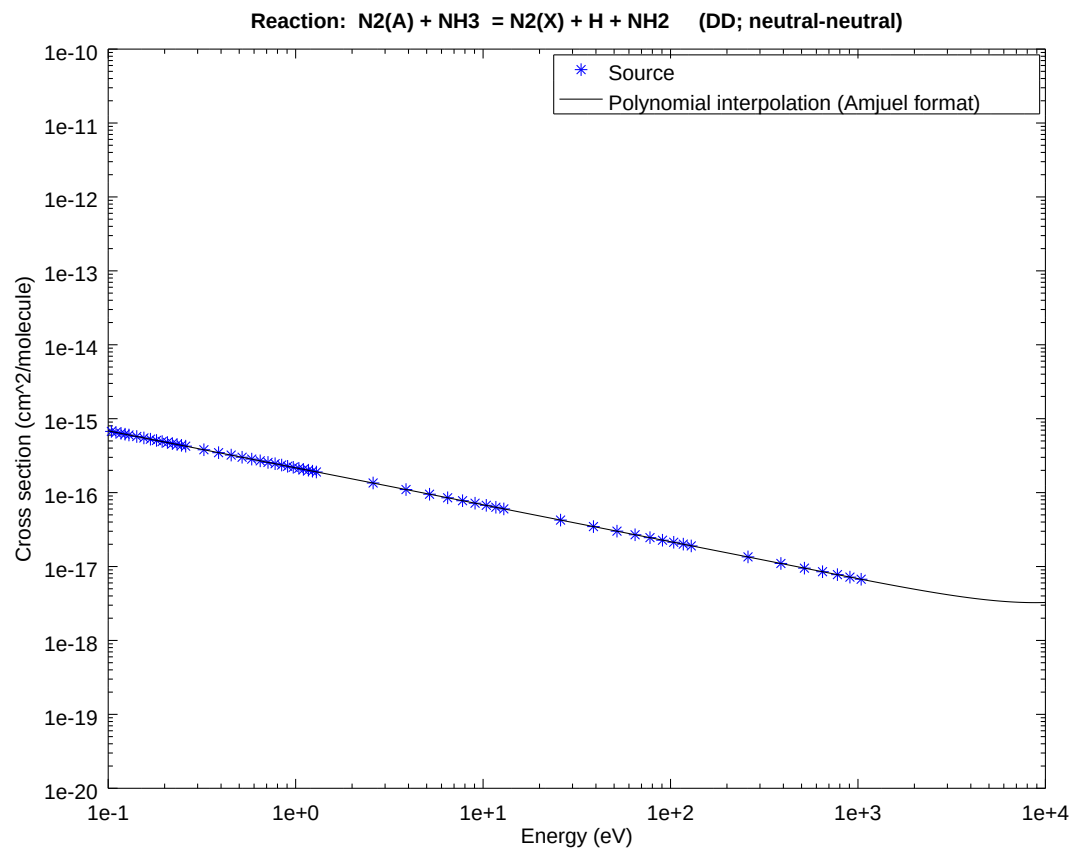
|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.525604931656E+01 | a1 | -1.169467047039E+00 | a2 | -3.016619218433E-04 |
| a3 | -1.131053596850E-04 | a4 | 5.075664503871E-05  | a5 | 5.186112097513E-06  |
| a6 | -2.519799999214E-06 | a7 | 4.025963385058E-08  | a8 | 1.886635246743E-08  |





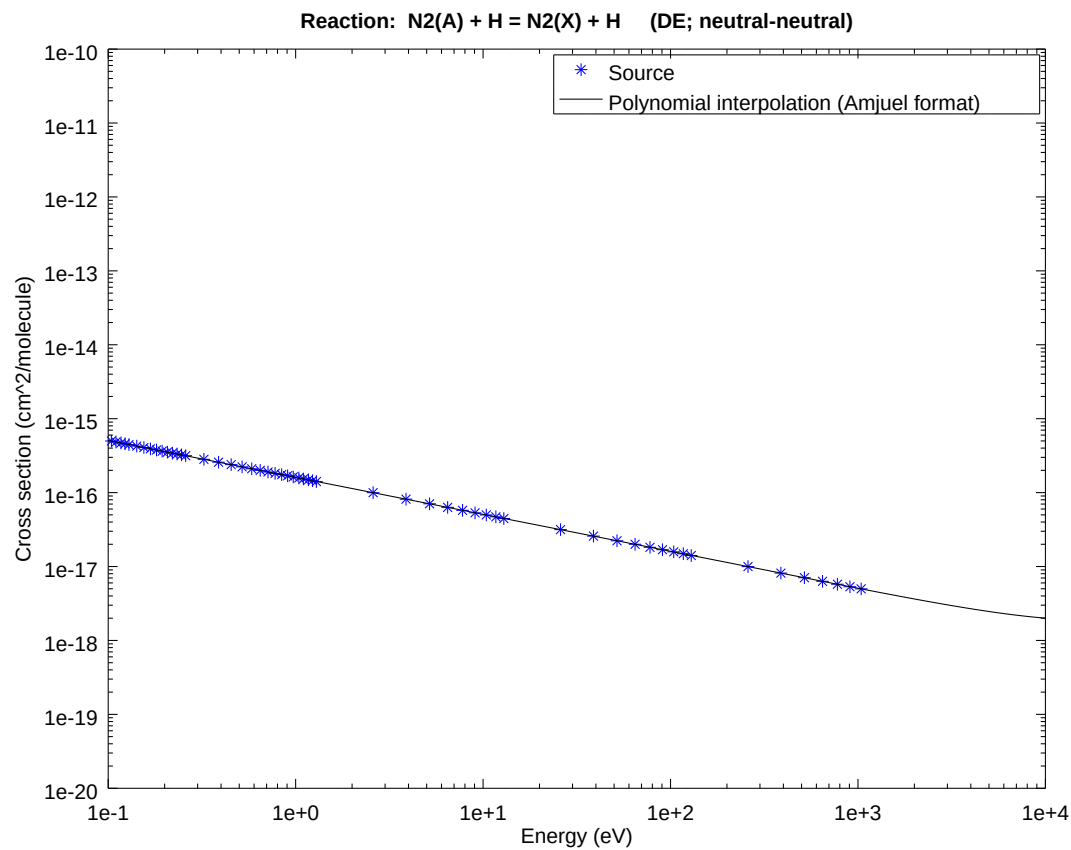
## 1.2 Reaction DD $\text{N}_2(\text{A}) + \text{NH}_3 = \text{N}_2(\text{X}) + \text{H} + \text{NH}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.606850050331E+01 | a1 | -4.985692234760E-01 | a2 | -2.554692077144E-04 |
| a3 | -4.069952082798E-04 | a4 | 5.028150025855E-05  | a5 | 3.019730055286E-05  |
| a6 | -3.697880512623E-06 | a7 | -6.049288920032E-07 | a8 | 7.786467574166E-08  |



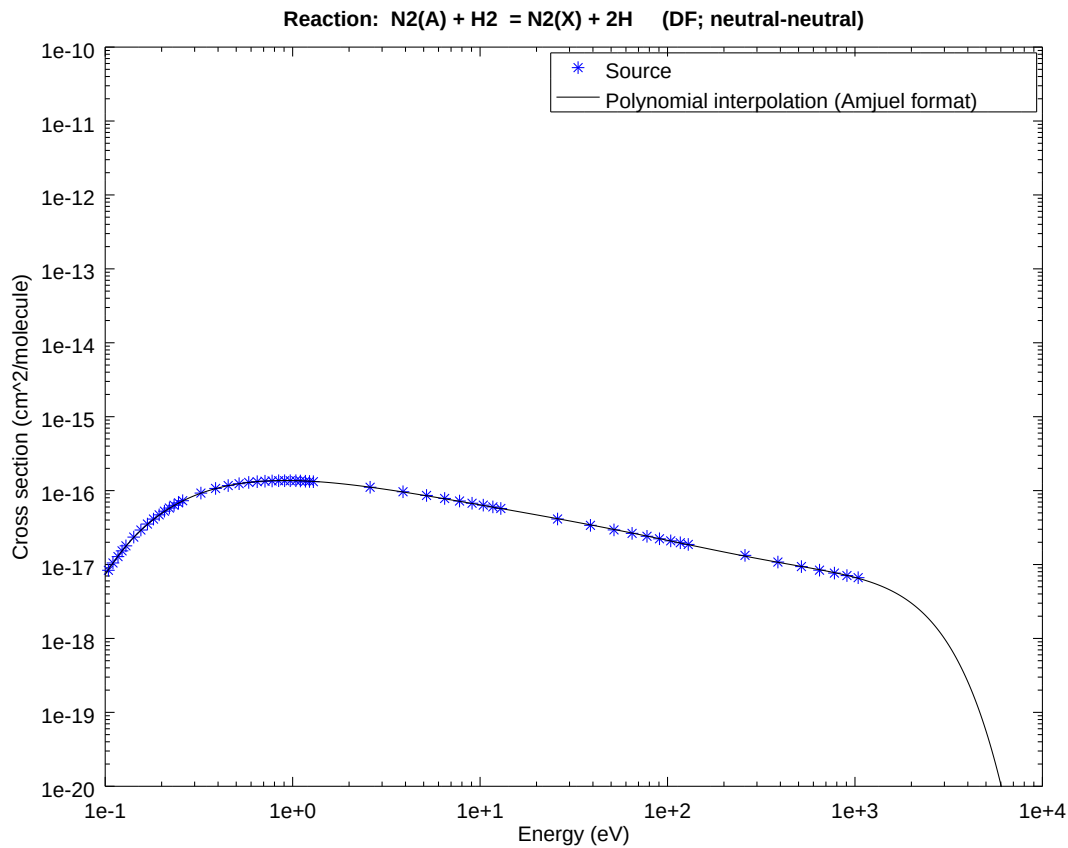
### 1.3 Reaction DE $\text{N}_2(\text{A}) + \text{H} = \text{N}_2(\text{X}) + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.636724579729E+01 | a1 | -4.996921680518E-01 | a2 | -3.004238048443E-04 |
| a3 | -7.027601268270E-05 | a4 | 4.639562577142E-05  | a5 | 4.000298458661E-06  |
| a6 | -2.200508309113E-06 | a7 | 1.127902891757E-08  | a8 | 1.890104031525E-08  |



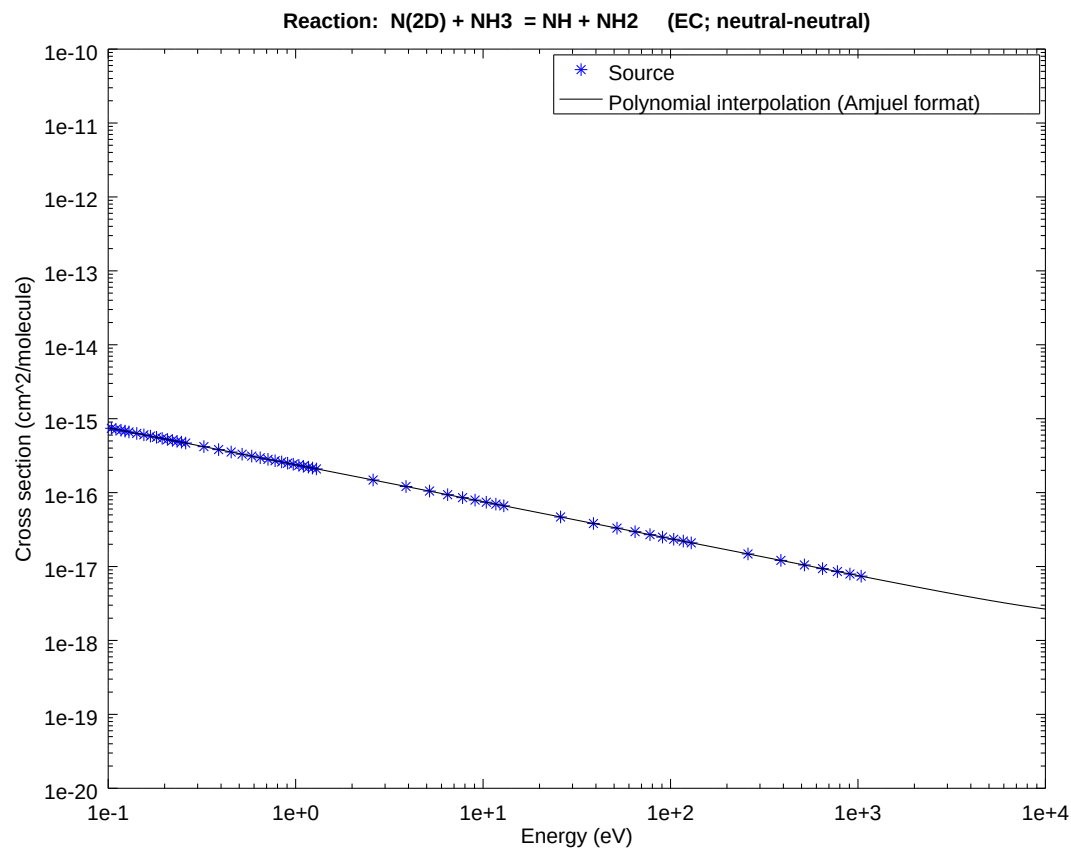
## 1.4 Reaction DF $\text{N}_2(\text{A}) + \text{H}_2 = \text{N}_2(\text{X}) + 2\text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.652800397725E+01 | a1 | -4.578607537346E-02 | a2 | -2.399741909925E-01 |
| a3 | 7.519012714366E-02  | a4 | -1.537136580981E-02 | a5 | 3.557144220075E-03  |
| a6 | -9.047956443652E-04 | a7 | 1.282478836314E-04  | a8 | -6.737601321279E-06 |



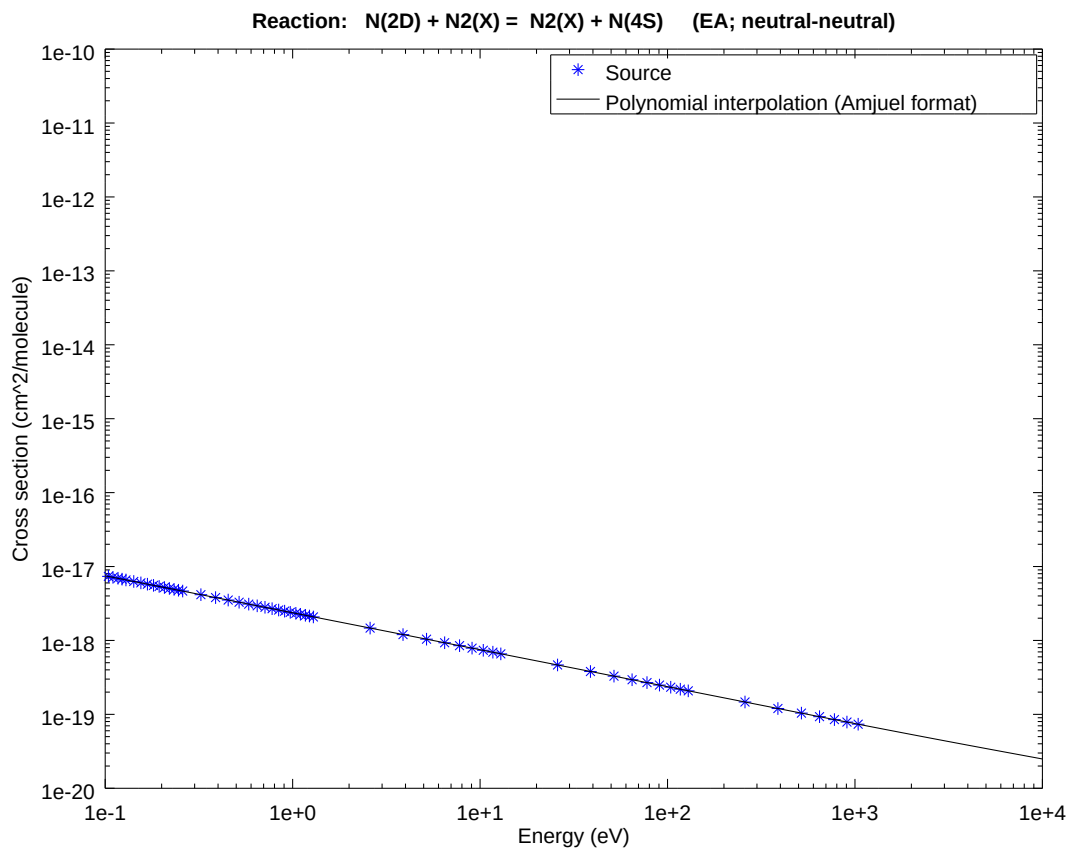
## 1.5 Reaction EC N(2D) + NH3 = NH + NH2

|    |                     |    |                     |    |                    |
|----|---------------------|----|---------------------|----|--------------------|
| a0 | -3.597312484632E+01 | a1 | -4.992860132291E-01 | a2 | 2.221724976694E-04 |
| a3 | -2.185431927439E-04 | a4 | -1.875235453436E-05 | a5 | 1.801683905709E-05 |
| a6 | -5.631533912406E-07 | a7 | -4.004005884577E-07 | a8 | 3.456994046962E-08 |



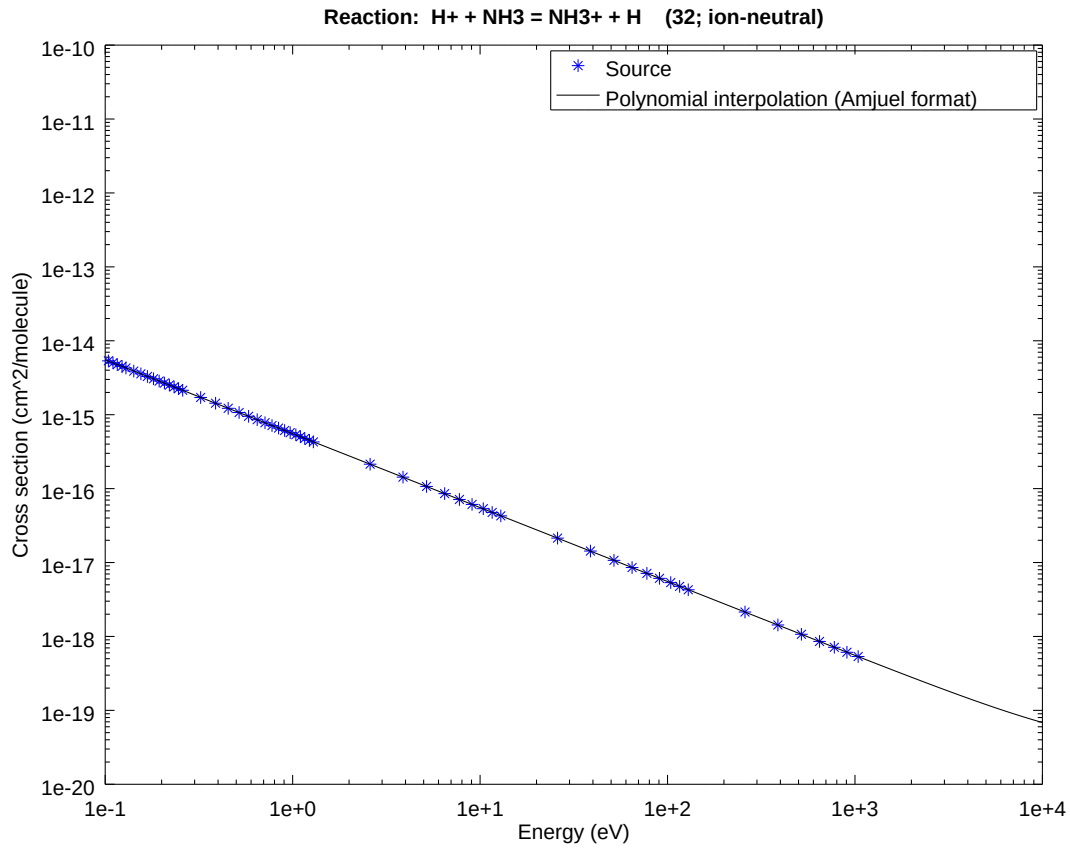
## 1.6 Reaction EA N(2D) + N2(X) = N2(X) + N(4S)

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -4.058426808527E+01 | a1 | -5.002334190407E-01 | a2 | -7.888387115786E-05 |
| a3 | 9.295612376230E-05  | a4 | 1.599541494452E-05  | a5 | -9.072793765241E-06 |
| a6 | -4.451575724006E-07 | a7 | 3.001864279963E-07  | a8 | -1.820461730557E-08 |



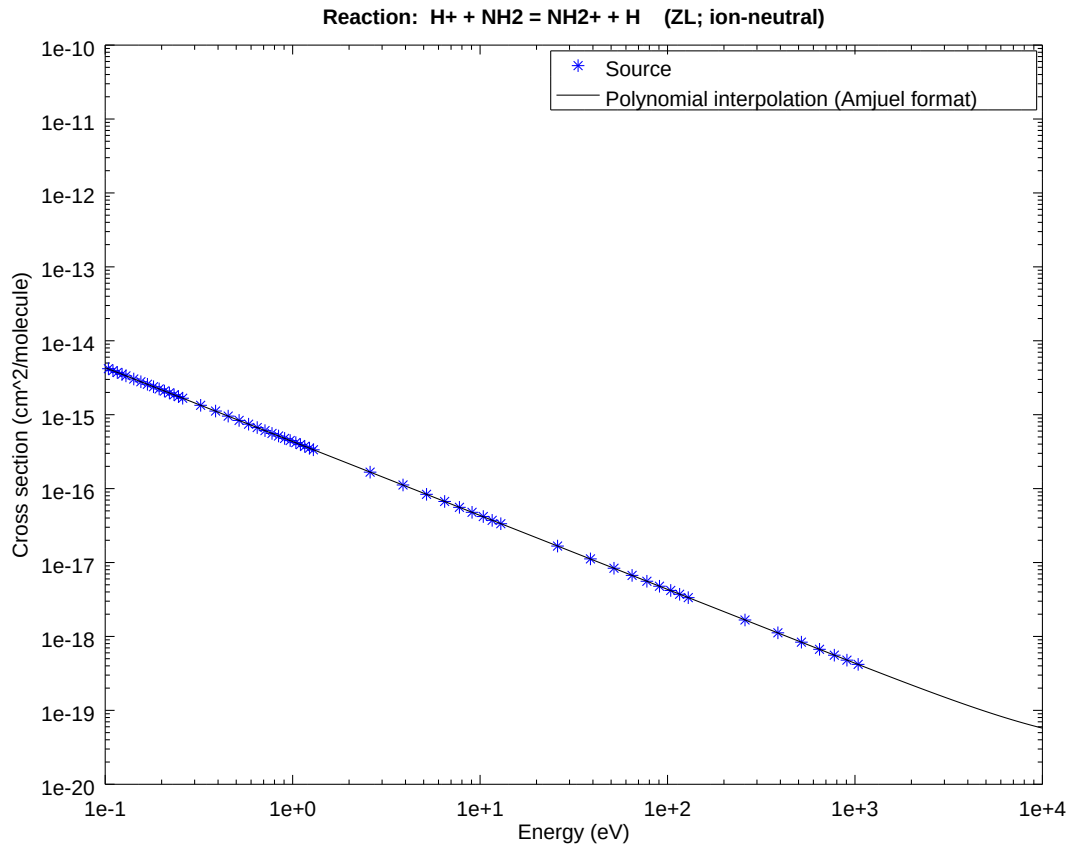
## 1.7 Reaction 32 $\text{H}^+ + \text{NH}_3 = \text{NH}_3^+ + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.512950843118E+01 | a1 | -9.995197673296E-01 | a2 | -1.089341894829E-04 |
| a3 | -1.324296798829E-04 | a4 | 1.991180950364E-05  | a5 | 1.021163226029E-05  |
| a6 | -1.537210572679E-06 | a7 | -1.883169541811E-07 | a8 | 2.892314580928E-08  |



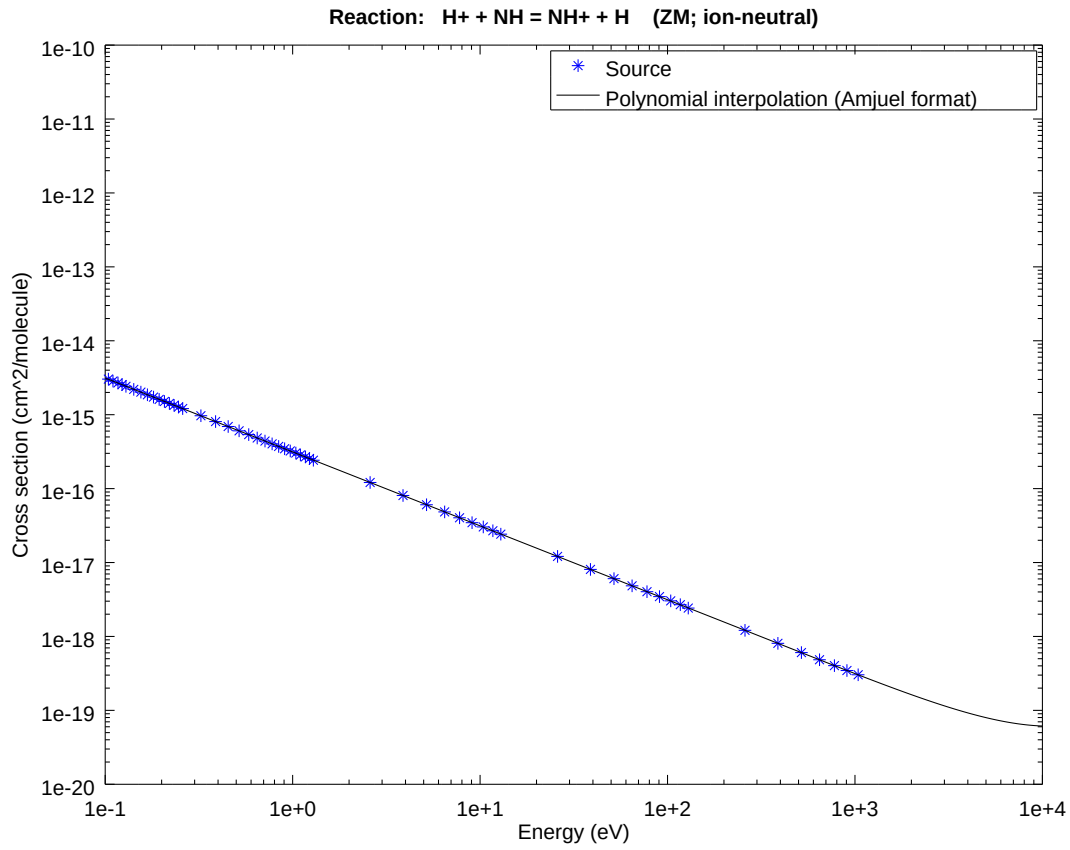
## 1.8 Reaction ZL $\text{H}^+ + \text{NH}_2 = \text{NH}_2^+ + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.537490331955E+01 | a1 | -9.993461448873E-01 | a2 | -1.949720016317E-04 |
| a3 | -1.729087654282E-04 | a4 | 3.569519275369E-05  | a5 | 1.165878099393E-05  |
| a6 | -2.244838297527E-06 | a7 | -1.767355711604E-07 | a8 | 3.403726702521E-08  |



## 1.9 Reaction ZM $\text{H}^+ + \text{NH} = \text{NH}^+ + \text{H}$

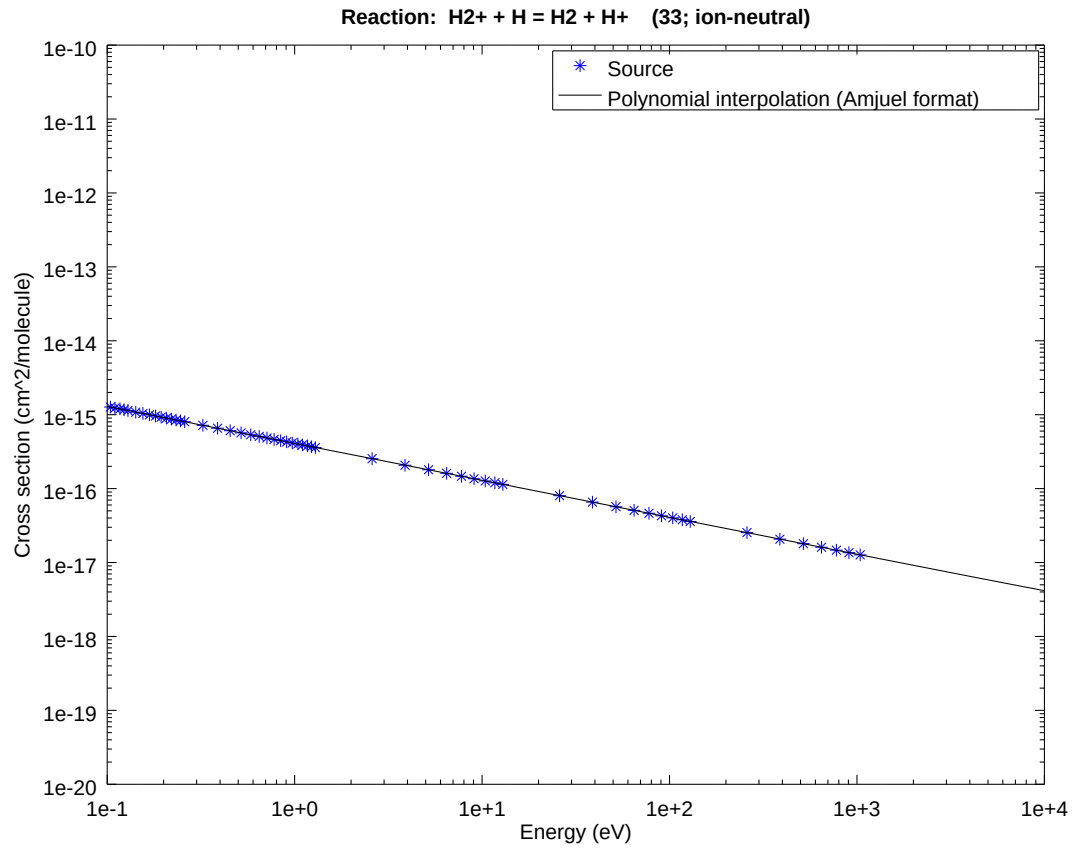
|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.569871167432E+01 | a1 | -9.983552723791E-01 | a2 | -2.774822166591E-04 |
| a3 | -4.556629385908E-04 | a4 | 5.969049346956E-05  | a5 | 3.538344846415E-05  |
| a6 | -4.659439894078E-06 | a7 | -7.556905829568E-07 | a8 | 1.034211511159E-07  |





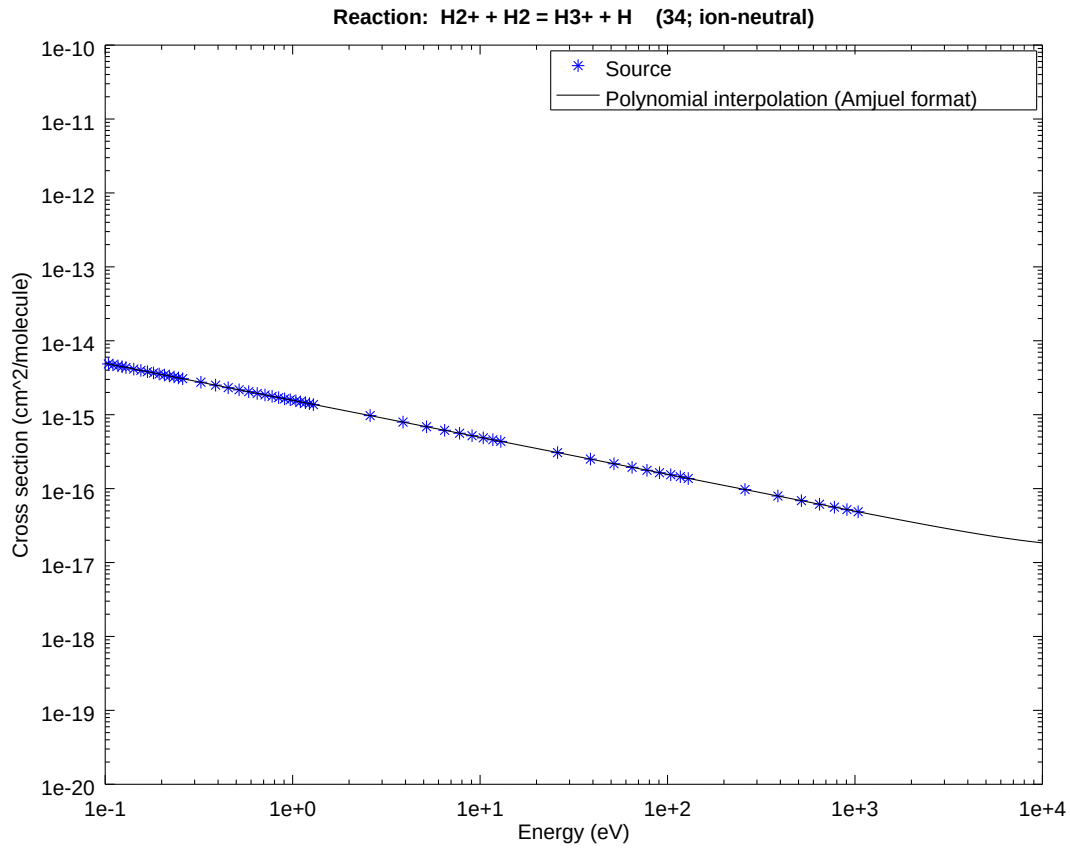
## 1.10 Reaction 33 $\text{H}_2^+ + \text{H} = \text{H}_2 + \text{H}^+$

|    |                     |    |                     |    |                    |
|----|---------------------|----|---------------------|----|--------------------|
| a0 | -3.543361362397E+01 | a1 | -4.992788117844E-01 | a2 | 4.583388077472E-04 |
| a3 | -1.407924001098E-04 | a4 | -4.505298269726E-05 | a5 | 1.169063046928E-05 |
| a6 | 6.572405609520E-07  | a7 | -2.867139255236E-07 | a8 | 1.628565543476E-08 |



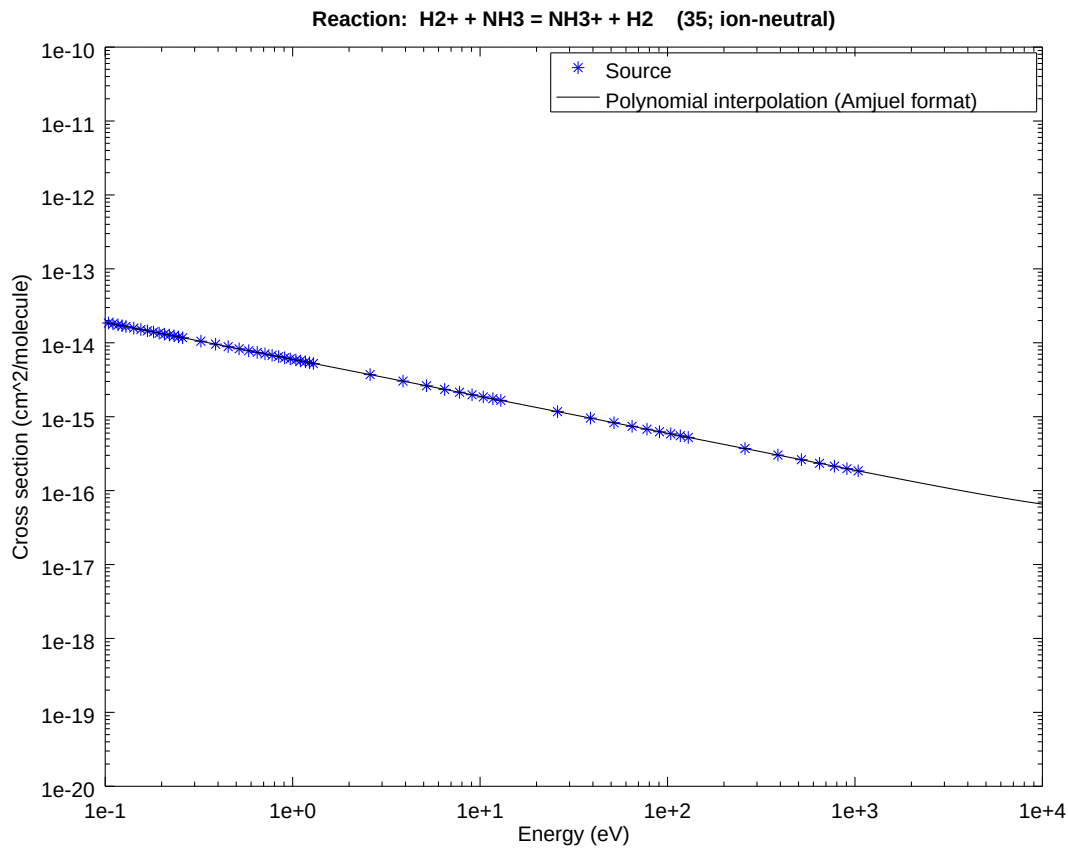
## 1.11 Reaction 34 $\text{H}_2^+ + \text{H}_2 = \text{H}_3^+ + \text{H}$

|    |                     |    |                     |    |                    |
|----|---------------------|----|---------------------|----|--------------------|
| a0 | -3.409202391649E+01 | a1 | -4.990685444208E-01 | a2 | 1.437669809837E-04 |
| a3 | -2.595121452718E-04 | a4 | 9.995622582894E-07  | a5 | 1.869798439979E-05 |
| a6 | -1.485708040220E-06 | a7 | -3.285779522997E-07 | a8 | 3.583715168109E-08 |



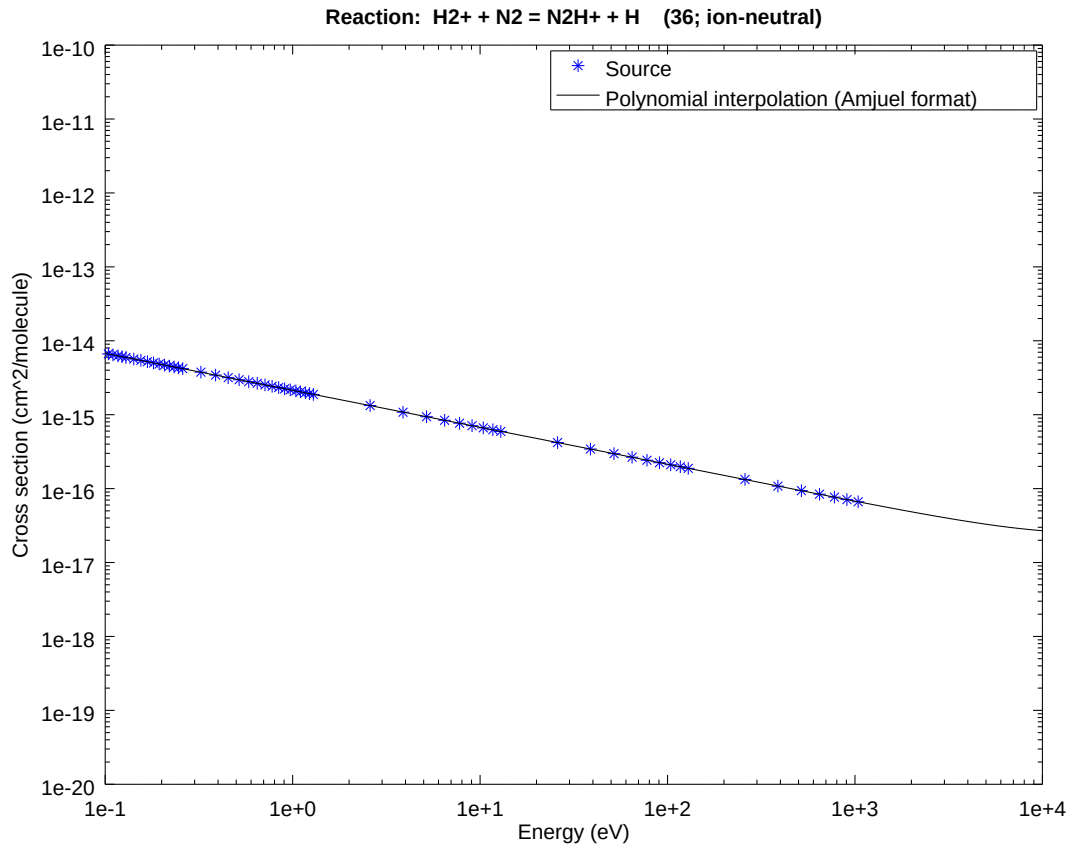
## 1.12 Reaction 35 $\text{H}_2^+ + \text{NH}_3 = \text{NH}_3^+ + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.275288968305E+01 | a1 | -4.994787050446E-01 | a2 | -2.188016275804E-04 |
| a3 | -1.033806506382E-04 | a4 | 3.316204729448E-05  | a5 | 4.858481757960E-06  |
| a6 | -1.522128427499E-06 | a7 | -1.004154955784E-08 | a8 | 1.187729688822E-08  |



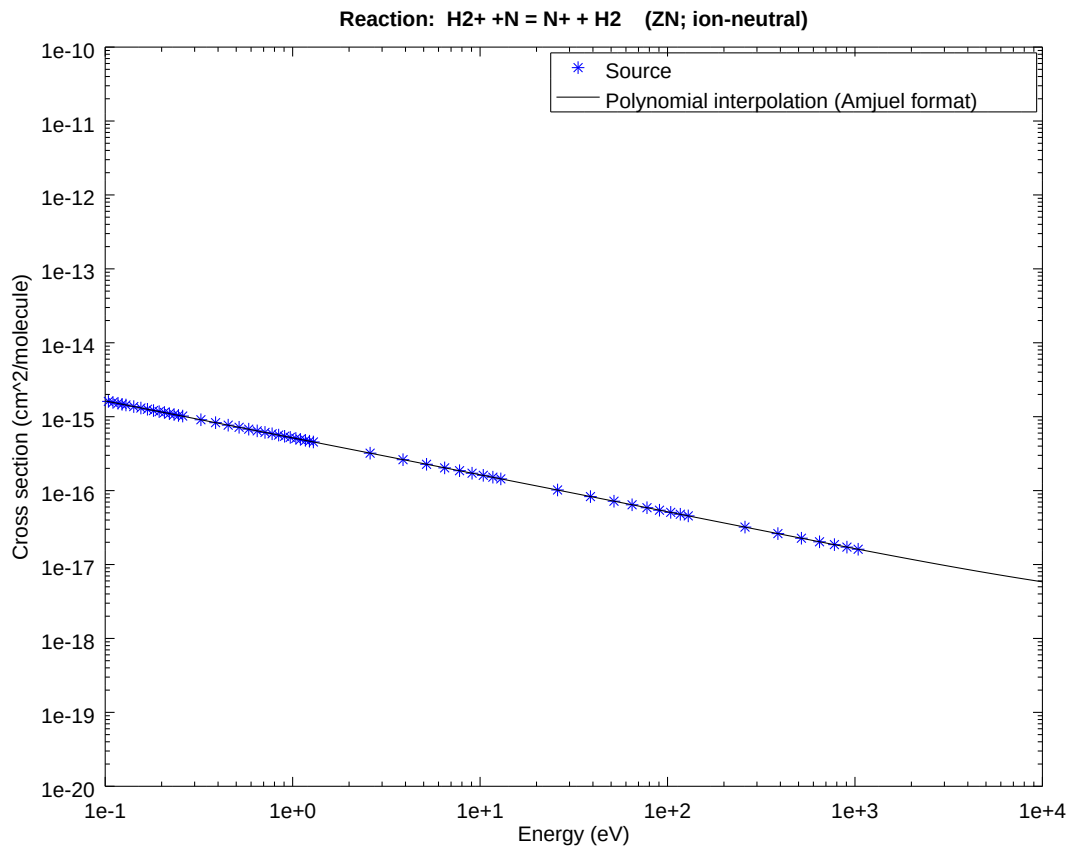
### 1.13 Reaction 36 $\text{H}_2^+ + \text{N}_2 = \text{N}_2\text{H}^+ + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.377920738035E+01 | a1 | -4.997687175351E-01 | a2 | -9.948208681396E-05 |
| a3 | -5.978514387256E-05 | a4 | 1.295890511041E-05  | a5 | 6.455654301945E-06  |
| a6 | -1.012966875438E-06 | a7 | -1.864498029942E-07 | a8 | 2.762522101525E-08  |



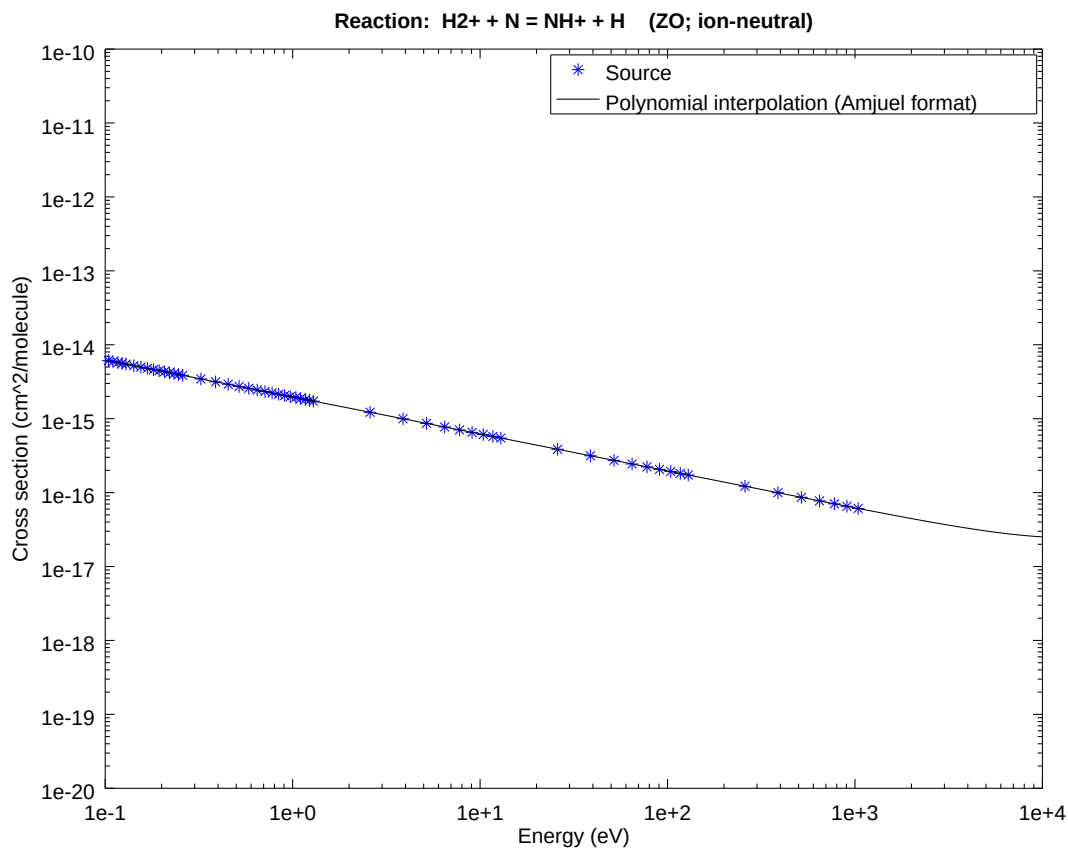
## 1.14 Reaction $\text{ZN H2}^+ + \text{N} = \text{N}^+ + \text{H2}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.519826157435E+01 | a1 | -4.998014584796E-01 | a2 | 1.530815192373E-04  |
| a3 | 5.036422855746E-05  | a4 | -5.083768450143E-06 | a5 | -5.745853671067E-06 |
| a6 | 4.837645055788E-09  | a7 | 1.667353837445E-07  | a8 | -8.813920612450E-09 |



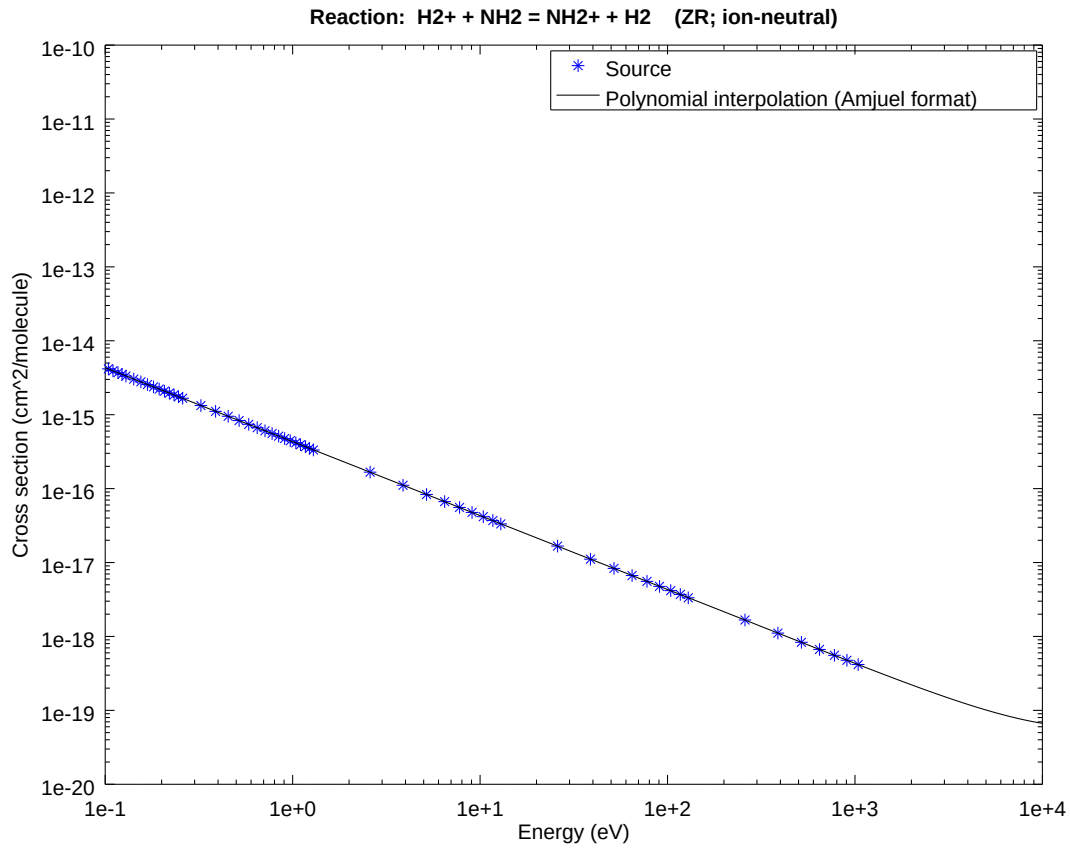
## 1.15 Reaction ZO H2+ + N = NH+ + H

|    |                     |    |                     |    |                    |
|----|---------------------|----|---------------------|----|--------------------|
| a0 | -3.386357554600E+01 | a1 | -4.994103923210E-01 | a2 | 2.706129685180E-04 |
| a3 | -2.095646572375E-04 | a4 | -2.749472868472E-05 | a5 | 2.142732365011E-05 |
| a6 | -3.458818819641E-07 | a7 | -6.382759584083E-07 | a8 | 5.716798847223E-08 |



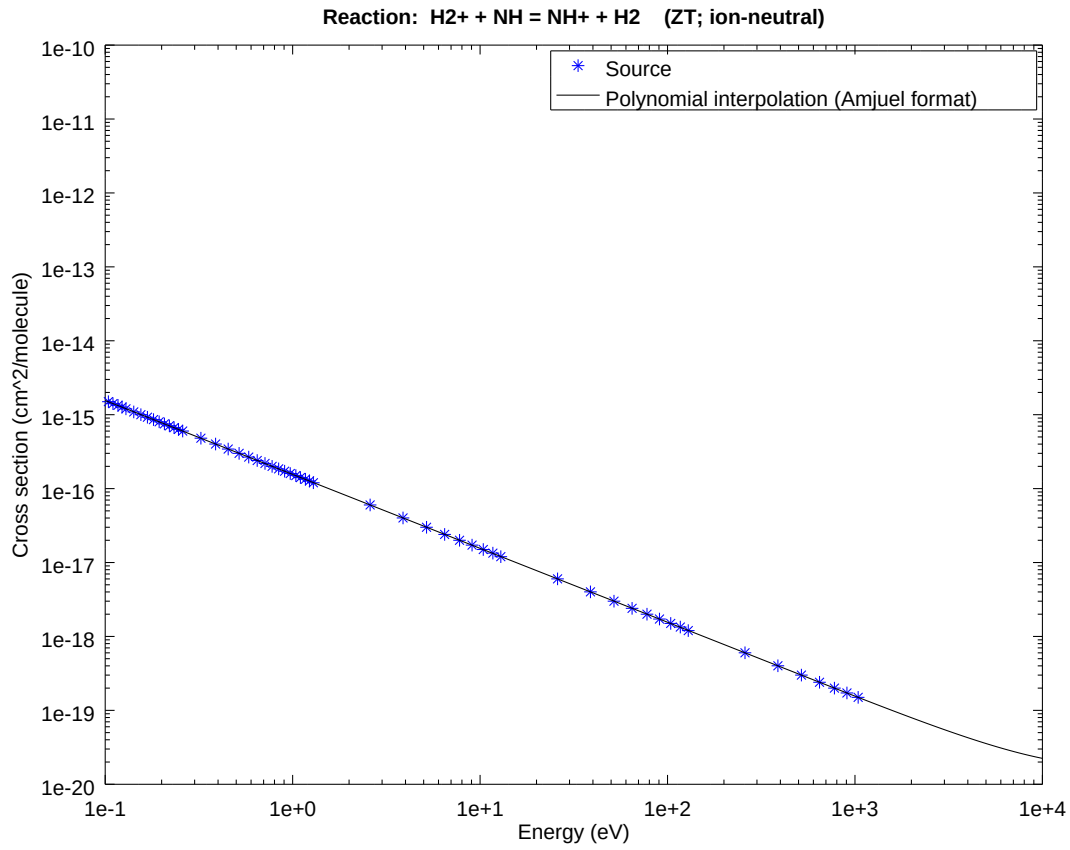
## 1.16 Reaction ZR $\text{H}_2^+ + \text{NH}_2 = \text{NH}_2^+ + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.537889371688E+01 | a1 | -9.988111947000E-01 | a2 | -2.580324837846E-04 |
| a3 | -2.911740540974E-04 | a4 | 5.386582495464E-05  | a5 | 1.997041706219E-05  |
| a6 | -3.537831334797E-06 | a7 | -3.296643009103E-07 | a8 | 5.745822253520E-08  |



## 1.17 Reaction ZT $\text{H}_2^+ + \text{NH} = \text{NH}^+ + \text{H}_2$

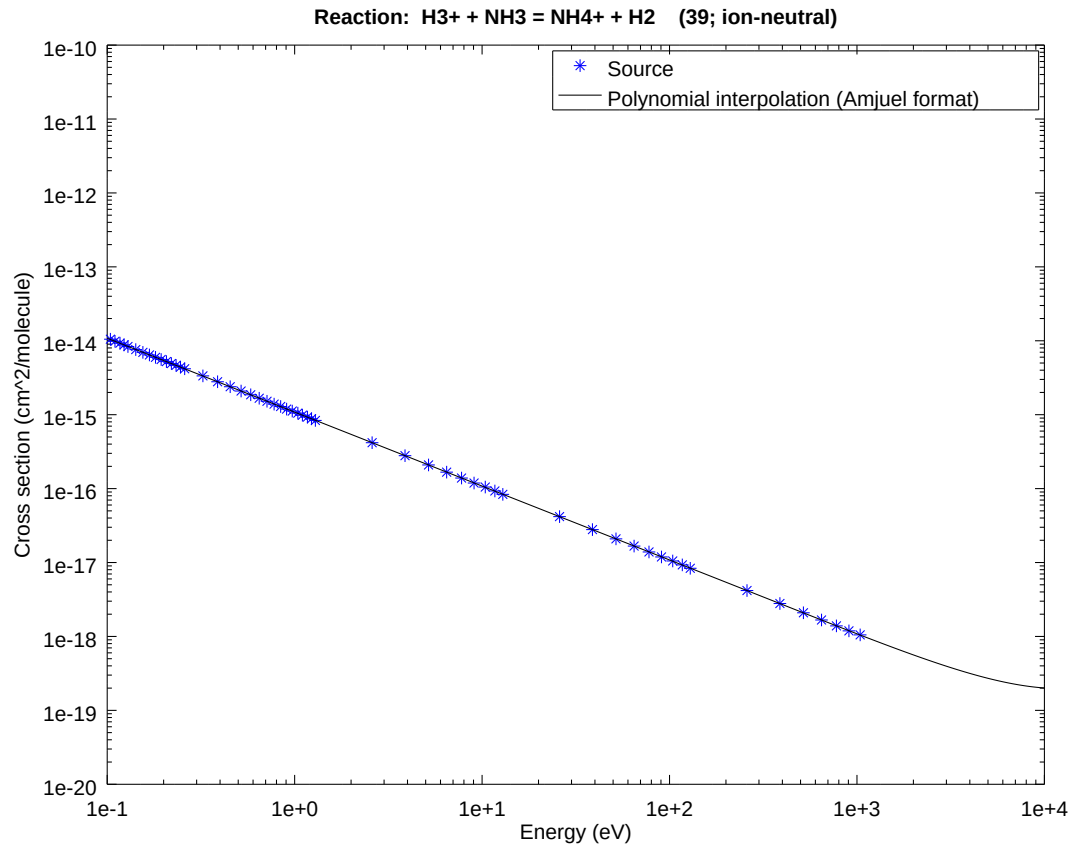
|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.639877685839E+01 | a1 | -9.991932271353E-01 | a2 | -3.786106055822E-04 |
| a3 | -2.000209526831E-04 | a4 | 6.490058121546E-05  | a5 | 1.204772346475E-05  |
| a6 | -3.491148657210E-06 | a7 | -1.039499856348E-07 | a8 | 3.854441560333E-08  |





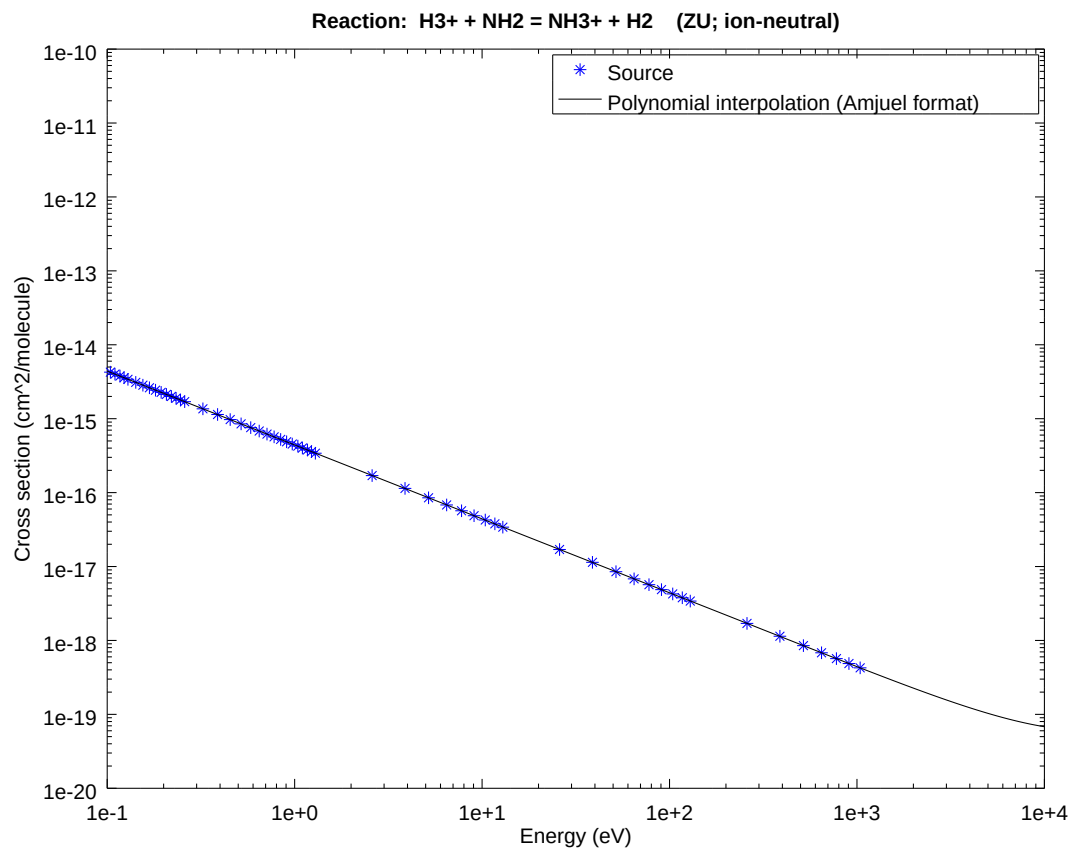
## 1.18 Reaction 39 $\text{H}_3^+ + \text{NH}_3 = \text{NH}_4^+ + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.445855504336E+01 | a1 | -9.991800597753E-01 | a2 | -4.749471155324E-04 |
| a3 | -2.864635224663E-04 | a4 | 7.700583560252E-05  | a5 | 2.201247421731E-05  |
| a6 | -4.531304197853E-06 | a7 | -4.179190244699E-07 | a8 | 7.678547276233E-08  |



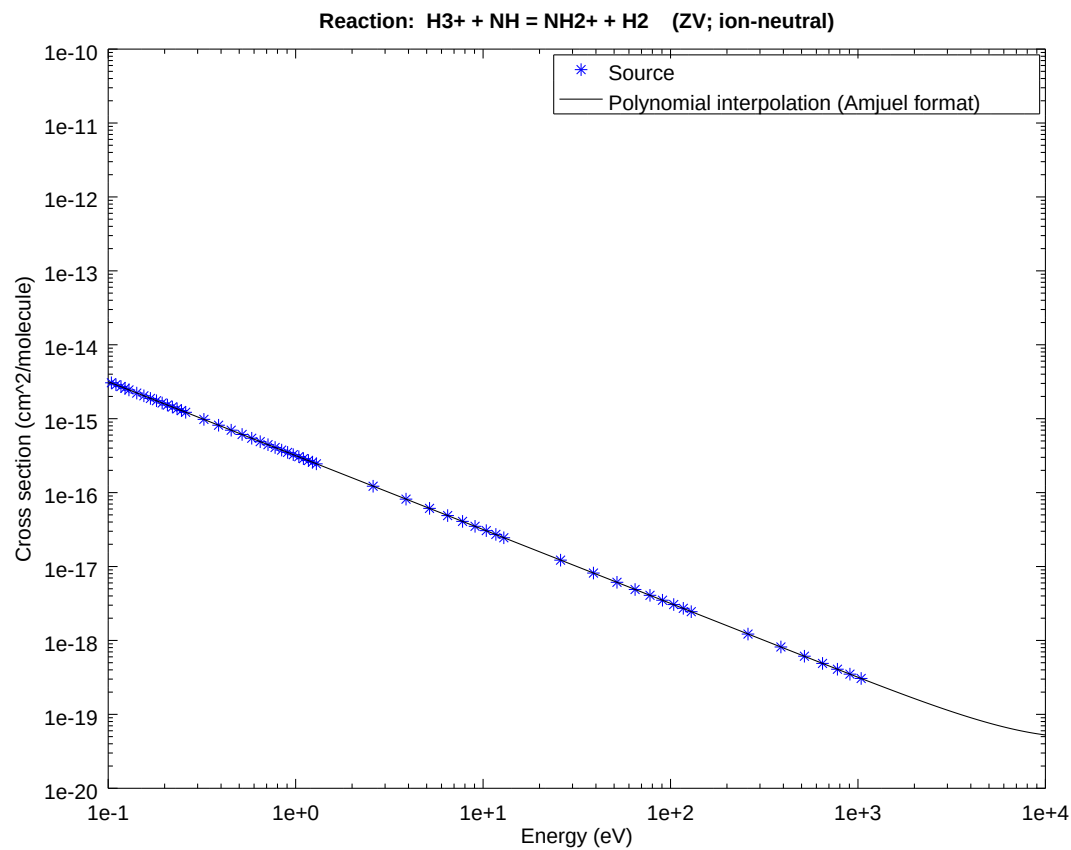
## 1.19 Reaction ZU $\text{H3}^+ + \text{NH2} = \text{NH3}^+ + \text{H2}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.535718875007E+01 | a1 | -9.988041211578E-01 | a2 | -1.327073962939E-04 |
| a3 | -3.024750238876E-04 | a4 | 3.537215304242E-05  | a5 | 2.306215109979E-05  |
| a6 | -2.938634729118E-06 | a7 | -4.885550353885E-07 | a8 | 6.619909023077E-08  |



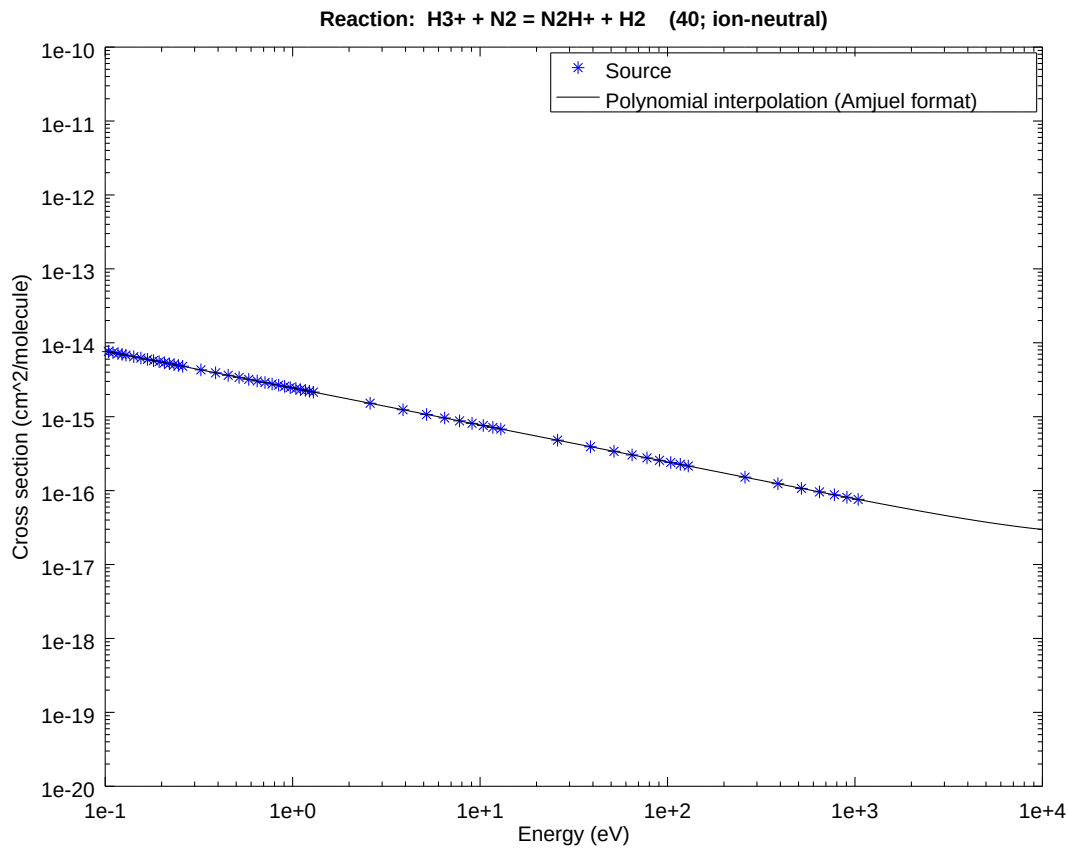
## 1.20 Reaction ZV $\text{H3}^+ + \text{NH} = \text{NH2}^+ + \text{H2}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.568779264120E+01 | a1 | -9.991262523277E-01 | a2 | -1.490267028808E-04 |
| a3 | -2.583679126152E-04 | a4 | 3.122615471999E-05  | a5 | 2.265333790401E-05  |
| a6 | -2.824905844315E-06 | a7 | -5.468812609031E-07 | a8 | 7.313382420176E-08  |



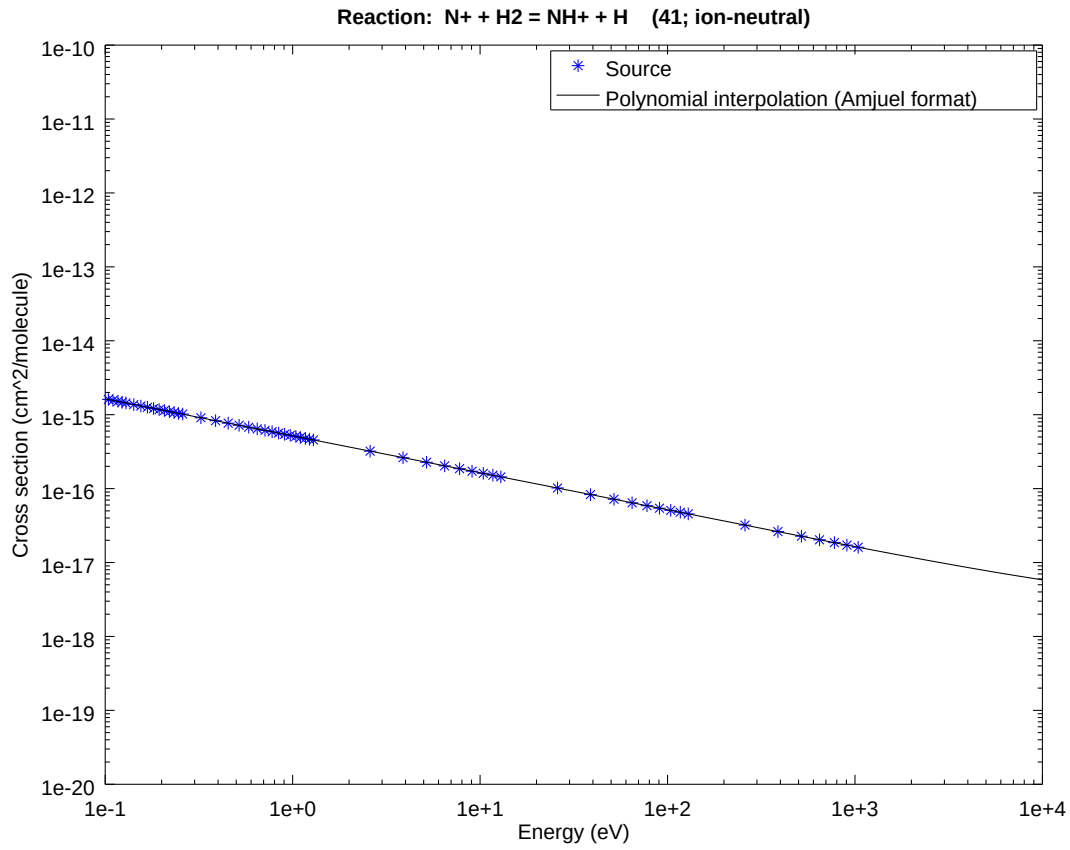
## 1.21 Reaction 40 $\text{H}_3^+ + \text{N}_2 = \text{N}_2\text{H}^+ + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.364424583043E+01 | a1 | -5.001053947537E-01 | a2 | -3.489472719550E-04 |
| a3 | 5.217830017246E-05  | a4 | 4.832027801732E-05  | a5 | -6.339876962418E-06 |
| a6 | -1.623185920202E-06 | a7 | 2.249465107832E-07  | a8 | -2.791634048550E-10 |



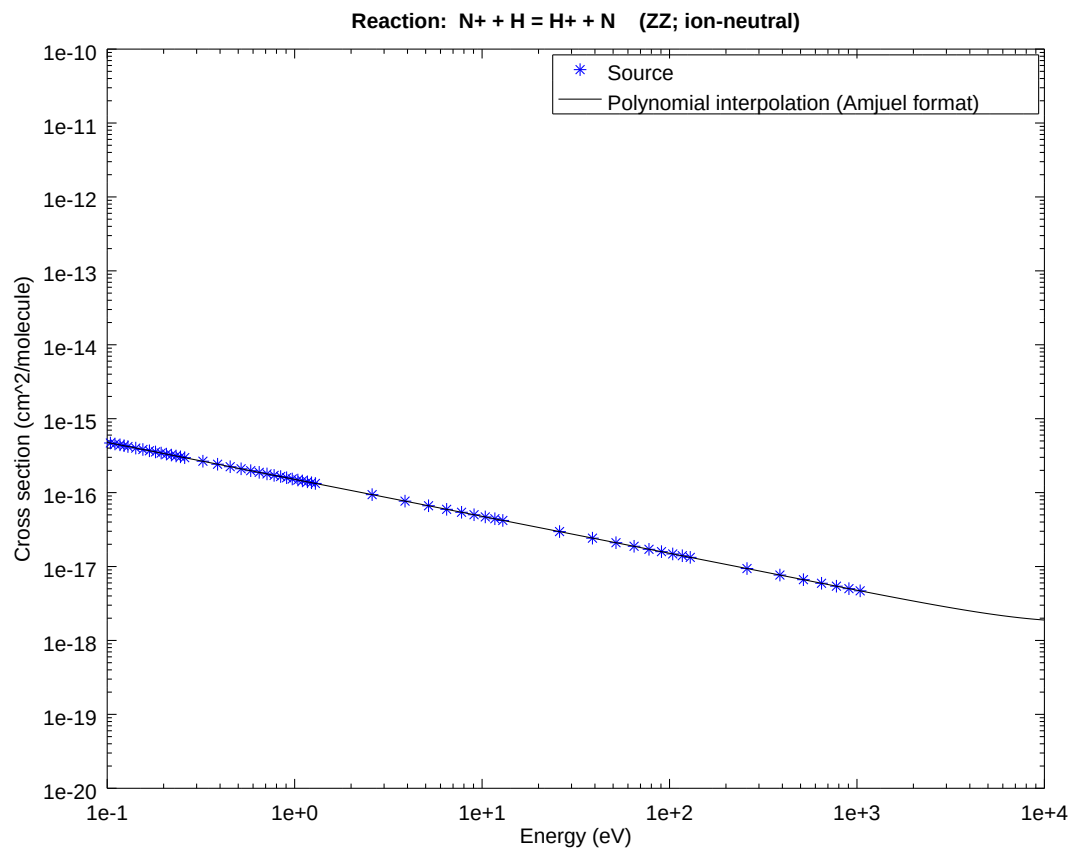
## 1.22 Reaction 41 $\text{N}^+ + \text{H}_2 = \text{NH}^+ + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.519826157435E+01 | a1 | -4.998014584796E-01 | a2 | 1.530815192373E-04  |
| a3 | 5.036422855746E-05  | a4 | -5.083768450143E-06 | a5 | -5.745853671067E-06 |
| a6 | 4.837645055788E-09  | a7 | 1.667353837445E-07  | a8 | -8.813920612450E-09 |



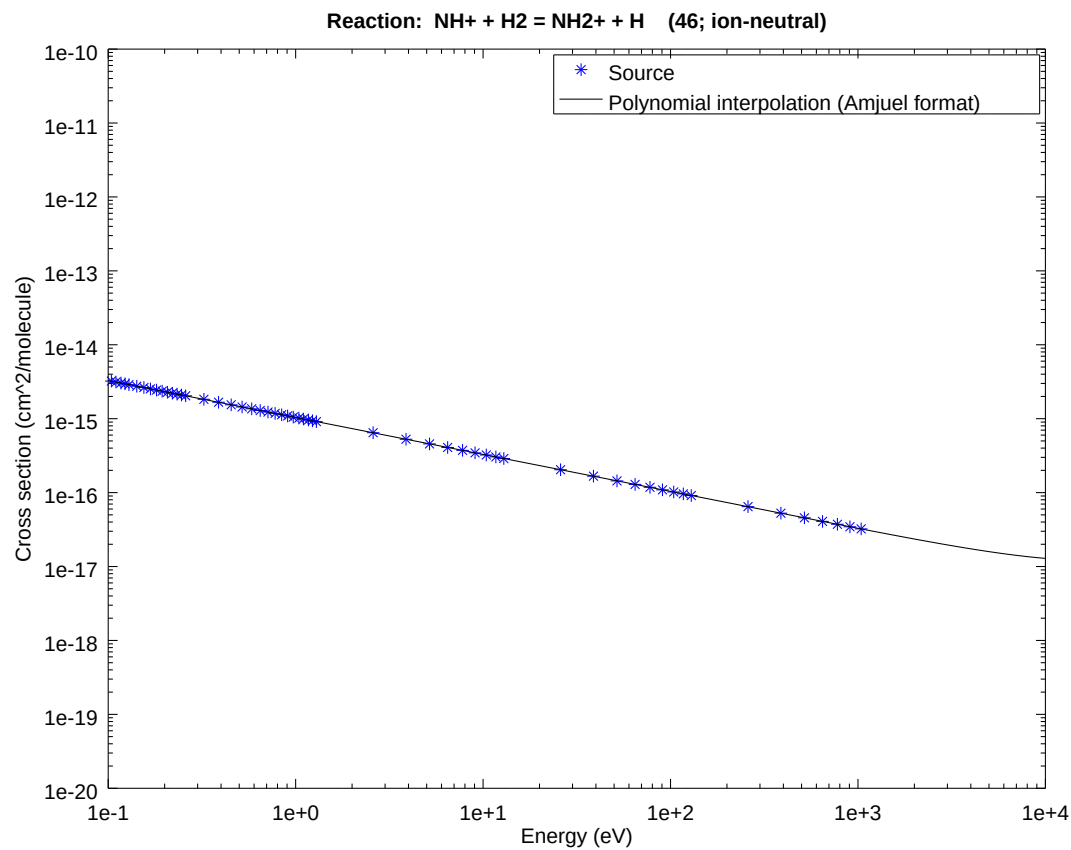
### 1.23 Reaction ZZ N+ + H = H+ + N

|    |                     |    |                     |    |                    |
|----|---------------------|----|---------------------|----|--------------------|
| a0 | -3.642877046760E+01 | a1 | -4.993284140026E-01 | a2 | 1.110120593205E-04 |
| a3 | -2.158746434780E-04 | a4 | -8.302489080848E-06 | a5 | 2.011967096965E-05 |
| a6 | -9.264918567094E-07 | a7 | -5.415644349055E-07 | a8 | 5.239164855073E-08 |



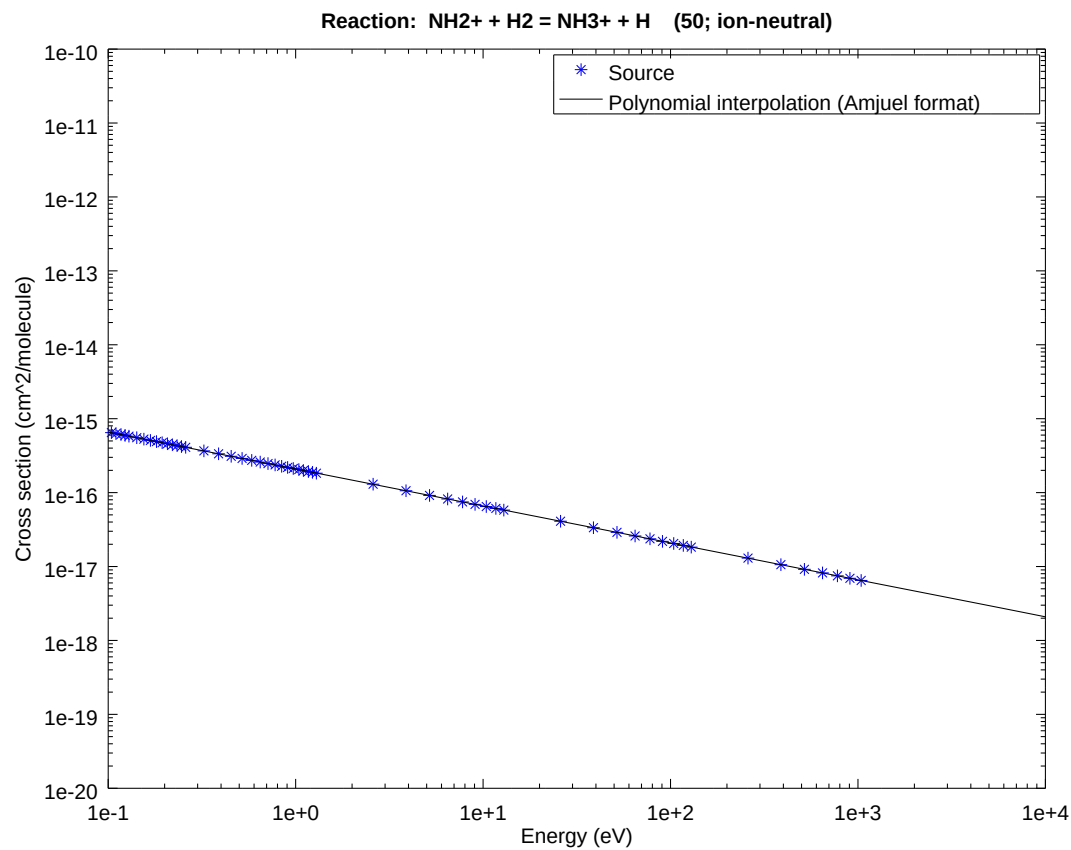
## 1.24 Reaction 46 $\text{NH}^+ + \text{H}_2 = \text{NH}_2^+ + \text{H}$

|    |                     |    |                     |    |                    |
|----|---------------------|----|---------------------|----|--------------------|
| a0 | -3.450120601495E+01 | a1 | -4.994452038861E-01 | a2 | 7.638532994153E-05 |
| a3 | -1.448123543562E-04 | a4 | 4.073634739478E-06  | a5 | 1.176427290163E-05 |
| a6 | -1.088326346202E-06 | a7 | -2.720391880305E-07 | a8 | 3.287797730196E-08 |



## 1.25 Reaction 50 $\text{NH}_2^+ + \text{H}_2 = \text{NH}_3^+ + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| a0 | -3.610585832408E+01 | a1 | -4.994904651358E-01 | a2 | -2.952907279827E-04 |
| a3 | -6.005065614576E-05 | a4 | 5.388044003298E-05  | a5 | -4.149821036921E-06 |
| a6 | -1.914629649127E-06 | a7 | 3.669195429192E-07  | a8 | -1.813946931009E-08 |

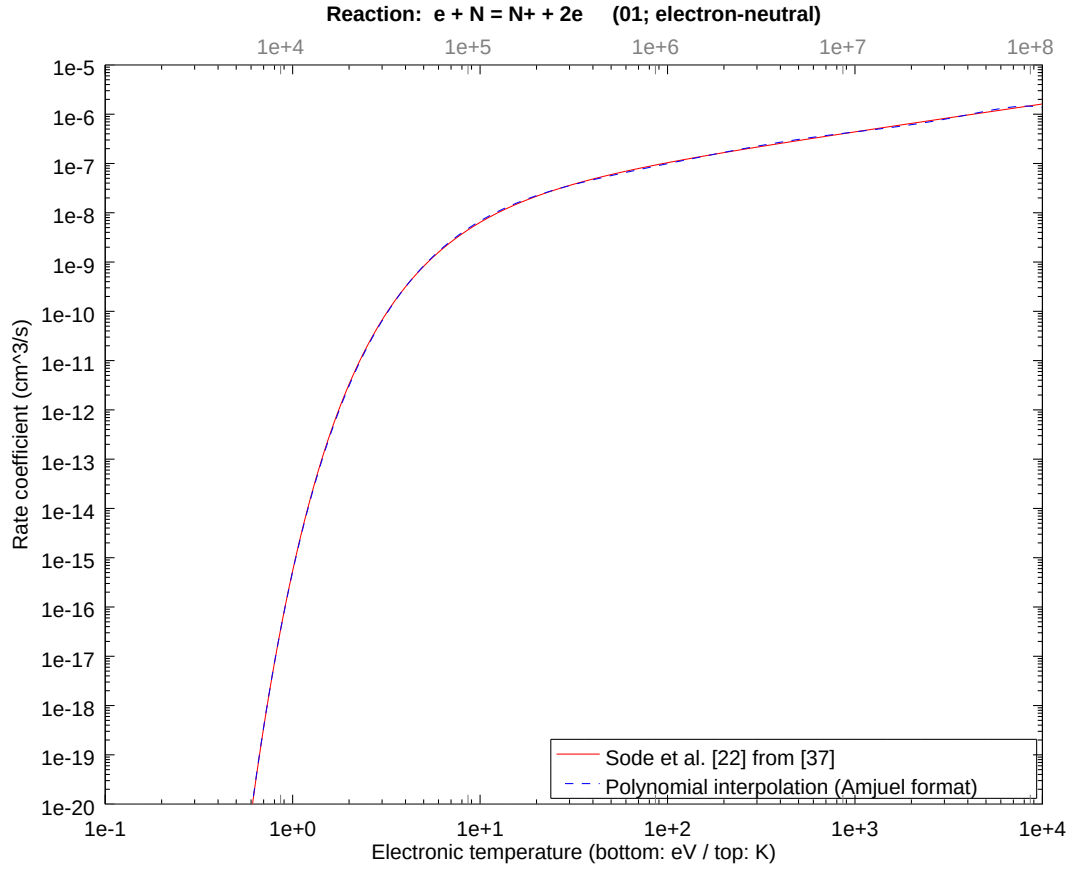




## 2 H.2 : Fits for rate coefficient

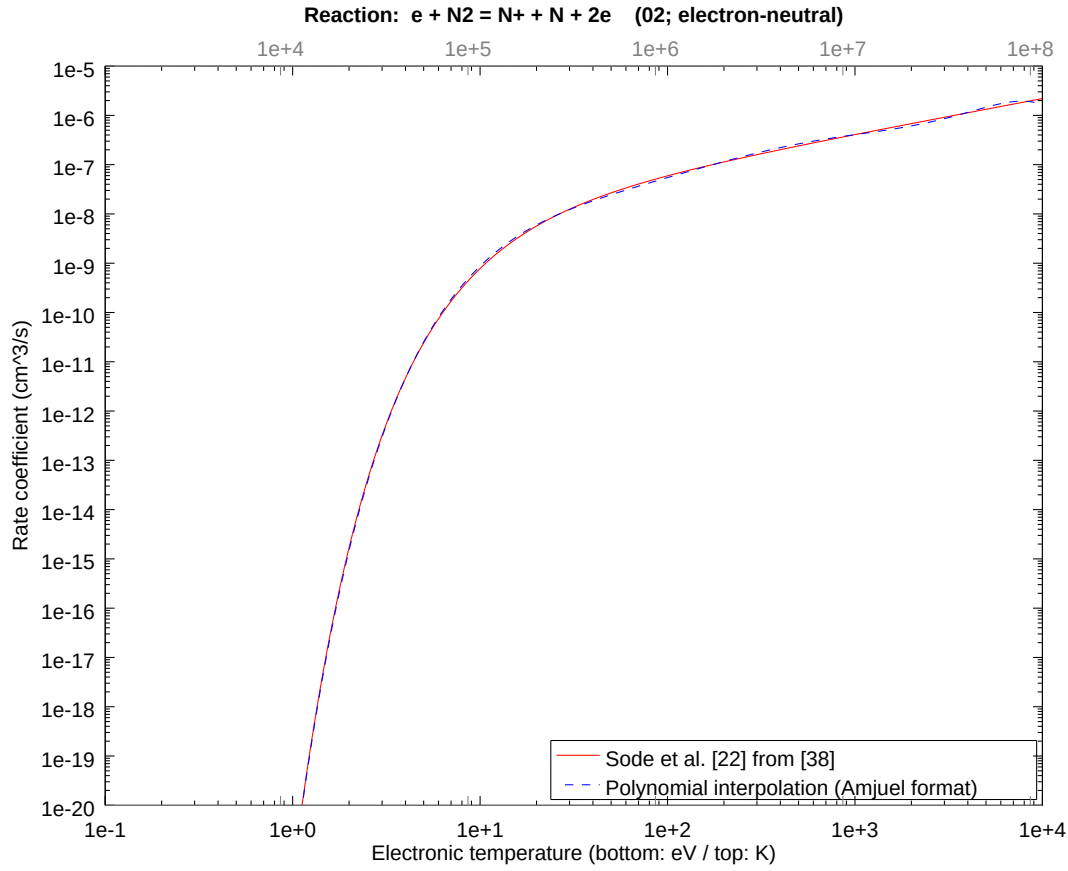
### 2.1 Reaction 01 $e + N = N^+ + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.515166129239E+01 | b1 | 1.719727980996E+01  | b2 | -8.252167519507E+00 |
| b3 | 2.662313955384E+00  | b4 | -6.041541298532E-01 | b5 | 9.629502687847E-02  |
| b6 | -1.020857933001E-02 | b7 | 6.402756602281E-04  | b8 | -1.782541563012E-05 |



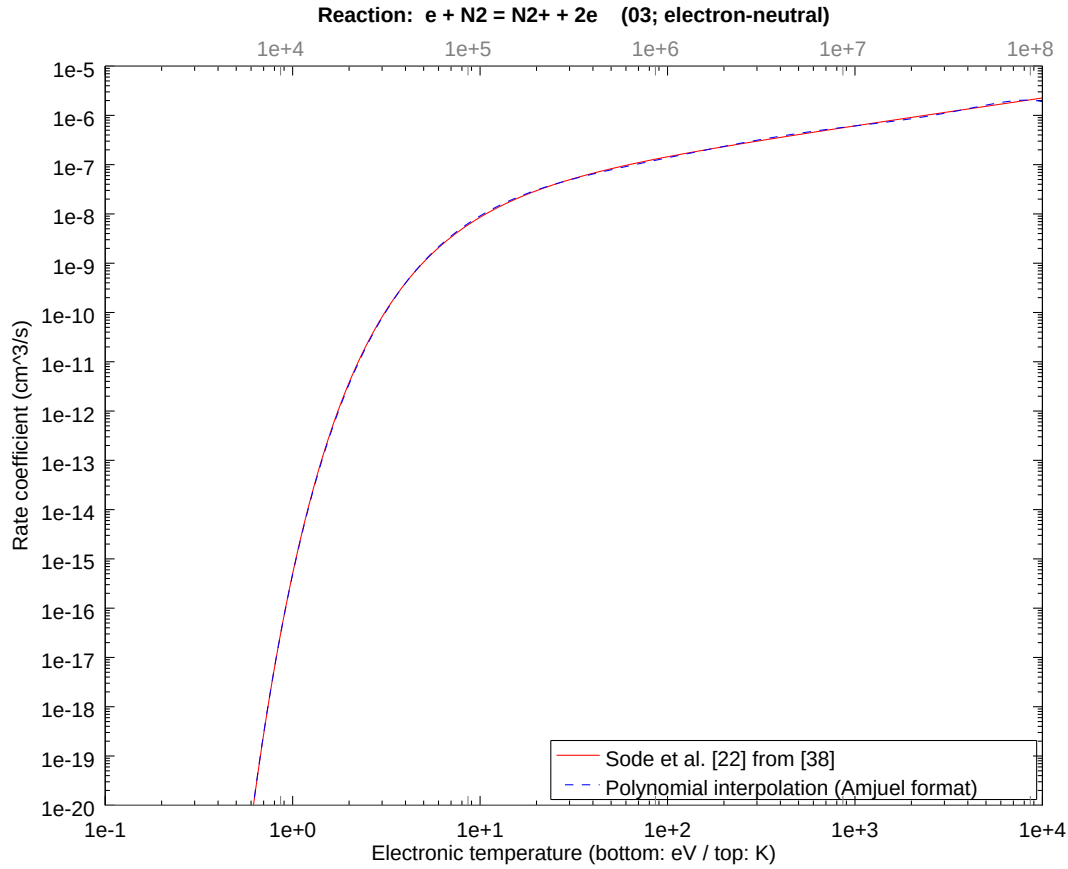
## 2.2 Reaction 02 $e + N_2 = N + N + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -4.936571937426E+01 | b1 | 3.038948278235E+01  | b2 | -1.471620030039E+01 |
| b3 | 4.747739952849E+00  | b4 | -1.077396110320E+00 | b5 | 1.717242045954E-01  |
| b6 | -1.820509555189E-02 | b7 | 1.141812116770E-03  | b8 | -3.178830122278E-05 |



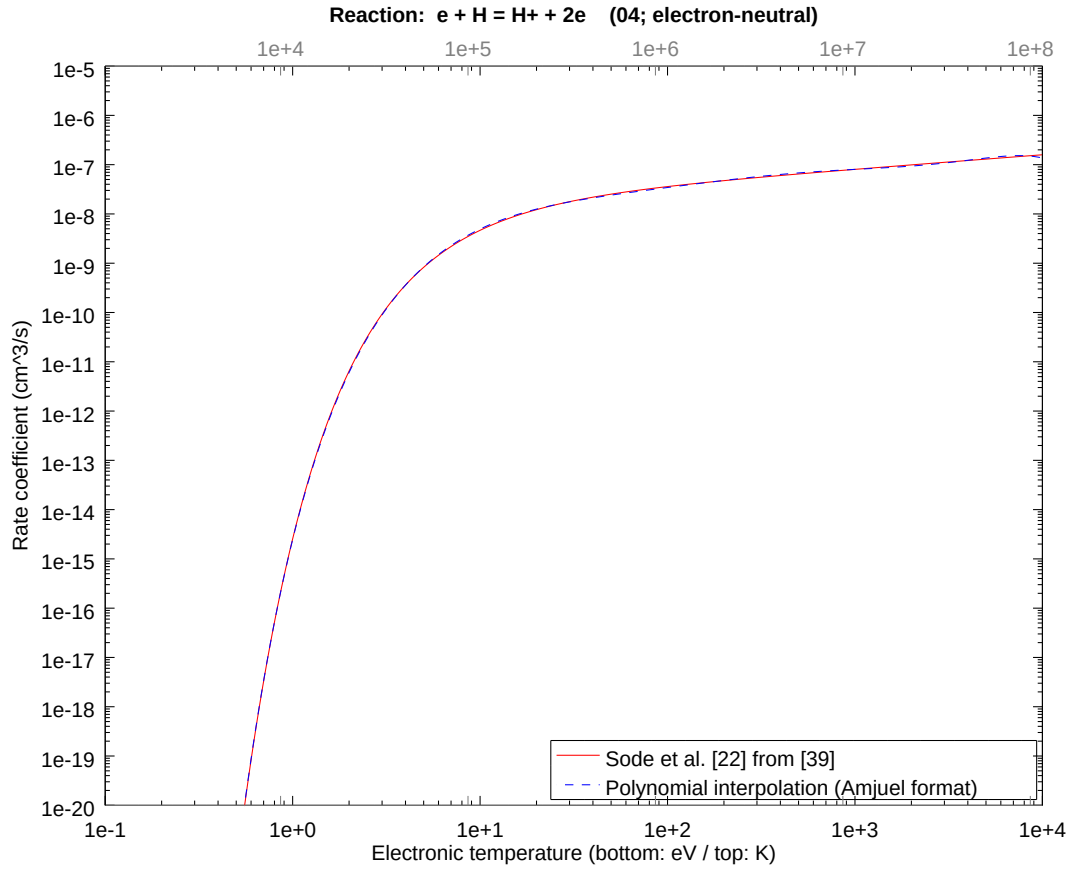
## 2.3 Reaction 03 $e + N_2 = N_2^+ + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.522668720188E+01 | b1 | 1.760672066963E+01  | b2 | -8.455252074309E+00 |
| b3 | 2.727833086339E+00  | b4 | -6.190222687031E-01 | b5 | 9.866483246191E-02  |
| b6 | -1.045981087414E-02 | b7 | 6.560327443037E-04  | b8 | -1.826409632692E-05 |



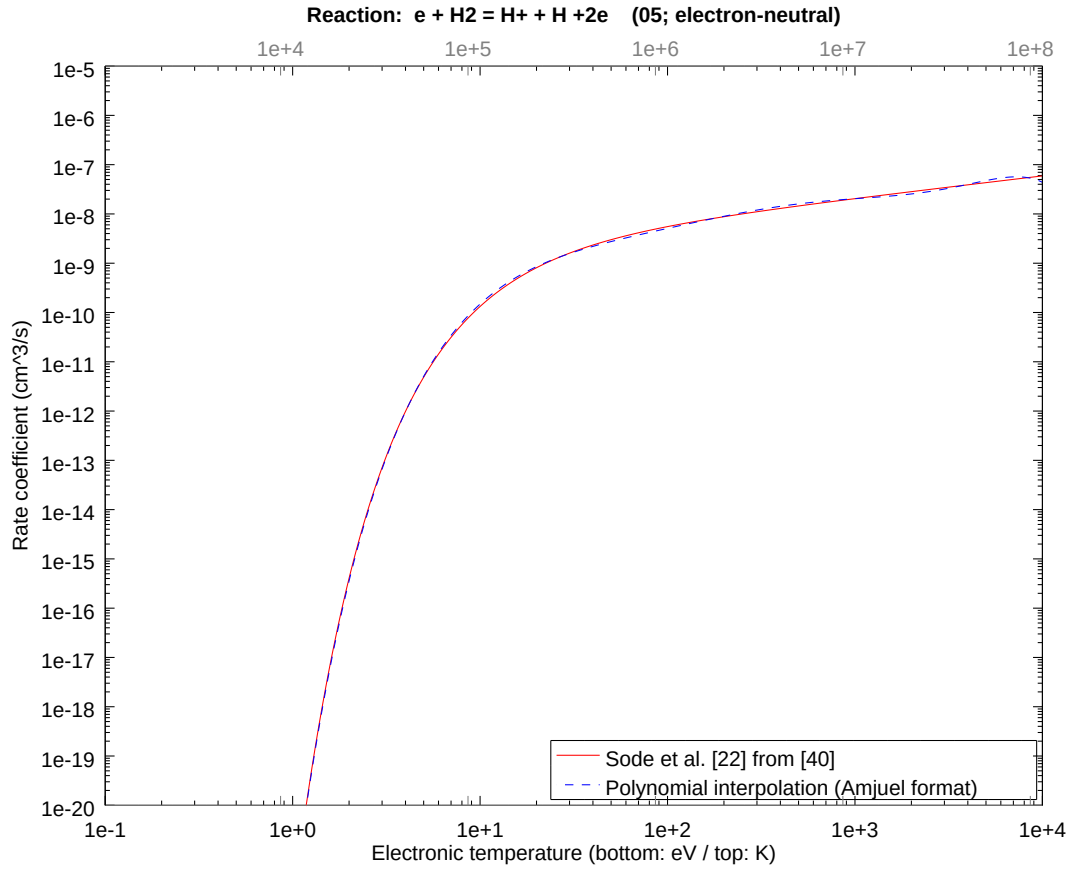
## 2.4 Reaction 04 $e + H = H^+ + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.360391213639E+01 | b1 | 1.554916179449E+01  | b2 | -7.568614627736E+00 |
| b3 | 2.441786148756E+00  | b4 | -5.541101503096E-01 | b5 | 8.831860808544E-02  |
| b6 | -9.362970718041E-03 | b7 | 5.872396211462E-04  | b8 | -1.634888060196E-05 |



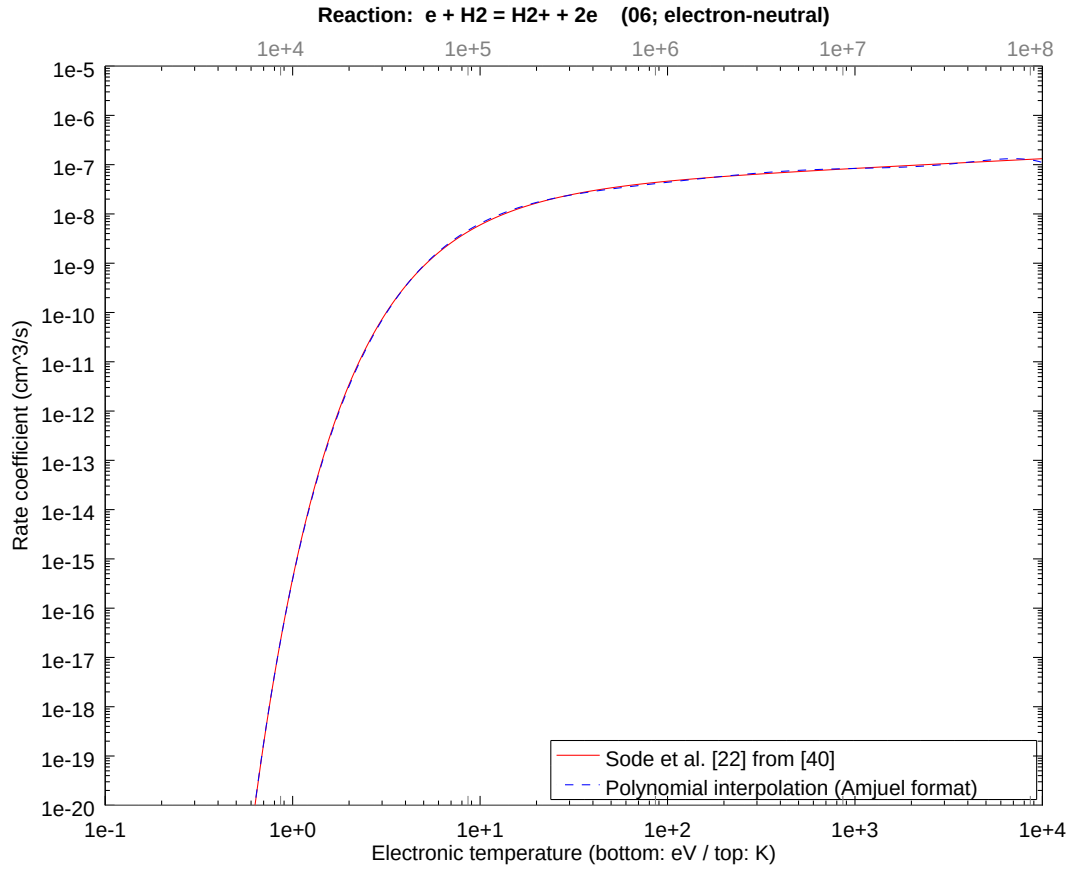
## 2.5 Reaction 05 $e + H_2 = H + H + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -5.072228356220E+01 | b1 | 3.034916911826E+01  | b2 | -1.483012578236E+01 |
| b3 | 4.784494587287E+00  | b4 | -1.085736773578E+00 | b5 | 1.730536077276E-01  |
| b6 | -1.834603032055E-02 | b7 | 1.150651456617E-03  | b8 | -3.203439039412E-05 |



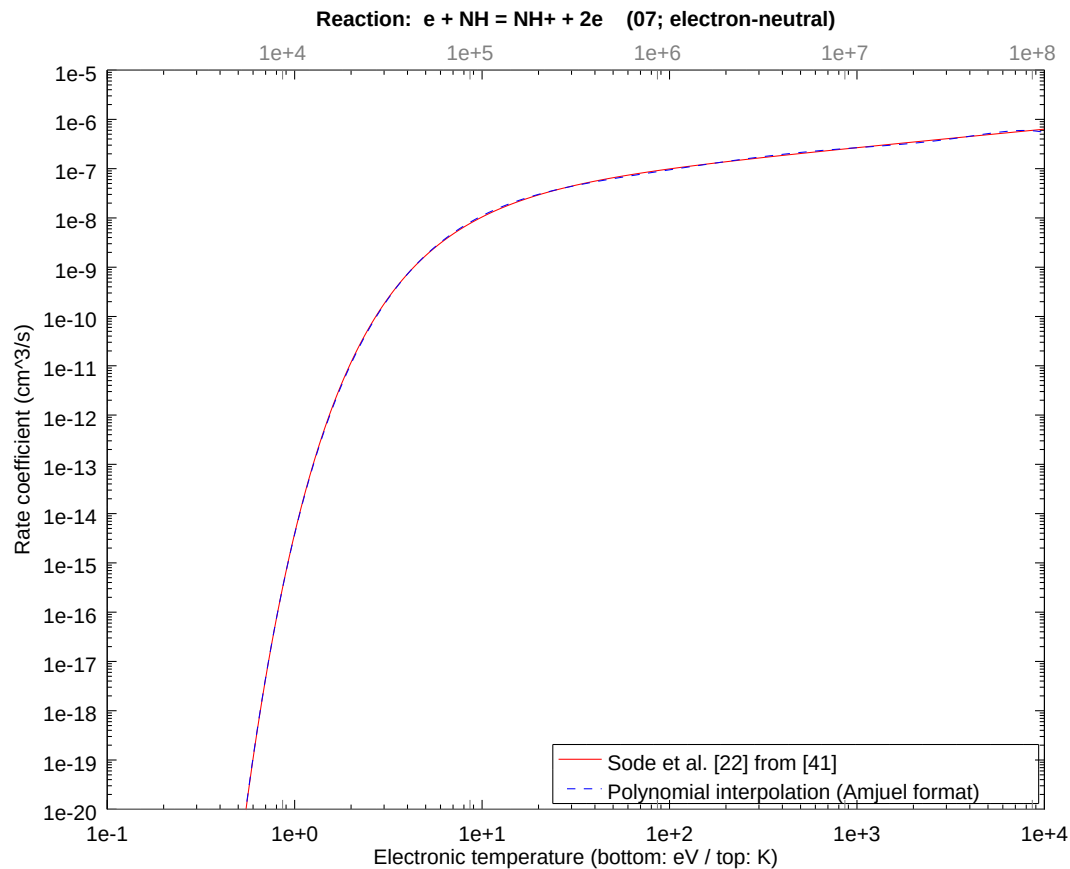
## 2.6 Reaction 06 $e + H_2 = H_2^+ + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.545606598594E+01 | b1 | 1.803562966411E+01  | b2 | -8.851514620264E+00 |
| b3 | 2.855675293080E+00  | b4 | -6.480332713371E-01 | b5 | 1.032888433564E-01  |
| b6 | -1.095001876514E-02 | b7 | 6.867782742069E-04  | b8 | -1.912005866211E-05 |



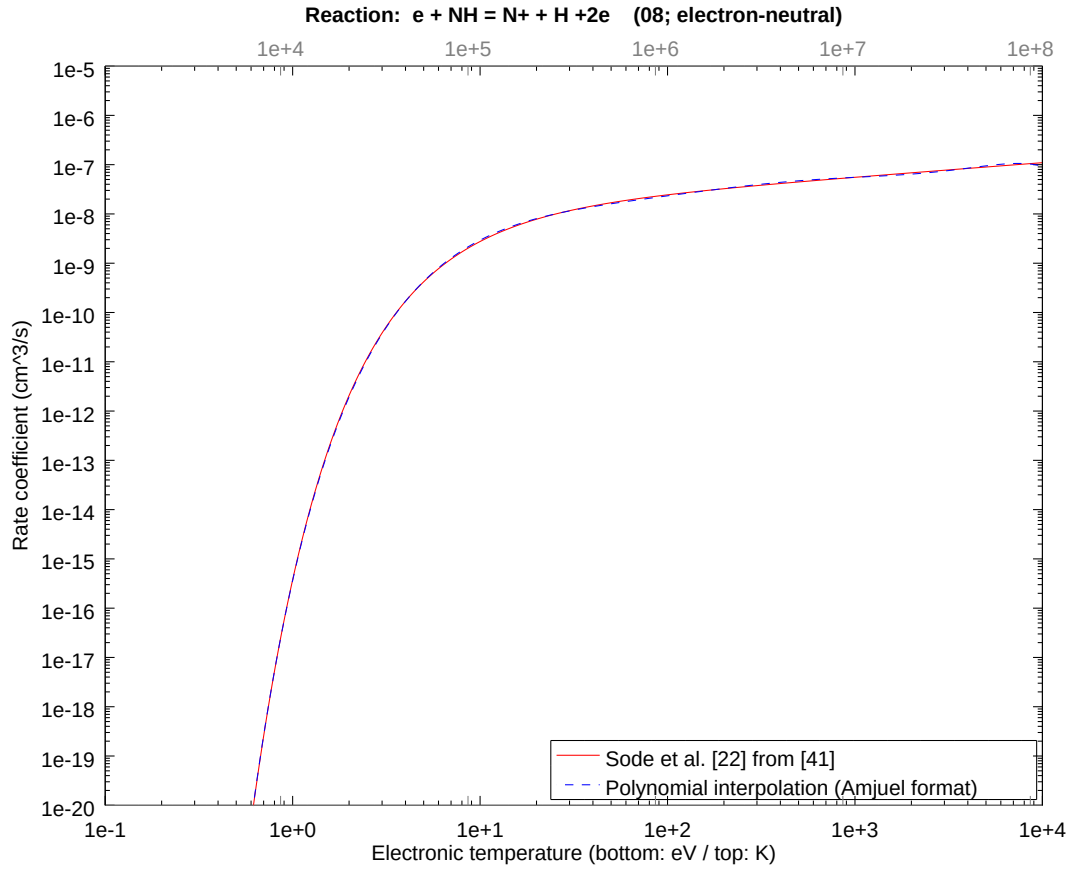
## 2.7 Reaction 07 $e + \text{NH} = \text{NH}^+ + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.316726492763E+01 | b1 | 1.583887540554E+01  | b2 | -7.672633546048E+00 |
| b3 | 2.475344728025E+00  | b4 | -5.617255385009E-01 | b5 | 8.953241094521E-02  |
| b6 | -9.491650289420E-03 | b7 | 5.953103227449E-04  | b8 | -1.657357071491E-05 |



## 2.8 Reaction 08 $e + \text{NH} = \text{N}^+ + \text{H} + 2e$

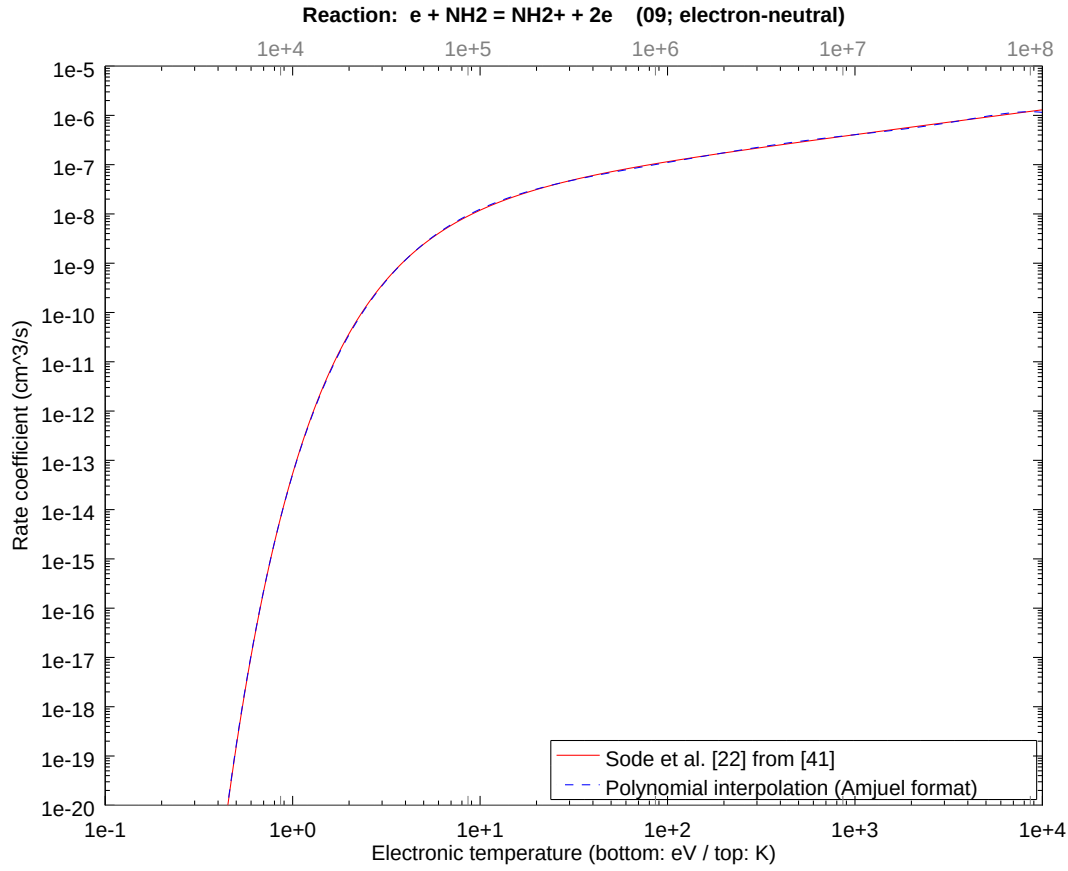
|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.551351217376E+01 | b1 | 1.708706160886E+01  | b2 | -8.331420028698E+00 |
| b3 | 2.687882396733E+00  | b4 | -6.099563303800E-01 | b5 | 9.721982905739E-02  |
| b6 | -1.030662090821E-02 | b7 | 6.464247662092E-04  | b8 | -1.799660809718E-05 |





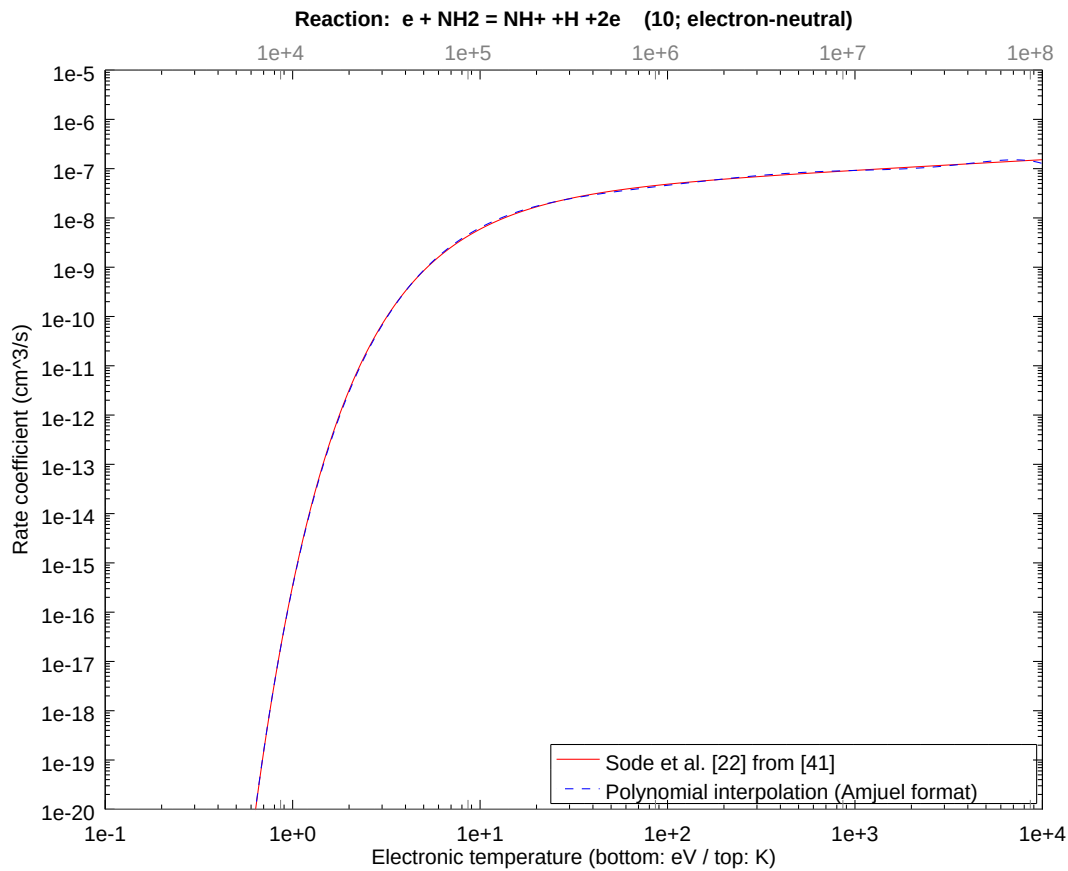
## 2.9 Reaction 09 $e + \text{NH}_2 = \text{NH}_2^+ + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.055713293864E+01 | b1 | 1.288308941438E+01  | b2 | -6.142069462298E+00 |
| b3 | 1.981554204488E+00  | b4 | -4.496705408271E-01 | b5 | 7.167216886514E-02  |
| b6 | -7.598222310453E-03 | b7 | 4.765557134956E-04  | b8 | -1.326741619531E-05 |



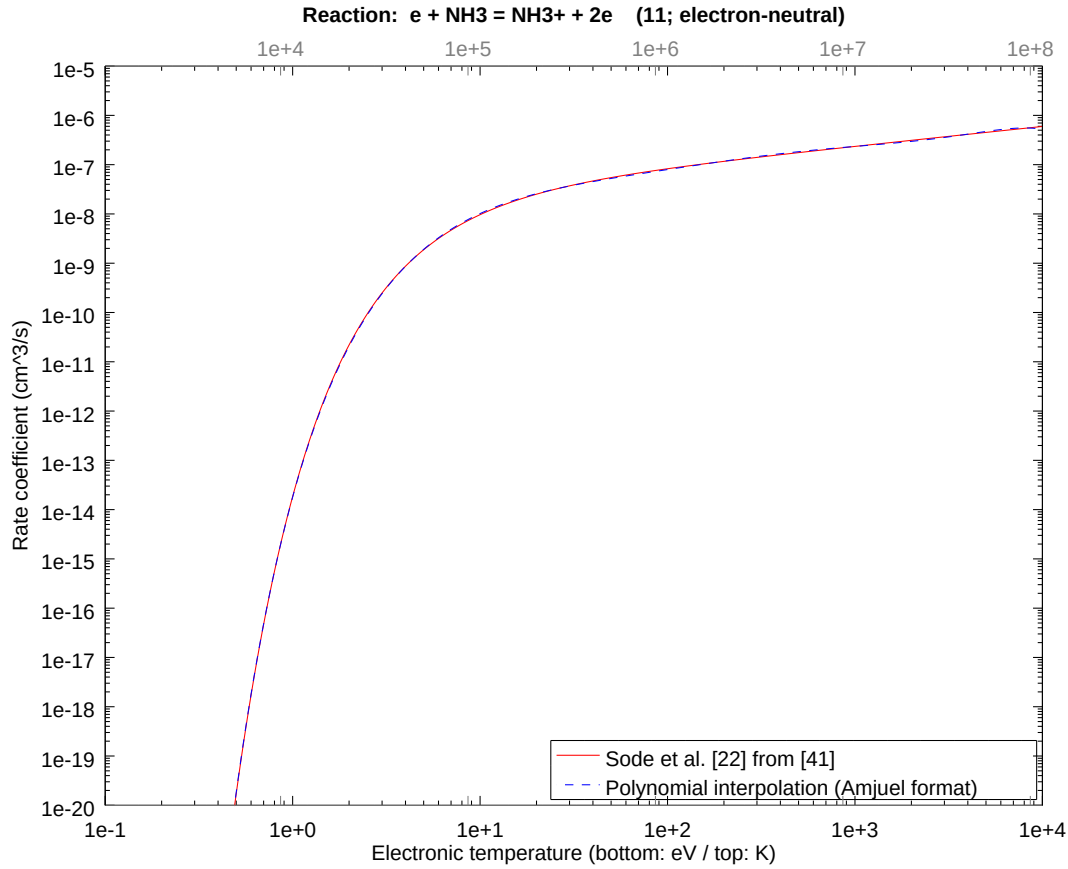
## 2.10 Reaction 10 $e + \text{NH}_2 = \text{NH}^+ + \text{H} + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.560050820383E+01 | b1 | 1.815549328842E+01  | b2 | -8.901047438508E+00 |
| b3 | 2.871655568923E+00  | b4 | -6.516596466663E-01 | b5 | 1.038668447182E-01  |
| b6 | -1.101129475151E-02 | b7 | 6.906214654439E-04  | b8 | -1.922705395397E-05 |



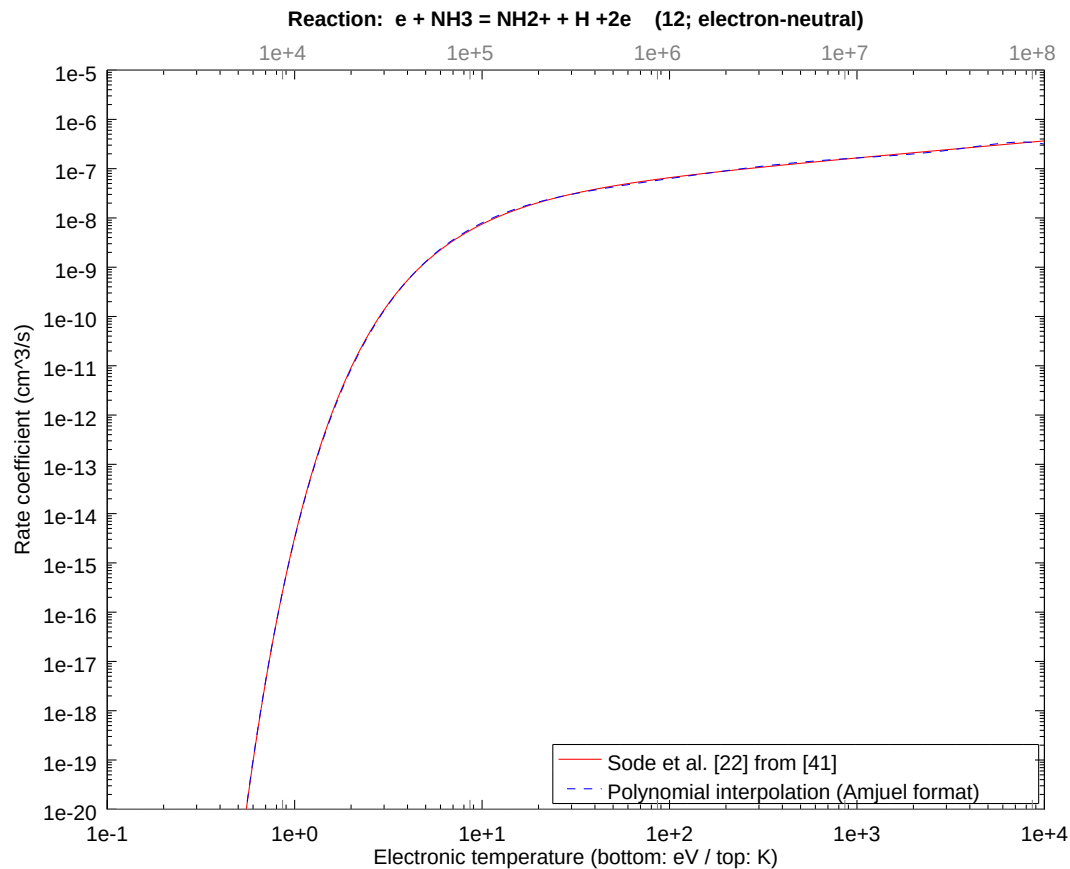
## 2.11 Reaction 11 $e + \text{NH}_3 = \text{NH}_3^+ + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.162391660432E+01 | b1 | 1.399143926852E+01  | b2 | -6.741416563055E+00 |
| b3 | 2.174915542184E+00  | b4 | -4.935496823111E-01 | b5 | 7.866598534312E-02  |
| b6 | -8.339661745585E-03 | b7 | 5.230583274737E-04  | b8 | -1.456205922726E-05 |



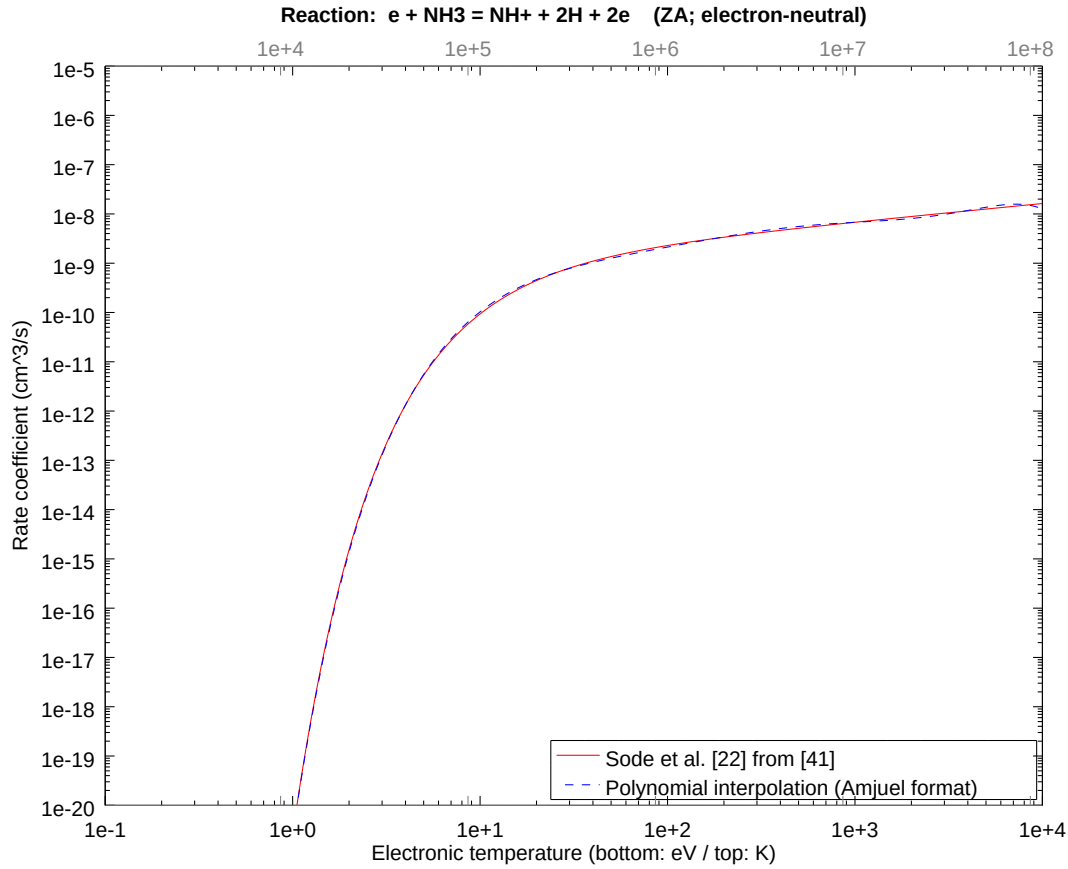
## 2.12 Reaction 12 $e + \text{NH}_3 = \text{NH}_2^+ + \text{H} + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.335920627886E+01 | b1 | 1.572898450609E+01  | b2 | -7.633007291453E+00 |
| b3 | 2.462560507351E+00  | b4 | -5.588244382375E-01 | b5 | 8.907000985578E-02  |
| b6 | -9.442629500327E-03 | b7 | 5.922357697553E-04  | b8 | -1.648797448143E-05 |



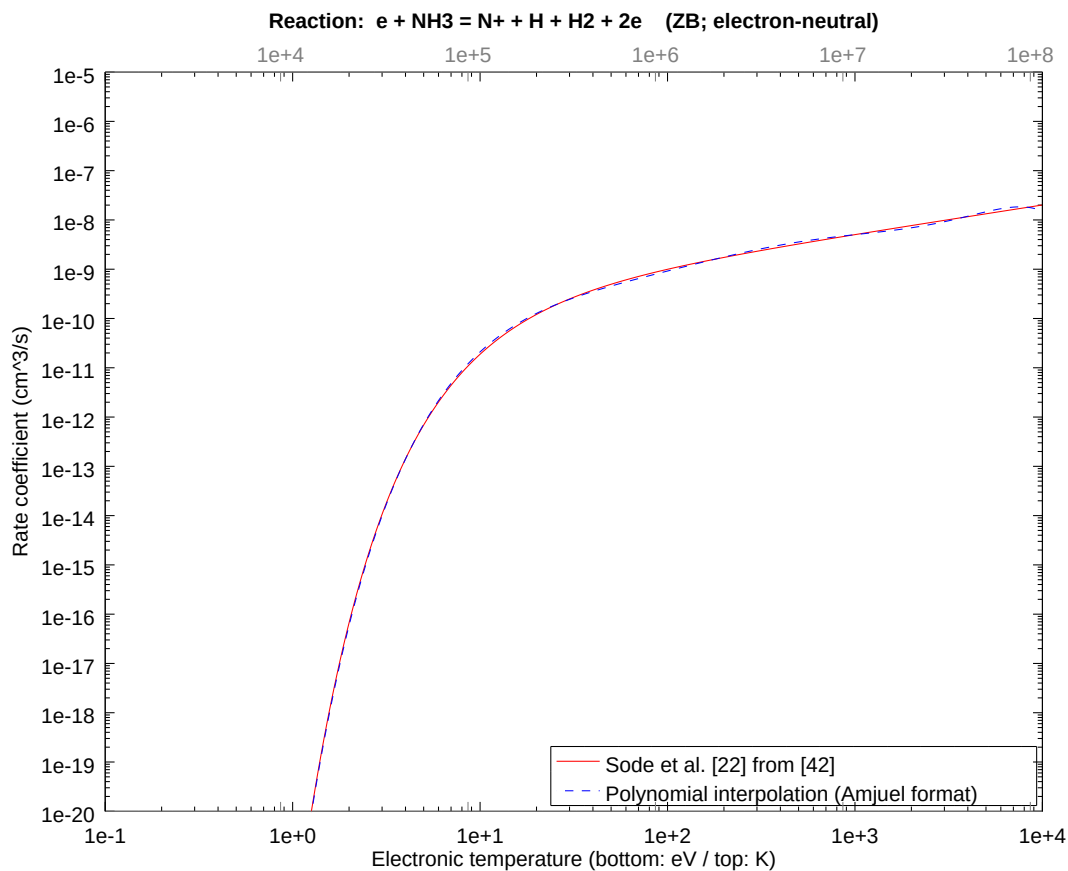
## 2.13 Reaction ZA $e + \text{NH}_3 = \text{NH}^+ + 2\text{H} + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -4.739696463165E+01 | b1 | 2.639446049505E+01  | b2 | -1.290825243448E+01 |
| b3 | 4.164459884593E+00  | b4 | -9.450334108027E-01 | b5 | 1.506271548891E-01  |
| b6 | -1.596852204922E-02 | b7 | 1.001535636588E-03  | b8 | -2.788297306850E-05 |



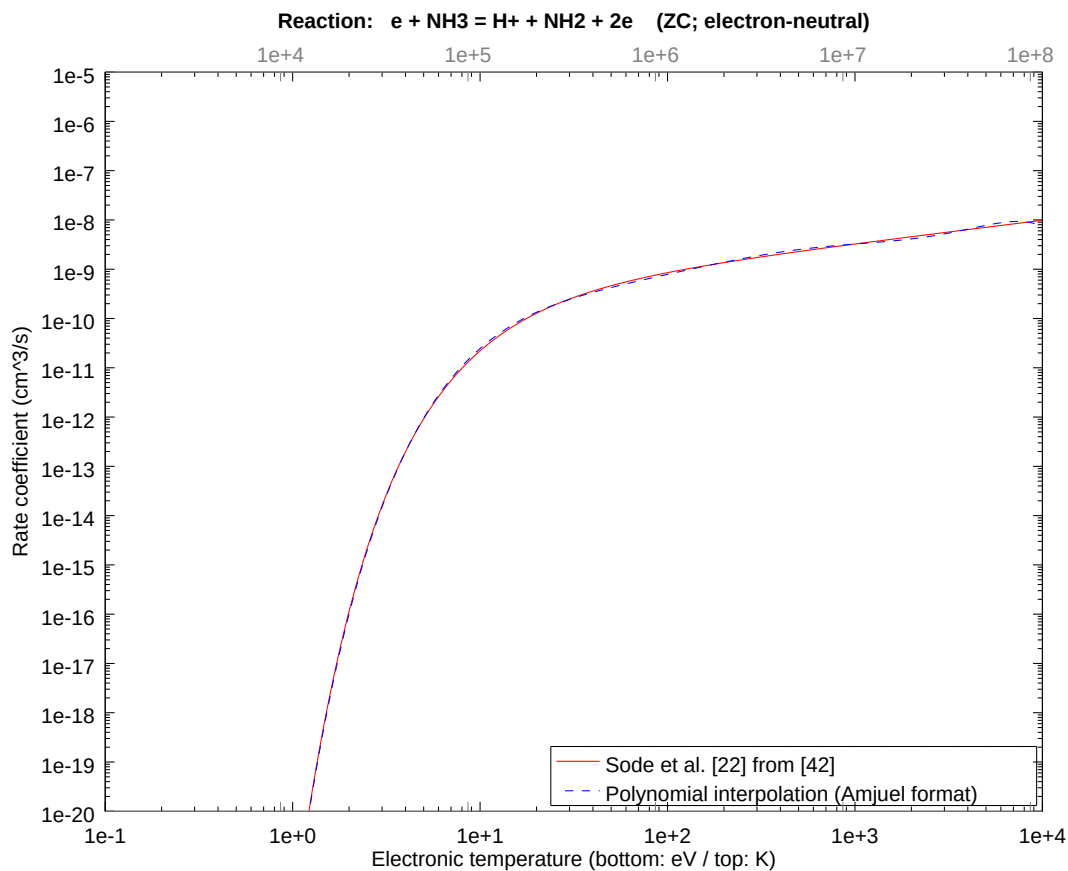
## 2.14 Reaction ZB $e + \text{NH}_3 = \text{N} + \text{H} + \text{H}_2 + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -5.215091634301E+01 | b1 | 2.955045104975E+01  | b2 | -1.436451729086E+01 |
| b3 | 4.634279994366E+00  | b4 | -1.051648845483E+00 | b5 | 1.676203949265E-01  |
| b6 | -1.777003604863E-02 | b7 | 1.114525458981E-03  | b8 | -3.102863465029E-05 |



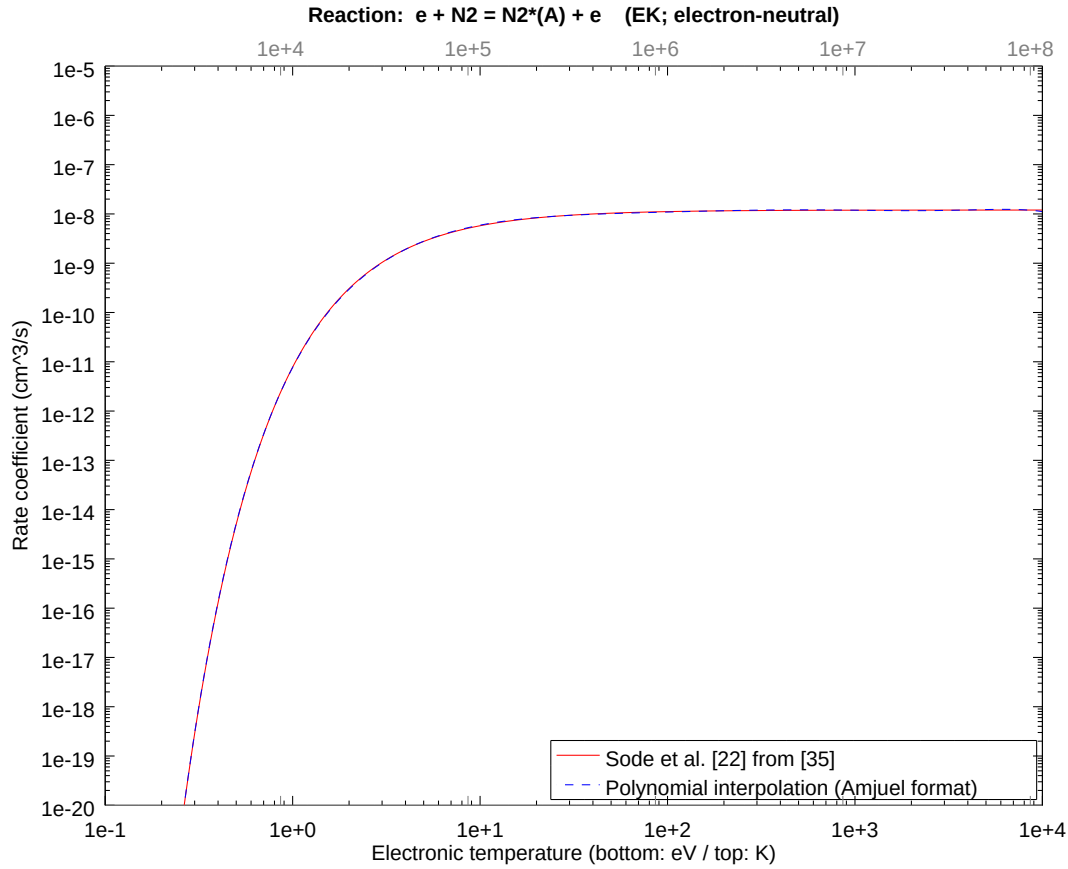
## 2.15 Reaction ZC $e + \text{NH}_3 = \text{H} + \text{NH}_2 + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -5.131076165811E+01 | b1 | 2.898106474036E+01  | b2 | -1.414161960876E+01 |
| b3 | 4.562368753075E+00  | b4 | -1.035330156501E+00 | b5 | 1.650193887983E-01  |
| b6 | -1.749429410995E-02 | b7 | 1.097231098411E-03  | b8 | -3.054715583677E-05 |



## 2.16 Reaction EK $e + N_2 = N_2^*(A) + e$

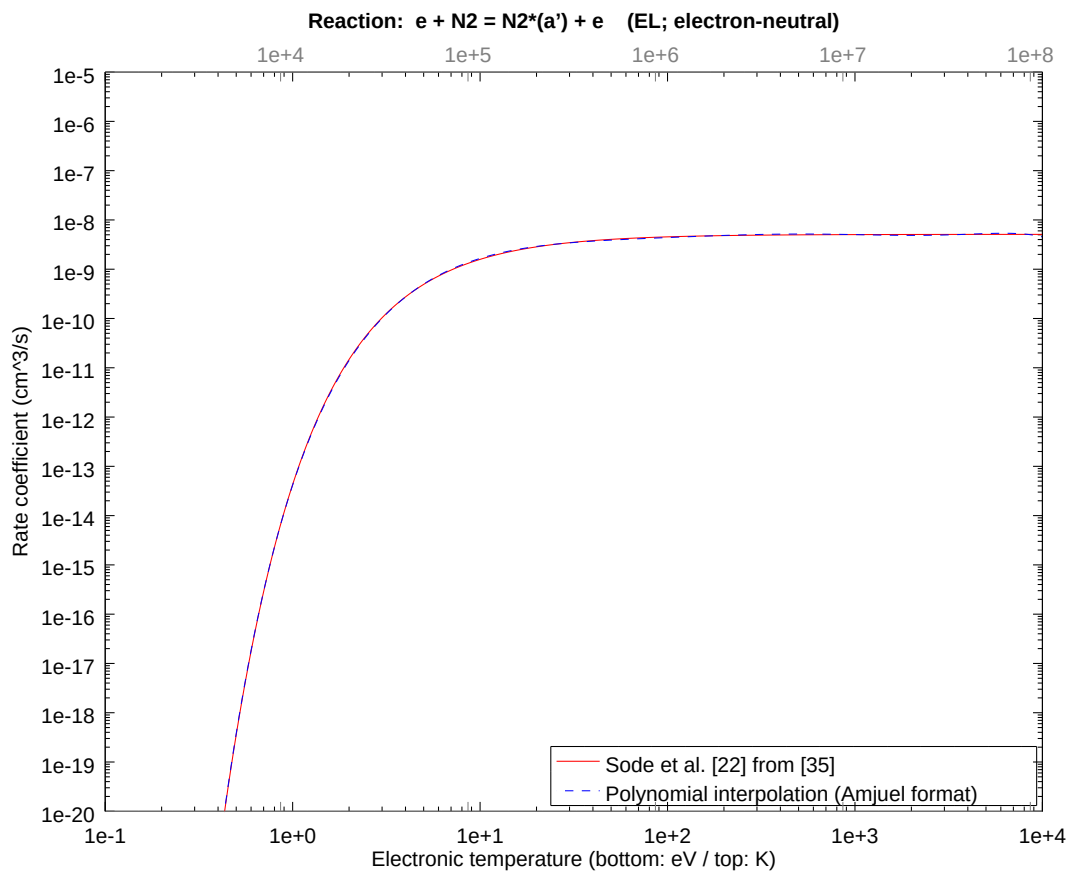
|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.557765860733E+01 | b1 | 7.329990024317E+00  | b2 | -3.635708859135E+00 |
| b3 | 1.172952246850E+00  | b4 | -2.661759491670E-01 | b5 | 4.242529995726E-02  |
| b6 | -4.497657399895E-03 | b7 | 2.820902368593E-04  | b8 | -7.853454425279E-06 |





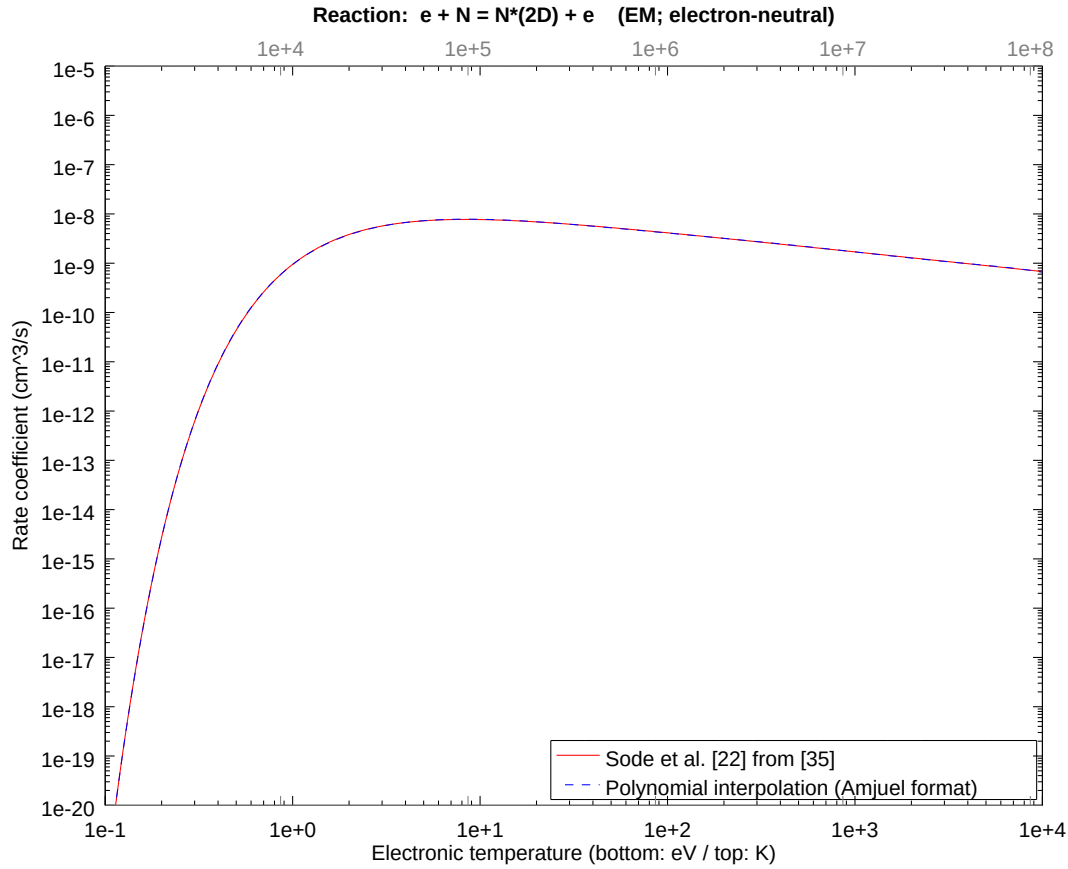
## 2.17 Reaction EL $e + N_2 = N_2^*(a') + e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.078290952363E+01 | b1 | 1.167405768178E+01  | b2 | -5.790386452764E+00 |
| b3 | 1.868094246005E+00  | b4 | -4.239232759895E-01 | b5 | 6.756835919627E-02  |
| b6 | -7.163162807193E-03 | b7 | 4.492690557066E-04  | b8 | -1.250774962282E-05 |



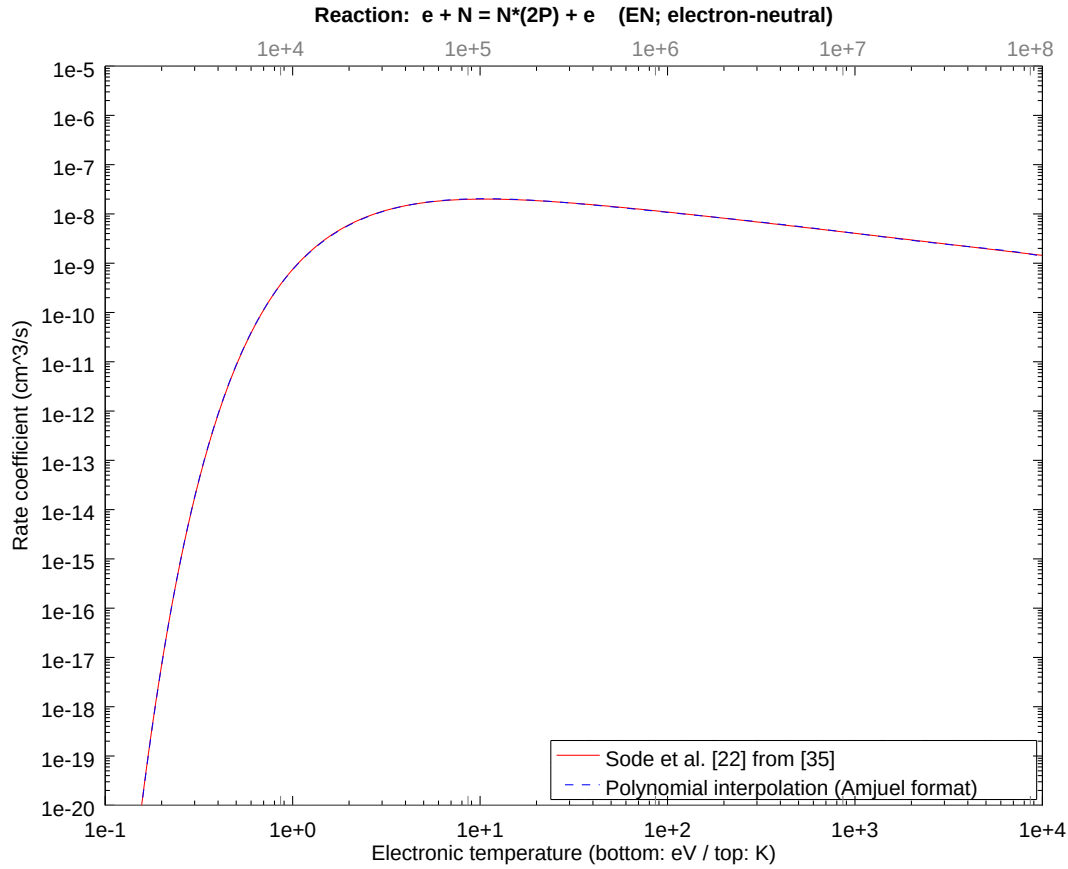
## 2.18 Reaction EM $e + N = N^*(2D) + e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.077710922402E+01 | b1 | 2.945431414368E+00  | b2 | -1.659349411186E+00 |
| b3 | 5.353392407288E-01  | b4 | -1.214835735300E-01 | b5 | 1.936304562086E-02  |
| b6 | -2.052745543553E-03 | b7 | 1.287469064688E-04  | b8 | -3.584342278573E-06 |



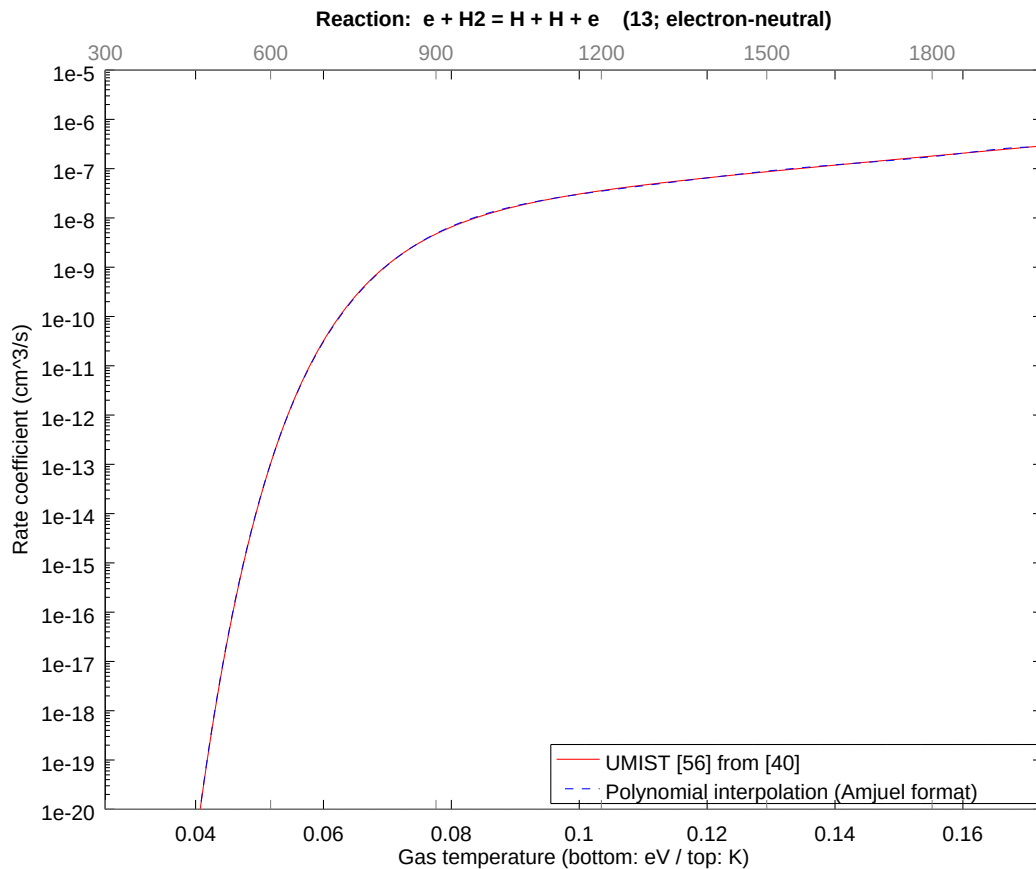
## 2.19 Reaction $\text{e} + \text{N} = \text{N}^*(2\text{P}) + \text{e}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.101194818558E+01 | b1 | 4.343453966856E+00  | b2 | -2.377575275729E+00 |
| b3 | 7.670532404472E-01  | b4 | -1.740660158042E-01 | b5 | 2.774406536719E-02  |
| b6 | -2.941247345987E-03 | b7 | 1.844731794181E-04  | b8 | -5.135774011098E-06 |



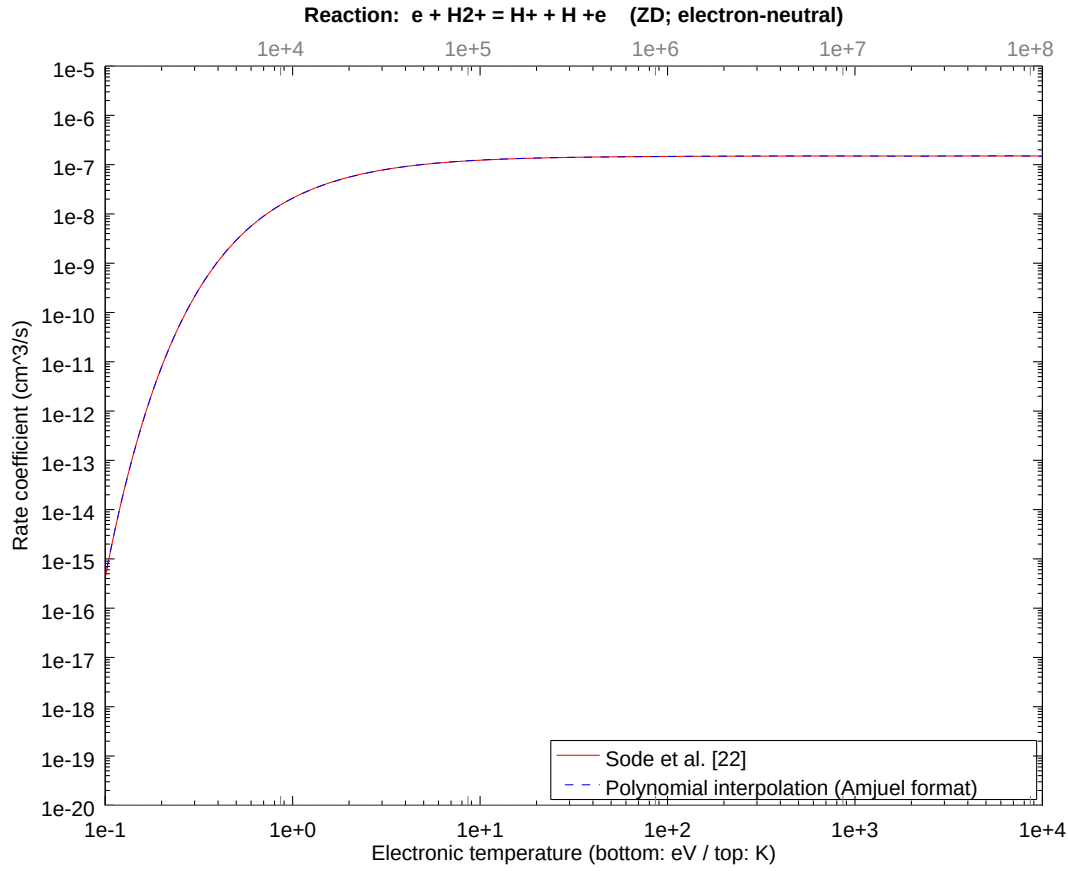
## 2.20 Reaction 13 $e + H_2 = H + H + e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.706801815573E+01 | b1 | 9.132179646174E+00  | b2 | -4.356001609304E+00 |
| b3 | 1.405333065127E+00  | b4 | -3.189097113803E-01 | b5 | 5.083043831321E-02  |
| b6 | -5.388716102176E-03 | b7 | 3.379768769551E-04  | b8 | -9.409350814540E-06 |



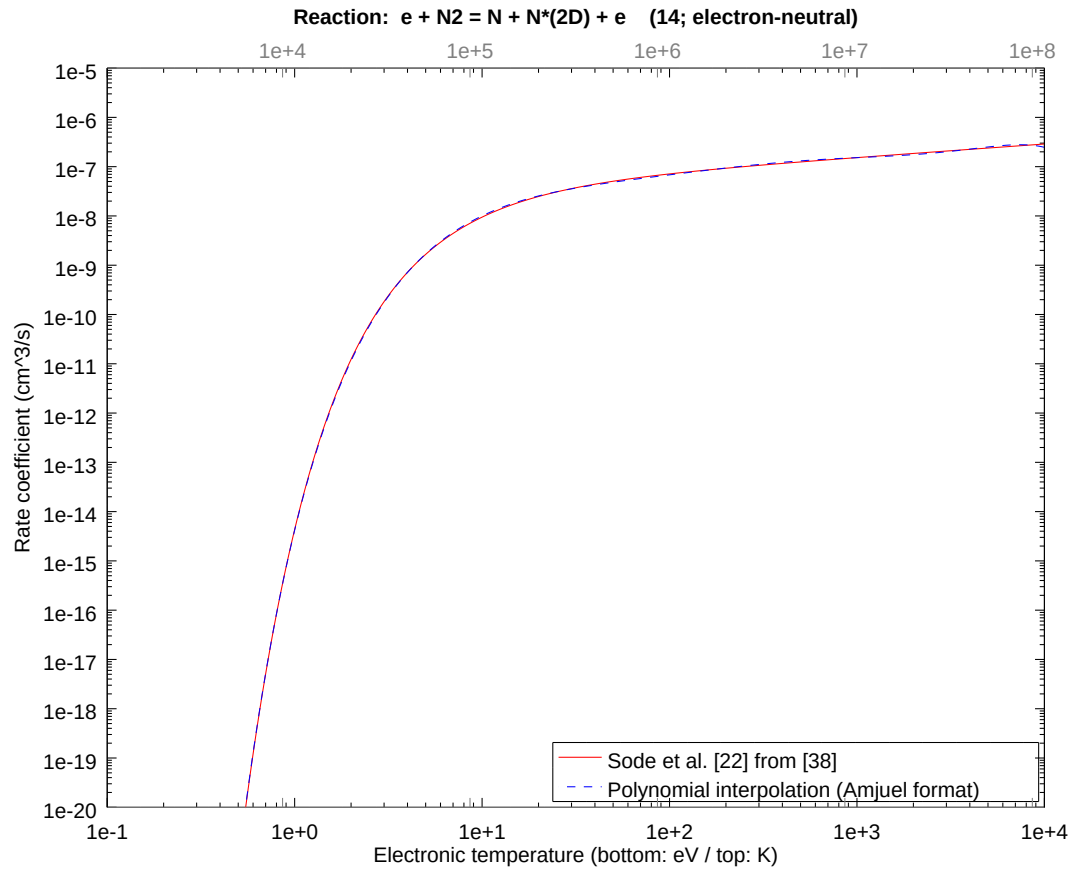
## 2.21 Reaction ZD $e + H_2^+ = H^+ + H + e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.768244251257E+01 | b1 | 1.967313398897E+00  | b2 | -9.757965194138E-01 |
| b3 | 3.148114341003E-01  | b4 | -7.143959398637E-02 | b5 | 1.138662682780E-02  |
| b6 | -1.207136931586E-03 | b7 | 7.571086738646E-05  | b8 | -2.107807250399E-06 |



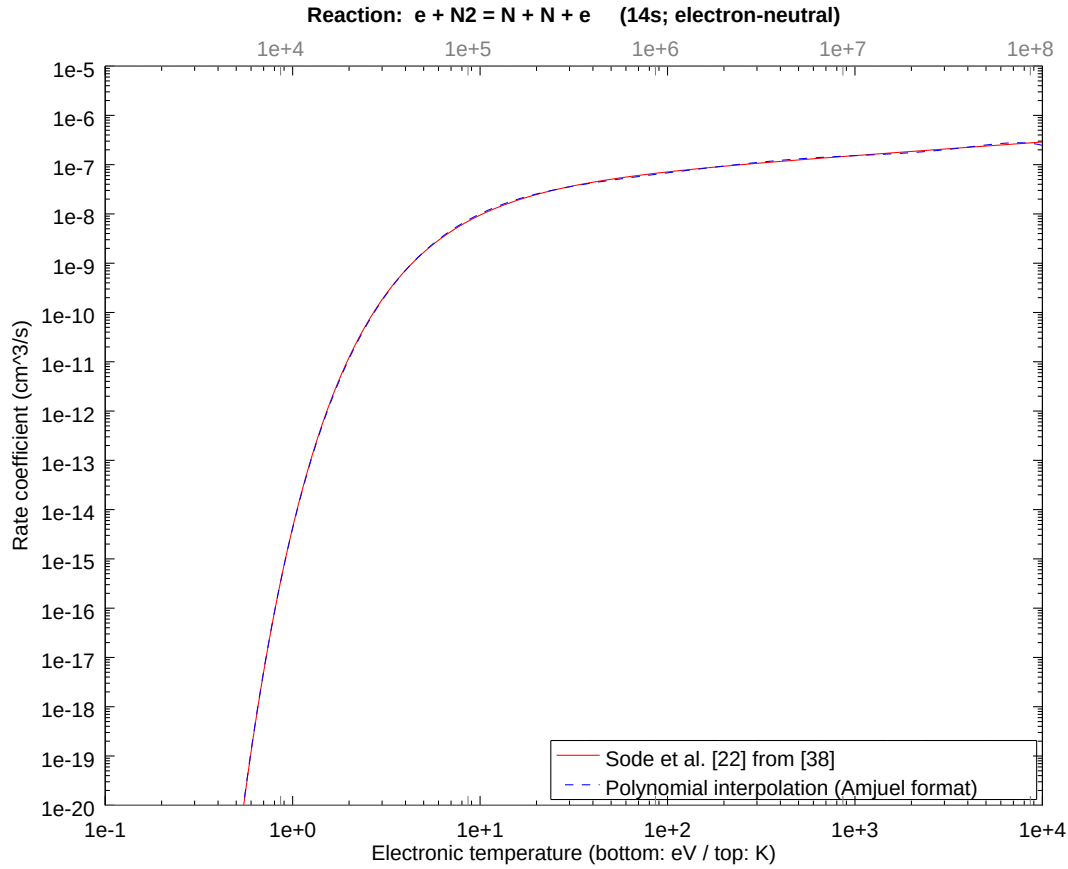
## 2.22 Reaction 14 $e + N_2 = N + N^*(2D) + e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.307372971713E+01 | b1 | 1.577882085526E+01  | b2 | -7.692446673346E+00 |
| b3 | 2.481736838363E+00  | b4 | -5.631760886327E-01 | b5 | 8.976361148995E-02  |
| b6 | -9.516160683972E-03 | b7 | 5.968475992401E-04  | b8 | -1.661636883167E-05 |



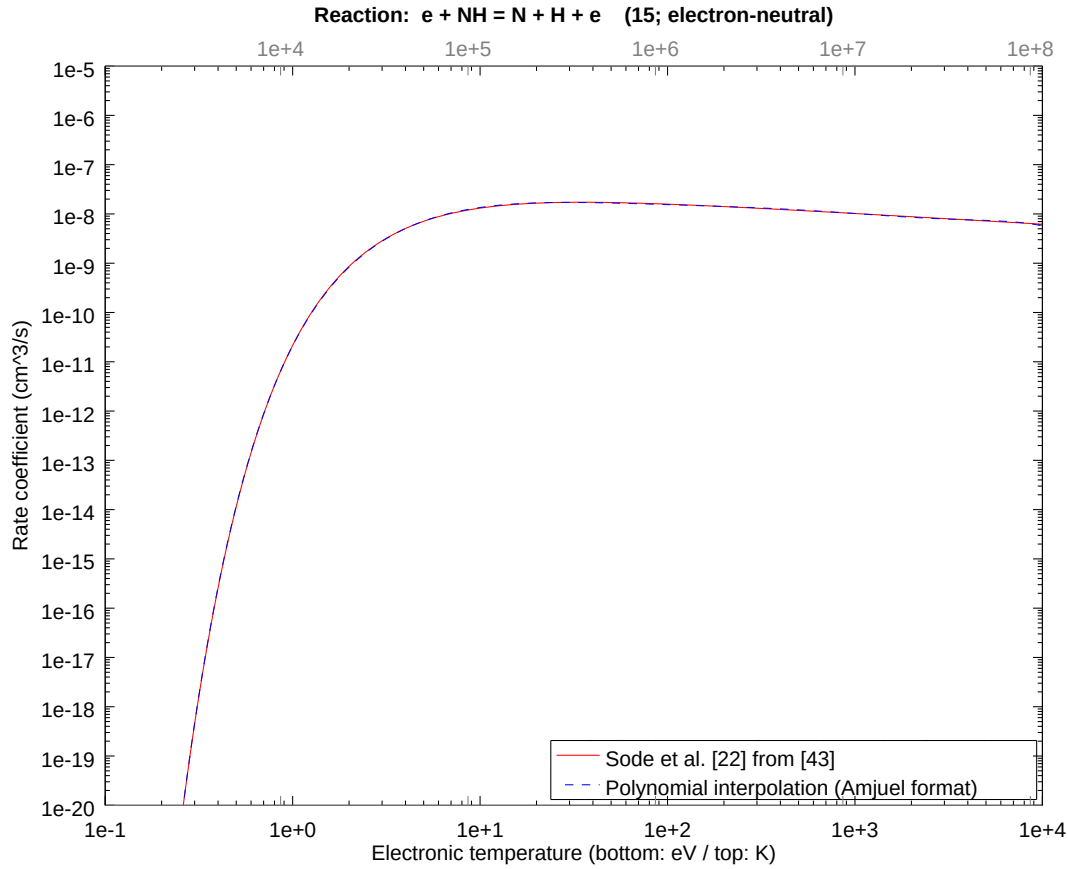
## 2.23 Reaction 14s $e + N_2 = N + N + e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.307372971713E+01 | b1 | 1.577882085526E+01  | b2 | -7.692446673346E+00 |
| b3 | 2.481736838363E+00  | b4 | -5.631760886327E-01 | b5 | 8.976361148995E-02  |
| b6 | -9.516160683972E-03 | b7 | 5.968475992401E-04  | b8 | -1.661636883167E-05 |



## 2.24 Reaction 15 $e + \text{NH} = \text{N} + \text{H} + e$

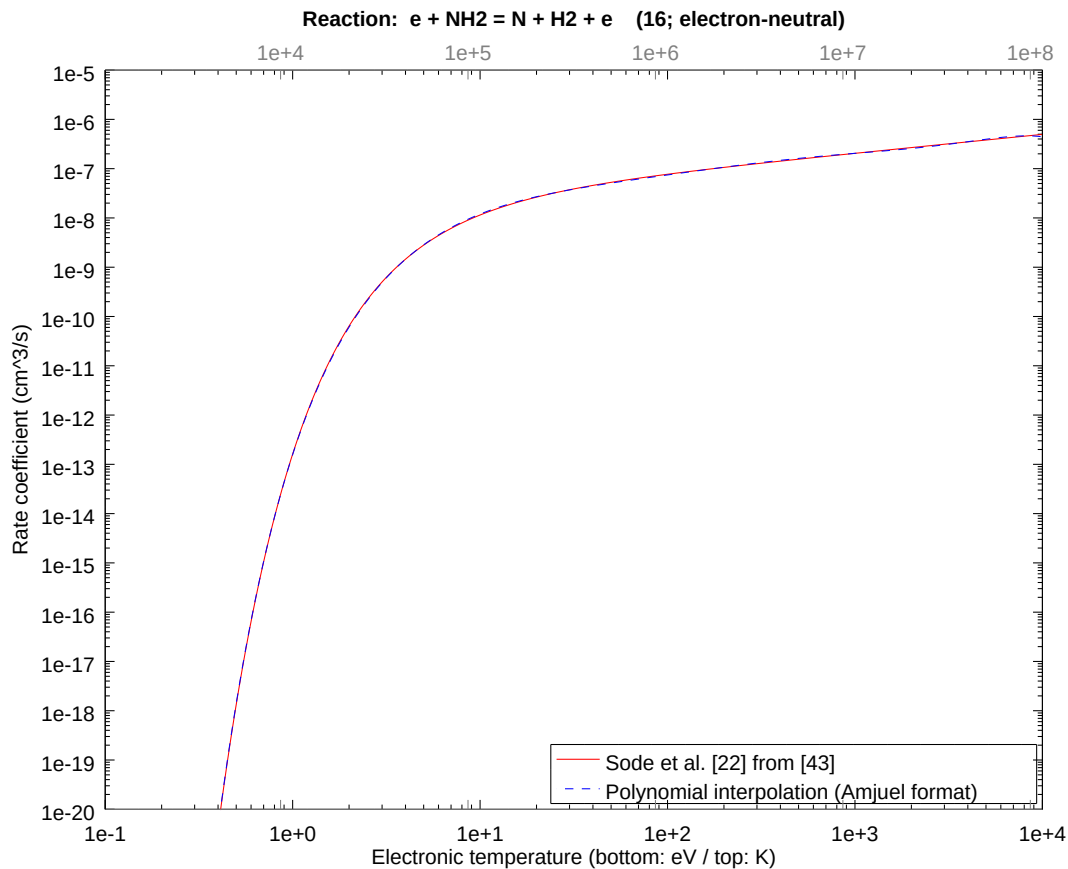
|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.456238424902E+01 | b1 | 7.459512709400E+00  | b2 | -3.809073722990E+00 |
| b3 | 1.228883212299E+00  | b4 | -2.788682628195E-01 | b5 | 4.444830472364E-02  |
| b6 | -4.712123352210E-03 | b7 | 2.955414061921E-04  | b8 | -8.227937946931E-06 |





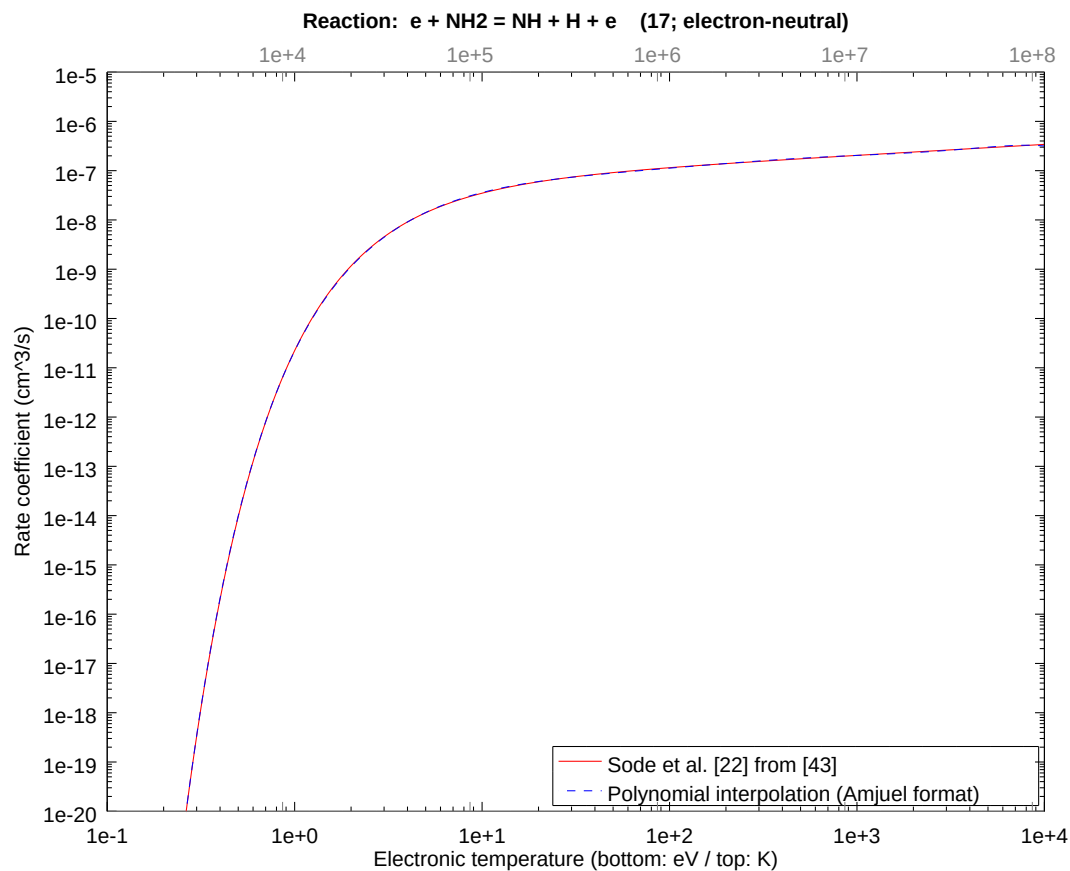
## 2.25 Reaction 16 $e + \text{NH}_2 = \text{N} + \text{H}_2 + e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.945412372396E+01 | b1 | 1.180439862101E+01  | b2 | -5.666554407153E+00 |
| b3 | 1.828143556398E+00  | b4 | -4.148573376663E-01 | b5 | 6.612335579172E-02  |
| b6 | -7.009972841258E-03 | b7 | 4.396610776123E-04  | b8 | -1.224026139310E-05 |



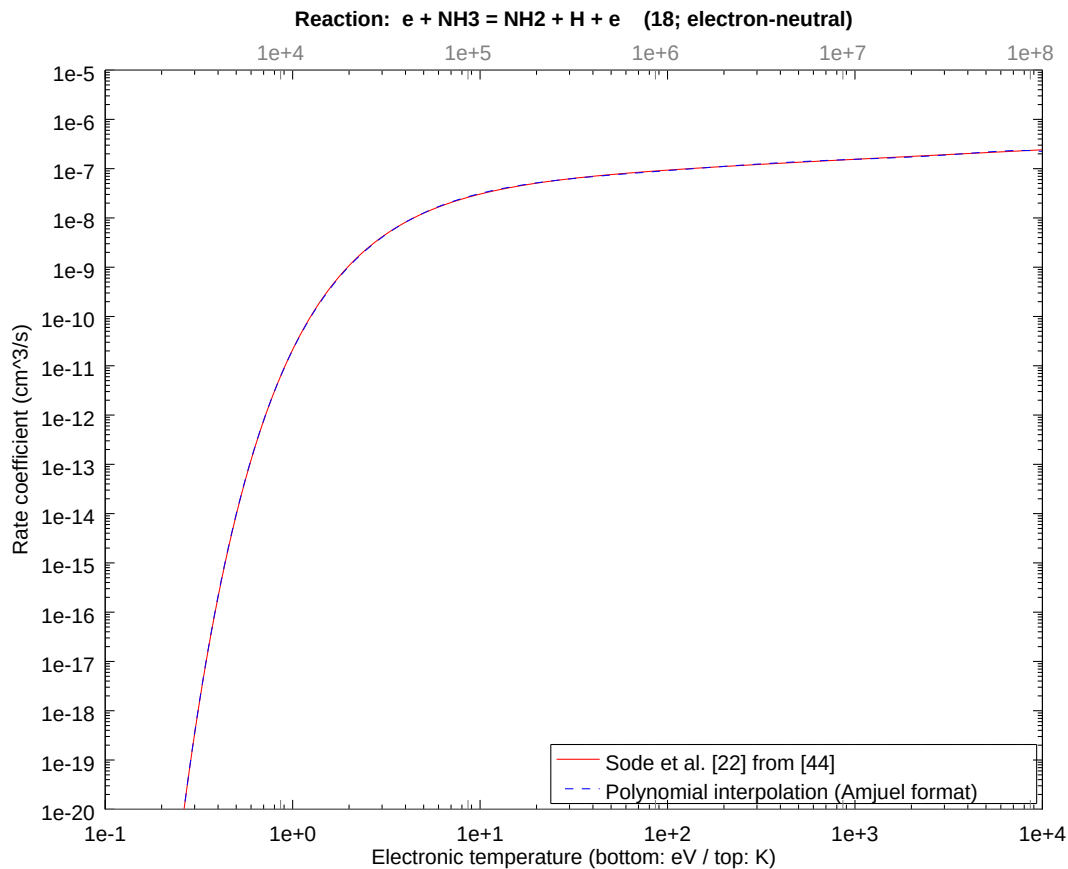
## 2.26 Reaction 17 $e + \text{NH}_2 = \text{NH} + \text{H} + e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.452587699670E+01 | b1 | 7.819621809953E+00  | b2 | -3.769447468395E+00 |
| b3 | 1.216098991625E+00  | b4 | -2.759671625561E-01 | b5 | 4.398590363420E-02  |
| b6 | -4.663102563113E-03 | b7 | 2.924668532021E-04  | b8 | -8.142341713424E-06 |



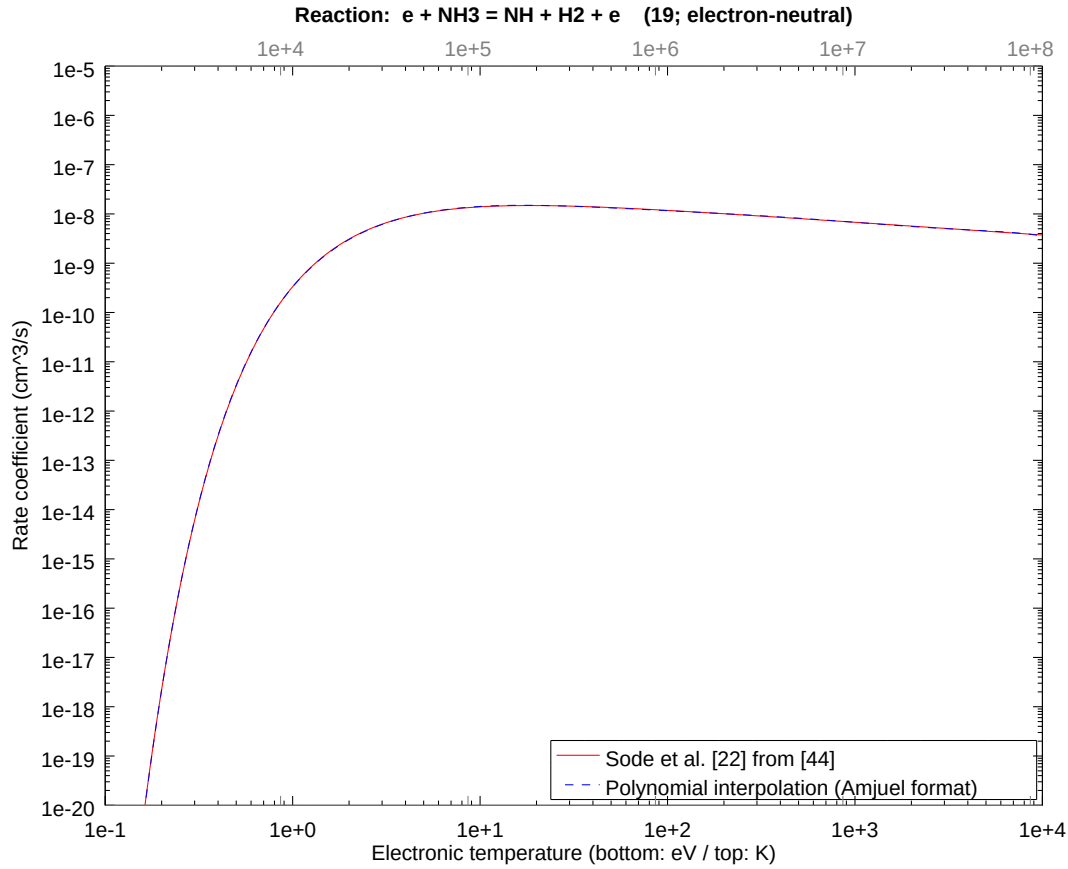
## 2.27 Reaction 18 $e + \text{NH}_3 = \text{NH}_2 + \text{H} + e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.457487177713E+01 | b1 | 7.769649085091E+00  | b2 | -3.759540904746E+00 |
| b3 | 1.212902936457E+00  | b4 | -2.752418874902E-01 | b5 | 4.387030336181E-02  |
| b6 | -4.650847365834E-03 | b7 | 2.916982149542E-04  | b8 | -8.120942655034E-06 |



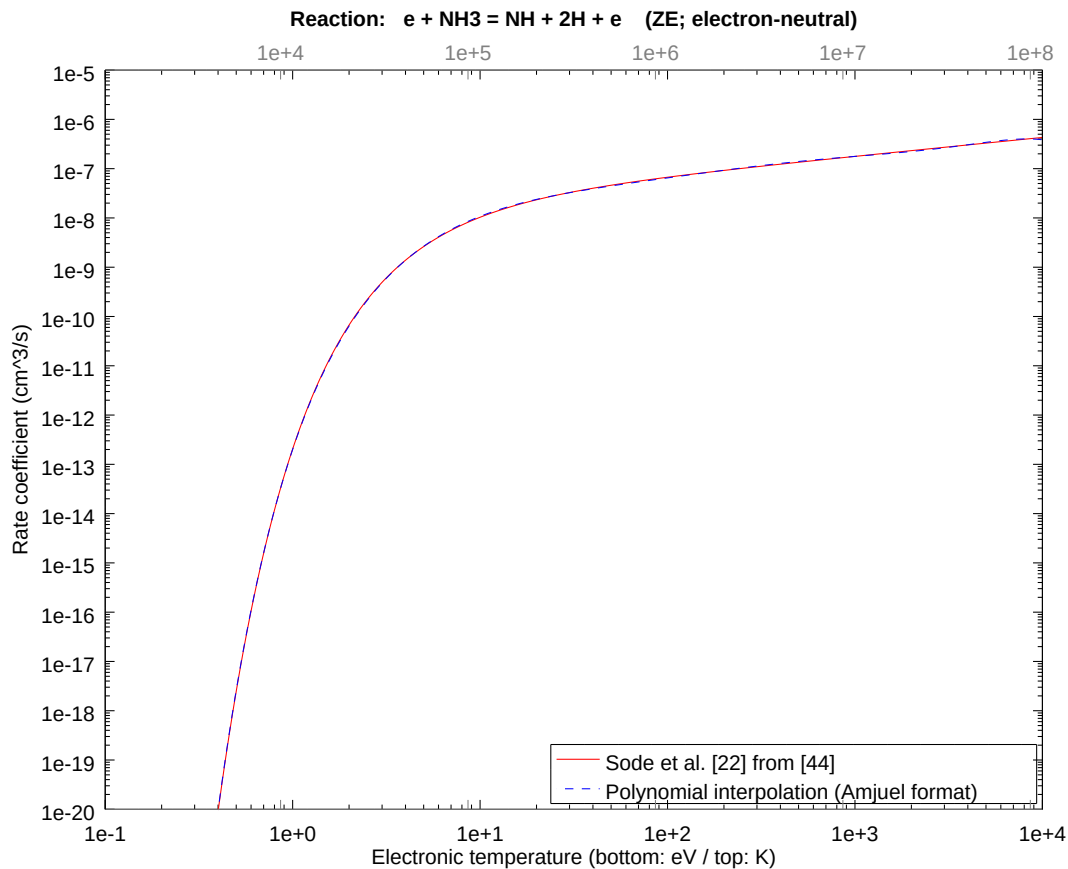
## 2.28 Reaction 19 $e + \text{NH}_3 = \text{NH} + \text{H}_2 + e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.181923467093E+01 | b1 | 4.543440329287E+00  | b2 | -2.382528557553E+00 |
| b3 | 7.686512680312E-01  | b4 | -1.744286533370E-01 | b5 | 2.780186550334E-02  |
| b6 | -2.947374944620E-03 | b7 | 1.848574985415E-04  | b8 | -5.146473540279E-06 |



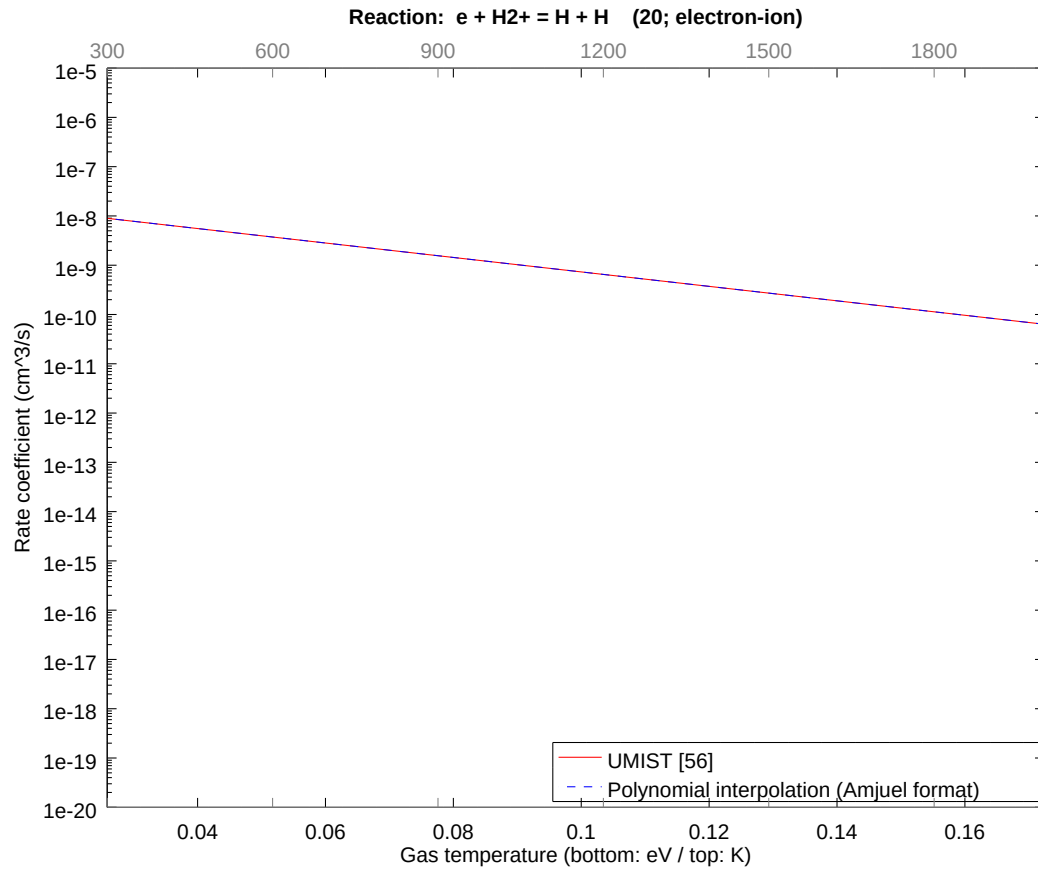
## 2.29 Reaction ZE $e + \text{NH}_3 = \text{NH} + 2\text{H} + e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.921726083740E+01 | b1 | 1.142491684863E+01  | b2 | -5.478329697824E+00 |
| b3 | 1.767418508196E+00  | b4 | -4.010771114152E-01 | b5 | 6.392695061682E-02  |
| b6 | -6.777124093035E-03 | b7 | 4.250569509084E-04  | b8 | -1.183367928389E-05 |



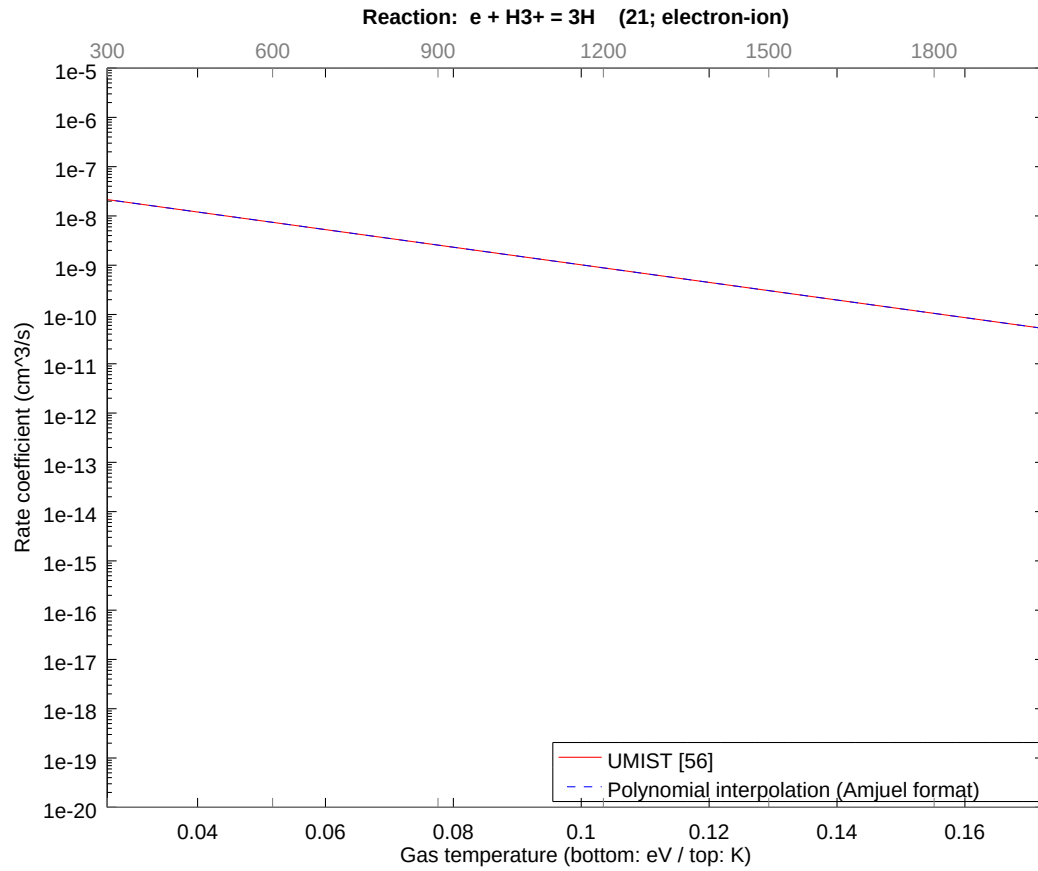
## 2.30 Reaction 20 $e + H_2^+ = H + H$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.952226532449E+01 | b1 | -4.300000000000E-01 | b2 | -1.587469235071E-13 |
| b3 | 1.803574051818E-13  | b4 | -9.282287176011E-14 | b5 | 2.531541920459E-14  |
| b6 | -3.802235874767E-15 | b7 | 2.986368614331E-16  | b8 | -9.626362404661E-18 |



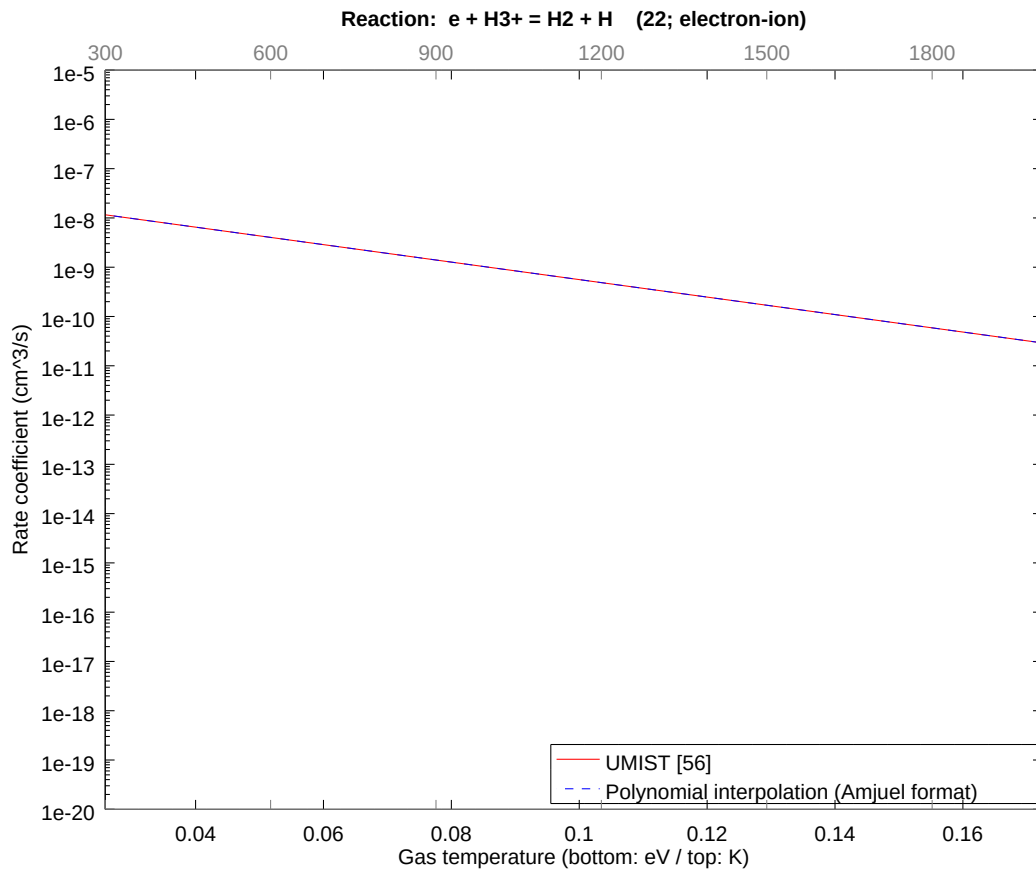
## 2.31 Reaction 21 $e + H_3^+ = 3H$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.886152596060E+01 | b1 | -5.234999999999E-01 | b2 | -2.872563377747E-13 |
| b3 | 2.579413169081E-13  | b4 | -1.149071316421E-13 | b5 | 2.727352466664E-14  |
| b6 | -3.423000284388E-15 | b7 | 2.054190354660E-16  | b8 | -4.189931360780E-18 |



## 2.32 Reaction 22 $e + H_3^+ = H_2 + H$

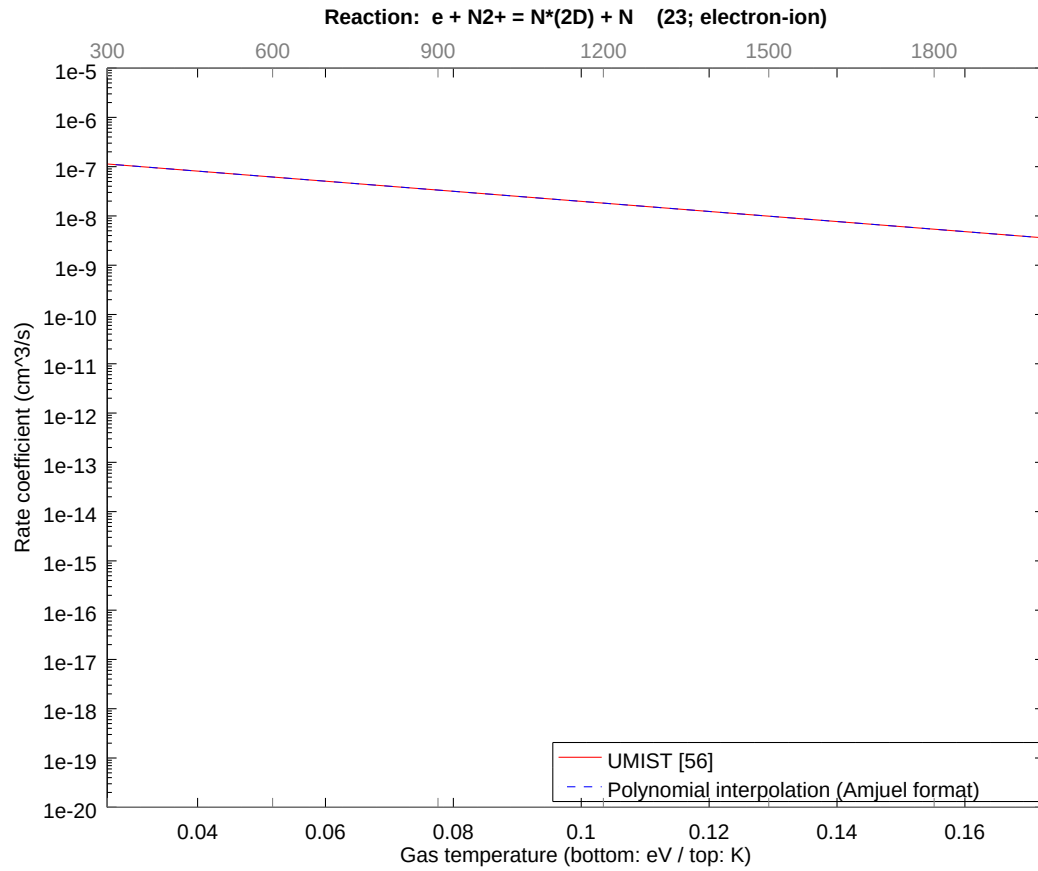
|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.947105509154E+01 | b1 | -5.199999999999E-01 | b2 | -2.998552999578E-13 |
| b3 | 3.234342294042E-13  | b4 | -1.651627757082E-13 | b5 | 4.386455652273E-14  |
| b6 | -6.153035641732E-15 | b7 | 4.226866835531E-16  | b8 | -1.071416049243E-17 |





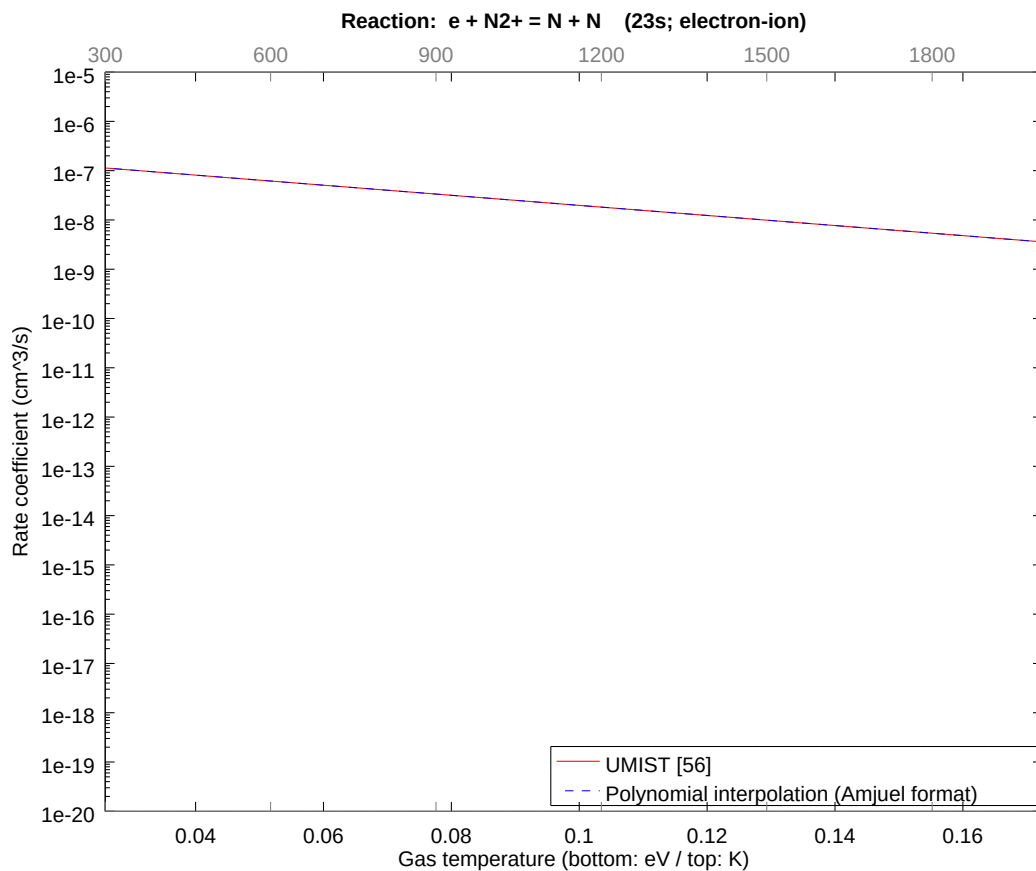
### 2.33 Reaction 23 $e + N_2^+ = N^*(2D) + N$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.668392429045E+01 | b1 | -2.999999999999E-01 | b2 | -2.771771680282E-13 |
| b3 | 2.629792332539E-13  | b4 | -1.225766242449E-13 | b5 | 3.048491854561E-14  |
| b6 | -4.083421077400E-15 | b7 | 2.730109335609E-16  | b8 | -6.918197945205E-18 |



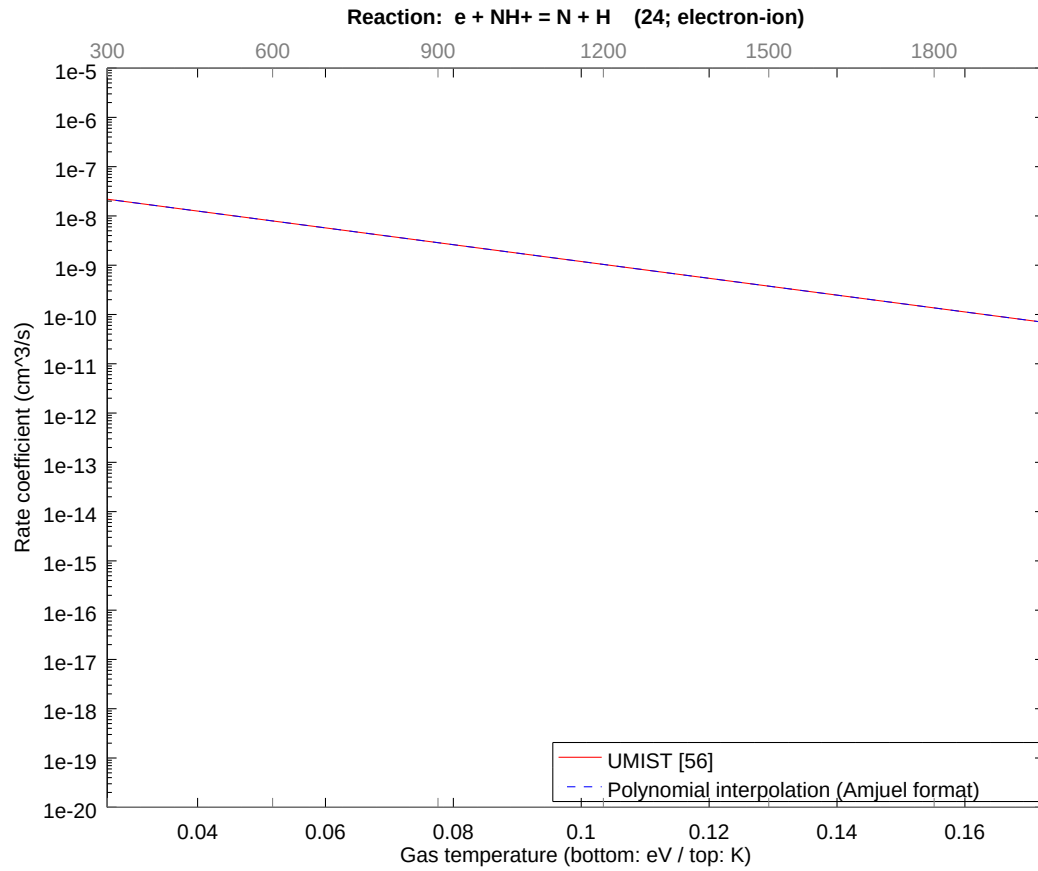
## 2.34 Reaction 23s $e + N_2^+ = N + N$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.668392429045E+01 | b1 | -2.999999999999E-01 | b2 | -2.771771680282E-13 |
| b3 | 2.629792332539E-13  | b4 | -1.225766242449E-13 | b5 | 3.048491854561E-14  |
| b6 | -4.083421077400E-15 | b7 | 2.730109335609E-16  | b8 | -6.918197945205E-18 |



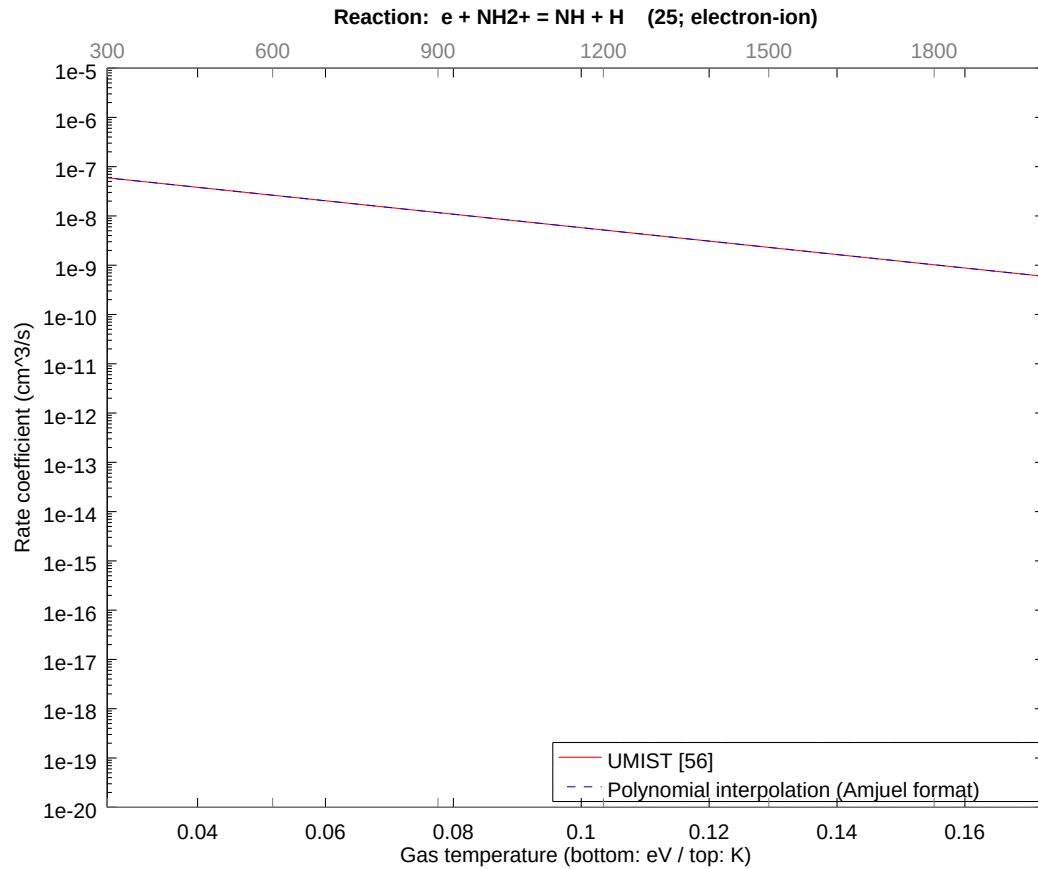
## 2.35 Reaction 24 $e + \text{NH}^+ = \text{N} + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.878949387217E+01 | b1 | -4.999999999999E-01 | b2 | -2.293011117324E-13 |
| b3 | 2.408124013321E-13  | b4 | -1.199145482507E-13 | b5 | 3.218723890284E-14  |
| b6 | -4.760804162503E-15 | b7 | 3.642519042223E-16  | b8 | -1.122439056266E-17 |



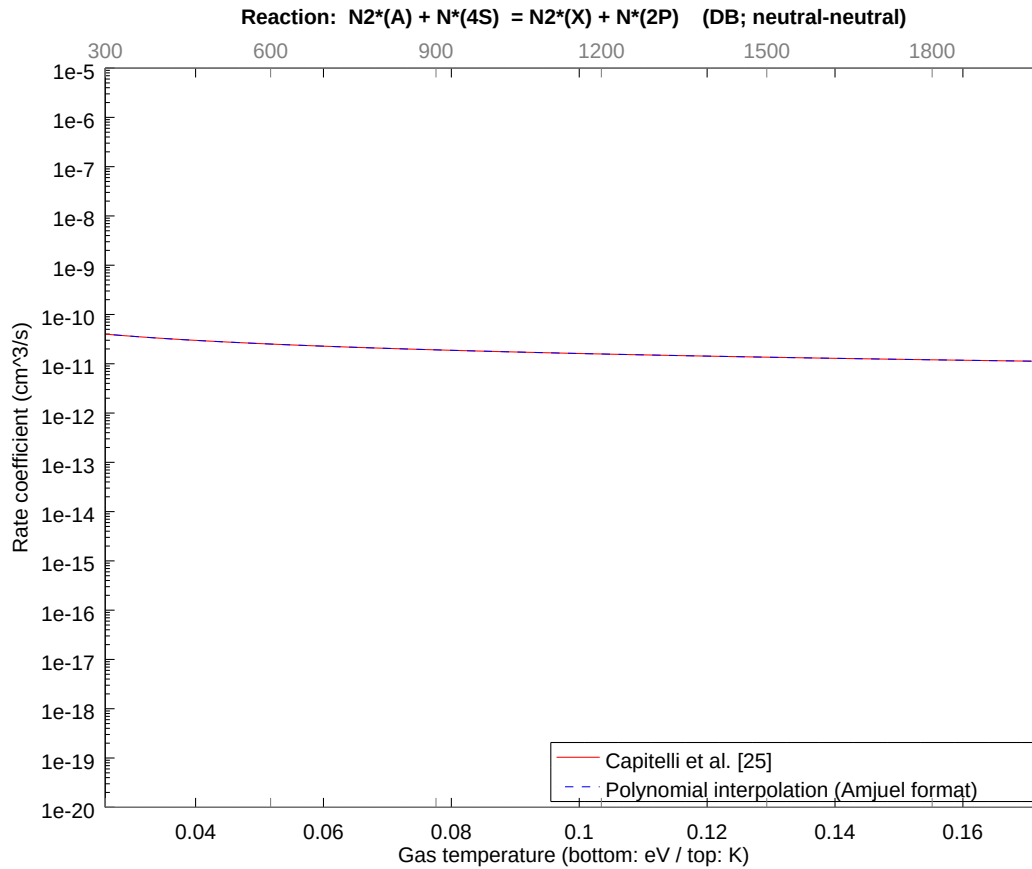
## 2.36 Reaction 25 $e + \text{NH}_2^+ = \text{NH} + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.756023554439E+01 | b1 | -3.999999999999E-01 | b2 | -2.796969604648E-13 |
| b3 | 2.529034005622E-13  | b4 | -1.145453104624E-13 | b5 | 2.765600058346E-14  |
| b6 | -3.495866866714E-15 | b7 | 2.041301376069E-16  | b8 | -3.538016461484E-18 |



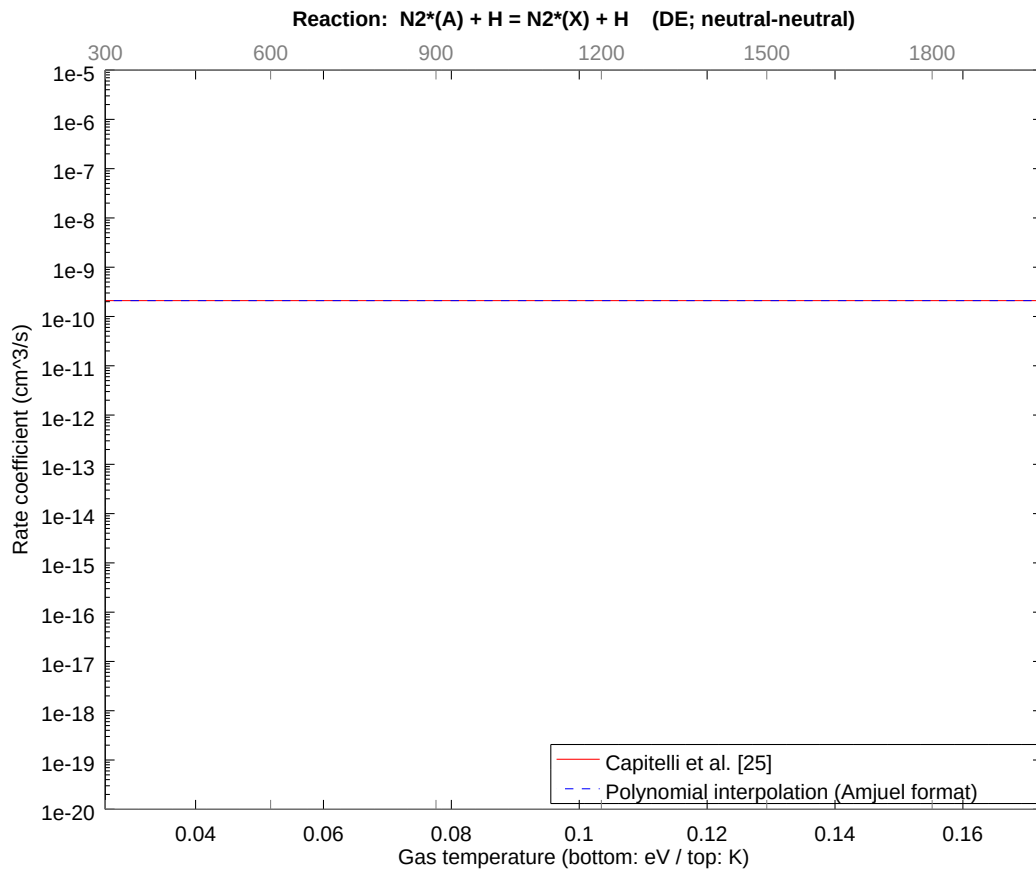
## 2.37 Reaction DB $\text{N}_2^*(\text{A}) + \text{N}^*(4\text{S}) = \text{N}_2^*(\text{X}) + \text{N}^*(2\text{P})$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.639089538605E+01 | b1 | -6.700000063335E-01 | b2 | -8.624699809858E-09 |
| b3 | -6.633570679721E-09 | b4 | -3.151551870313E-09 | b5 | -9.470599538949E-10 |
| b6 | -1.758156215834E-10 | b7 | -1.843916759489E-11 | b8 | -8.367108693434E-13 |



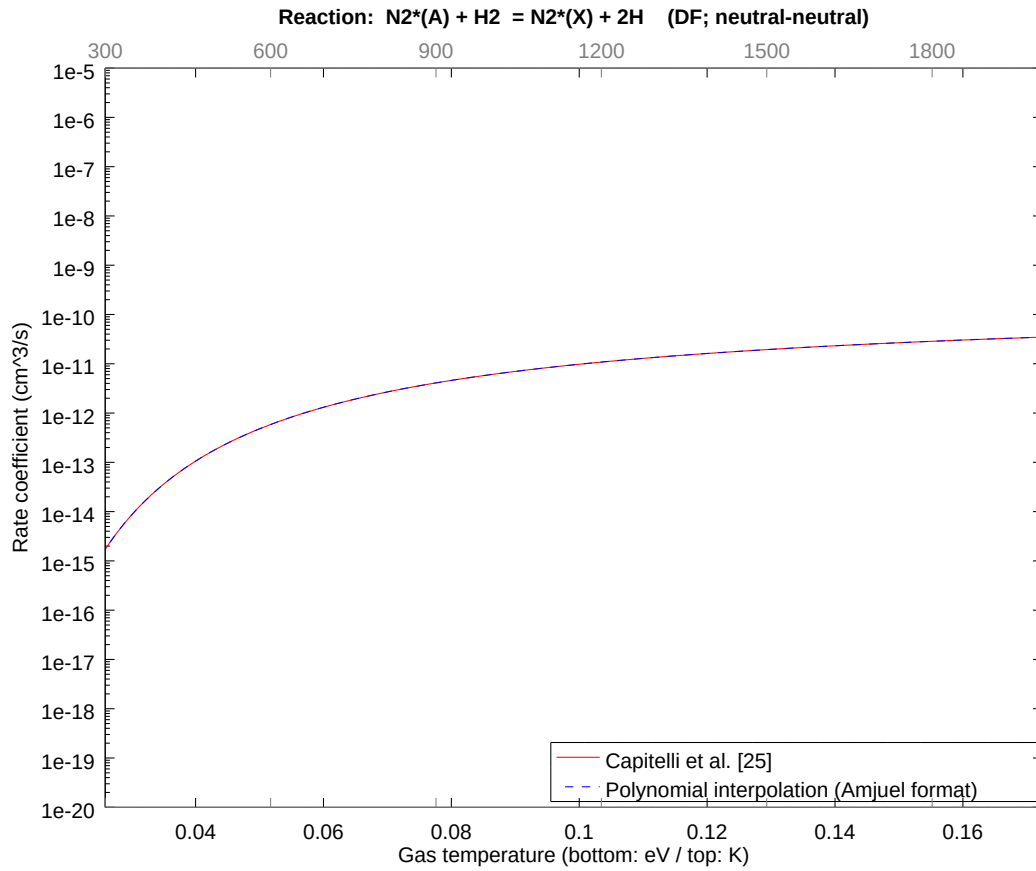
## 2.38 Reaction DE $\text{N}_2^*(\text{A}) + \text{H} = \text{N}_2^*(\text{X}) + \text{H}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.228391358521E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



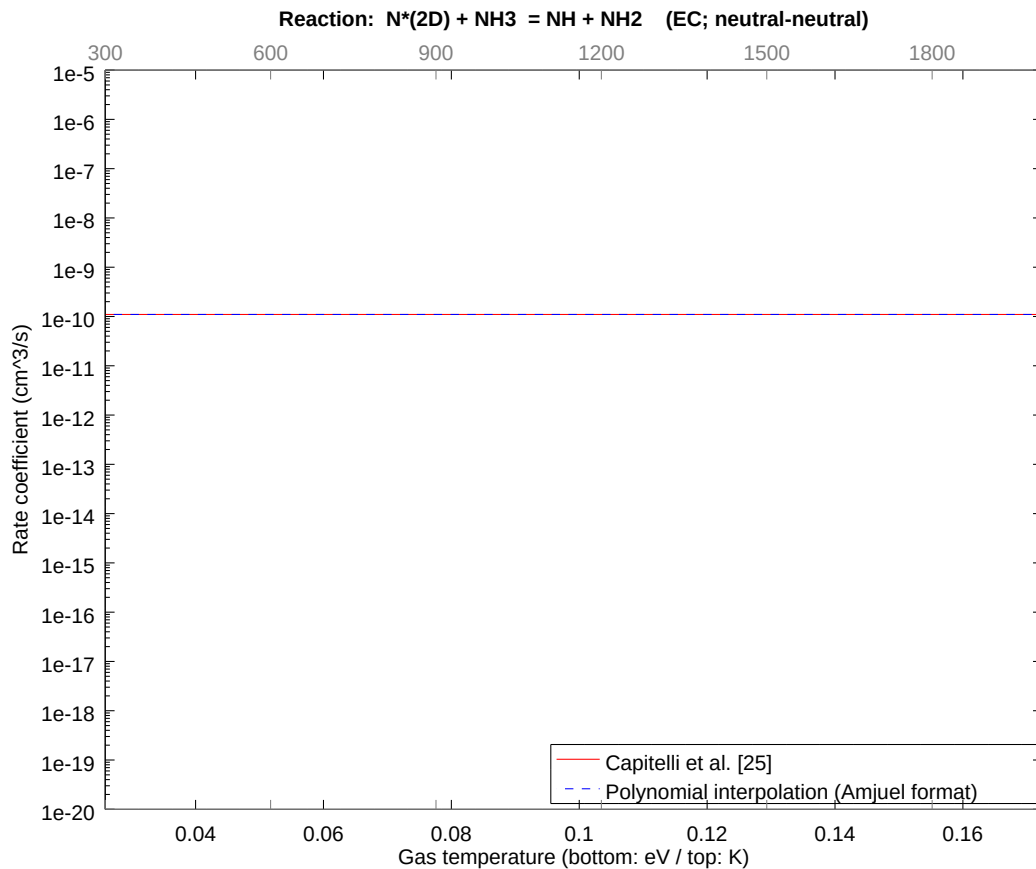
## 2.39 Reaction DF $\text{N}_2^*(\text{A}) + \text{H}_2 = \text{N}_2^*(\text{X}) + 2\text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.268826321125E+01 | b1 | 1.143045937131E-01  | b2 | -4.372098380053E-01 |
| b3 | -2.007521826120E-01 | b4 | -1.512801491418E-01 | b5 | -4.722928220100E-02 |
| b6 | -1.184186593315E-02 | b7 | -1.507809868440E-03 | b8 | -1.121565090195E-04 |



## 2.40 Reaction EC N\*(2D) + NH3 = NH + NH2

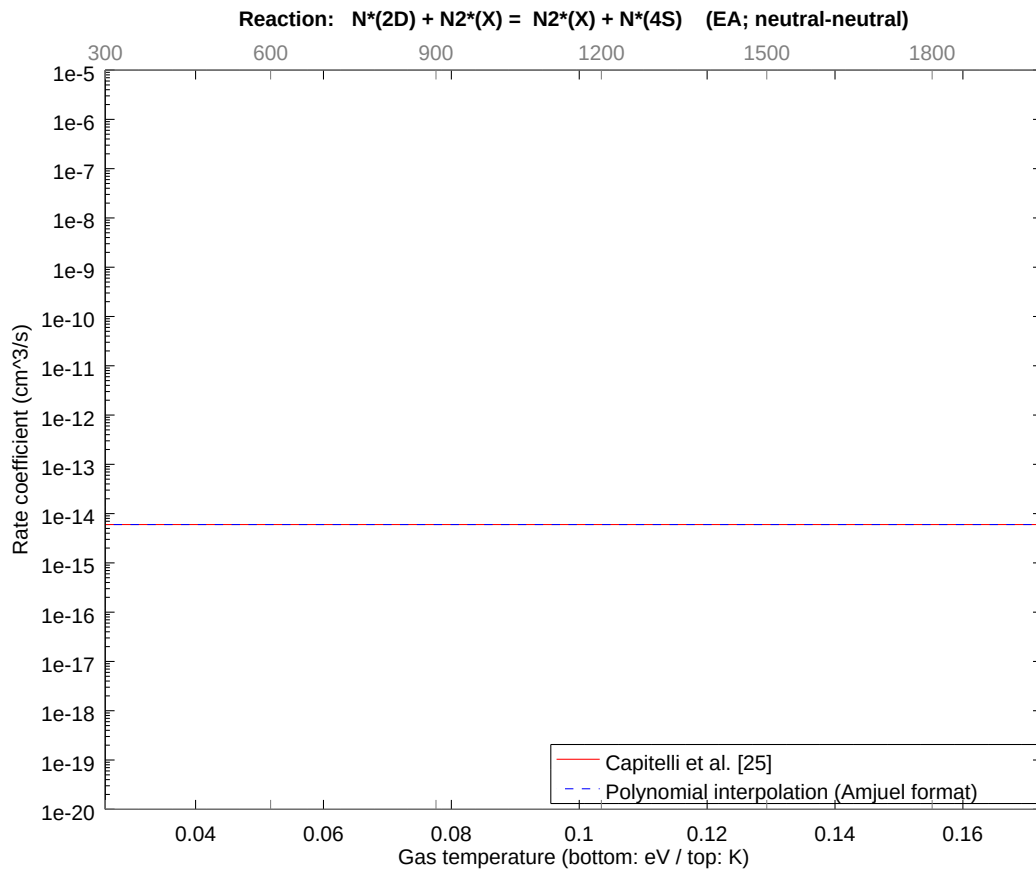
|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.293054075014E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |





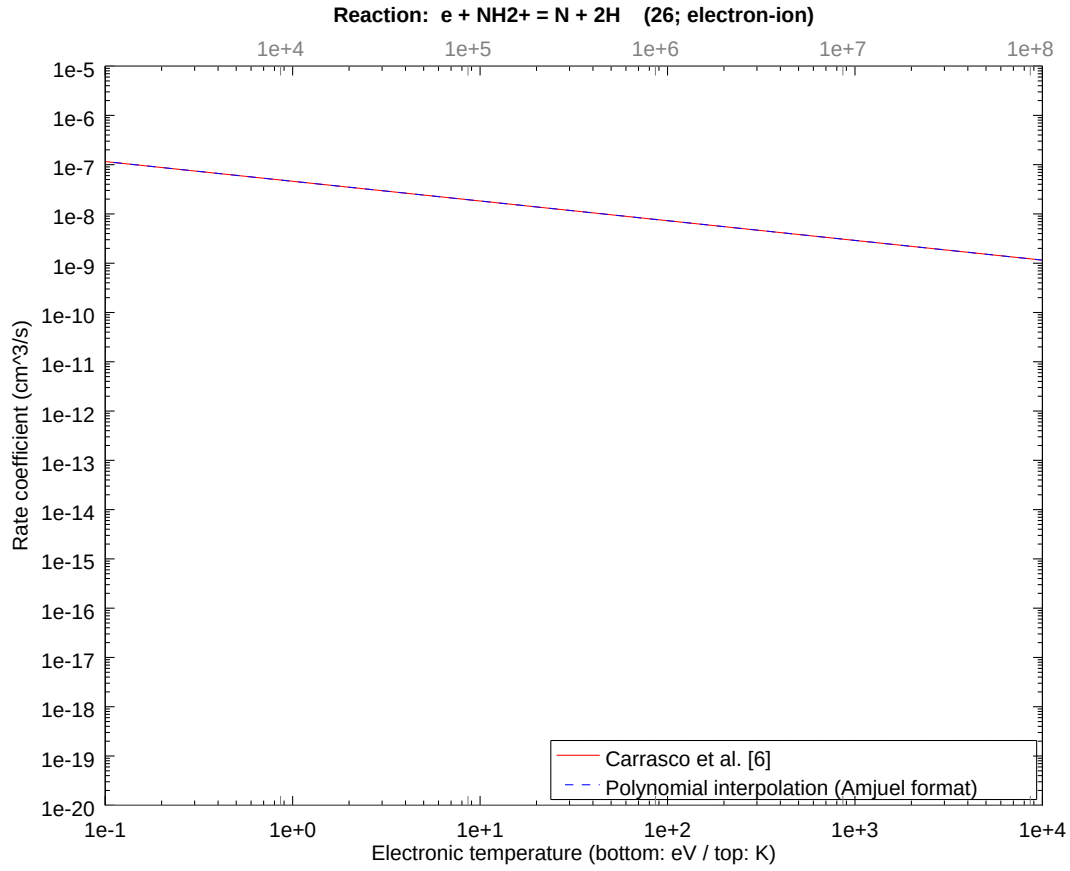
## 2.41 Reaction EA $\text{N}^*(2\text{D}) + \text{N}_2^*(\text{X}) = \text{N}_2^*(\text{X}) + \text{N}^*(4\text{S})$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -3.274701692568E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



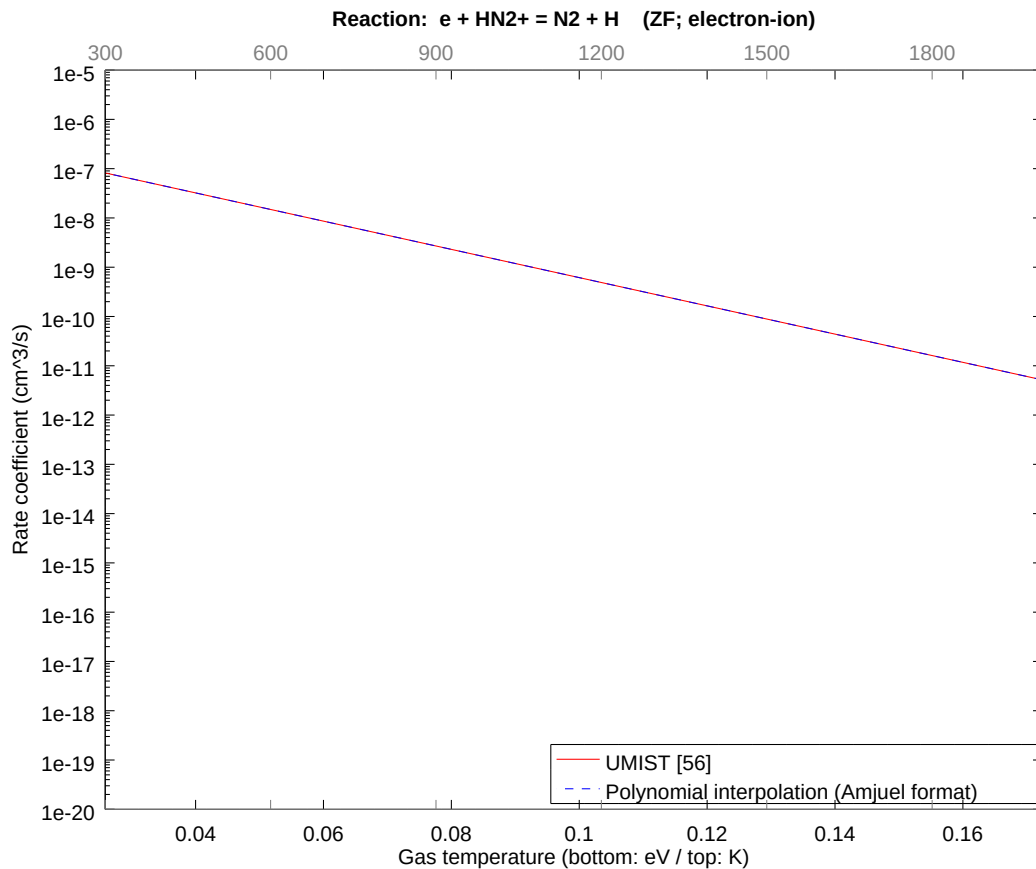
## 2.42 Reaction 26 $e + \text{NH}_2^+ = \text{N} + 2\text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.689486230264E+01 | b1 | -3.999999999999E-01 | b2 | -1.700859894719E-13 |
| b3 | 1.531526569142E-13  | b4 | -6.835402821794E-14 | b5 | 1.632721124296E-14  |
| b6 | -2.084464965884E-15 | b7 | 1.301523037146E-16  | b8 | -2.935865203087E-18 |



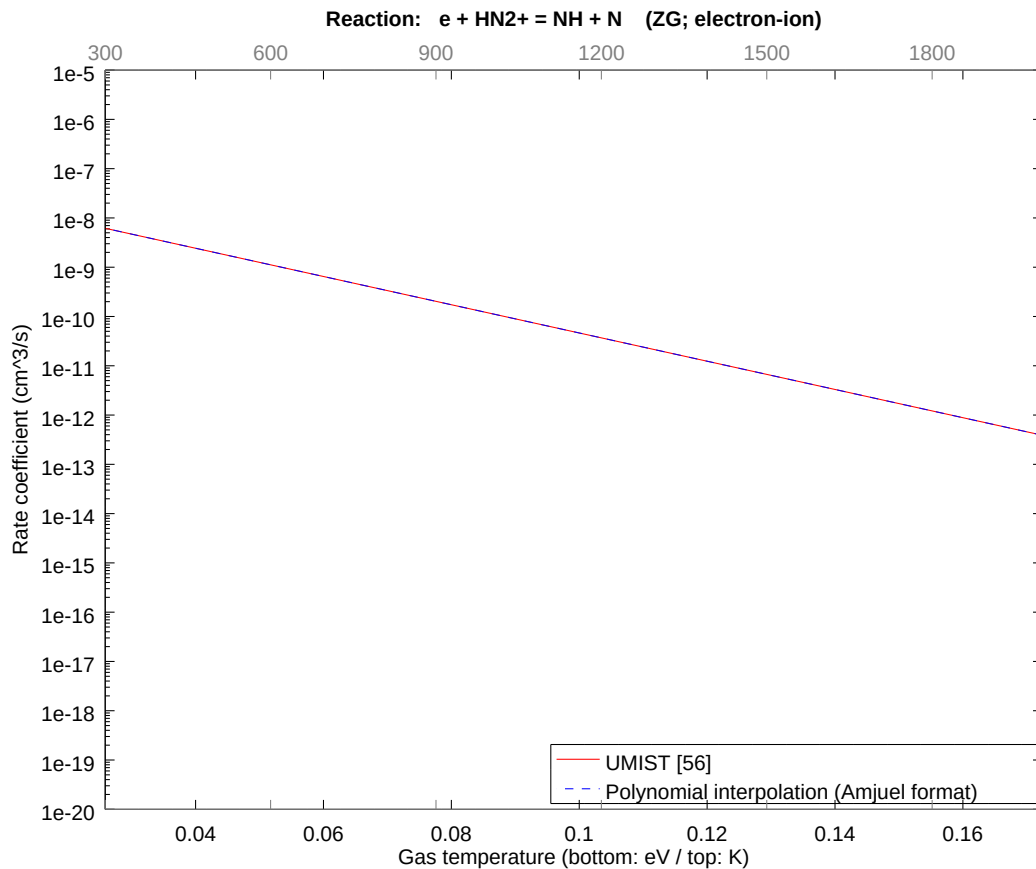
## 2.43 Reaction ZF $e + \text{HN}_2^+ = \text{N}_2 + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.825208158533E+01 | b1 | -8.399999999999E-01 | b2 | -2.570188285353E-13 |
| b3 | 2.387972347938E-13  | b4 | -1.197590584835E-13 | b5 | 3.498220922755E-14  |
| b6 | -5.855404126120E-15 | b7 | 5.157686837520E-16  | b8 | -1.843558700595E-17 |



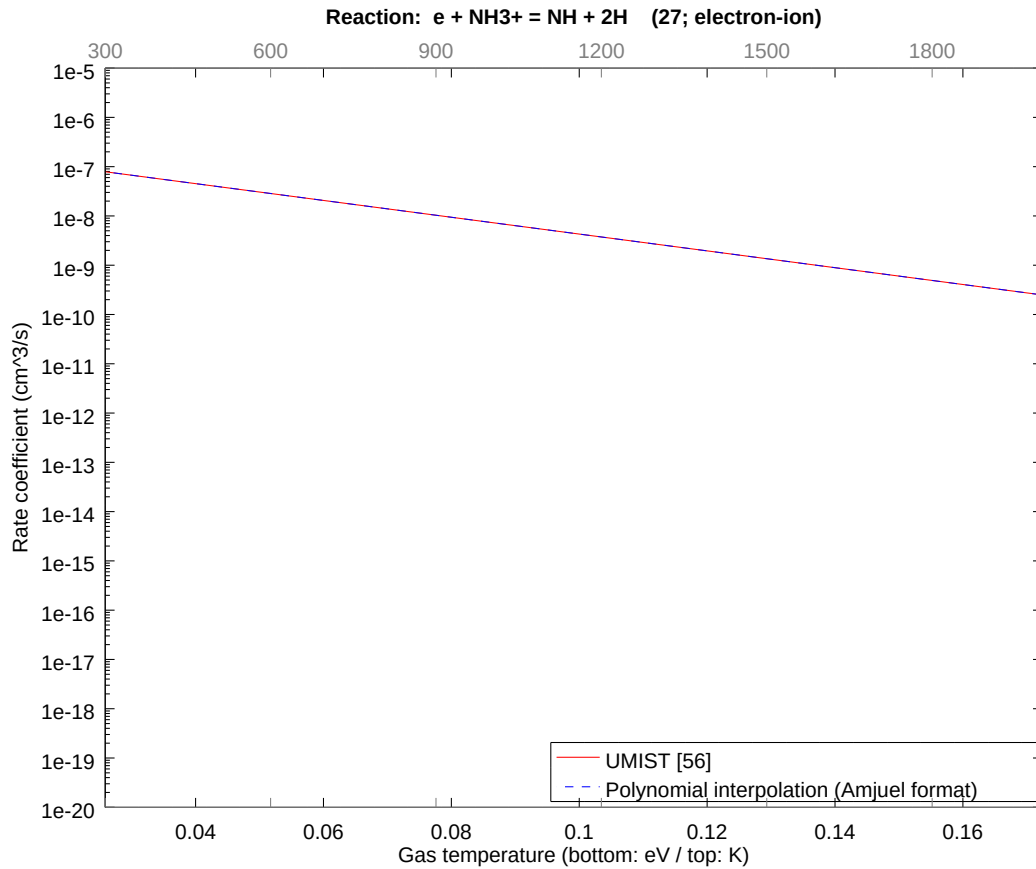
## 2.44 Reaction ZG $e + \text{HN}_2^+ = \text{NH} + \text{N}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.083843485145E+01 | b1 | -8.400000000000E-01 | b2 | -2.167021495493E-13 |
| b3 | 2.609640667156E-13  | b4 | -1.437672754782E-13 | b5 | 4.085615492632E-14  |
| b6 | -6.080022815360E-15 | b7 | 4.379396775936E-16  | b8 | -1.142827298318E-17 |



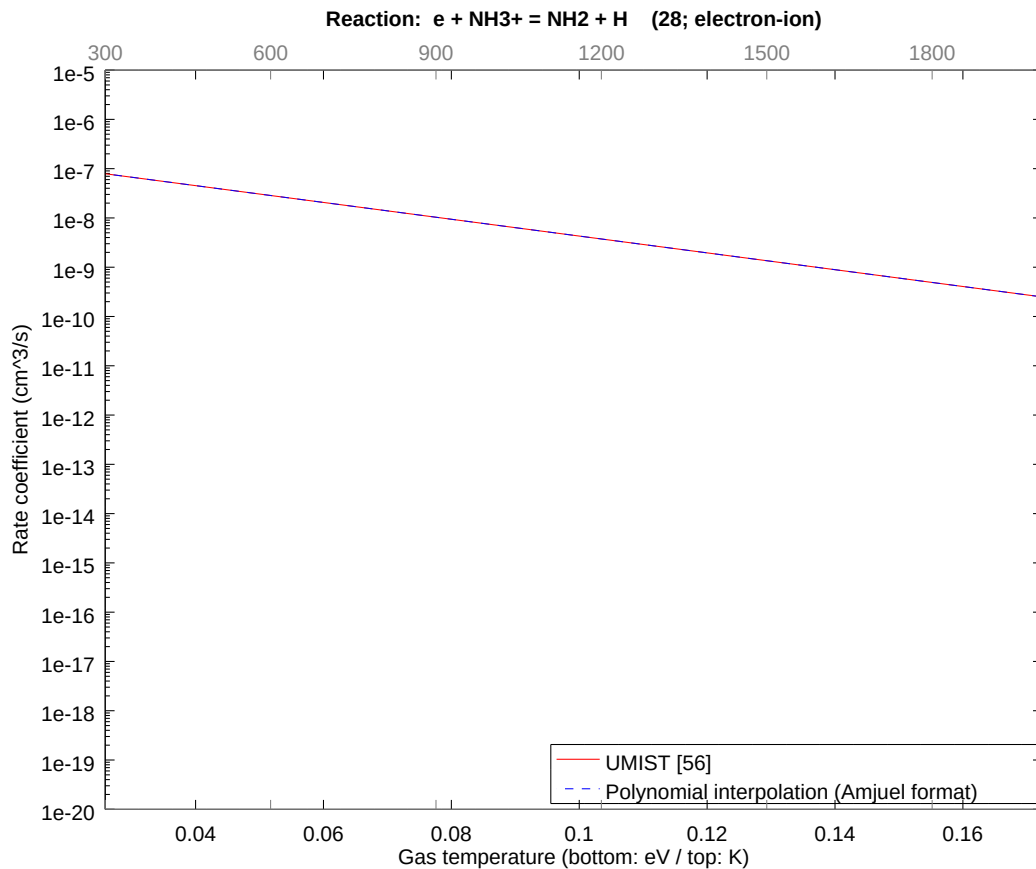
## 2.45 Reaction 27 $e + \text{NH}_3^+ = \text{NH} + 2\text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.750726887094E+01 | b1 | -4.999999999999E-01 | b2 | -2.167021495493E-13 |
| b3 | 1.874104880660E-13  | b4 | -7.299378089273E-14 | b5 | 1.193915327230E-14  |
| b6 | -1.915355154430E-16 | b7 | -1.484104869728E-16 | b8 | 1.130279707903E-17  |



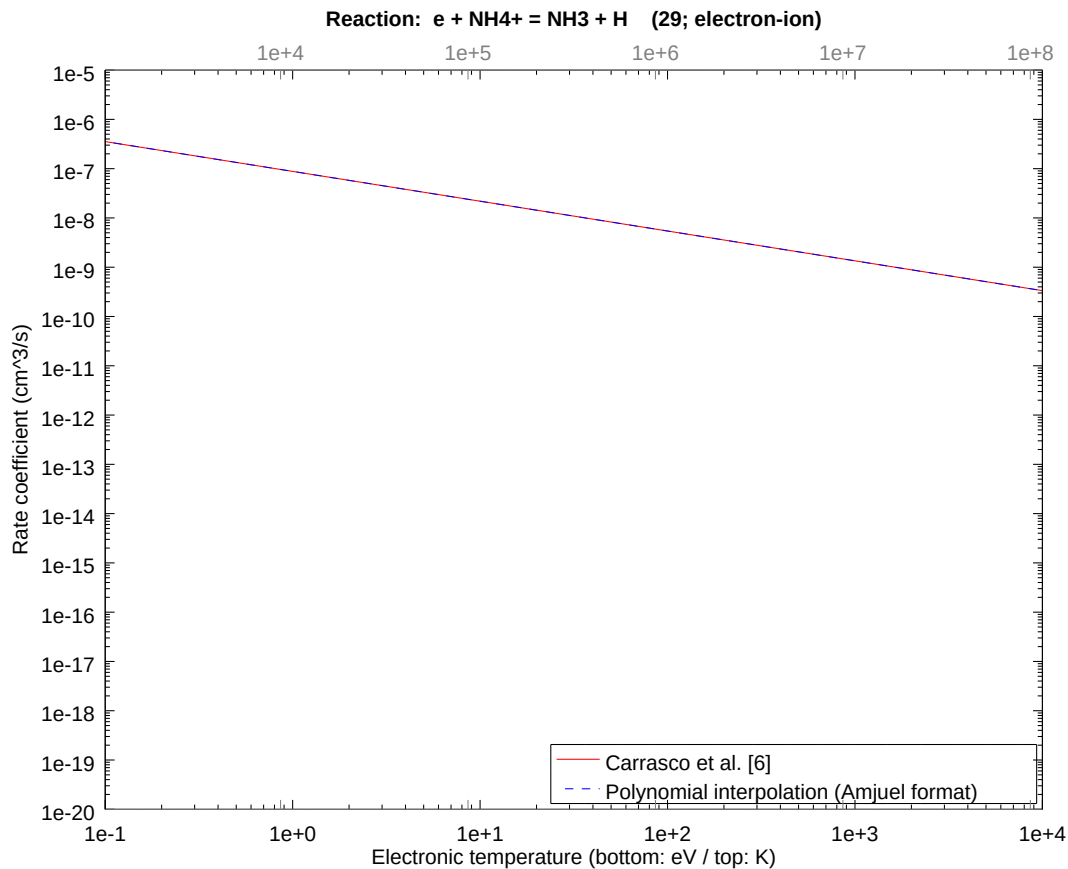
## 2.46 Reaction 28 $e + \text{NH}_3^+ = \text{NH}_2 + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.750726887094E+01 | b1 | -4.999999999999E-01 | b2 | -2.167021495493E-13 |
| b3 | 1.874104880660E-13  | b4 | -7.299378089273E-14 | b5 | 1.193915327230E-14  |
| b6 | -1.915355154430E-16 | b7 | -1.484104869728E-16 | b8 | 1.130279707903E-17  |



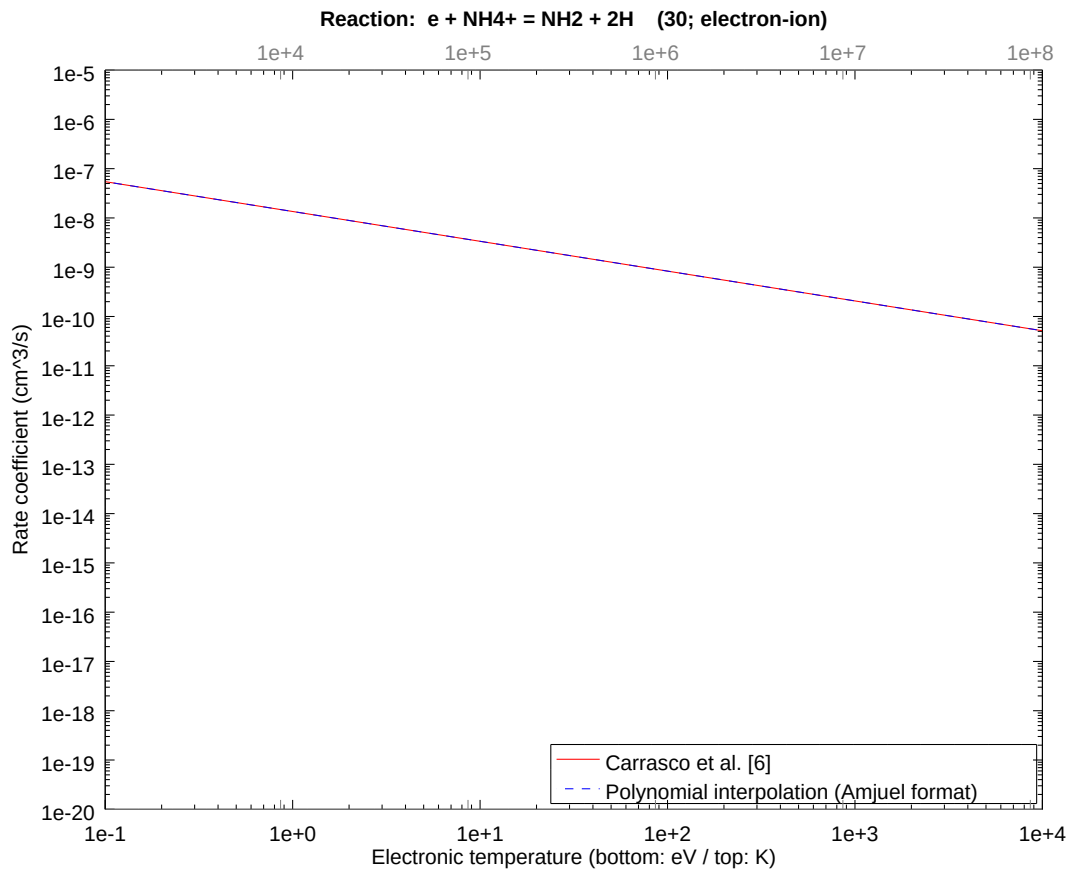
## 2.47 Reaction 29 $e + \text{NH}_4^+ = \text{NH}_3 + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.624544842816E+01 | b1 | -6.049999999999E-01 | b2 | -2.670979982817E-13 |
| b3 | 2.518958172930E-13  | b4 | -1.165360681583E-13 | b5 | 2.866495928141E-14  |
| b6 | -3.733343301059E-15 | b7 | 2.320383691723E-16  | b8 | -4.816504508900E-18 |



## 2.48 Reaction 30 $e + \text{NH}_4^+ = \text{NH}_2 + 2\text{H}$

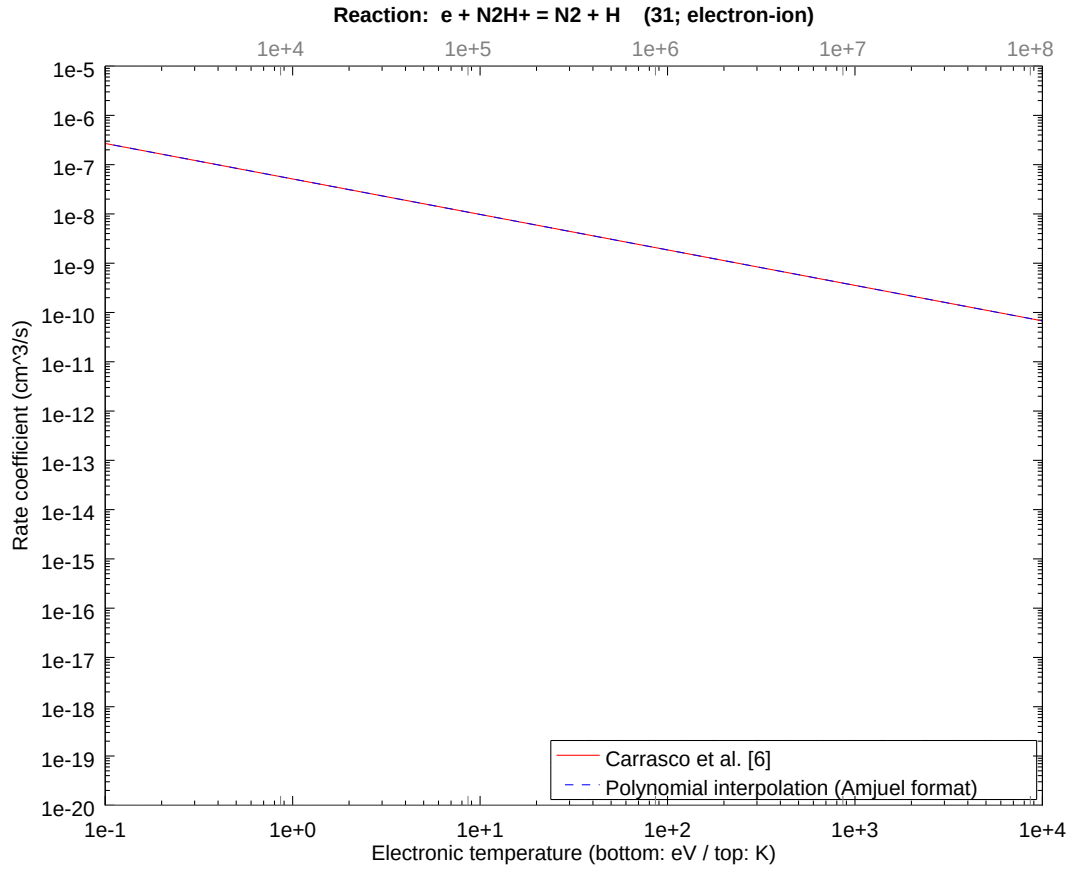
|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.811912501986E+01 | b1 | -6.049999999999E-01 | b2 | -2.015833949296E-13 |
| b3 | 2.055469869111E-13  | b4 | -9.709644291686E-14 | b5 | 2.408296093321E-14  |
| b6 | -3.189943660088E-15 | b7 | 2.092960714398E-16  | b8 | -5.157913154542E-18 |





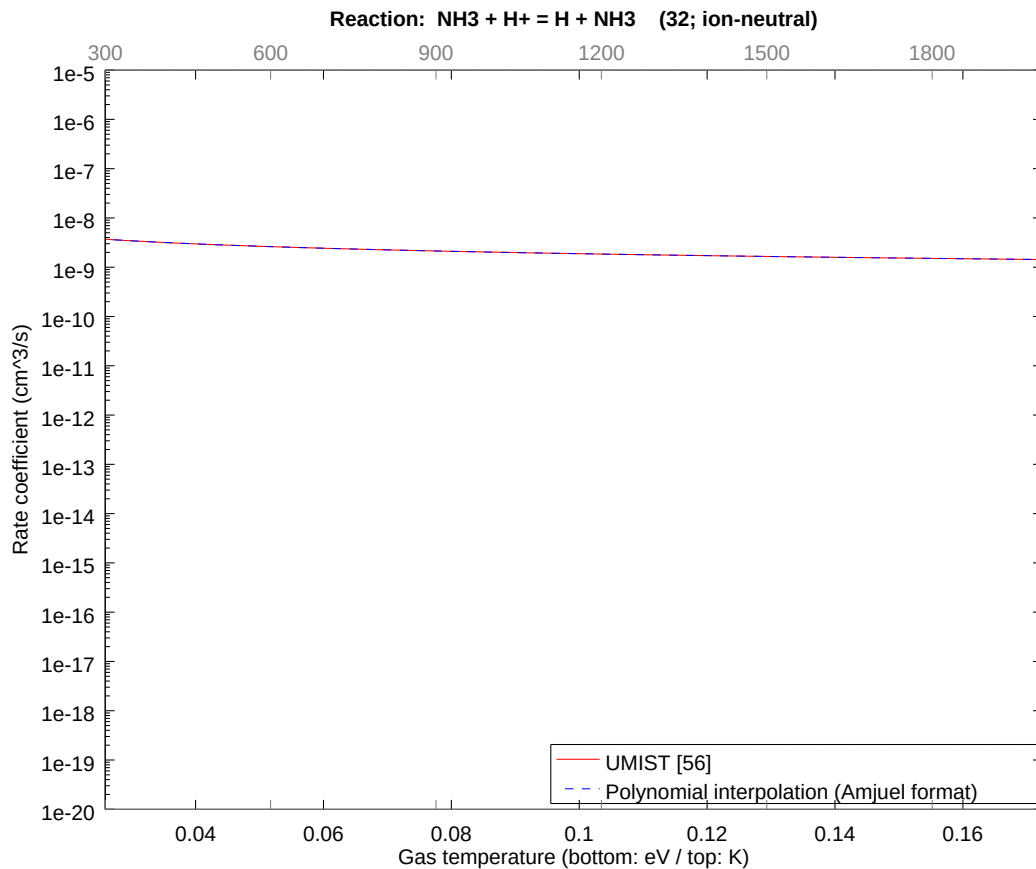
## 2.49 Reaction 31 $e + N_2H^+ = N_2 + H$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.678575516040E+01 | b1 | -7.199999999999E-01 | b2 | -3.880480352395E-13 |
| b3 | 3.969878080538E-13  | b4 | -2.019917061732E-13 | b5 | 5.646344667153E-14  |
| b6 | -8.799268909270E-15 | b7 | 7.147944647556E-16  | b8 | -2.350156970186E-17 |



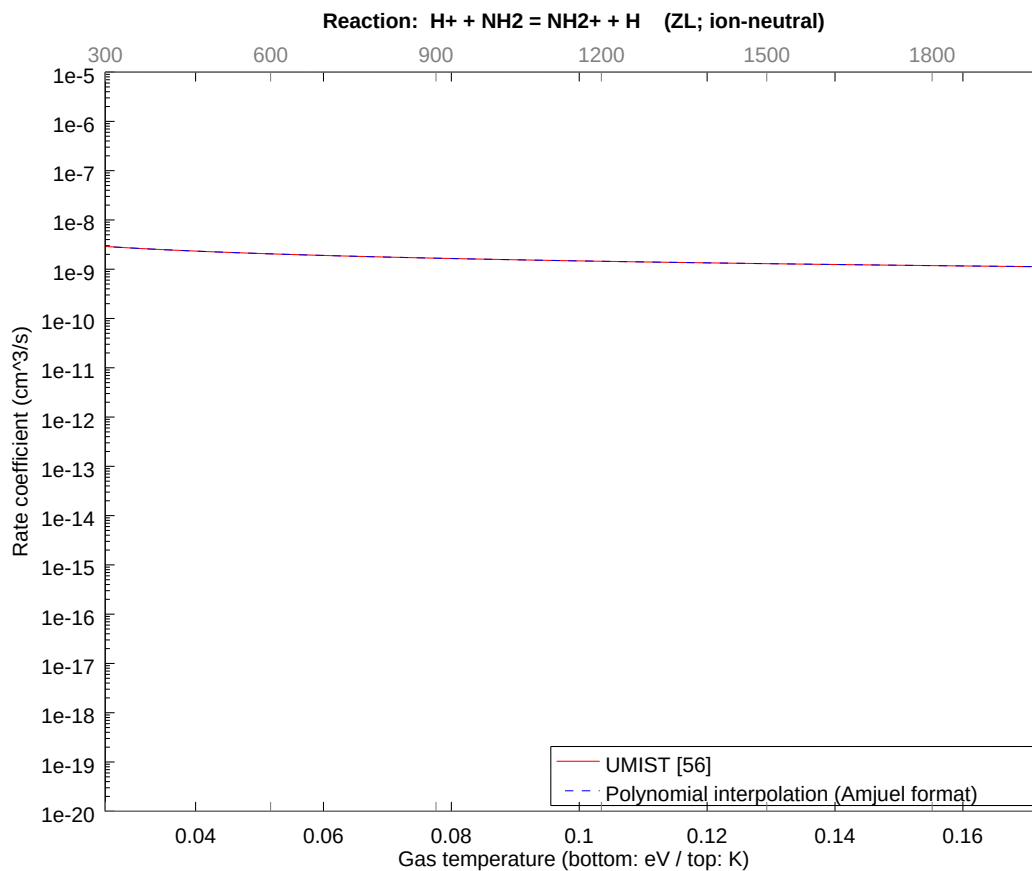
## 2.50 Reaction 32 $\text{NH}_3 + \text{H}^+ = \text{H} + \text{NH}_3$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.124236116983E+01 | b1 | -5.000000049540E-01 | b2 | -6.581921337736E-09 |
| b3 | -4.941457323182E-09 | b4 | -2.292305524667E-09 | b5 | -6.727088691196E-10 |
| b6 | -1.219481205389E-10 | b7 | -1.248479911283E-11 | b8 | -5.526669688591E-13 |



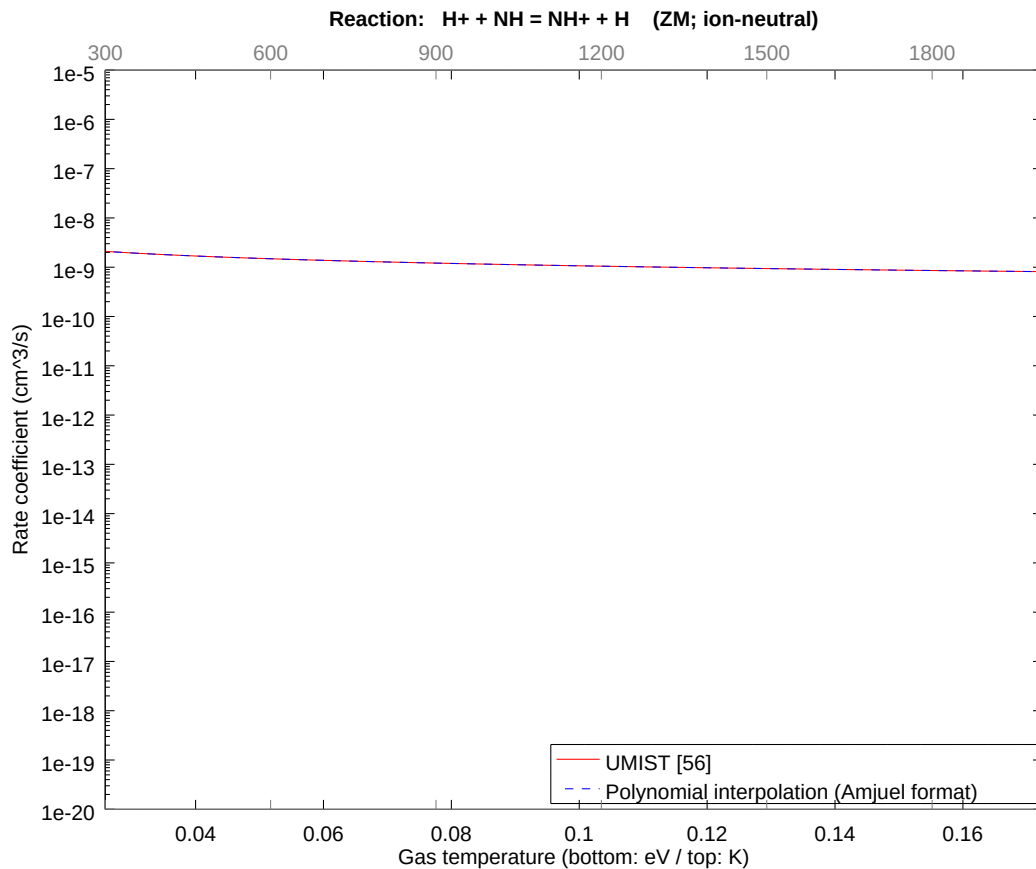
## 2.51 Reaction ZL $\text{H}^+ + \text{NH}_2 = \text{NH}_2^+ + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.148598325252E+01 | b1 | -5.000000052634E-01 | b2 | -7.274686331173E-09 |
| b3 | -5.686468493740E-09 | b4 | -2.749065636477E-09 | b5 | -8.416173449789E-10 |
| b6 | -1.593528795054E-10 | b7 | -1.706404308800E-11 | b8 | -7.914454258785E-13 |



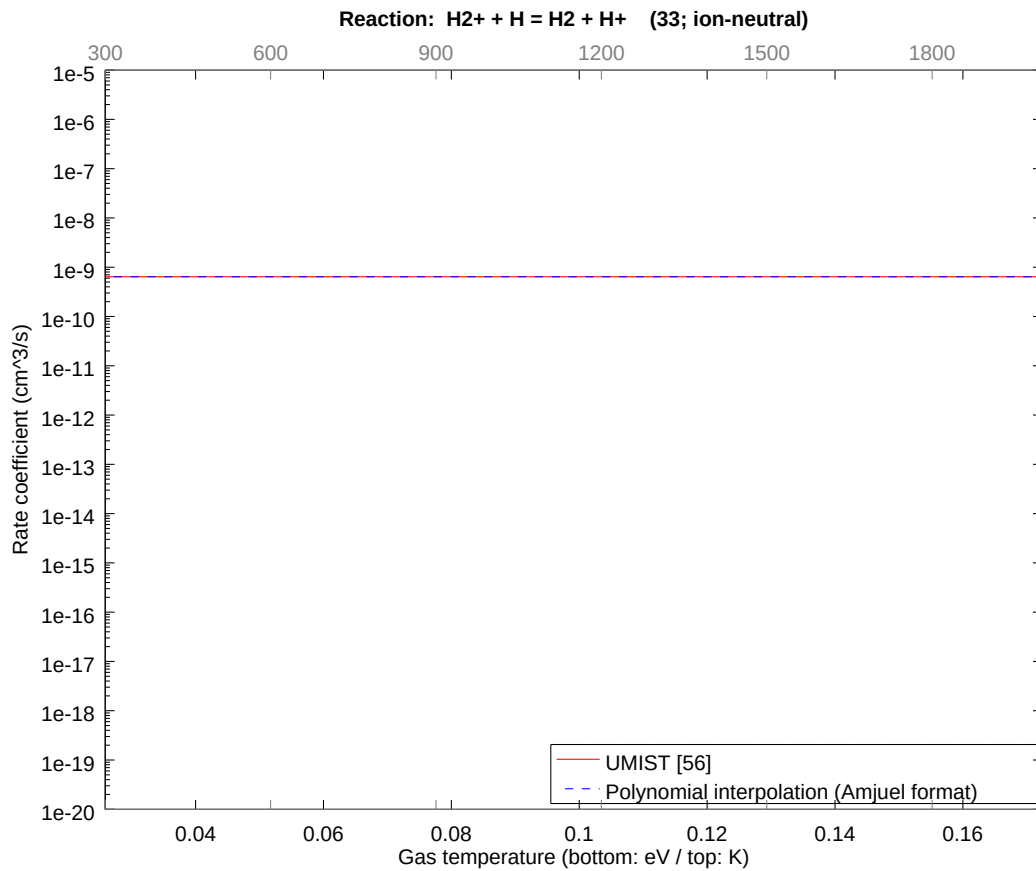
## 2.52 Reaction ZM $\text{H}^+ + \text{NH} = \text{NH}^+ + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.180875664609E+01 | b1 | -5.000000092849E-01 | b2 | -1.261517857558E-08 |
| b3 | -9.699489564652E-09 | b4 | -4.615551314815E-09 | b5 | -1.391879349947E-09 |
| b6 | -2.597688881768E-10 | b7 | -2.743376818213E-11 | b8 | -1.255322335567E-12 |



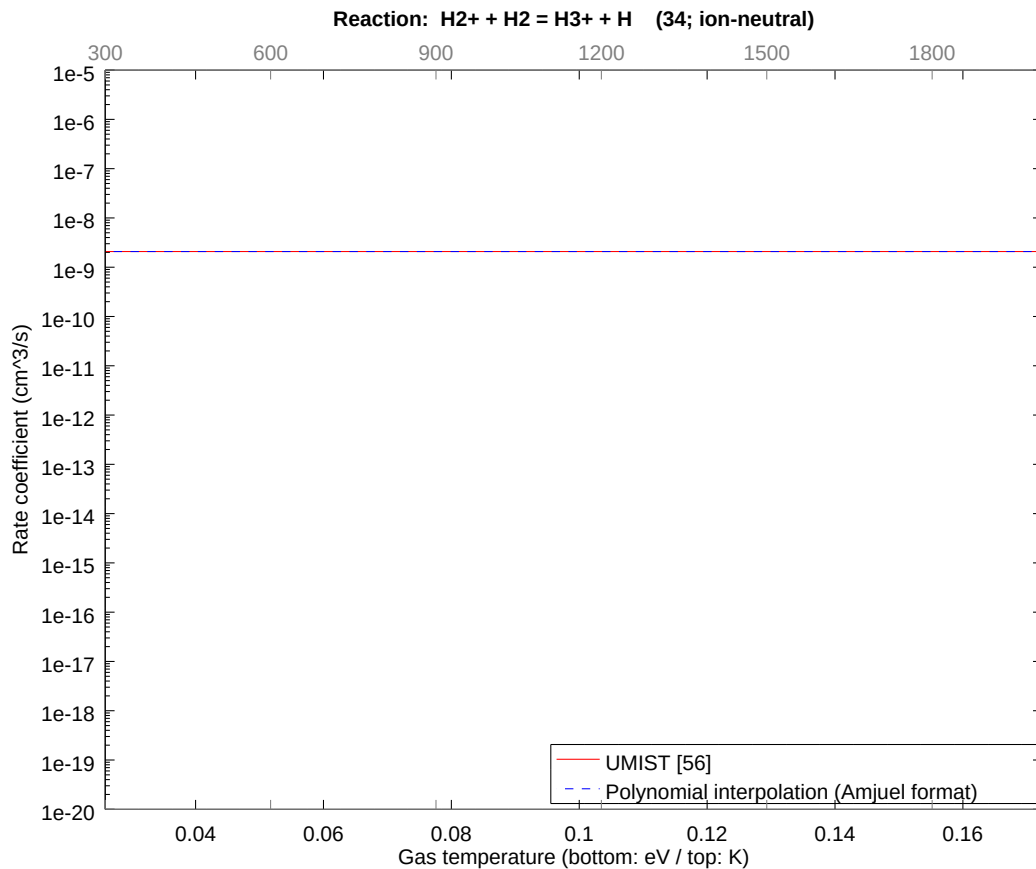
## 2.53 Reaction 33 $\text{H}_2^+ + \text{H} = \text{H}_2 + \text{H}^+$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.116955293957E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



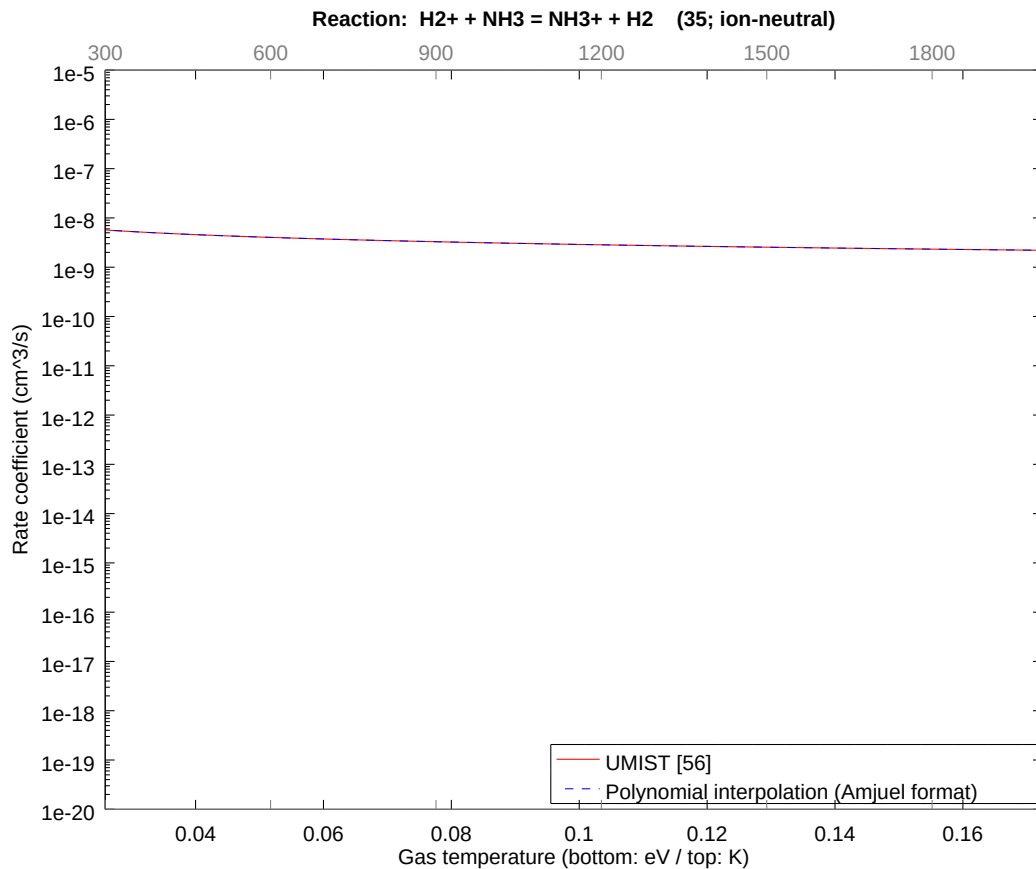
## 2.54 Reaction 34 $\text{H}_2^+ + \text{H}_2 = \text{H}_3^+ + \text{H}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -1.999089794323E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



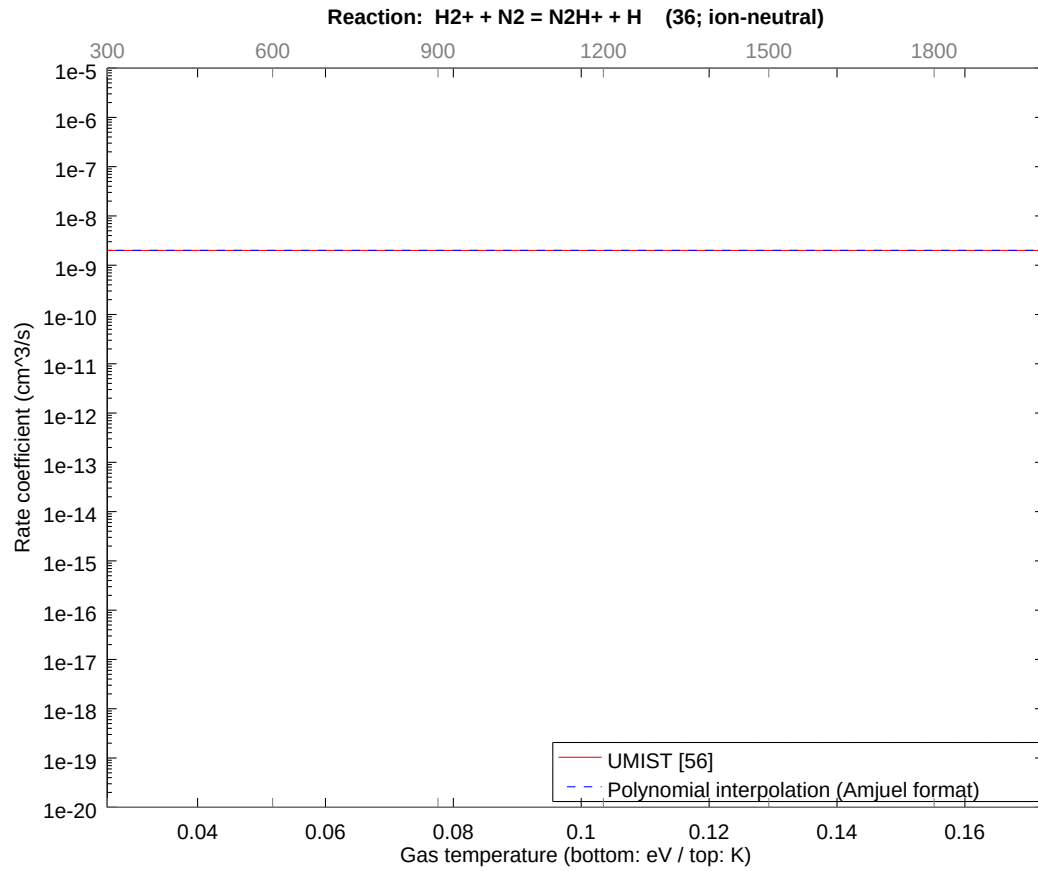
## 2.55 Reaction 35 $\text{H}_2^+ + \text{NH}_3 = \text{NH}_3^+ + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.081022781504E+01 | b1 | -5.000000063682E-01 | b2 | -8.716351898605E-09 |
| b3 | -6.744375376163E-09 | b4 | -3.226162467434E-09 | b5 | -9.769045577405E-10 |
| b6 | -1.828822359175E-10 | b7 | -1.935531934856E-11 | b8 | -8.868789870933E-13 |



## 2.56 Reaction 36 $\text{H}_2^+ + \text{N}_2 = \text{N}_2\text{H}^+ + \text{H}$

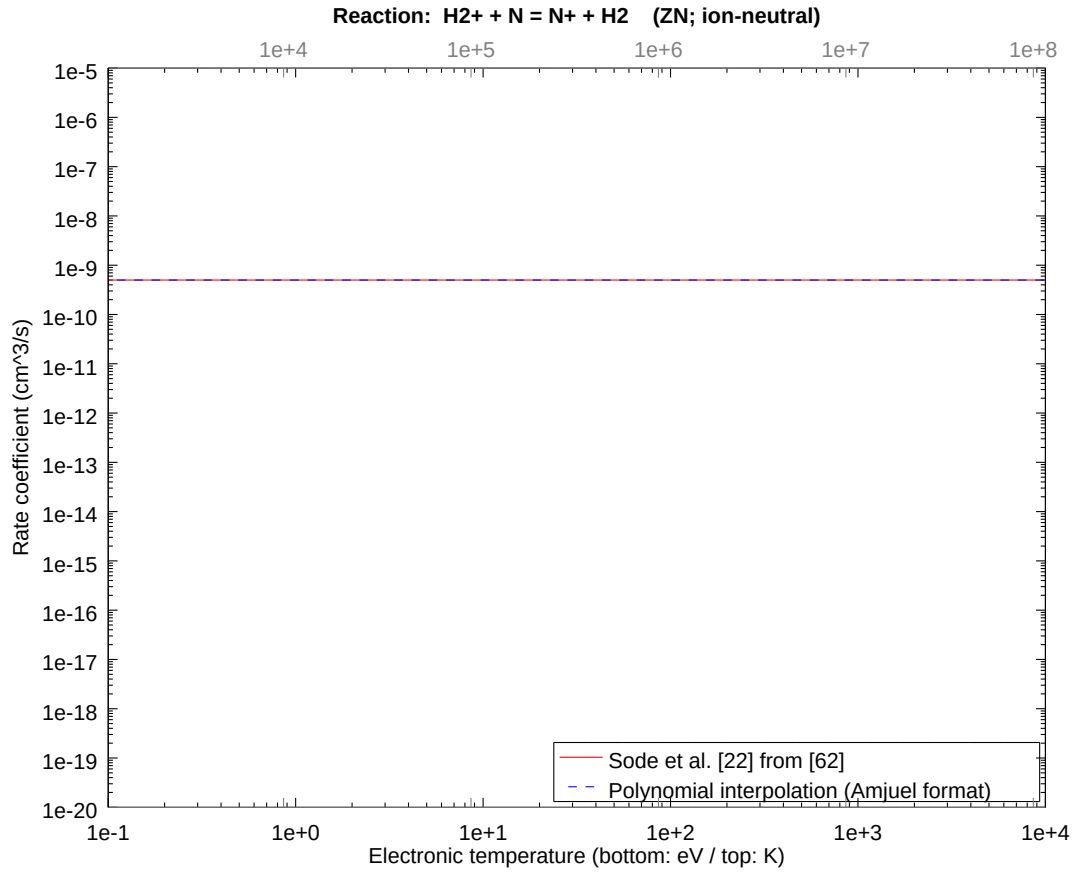
|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.003011865639E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |





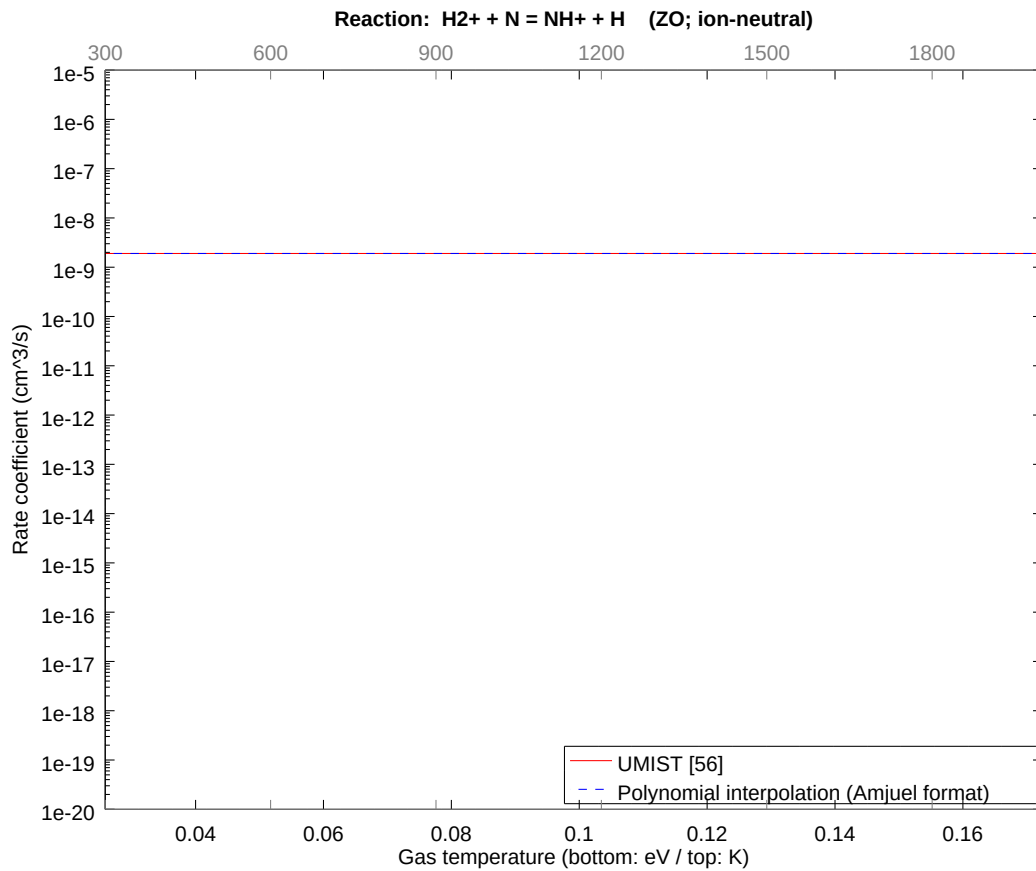
## 2.57 Reaction $\text{ZN H2}^+ + \text{N} = \text{N}^+ + \text{H2}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.141641301751E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



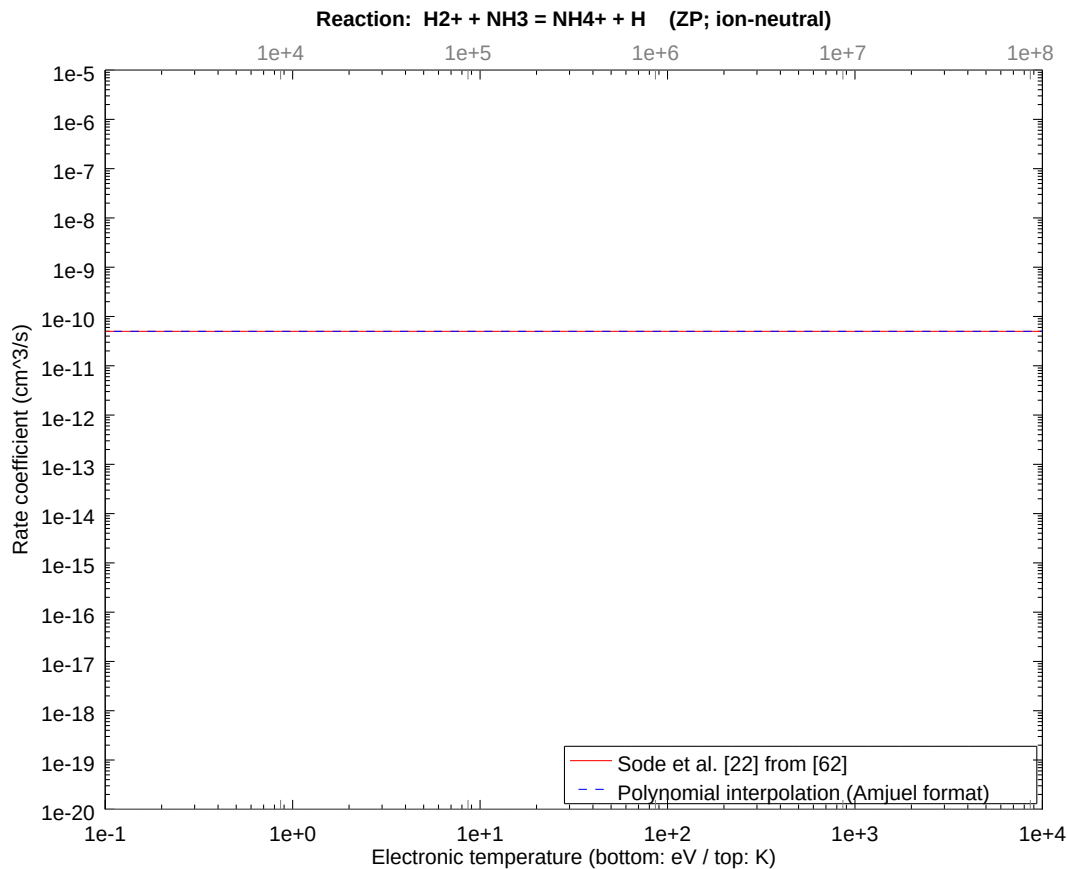
## 2.58 Reaction ZO H2+ + N = NH+ + H

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.008141195077E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



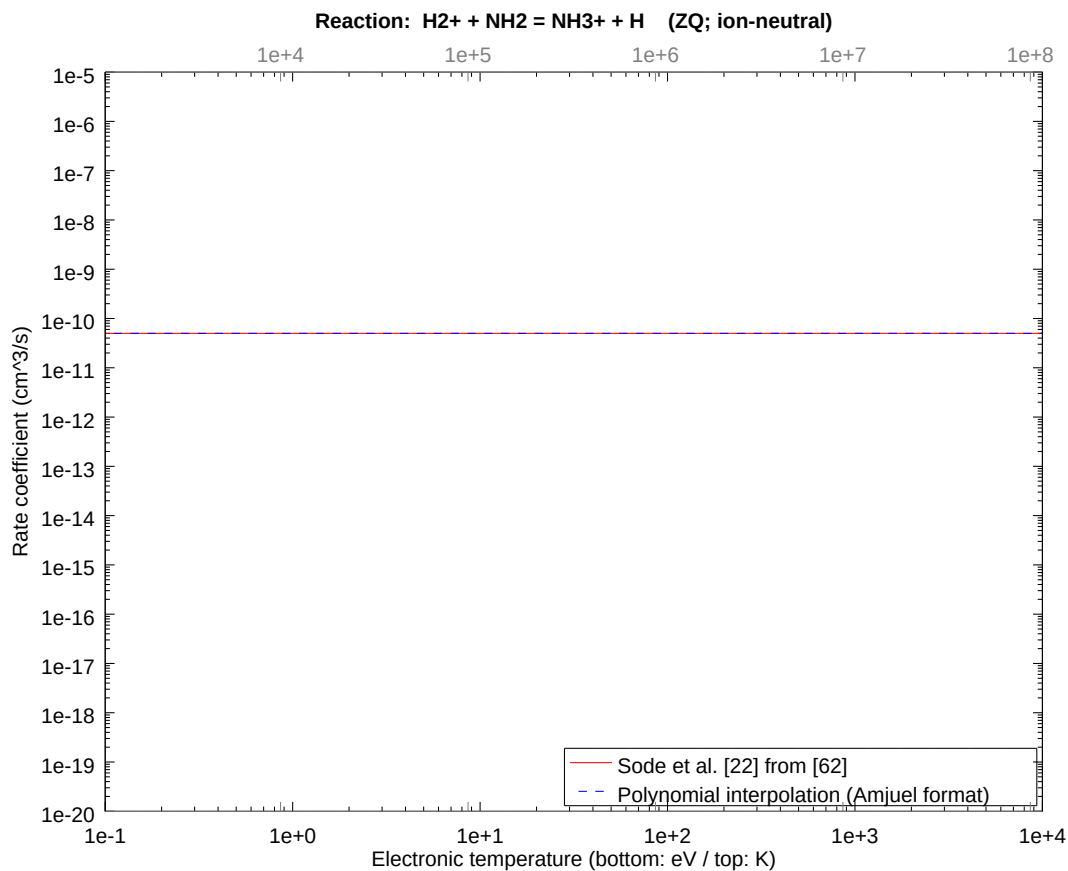
## 2.59 Reaction ZP $\text{H}_2^+ + \text{NH}_3 = \text{NH}_4^+ + \text{H}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.371899811050E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



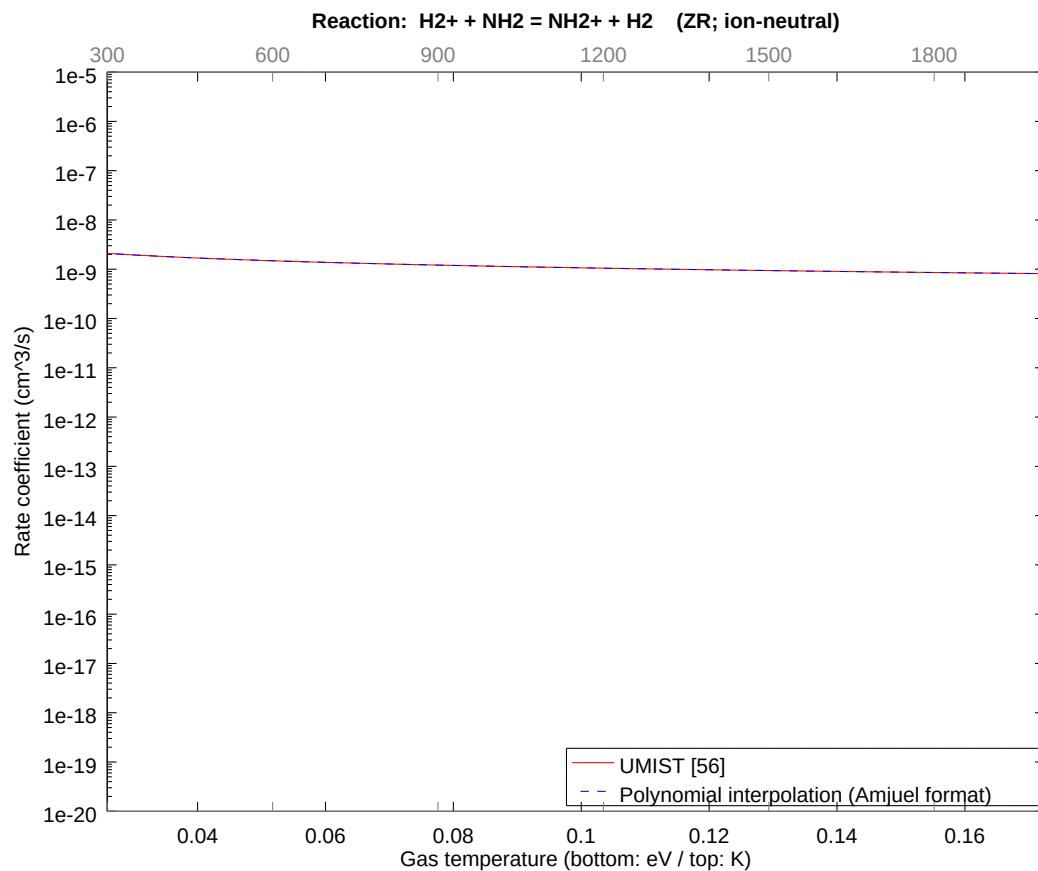
## 2.60 Reaction ZQ $\text{H}_2^+ + \text{NH}_2 = \text{NH}_3^+ + \text{H}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.371899811050E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



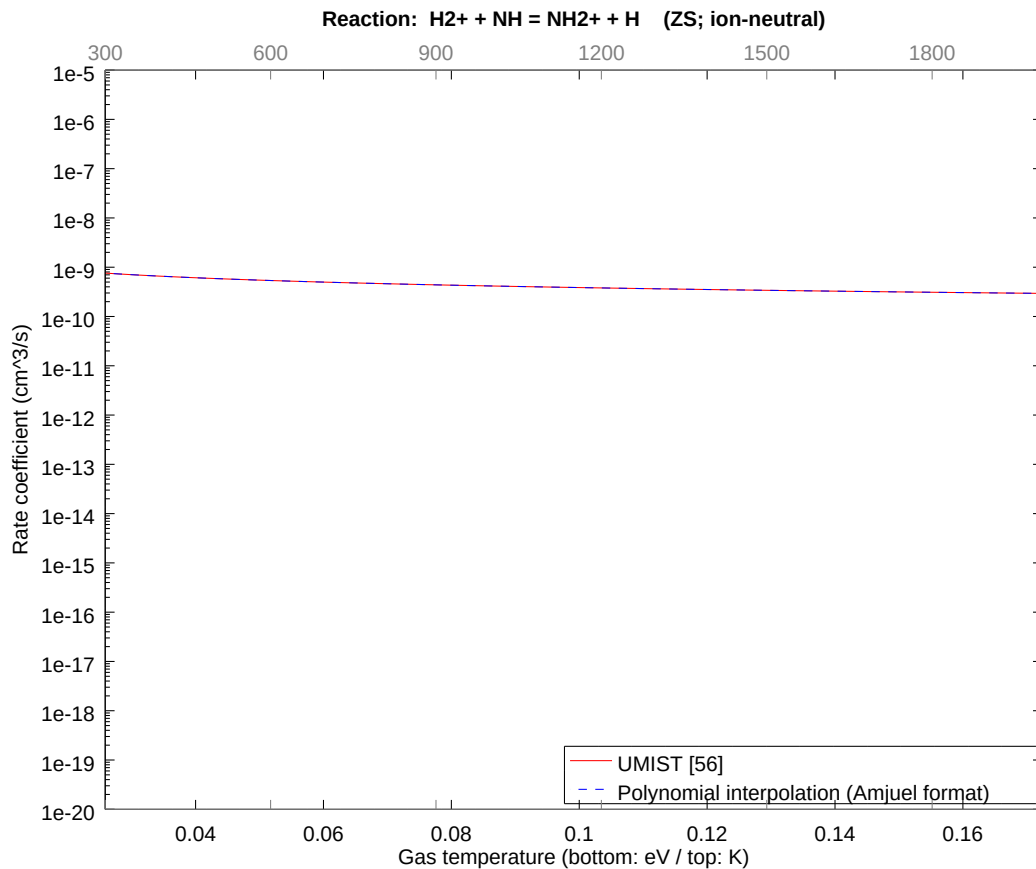
## 2.61 Reaction ZR $\text{H}_2^+ + \text{NH}_2 = \text{NH}_2^+ + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.180875664609E+01 | b1 | -5.000000092849E-01 | b2 | -1.261517857558E-08 |
| b3 | -9.699489564652E-09 | b4 | -4.615551314815E-09 | b5 | -1.391879349947E-09 |
| b6 | -2.597688881768E-10 | b7 | -2.743376818213E-11 | b8 | -1.255322335567E-12 |



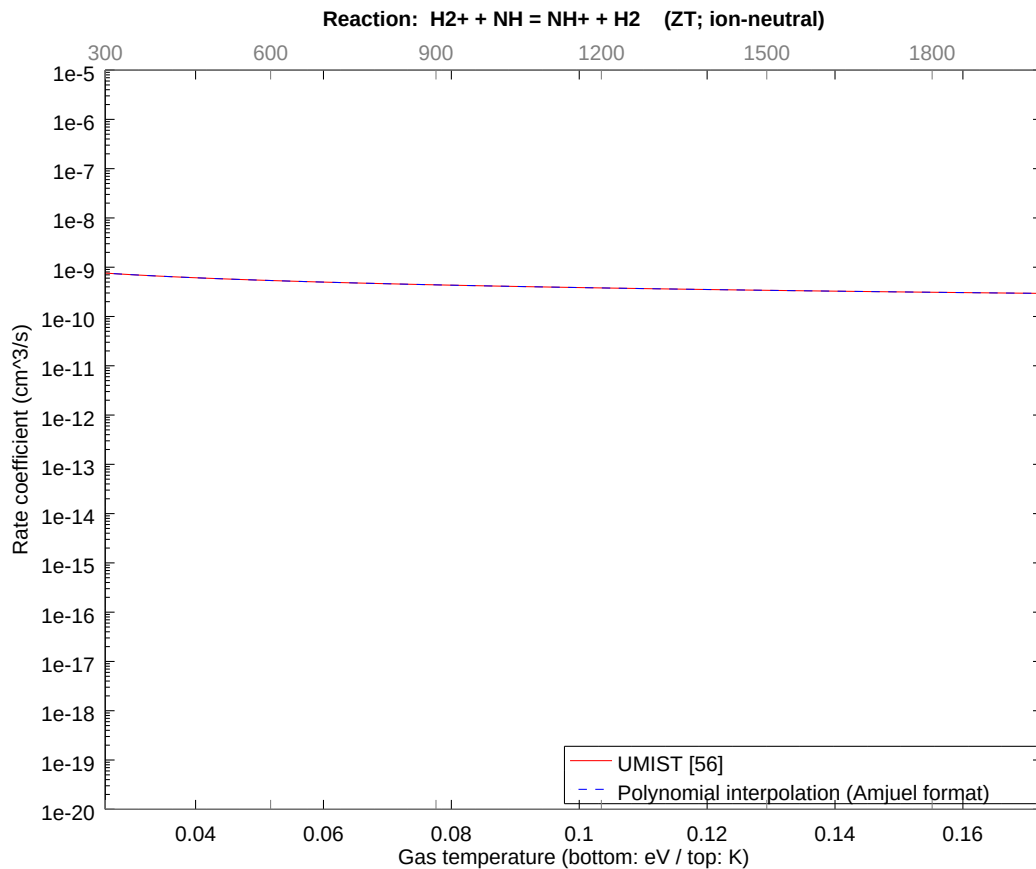
## 2.62 Reaction ZS H2+ + NH = NH2+ + H

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.282513083456E+01 | b1 | -5.000000030980E-01 | b2 | -4.172533795604E-09 |
| b3 | -3.178279948201E-09 | b4 | -1.497002555752E-09 | b5 | -4.463695735110E-10 |
| b6 | -8.227558117594E-11 | b7 | -8.571561358925E-12 | b8 | -3.865197085583E-13 |



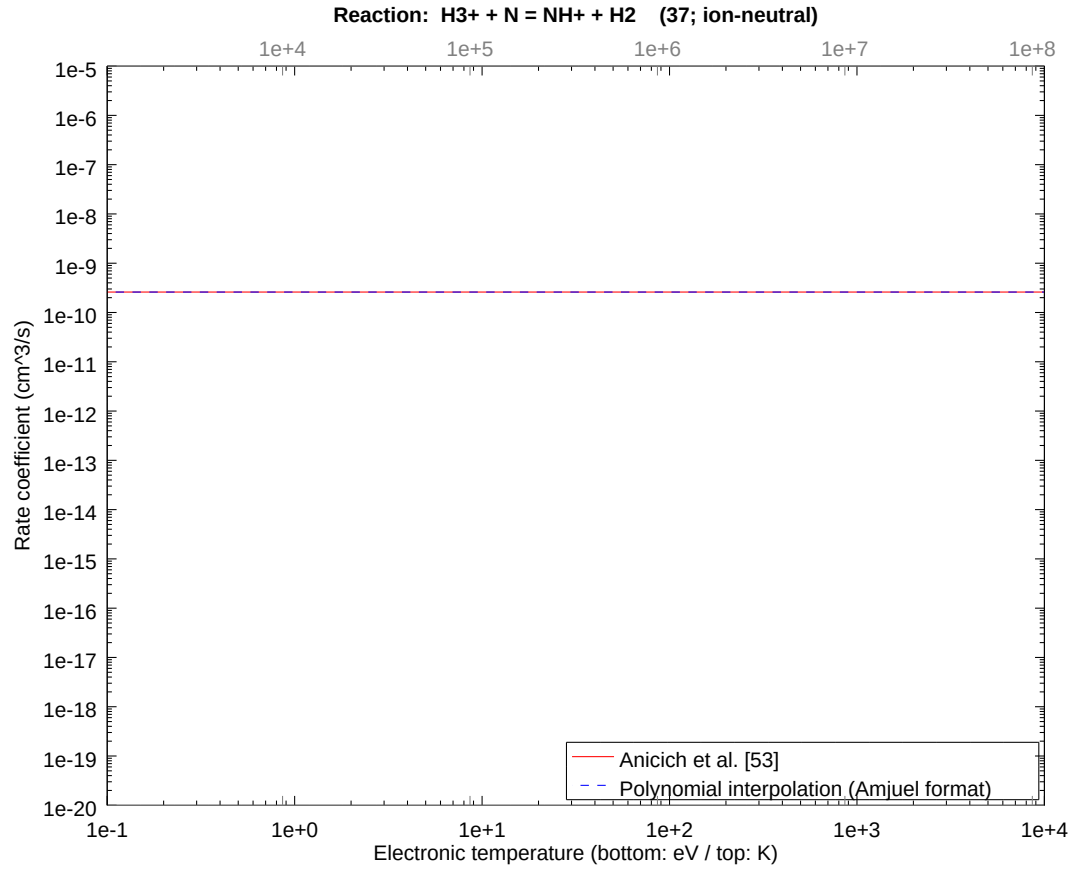
## 2.63 Reaction ZT $\text{H}_2^+ + \text{NH} = \text{NH}^+ + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.282513083456E+01 | b1 | -5.000000030980E-01 | b2 | -4.172533795604E-09 |
| b3 | -3.178279948201E-09 | b4 | -1.497002555752E-09 | b5 | -4.463695735110E-10 |
| b6 | -8.227558117594E-11 | b7 | -8.571561358925E-12 | b8 | -3.865197085583E-13 |



## 2.64 Reaction 37 $\text{H3}^+ + \text{N} = \text{NH}^+ + \text{H2}$

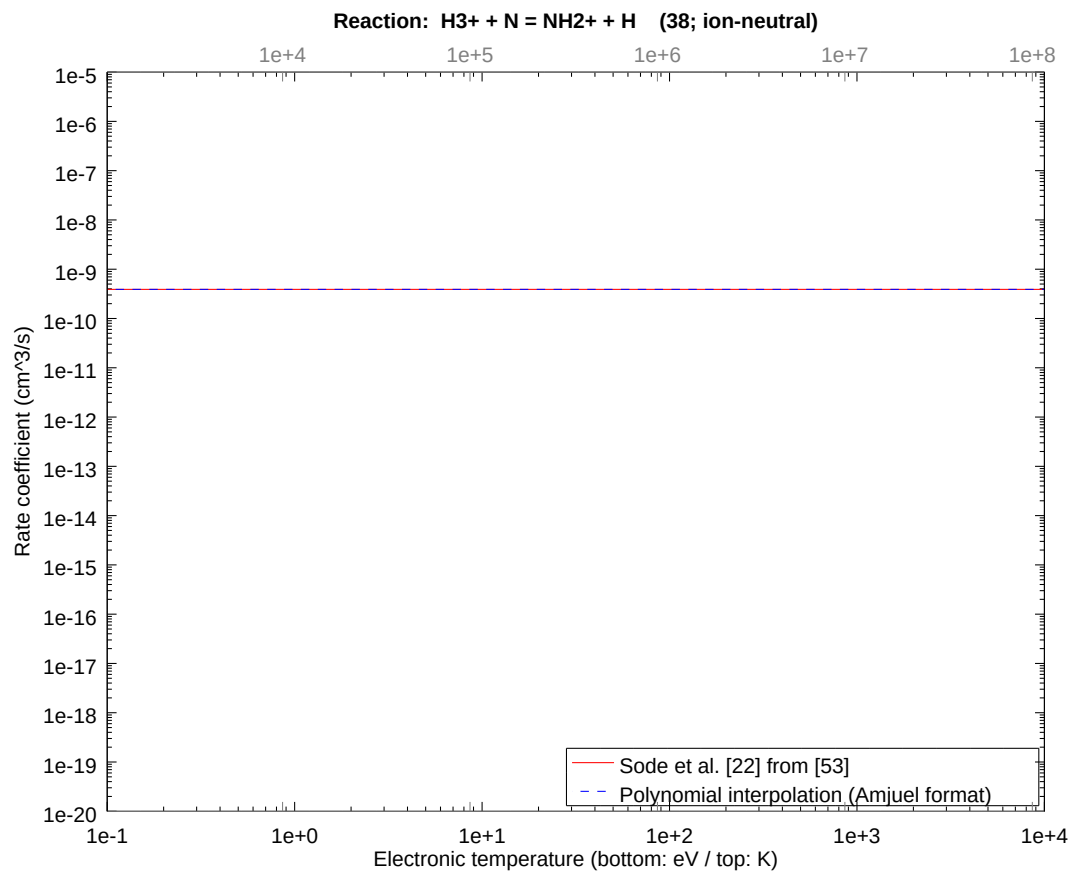
|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.207033948491E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |





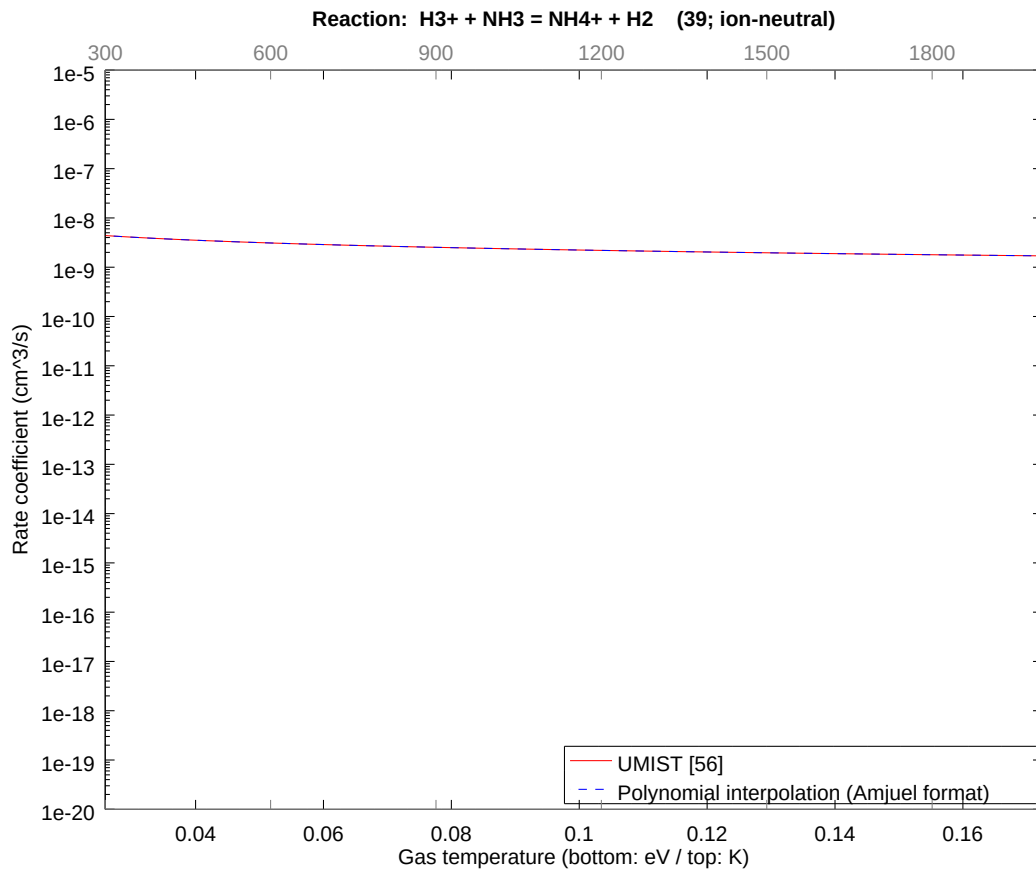
## 2.65 Reaction 38 $\text{H3}^+ + \text{N} = \text{NH2}^+ + \text{H}$ (38; ion-neutral)

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.166487437680E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



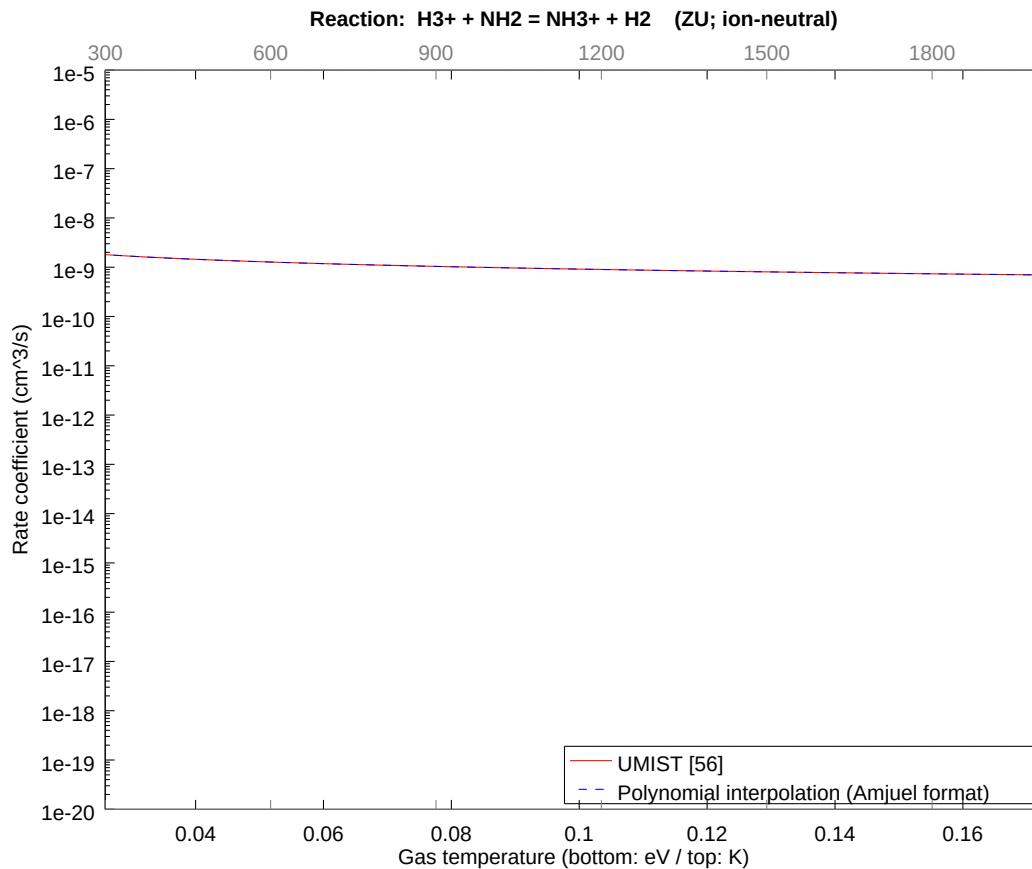
## 2.66 Reaction 39 $\text{H3}^+ + \text{NH3} = \text{NH4}^+ + \text{H2}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.107136476258E+01 | b1 | -5.000000055727E-01 | b2 | -7.437148651905E-09 |
| b3 | -5.609819632118E-09 | b4 | -2.615584427083E-09 | b5 | -7.718869172379E-10 |
| b6 | -1.408075560704E-10 | b7 | -1.451862542892E-11 | b8 | -6.479800800281E-13 |



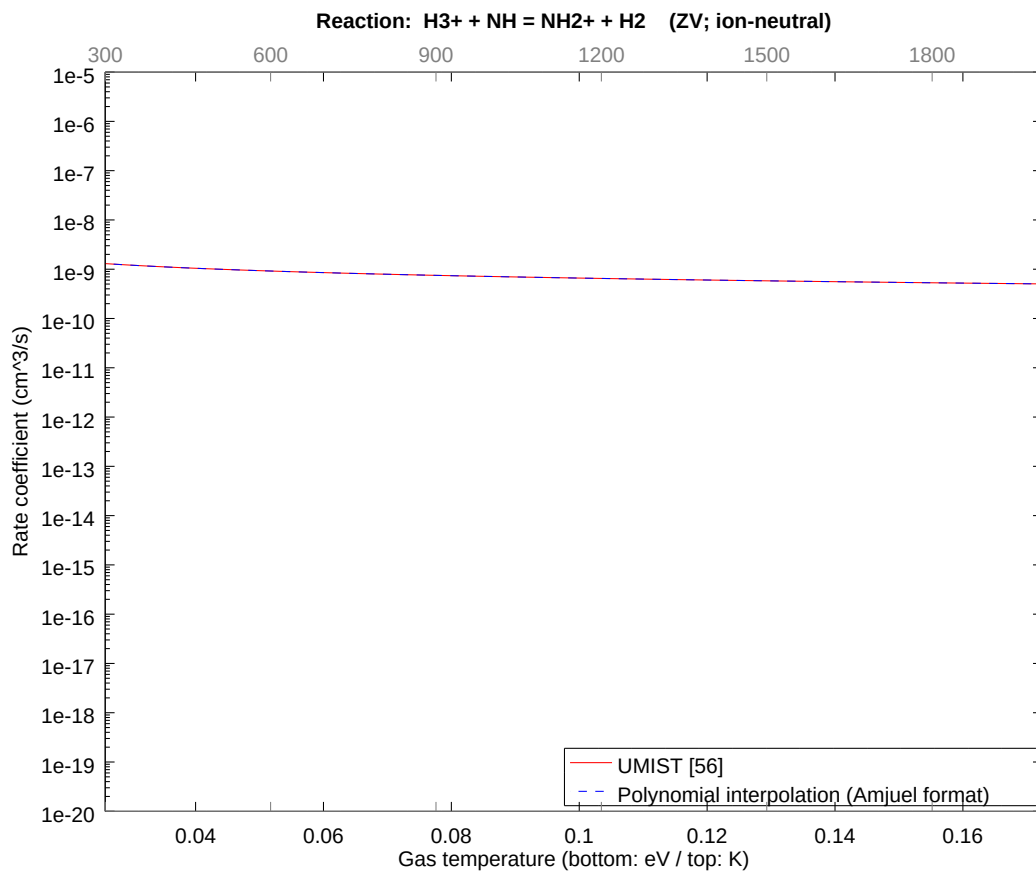
## 2.67 Reaction ZU $\text{H3}^+ + \text{NH2} = \text{NH3}^+ + \text{H2}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.196290732349E+01 | b1 | -5.000000017280E-01 | b2 | -2.428579288053E-09 |
| b3 | -1.921777919375E-09 | b4 | -9.364377031497E-10 | b5 | -2.877434523747E-10 |
| b6 | -5.445759602353E-11 | b7 | -5.805294394799E-12 | b8 | -2.669522116361E-13 |



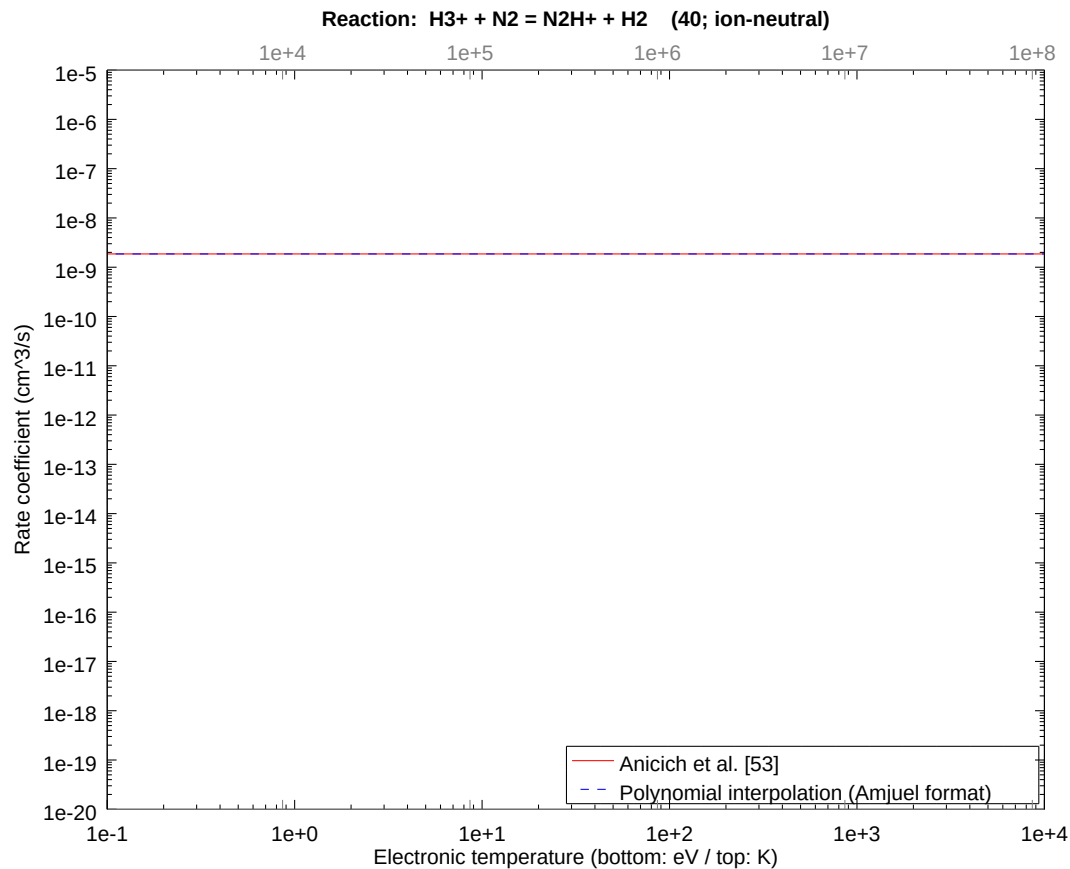
## 2.68 Reaction ZV $\text{H3}^+ + \text{NH} = \text{NH2}^+ + \text{H2}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.228832972476E+01 | b1 | -5.000000042028E-01 | b2 | -5.569379050783E-09 |
| b3 | -4.161520963593E-09 | b4 | -1.916979254164E-09 | b5 | -5.572768434265E-10 |
| b6 | -9.981865098488E-11 | b7 | -1.007015872766E-11 | b8 | -4.380051097582E-13 |



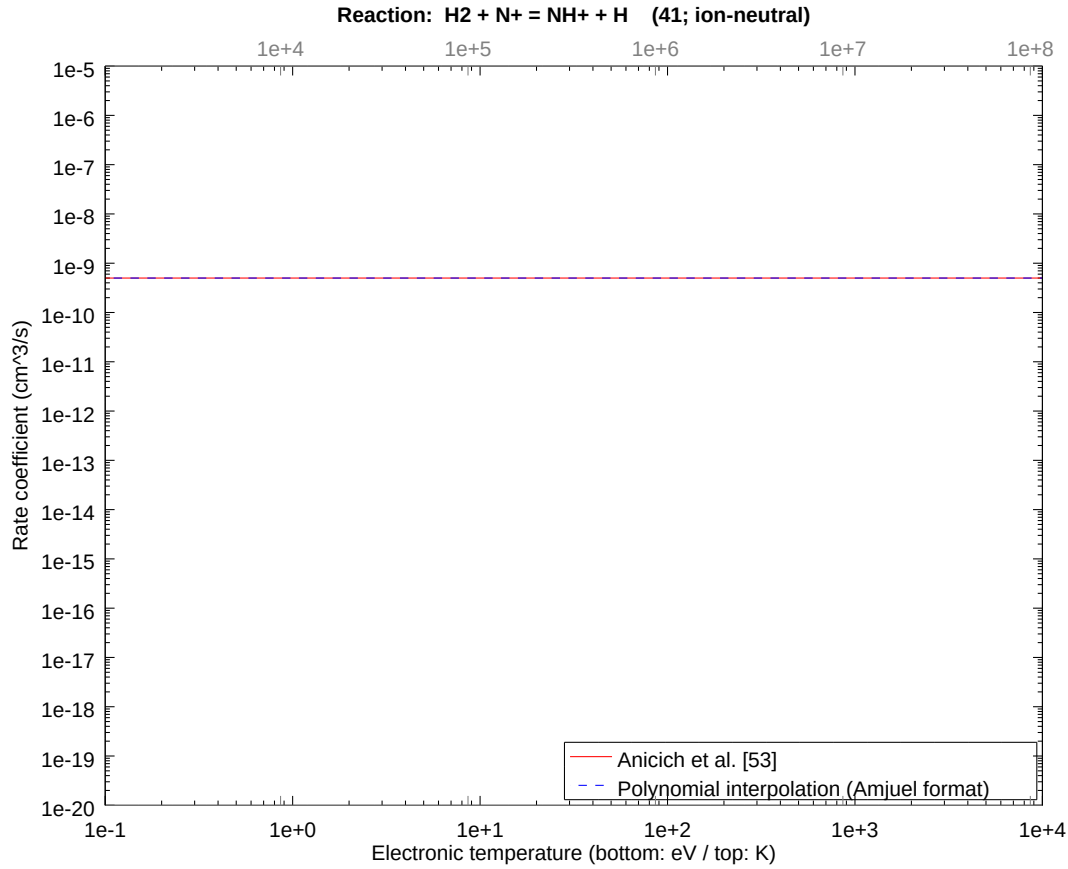
## 2.69 Reaction 40 $\text{H3}^+ + \text{N2} = \text{N2H}^+ + \text{H2}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.010268934922E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



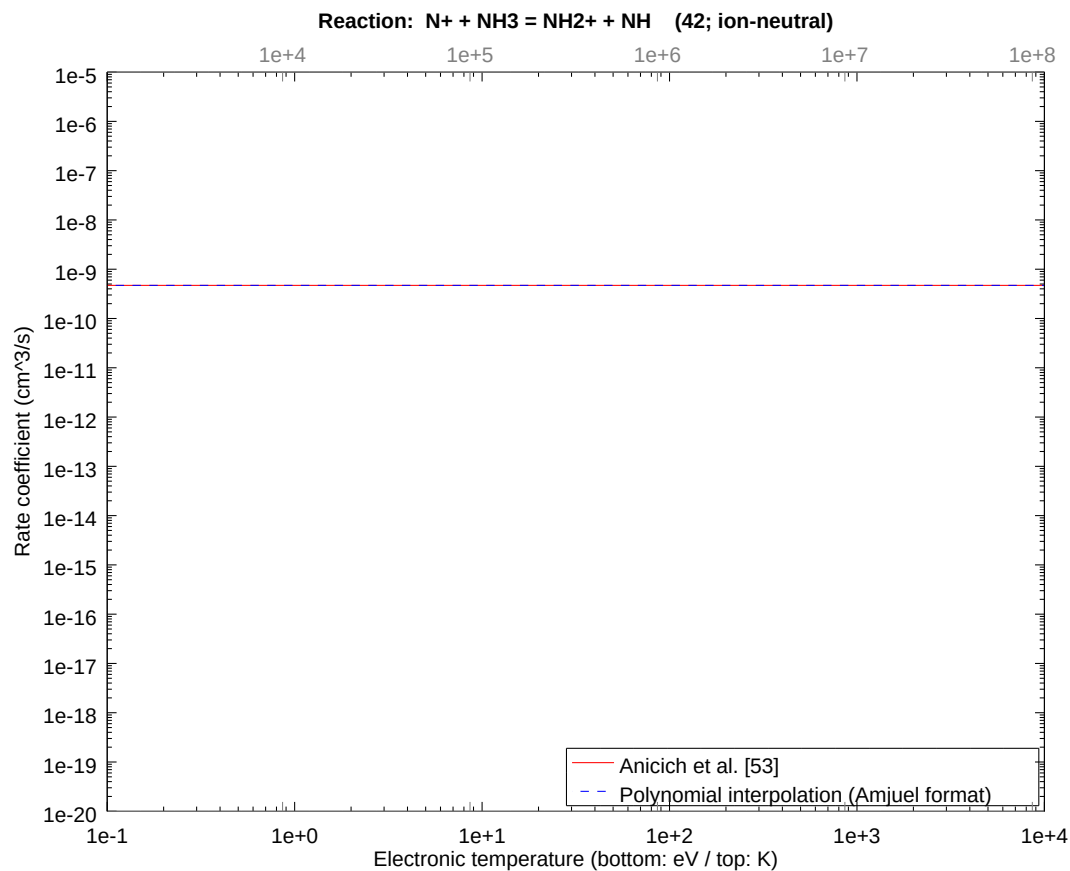
## 2.70 Reaction 41 $\text{H}_2 + \text{N}^+ = \text{NH}^+ + \text{H}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.141641301751E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



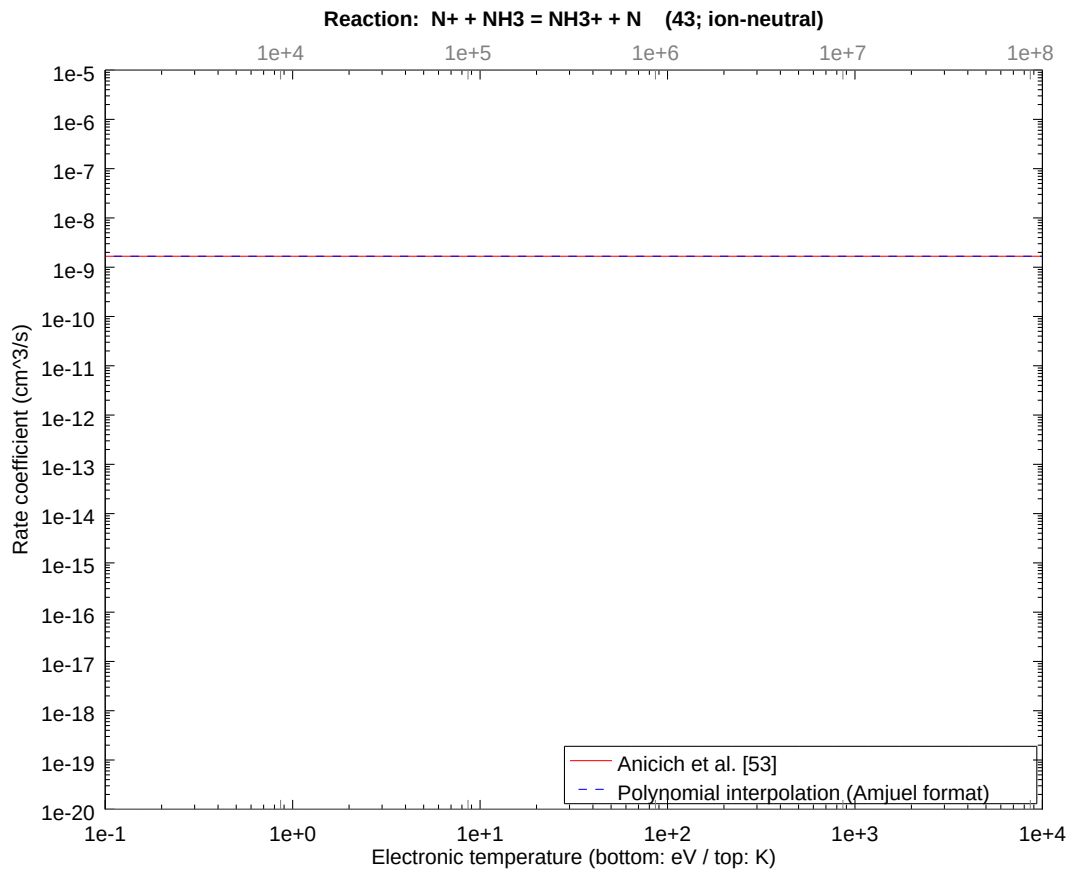
## 2.71 Reaction 42 $\text{N}^+ + \text{NH}_3 = \text{NH}_2^+ + \text{NH}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.147828842122E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



## 2.72 Reaction 43 $\text{N}^+ + \text{NH}_3 = \text{NH}_3^+ + \text{N}$

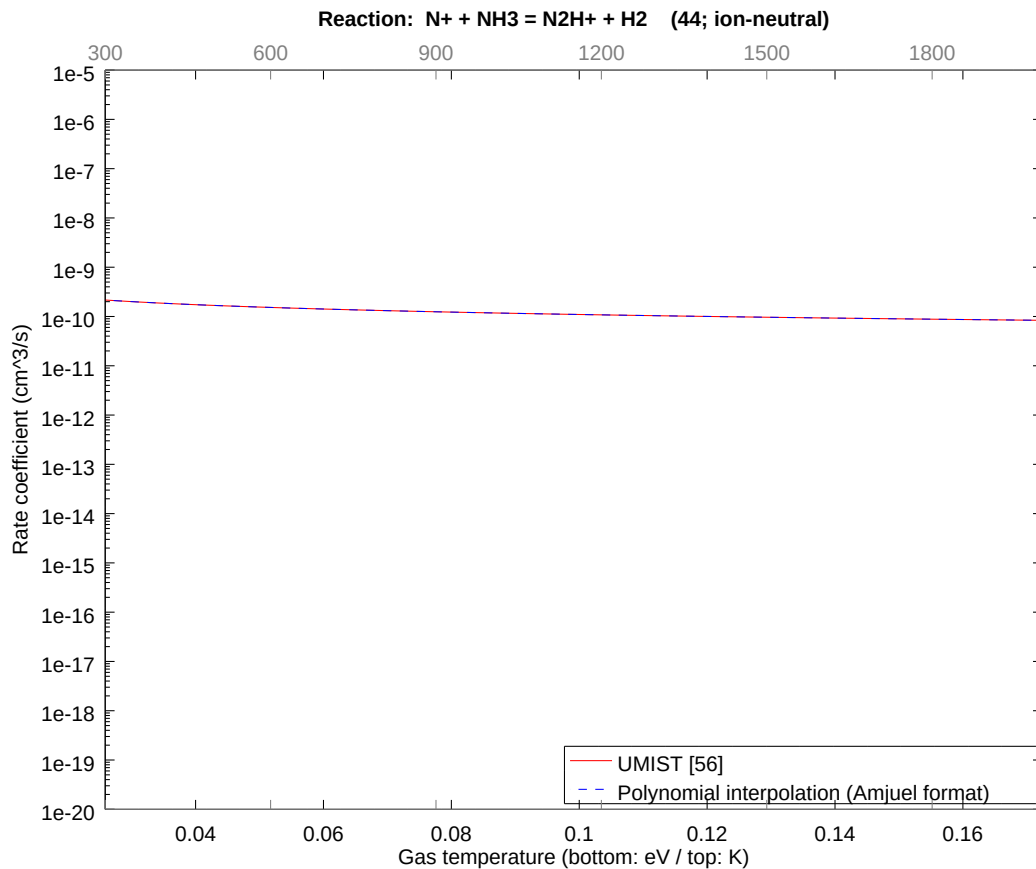
|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.021044221052E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |





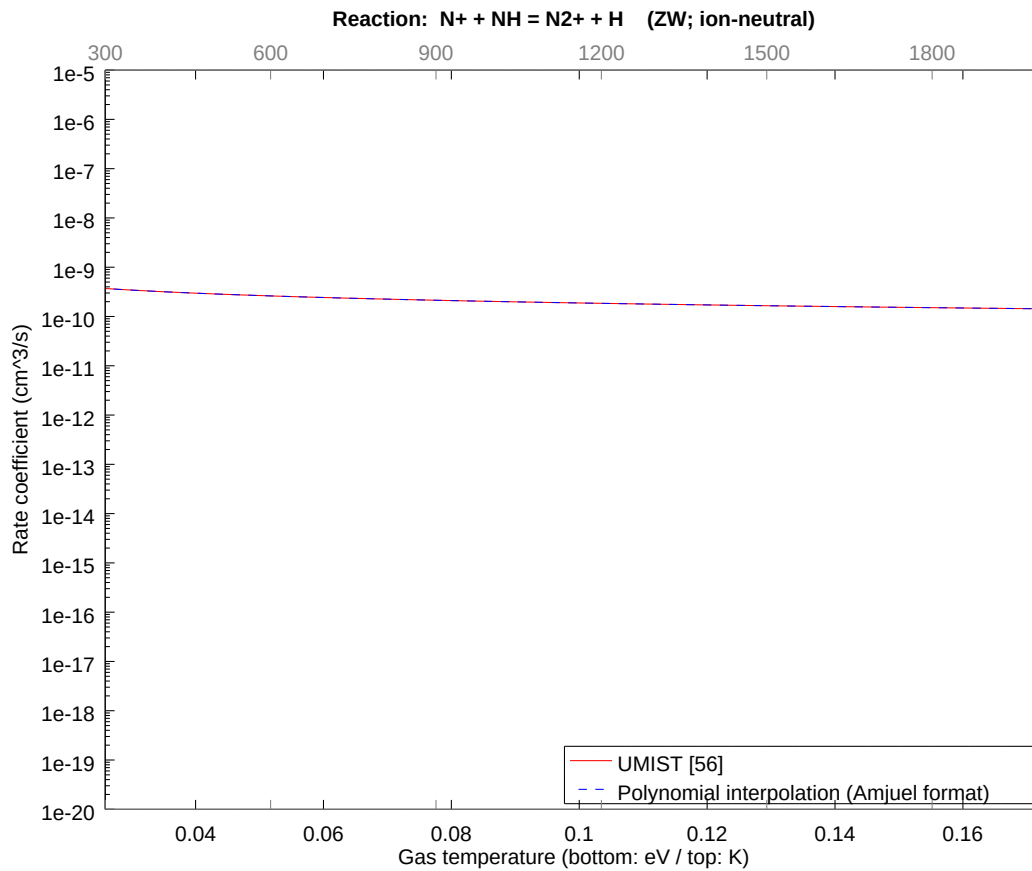
## 2.73 Reaction 44 $\text{N}^+ + \text{NH}_3 = \text{N}_2\text{H}^+ + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.408317086045E+01 | b1 | -5.000000039818E-01 | b2 | -5.322487393845E-09 |
| b3 | -4.009817581528E-09 | b4 | -1.860909331983E-09 | b5 | -5.443952121280E-10 |
| b6 | -9.796847626584E-11 | b7 | -9.908533882743E-12 | b8 | -4.308943182685E-13 |



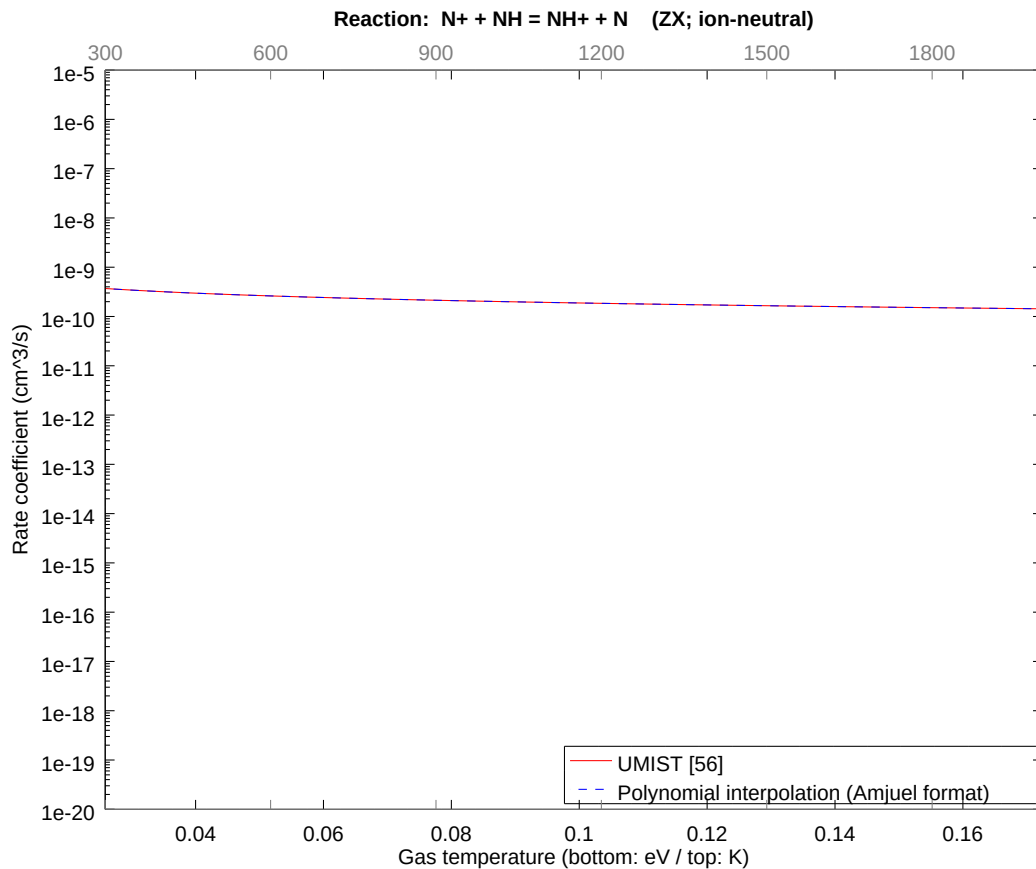
## 2.74 Reaction ZW $\text{N}^+ + \text{NH} = \text{N}_2^+ + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.354494626224E+01 | b1 | -5.000000032306E-01 | b2 | -4.347944933015E-09 |
| b3 | -3.299966277679E-09 | b4 | -1.544289864373E-09 | b5 | -4.561831738007E-10 |
| b6 | -8.305450654833E-11 | b7 | -8.519342746733E-12 | b8 | -3.768628601479E-13 |



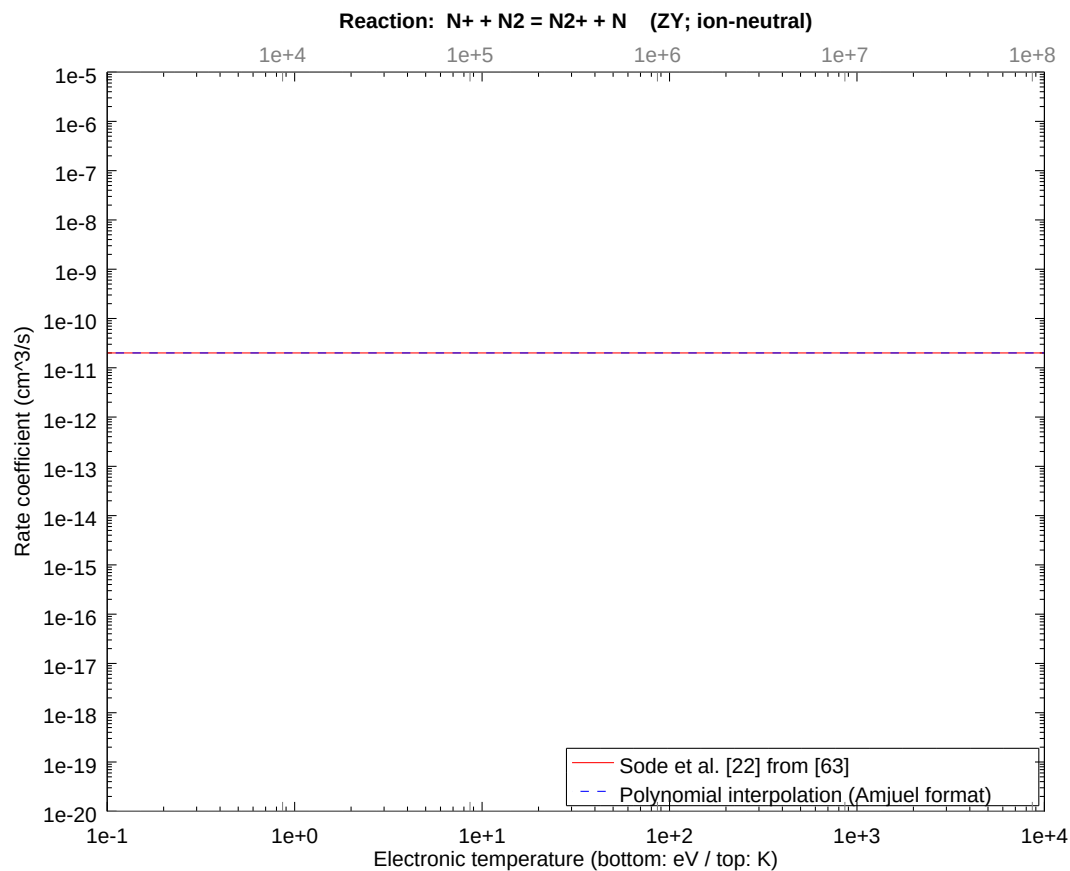
## 2.75 Reaction ZX $\text{N}^+ + \text{NH} = \text{NH}^+ + \text{N}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.354494626224E+01 | b1 | -5.000000032306E-01 | b2 | -4.347944933015E-09 |
| b3 | -3.299966277679E-09 | b4 | -1.544289864373E-09 | b5 | -4.561831738007E-10 |
| b6 | -8.305450654833E-11 | b7 | -8.519342746733E-12 | b8 | -3.768628601479E-13 |



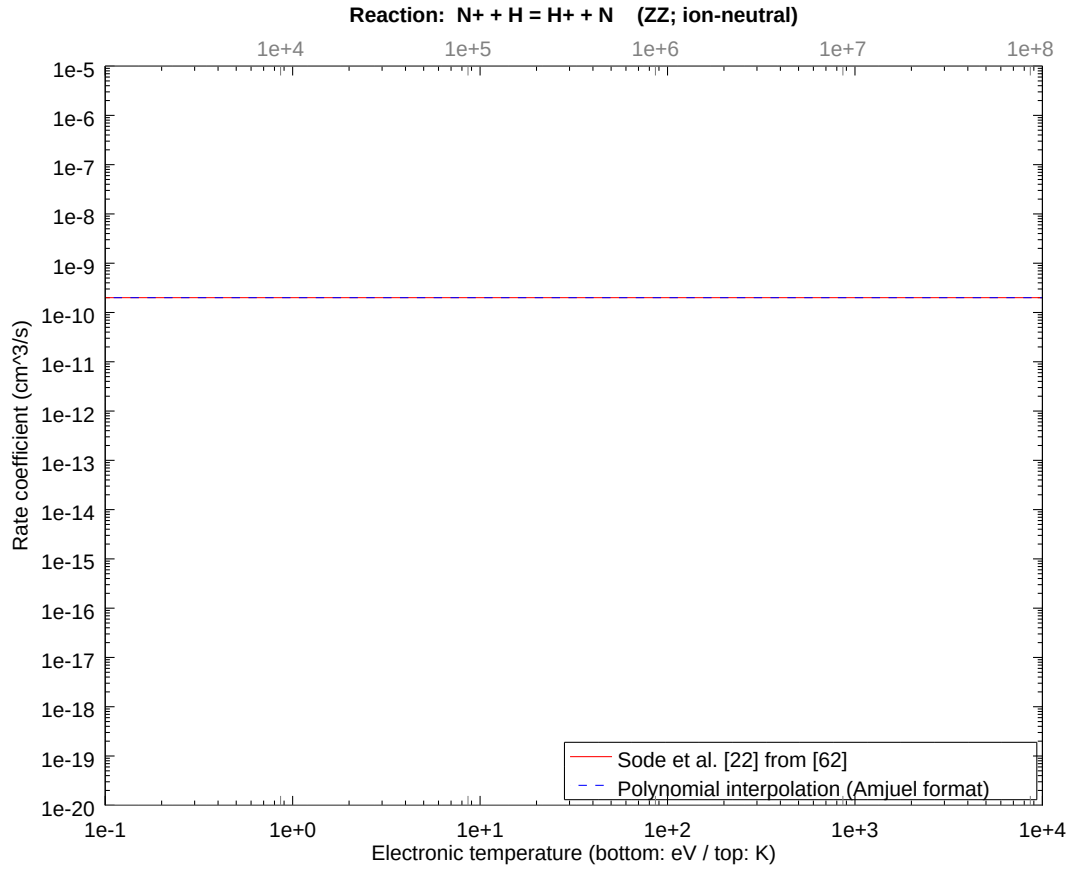
## 2.76 Reaction ZY $\text{N}^+ + \text{N}_2 = \text{N}_2^+ + \text{N}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.463528884237E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



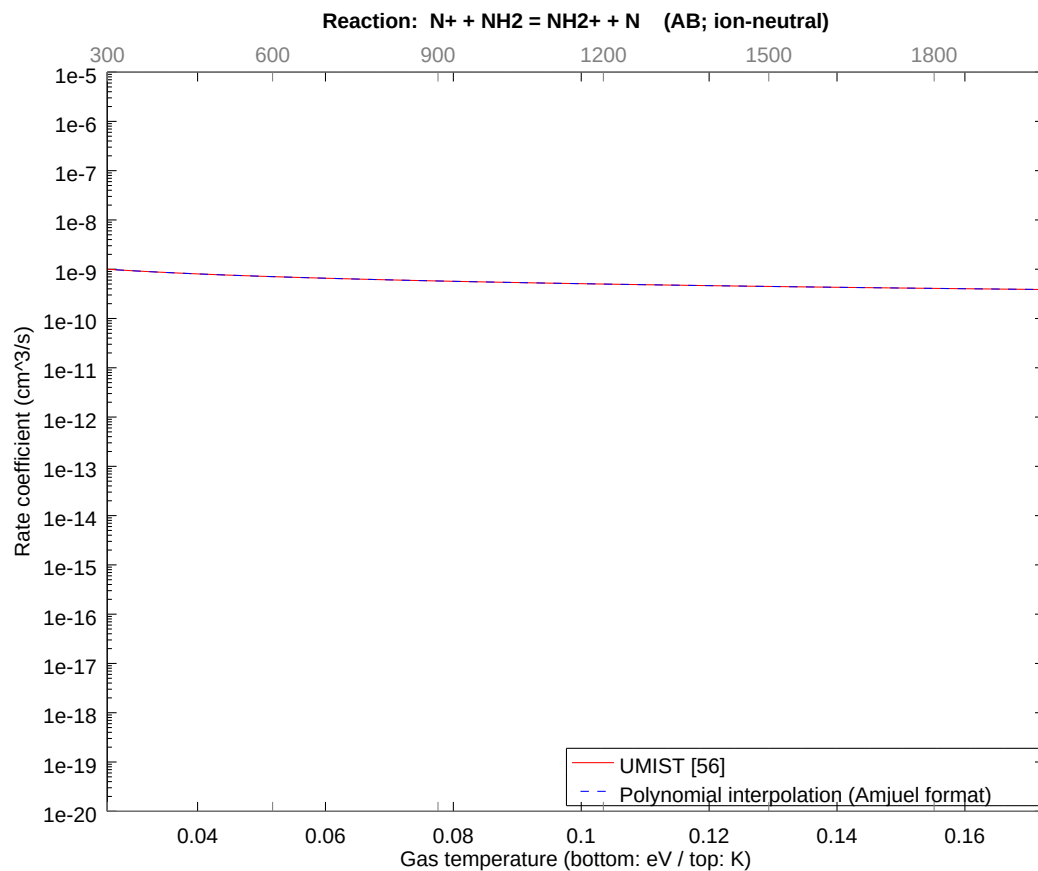
## 2.77 Reaction ZZ N+ + H = H+ + N

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.233270374938E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



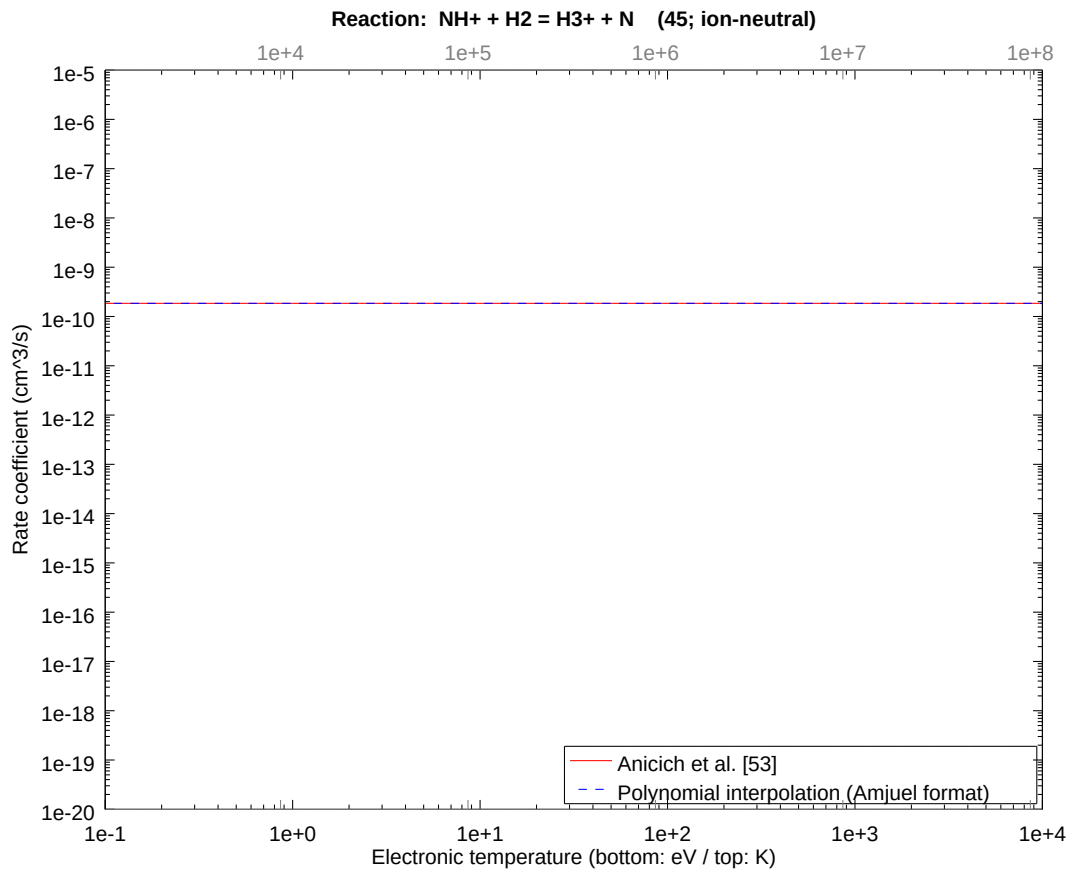
## 2.78 Reaction AB $\text{N}^+ + \text{NH}_2 = \text{NH}_2^+ + \text{N}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.255069399034E+01 | b1 | -5.000000077382E-01 | b2 | -1.048331289759E-08 |
| b3 | -8.030169368750E-09 | b4 | -3.804180693869E-09 | b5 | -1.141479368811E-09 |
| b6 | -2.119003834735E-10 | b7 | -2.225565586486E-11 | b8 | -1.012819638805E-12 |



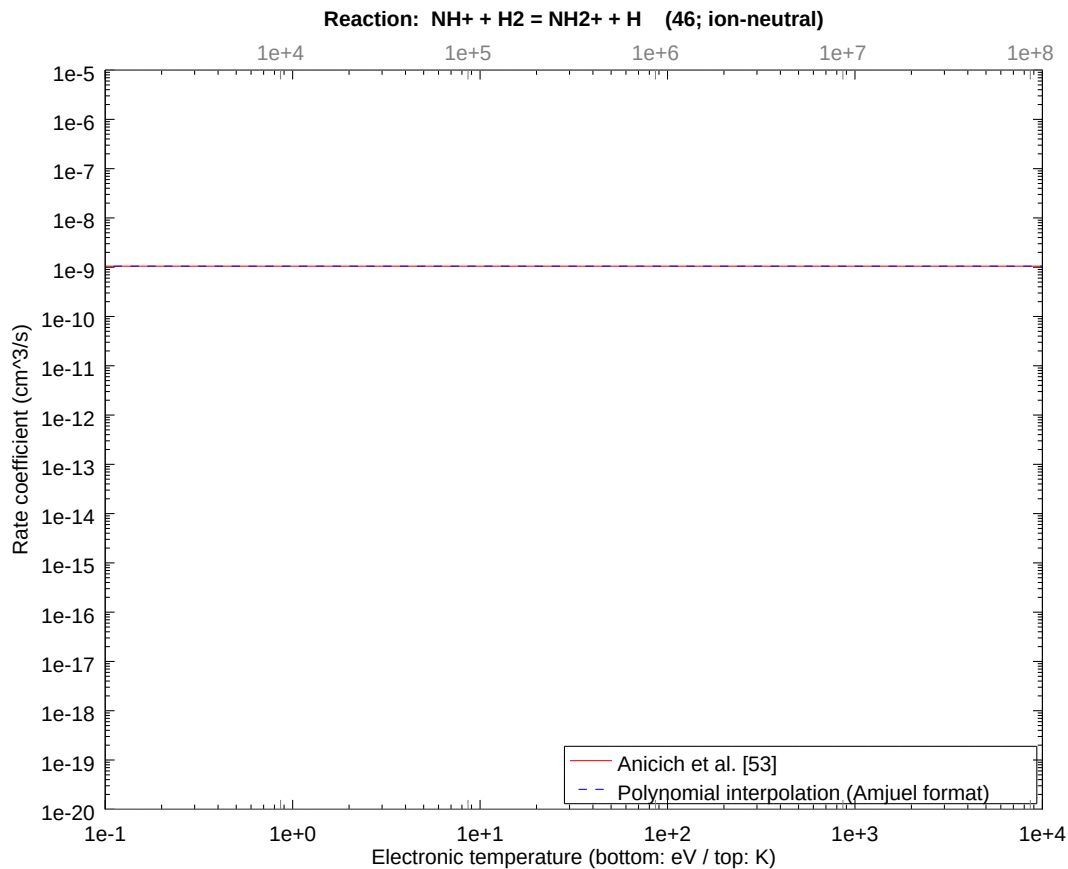
## 2.79 Reaction 45 $\text{NH}^+ + \text{H}_2 = \text{H}_3^+ + \text{N}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.241066529085E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



## 2.80 Reaction 46 $\text{NH}^+ + \text{H}_2 = \text{NH}_2^+ + \text{H}$

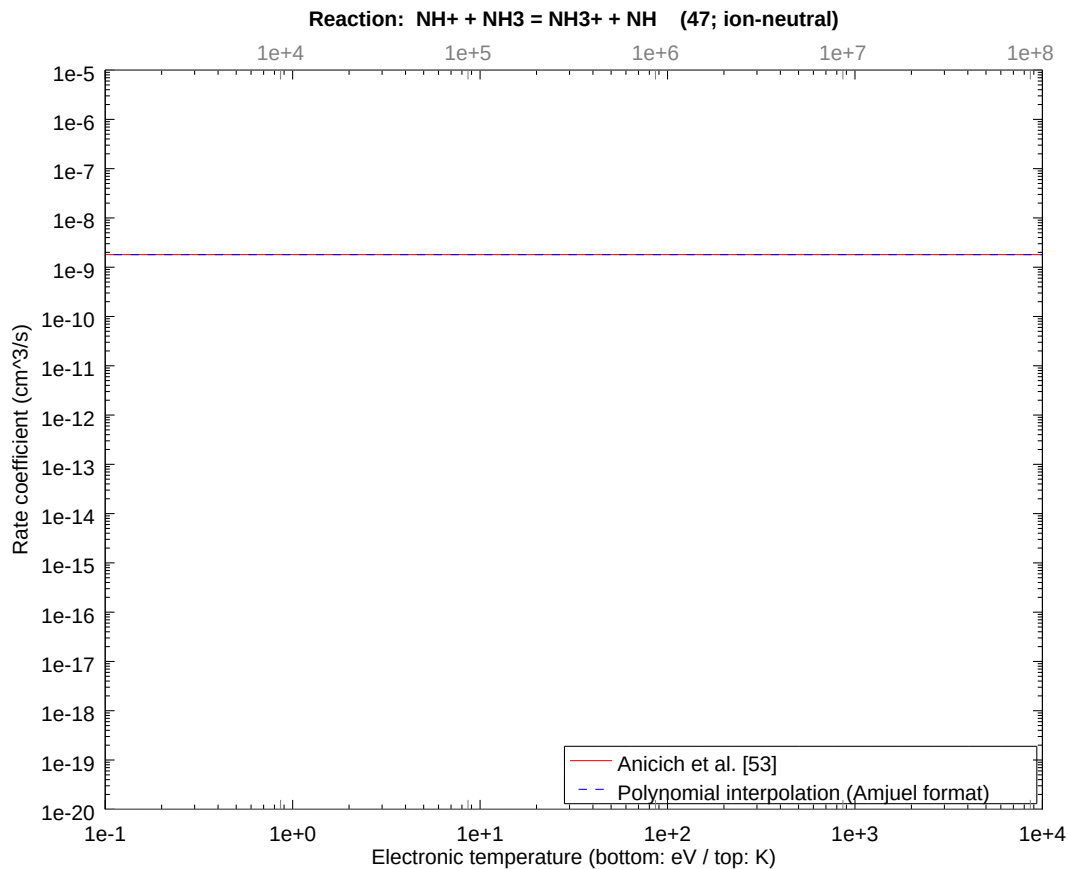
|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.067447567278E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |





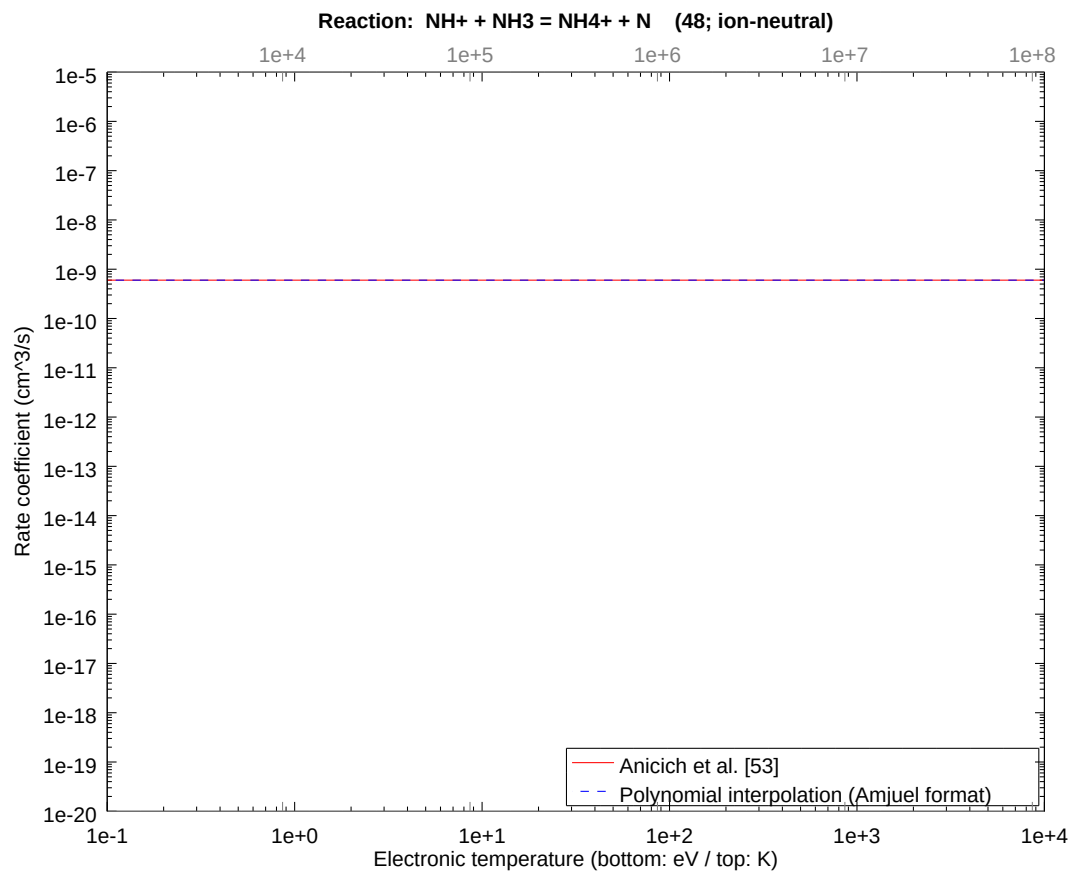
## 2.81 Reaction 47 $\text{NH}^+ + \text{NH}_3 = \text{NH}_3^+ + \text{NH}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.013547917204E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



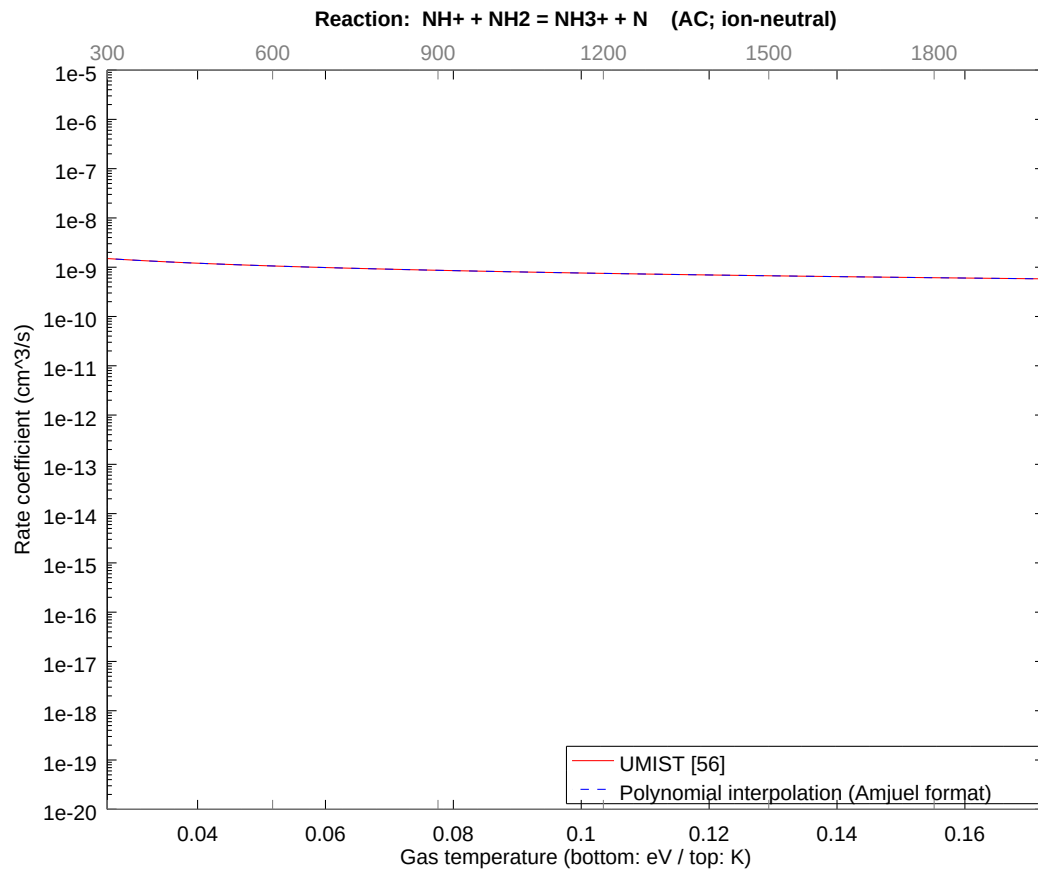
## 2.82 Reaction 48 $\text{NH}^+ + \text{NH}_3 = \text{NH}_4^+ + \text{N}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.123409146071E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



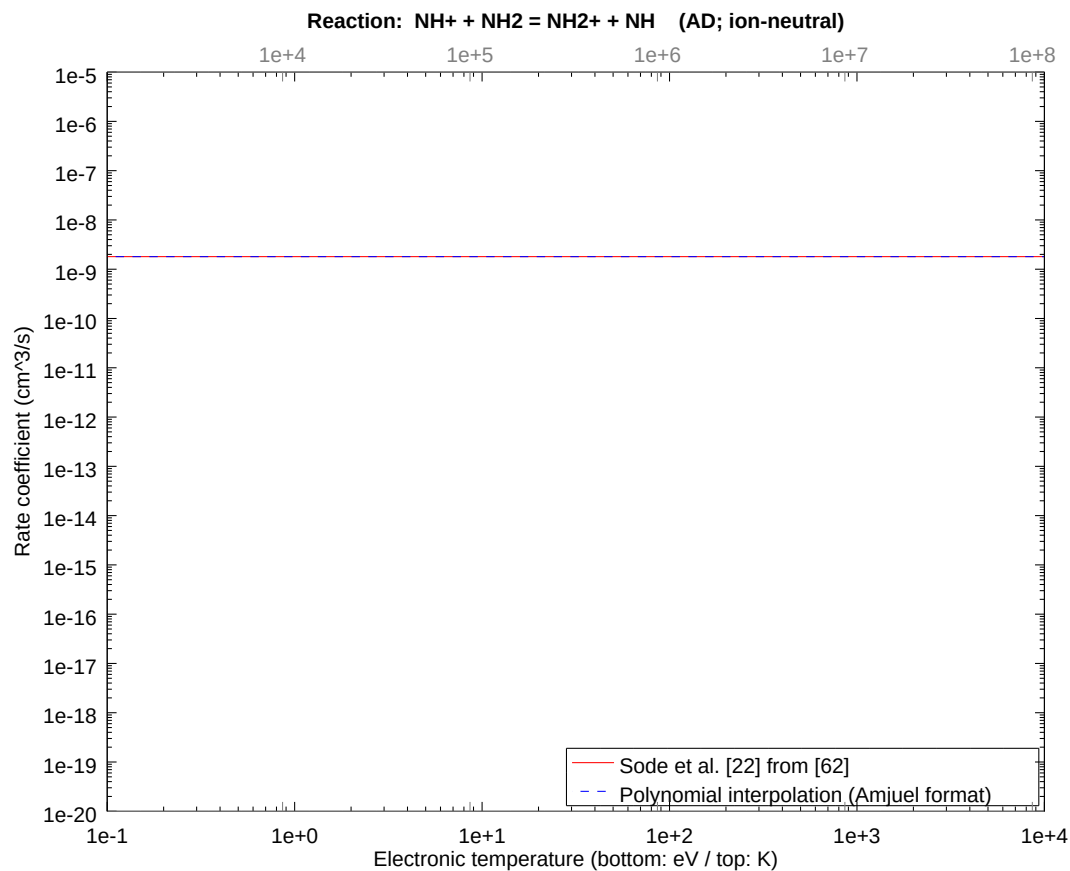
## 2.83 Reaction AC $\text{NH}^+ + \text{NH}_2 = \text{NH}_3^+ + \text{N}$

|                        |                        |                        |
|------------------------|------------------------|------------------------|
| b0 -2.214522888071E+01 | b1 -5.000000029212E-01 | b2 -3.860018050693E-09 |
| b3 -2.872424232405E-09 | b4 -1.316010331162E-09 | b5 -3.799779029463E-10 |
| b6 -6.750141857656E-11 | b7 -6.743736474333E-12 | b8 -2.900439964921E-13 |



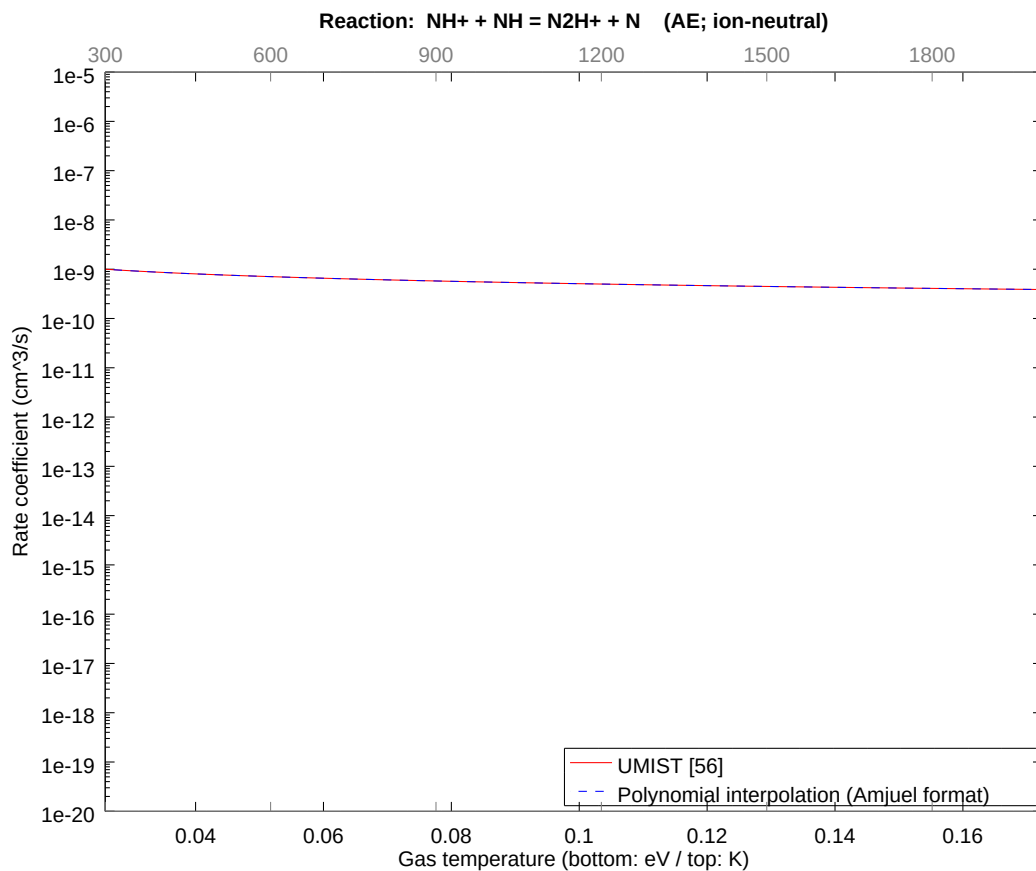
## 2.84 Reaction AD $\text{NH}^+ + \text{NH}_2 = \text{NH}_2^+ + \text{NH}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.013547917204E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



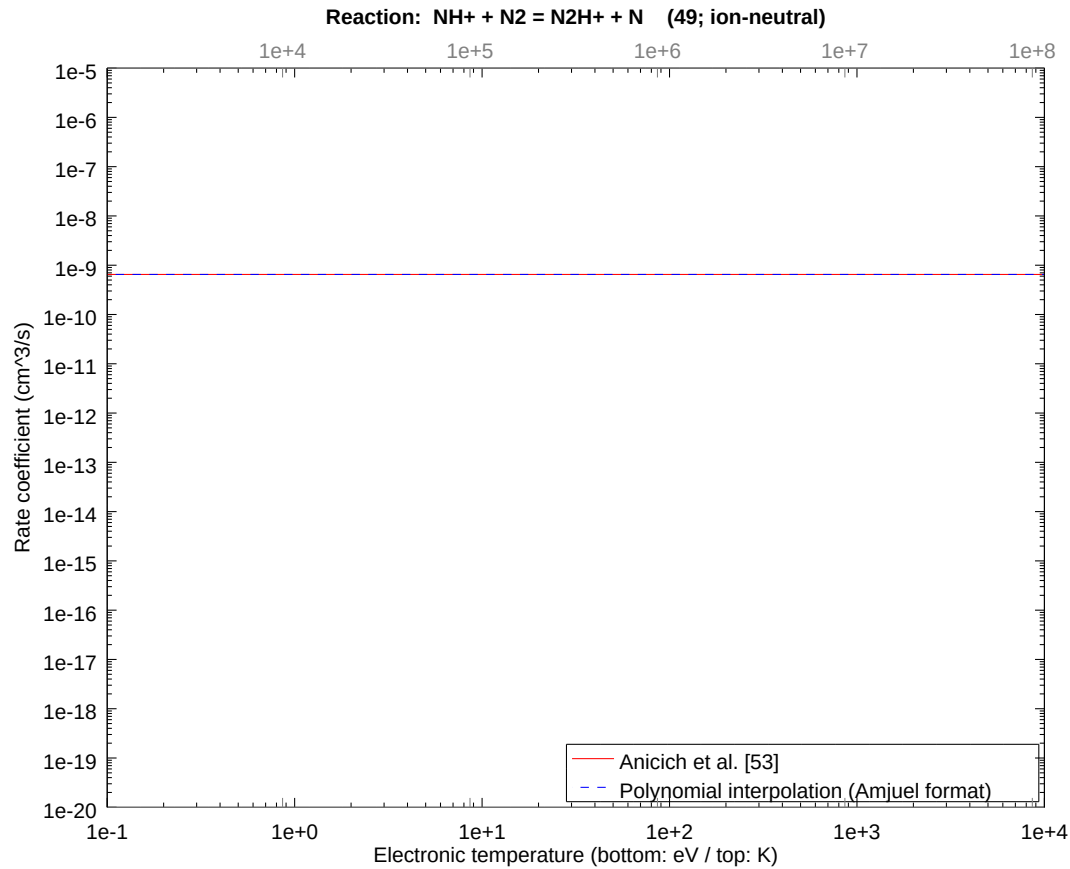
## 2.85 Reaction AE $\text{NH}^+ + \text{NH} = \text{N}_2\text{H}^+ + \text{N}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.255069399034E+01 | b1 | -5.000000077382E-01 | b2 | -1.048331289759E-08 |
| b3 | -8.030169368750E-09 | b4 | -3.804180693869E-09 | b5 | -1.141479368811E-09 |
| b6 | -2.119003834735E-10 | b7 | -2.225565586486E-11 | b8 | -1.012819638805E-12 |



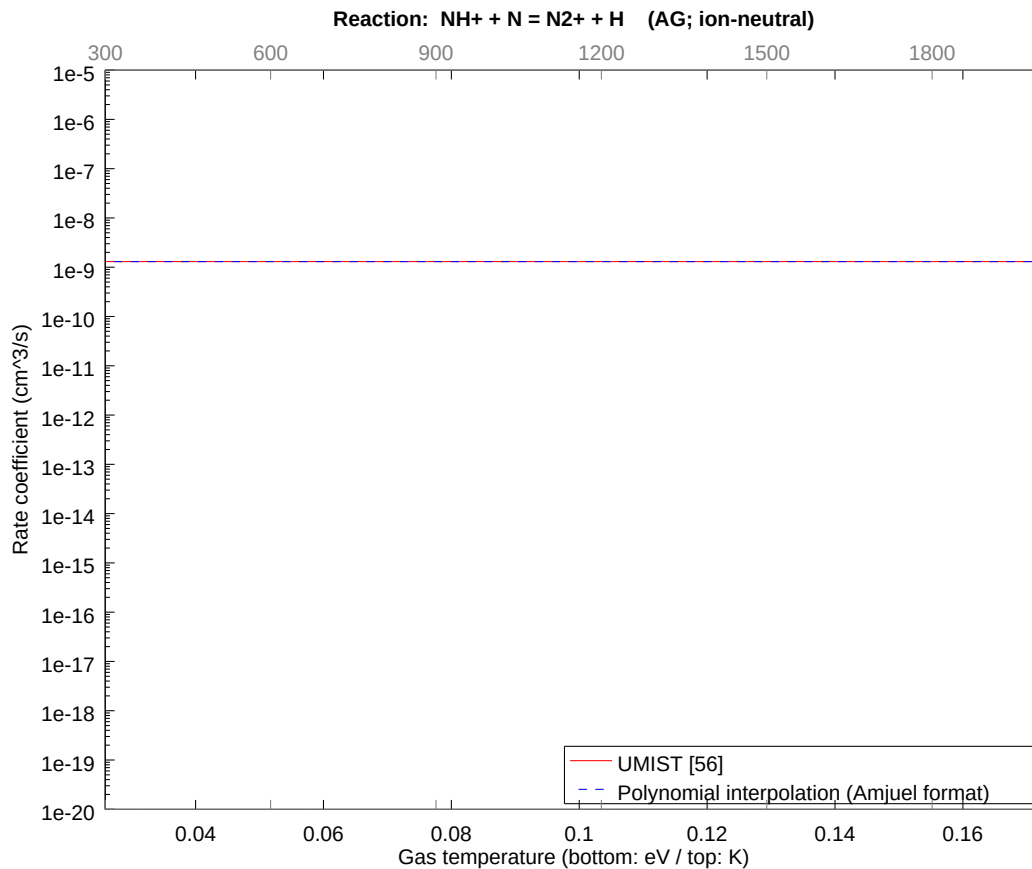
## 2.86 Reaction 49 $\text{NH}^+ + \text{N}_2 = \text{N}_2\text{H}^+ + \text{N}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.115404875304E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



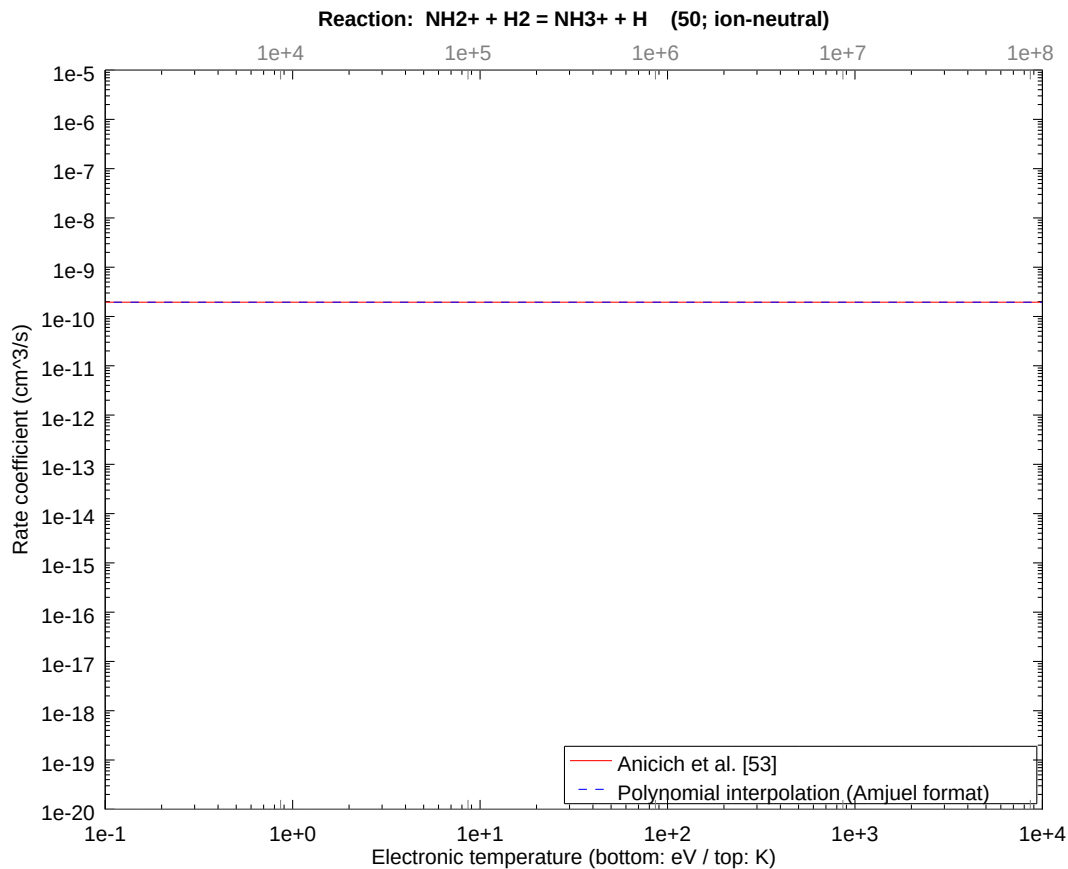
## 2.87 Reaction AG $\text{NH}^+ + \text{N} = \text{N}_2^+ + \text{H}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.046090157248E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



## 2.88 Reaction 50 $\text{NH}_2^+ + \text{H}_2 = \text{NH}_3^+ + \text{H}$

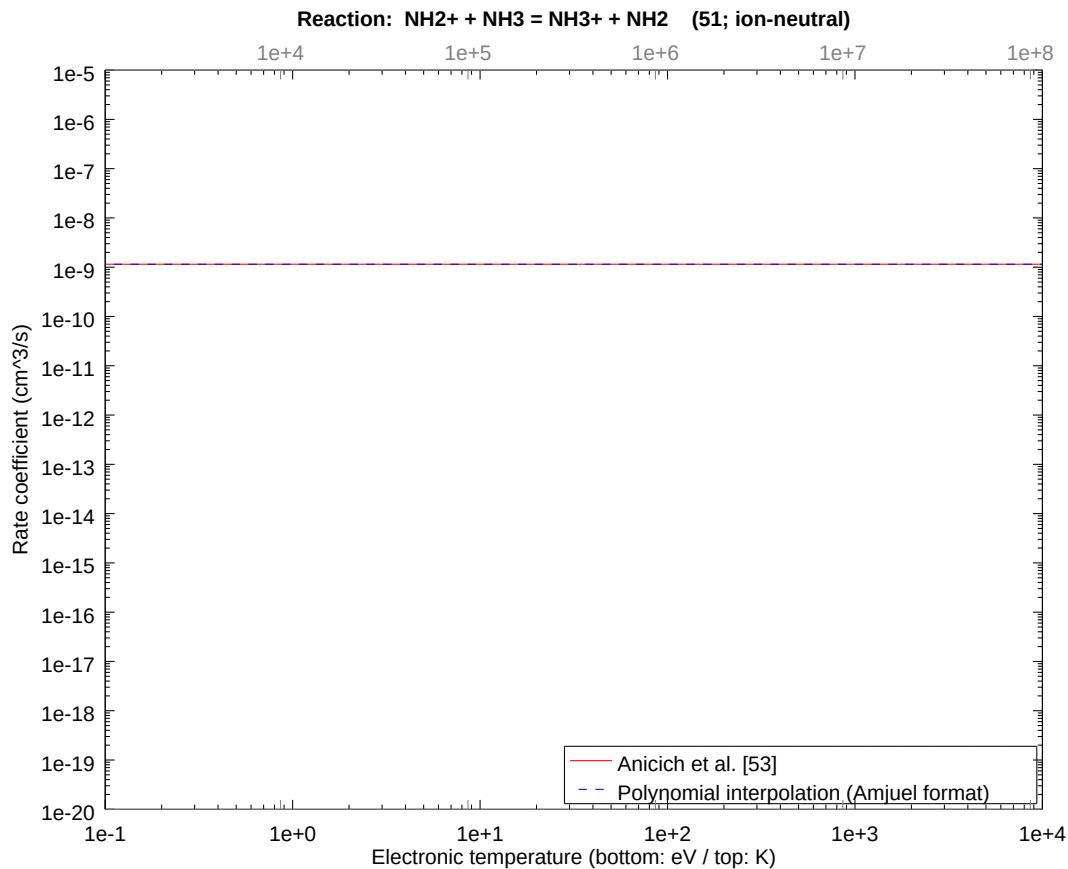
|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.235802155736E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |





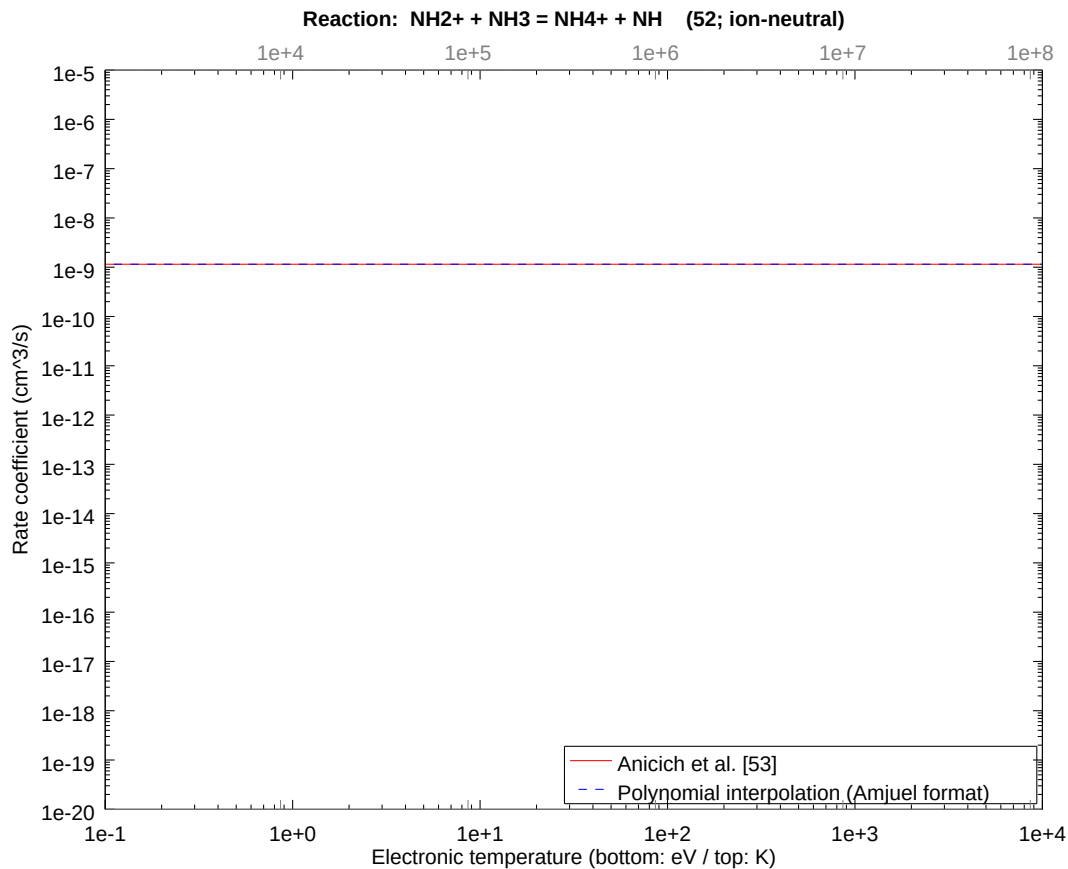
## 2.89 Reaction 51 $\text{NH}_2^+ + \text{NH}_3 = \text{NH}_3^+ + \text{NH}_2$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.058350389457E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



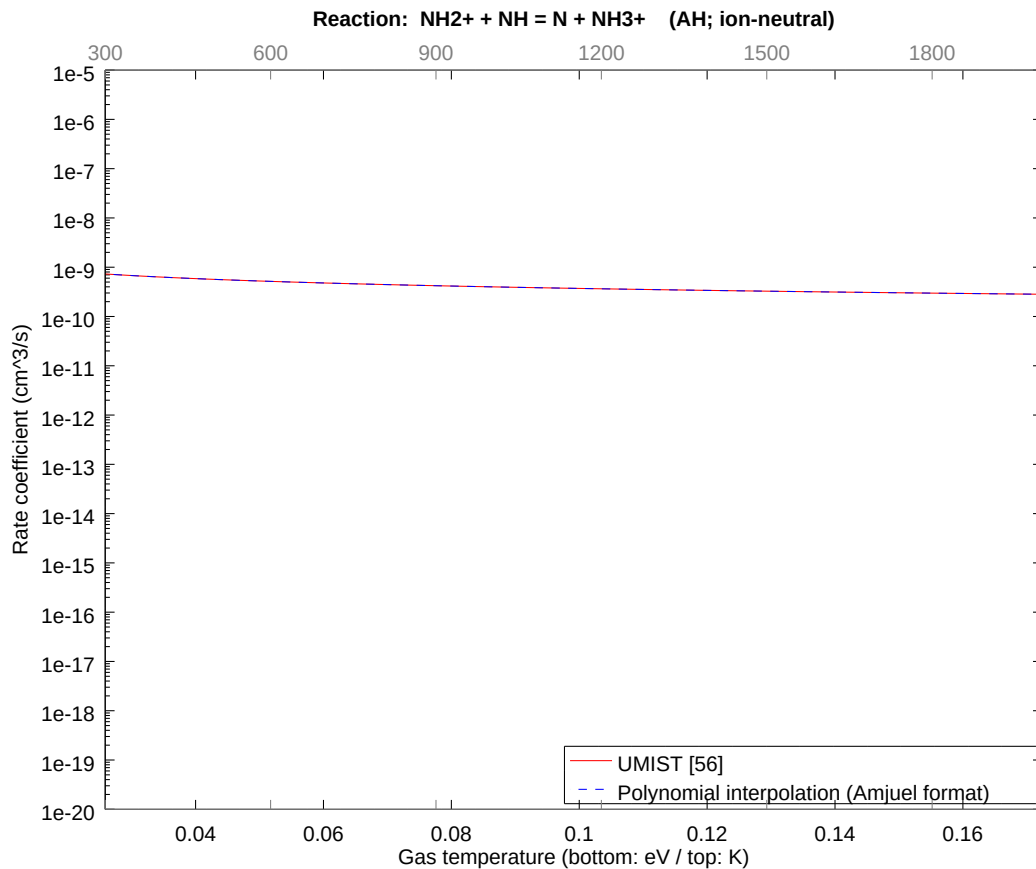
## 2.90 Reaction 52 $\text{NH}_2^+ + \text{NH}_3 = \text{NH}_4^+ + \text{NH}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.058350389457E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



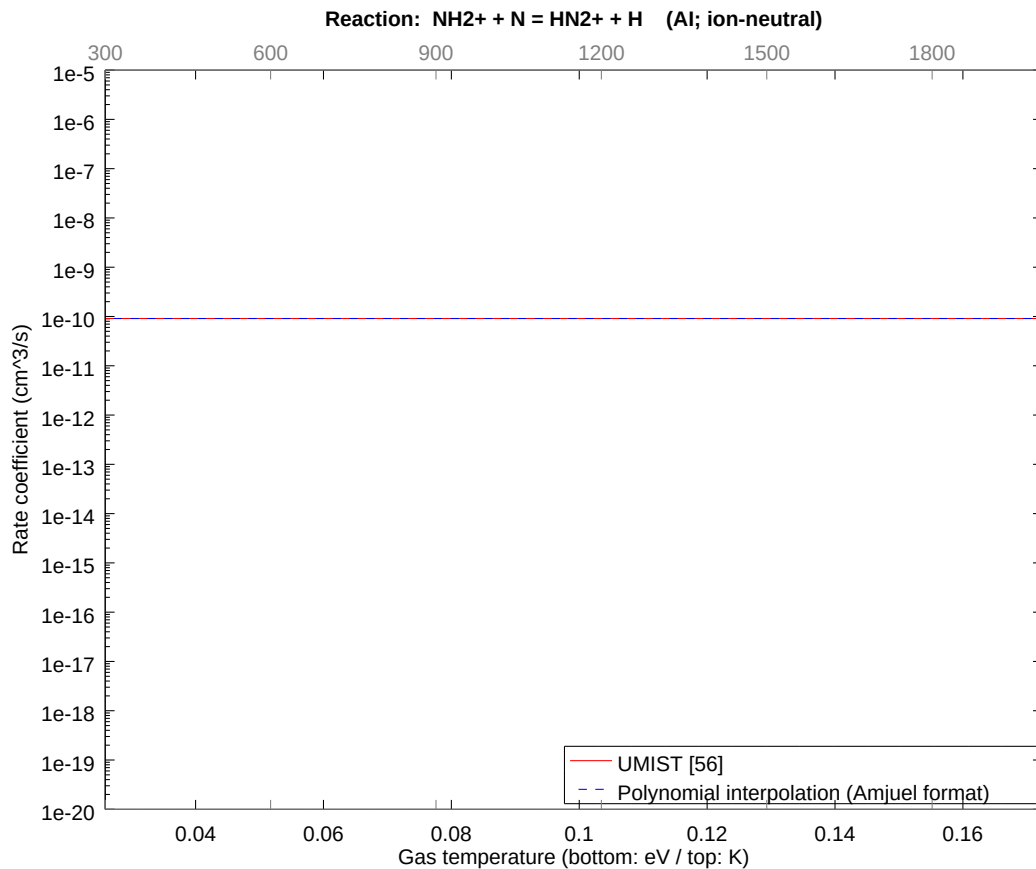
## 2.91 Reaction $\text{AH NH}_2^+ + \text{NH} = \text{N} + \text{NH}_3^+$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.286540473346E+01 | b1 | -5.000000022583E-01 | b2 | -2.903991793554E-09 |
| b3 | -2.092602780058E-09 | b4 | -9.224932782956E-10 | b5 | -2.541818400592E-10 |
| b6 | -4.262482993526E-11 | b7 | -3.961221014971E-12 | b8 | -1.552398934101E-13 |



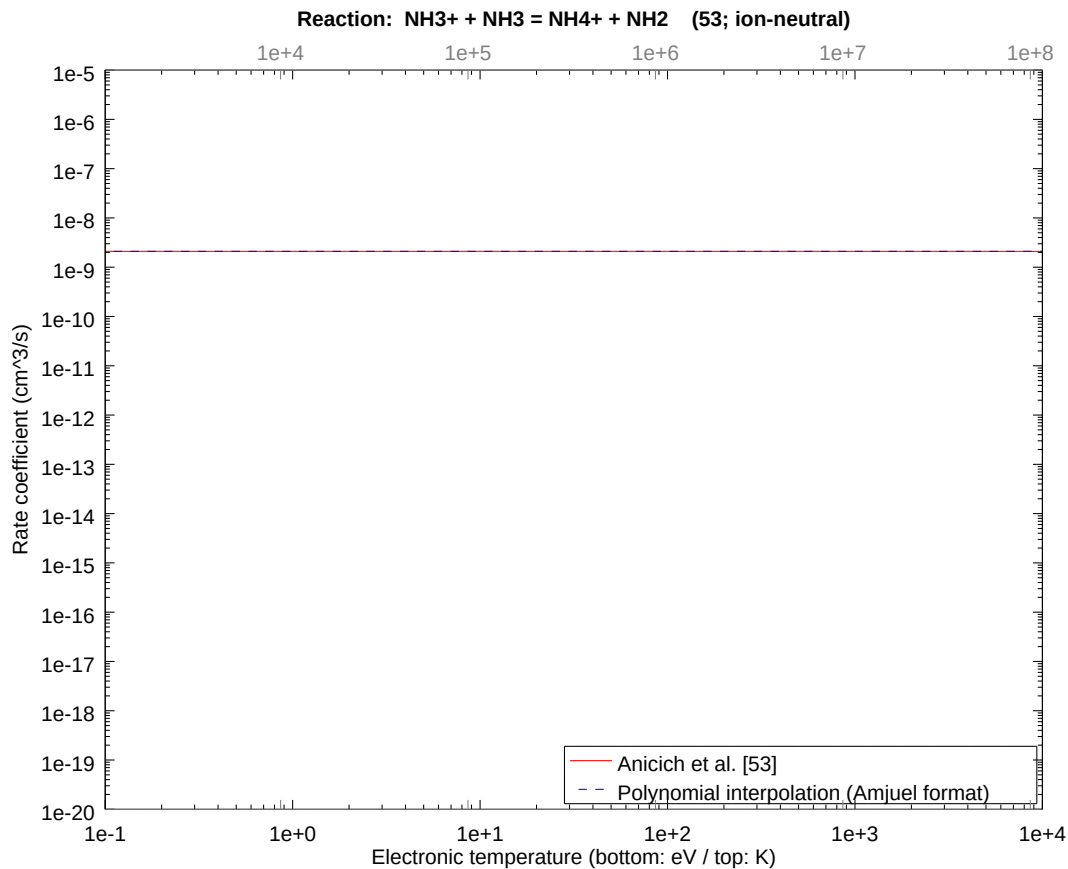
## 2.92 Reaction $\text{Al NH}_2^+ + \text{N} = \text{HN}_2^+ + \text{H}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.312016160941E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



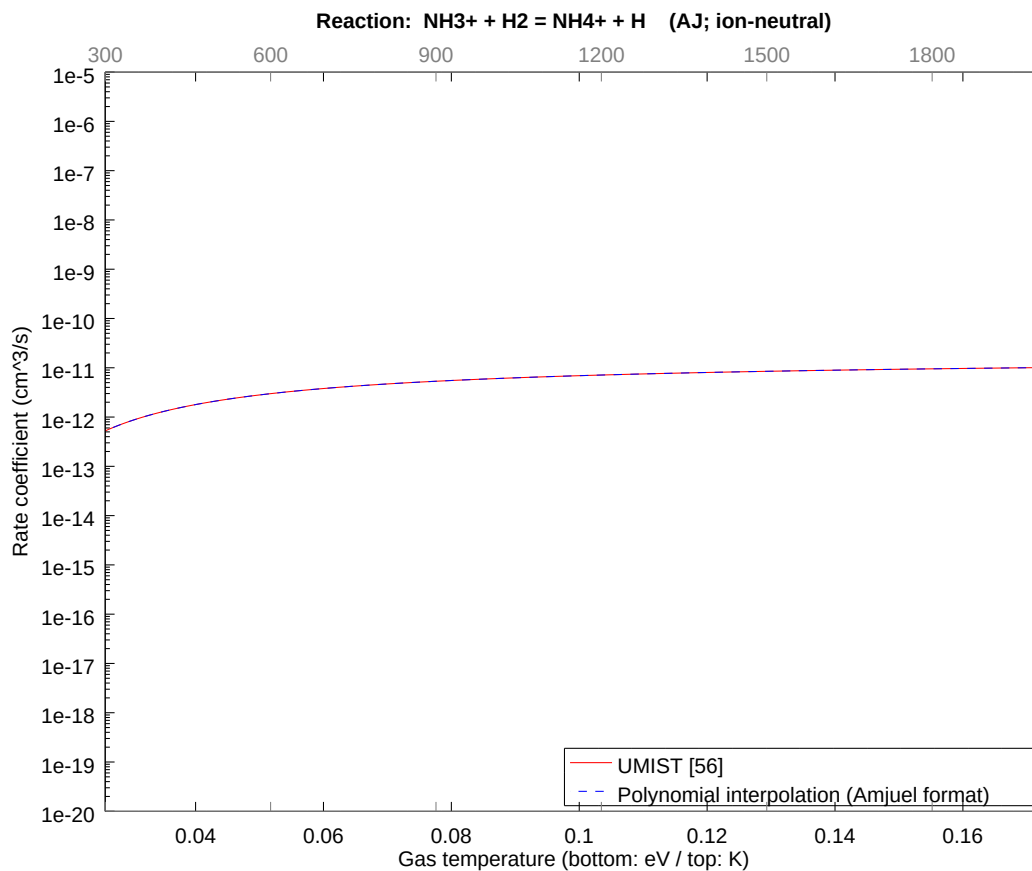
## 2.93 Reaction 53 $\text{NH}_3^+ + \text{NH}_3 = \text{NH}_4^+ + \text{NH}_2$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -1.998132849222E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



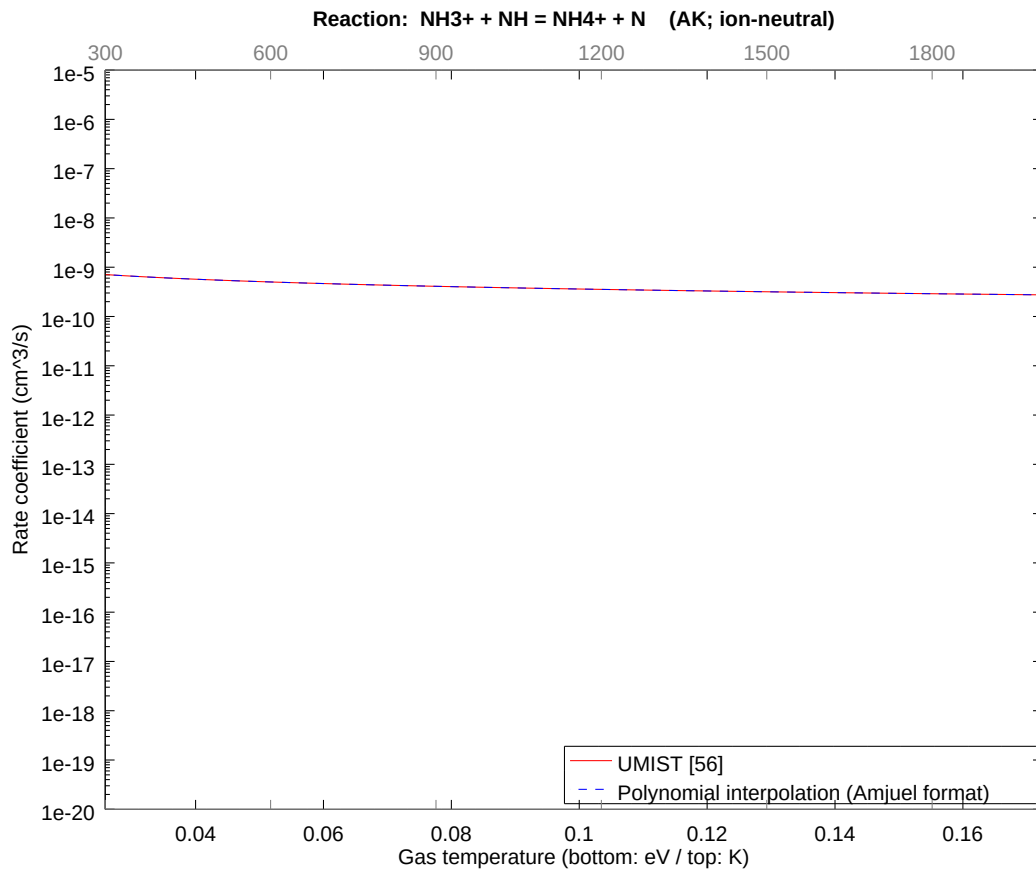
## 2.94 Reaction AJ $\text{NH}_3^+ + \text{H}_2 = \text{NH}_4^+ + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.490386608232E+01 | b1 | 3.409541987533E-02  | b2 | -1.304134590088E-01 |
| b3 | -5.988151600689E-02 | b4 | -4.512471106138E-02 | b5 | -1.408782129630E-02 |
| b6 | -3.532259647172E-03 | b7 | -4.497581656640E-04 | b8 | -3.345468541418E-05 |



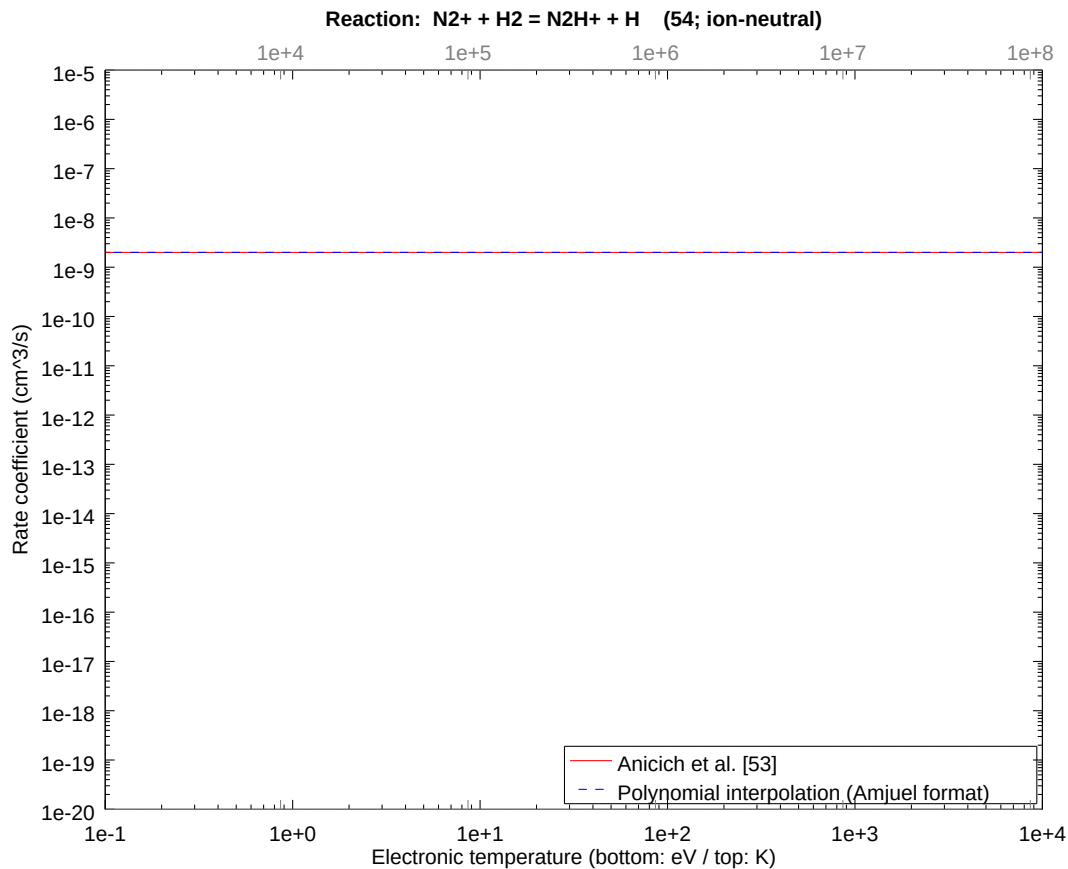
## 2.95 Reaction AK $\text{NH}_3^+ + \text{NH} = \text{NH}_4^+ + \text{N}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.289318429743E+01 | b1 | -5.000000018164E-01 | b2 | -2.303568193971E-09 |
| b3 | -1.639801354397E-09 | b4 | -7.151531646047E-10 | b5 | -1.951520773247E-10 |
| b6 | -3.242435734298E-11 | b7 | -2.983863276617E-12 | b8 | -1.155432594465E-13 |



## 2.96 Reaction 54 $\text{N}_2^+ + \text{H}_2 = \text{N}_2\text{H}^+ + \text{H}$

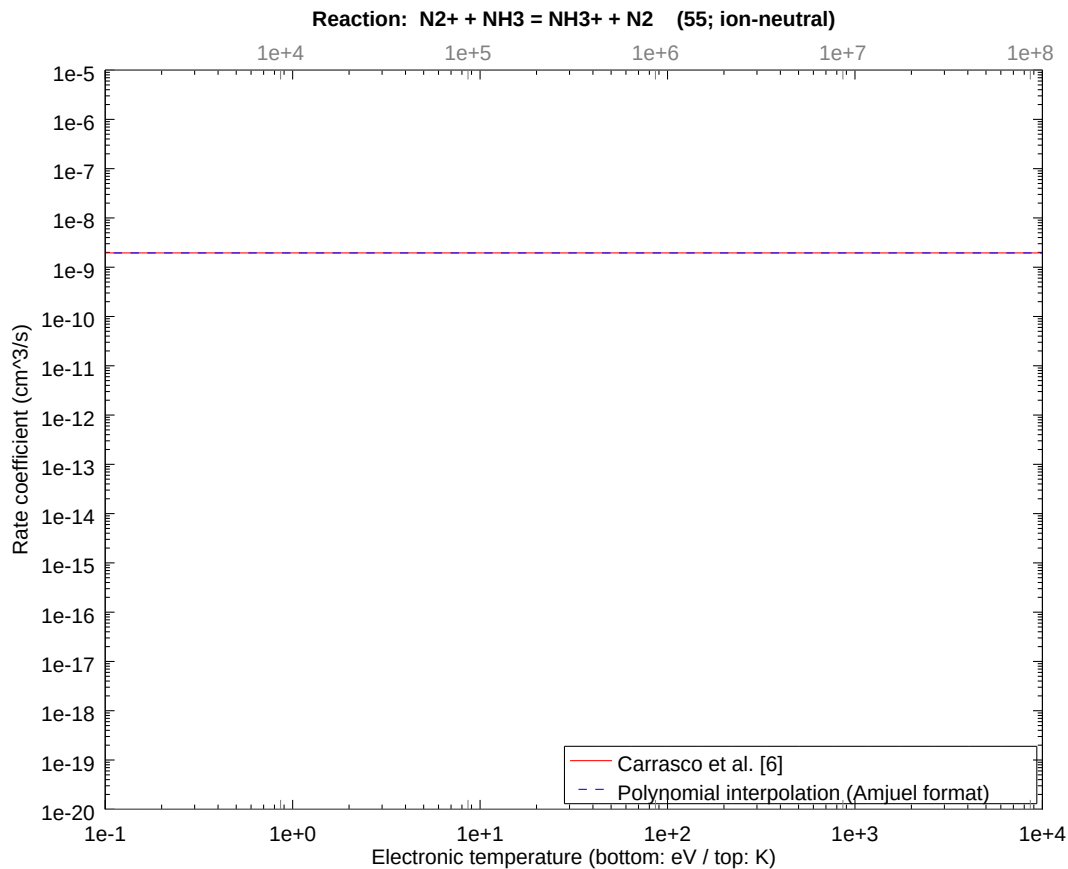
|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.003011865639E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |





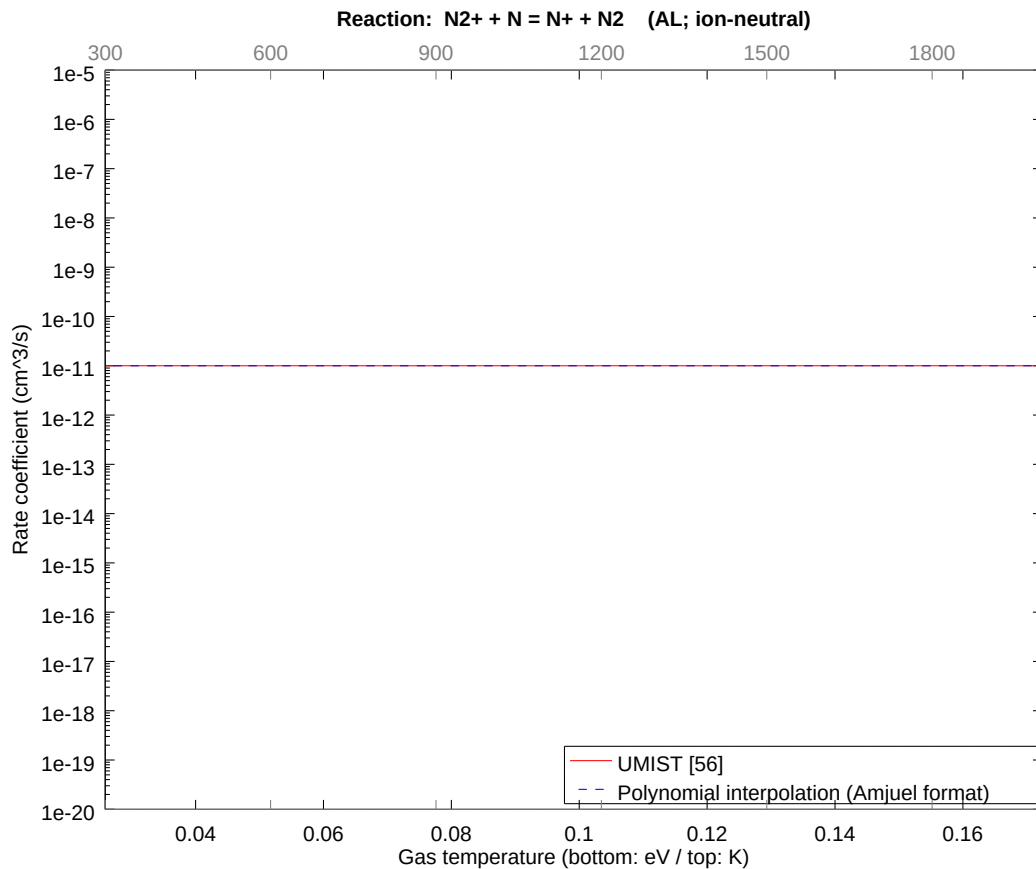
## 2.97 Reaction 55 $\text{N}_2^+ + \text{NH}_3 = \text{NH}_3^+ + \text{N}_2$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.005543646437E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



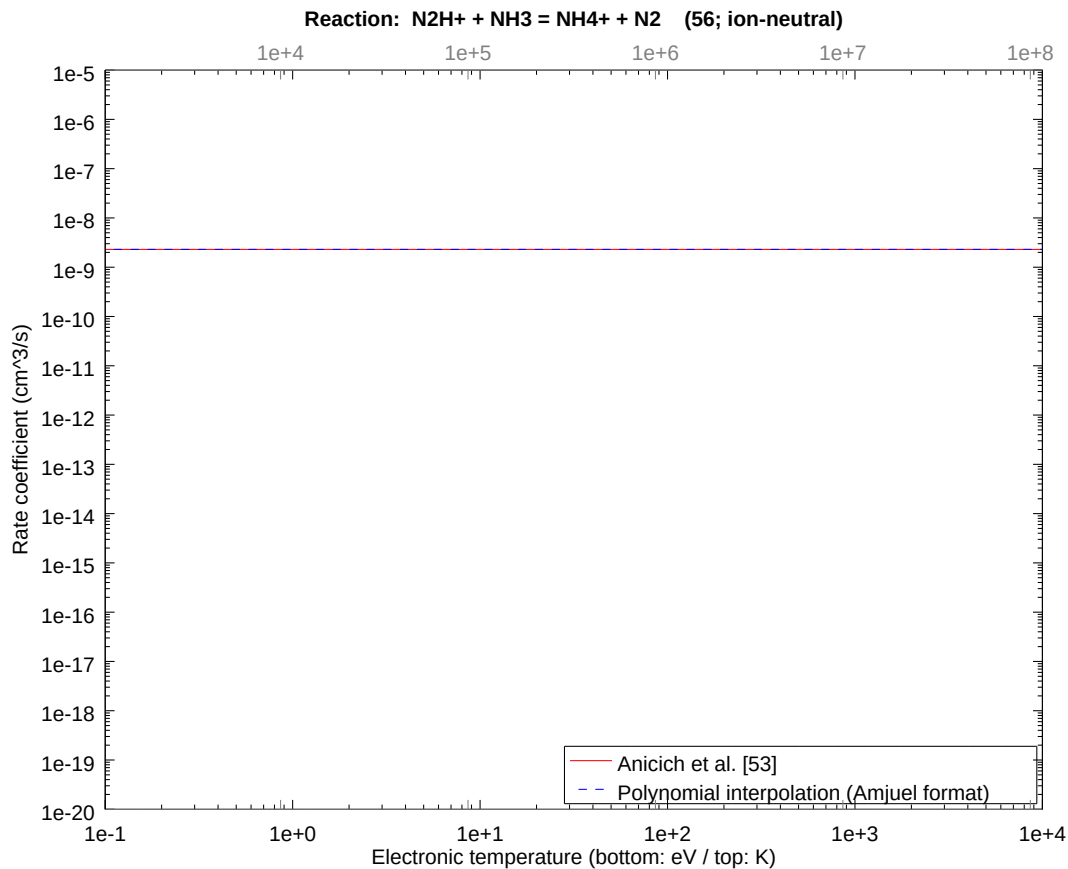
## 2.98 Reaction AL N2+ + N = N+ + N2

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.532843602293E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



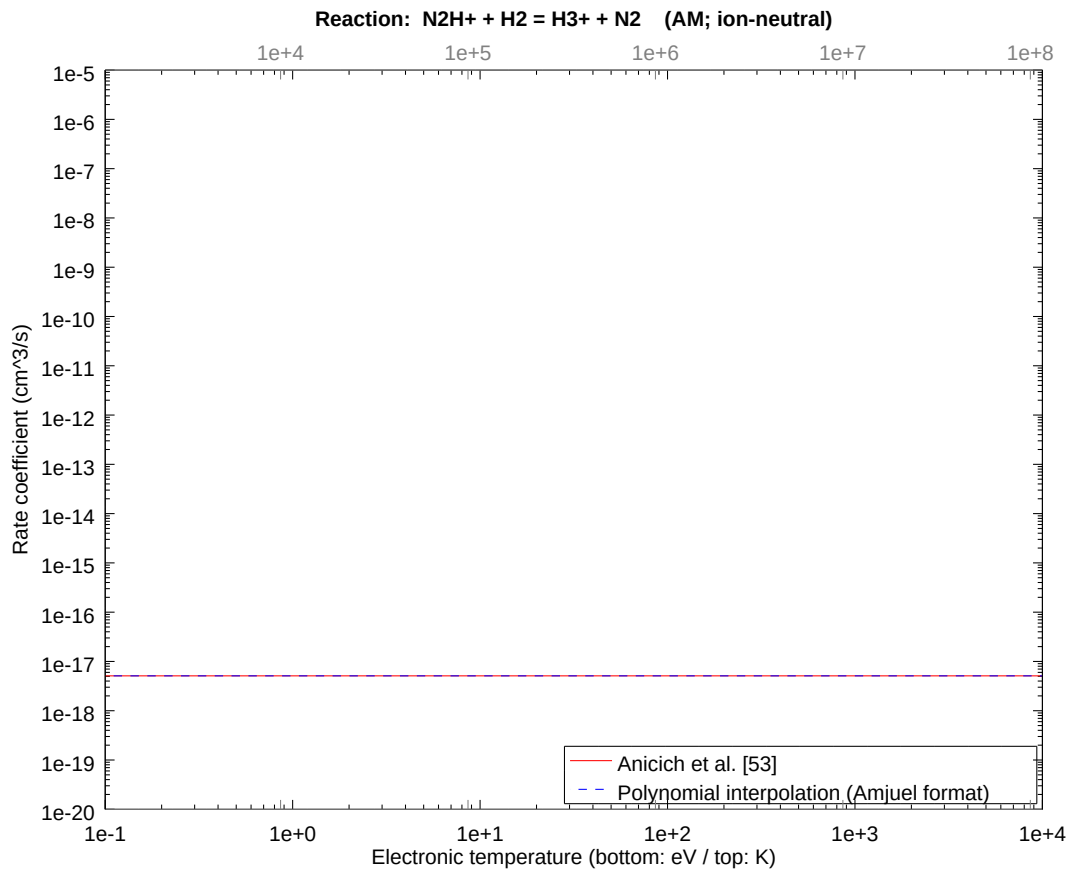
## 2.99 Reaction 56 $\text{N}_2\text{H}^+ + \text{NH}_3 = \text{NH}_4^+ + \text{N}_2$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -1.989035671401E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



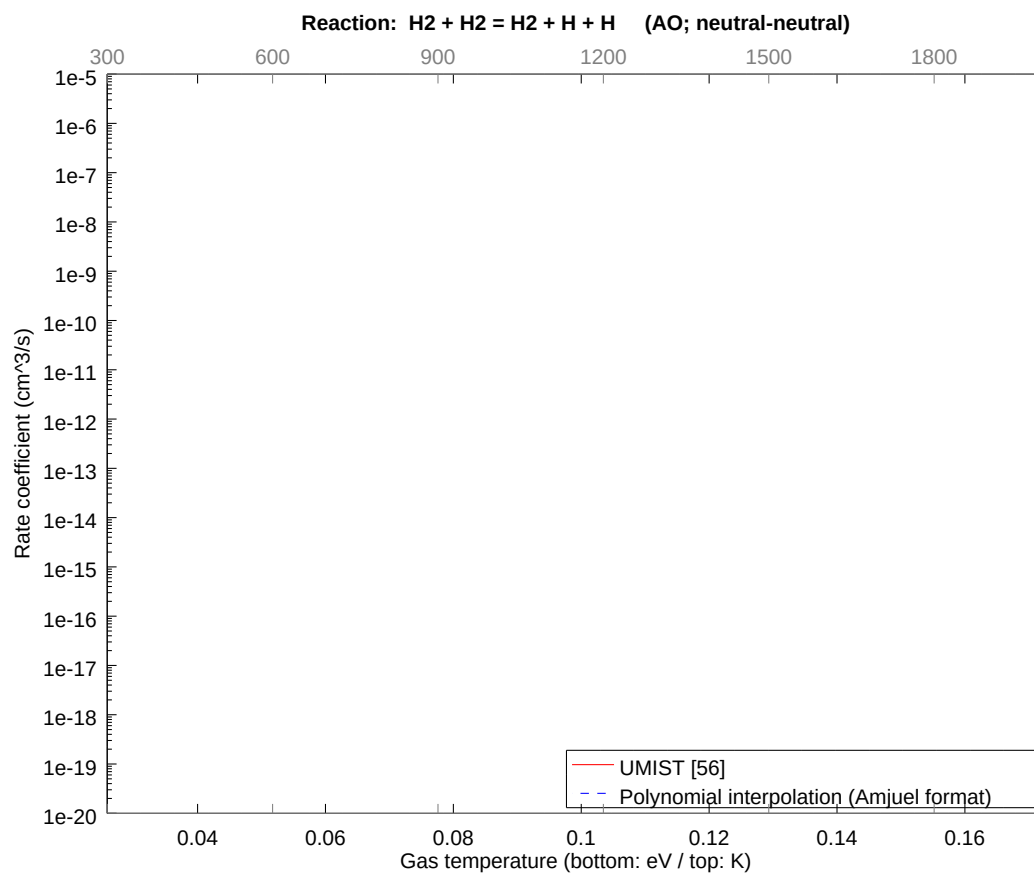
## 2.100 Reaction AM $\text{N}_2\text{H}^+ + \text{H}_2 = \text{H}_3^+ + \text{N}_2$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -3.981729113416E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



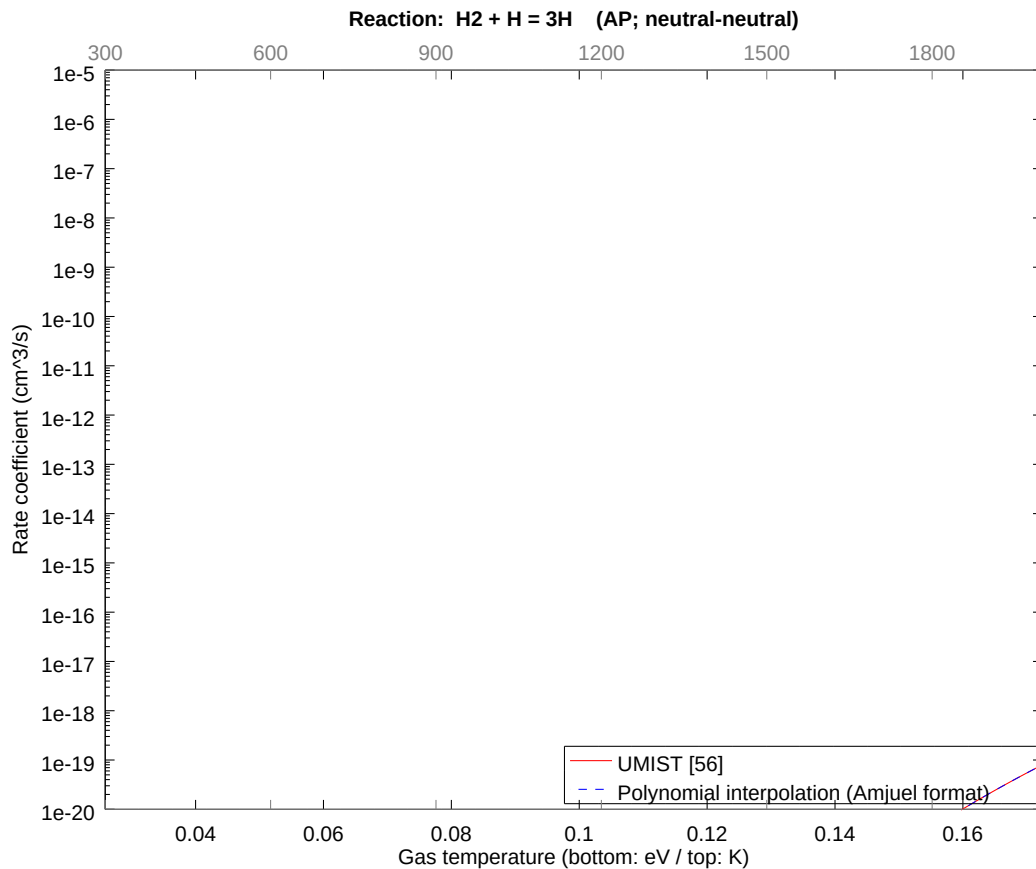
## 2.101 Reaction AO $\text{H}_2 + \text{H}_2 = \text{H}_2 + \text{H} + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.696426662024E+01 | b1 | 2.746576251943E+00  | b2 | -1.050552760935E+01 |
| b3 | -4.823787997126E+00 | b4 | -3.635045792807E+00 | b5 | -1.134852158008E+00 |
| b6 | -2.845431171969E-01 | b7 | -3.623051668352E-02 | b8 | -2.694960668238E-03 |



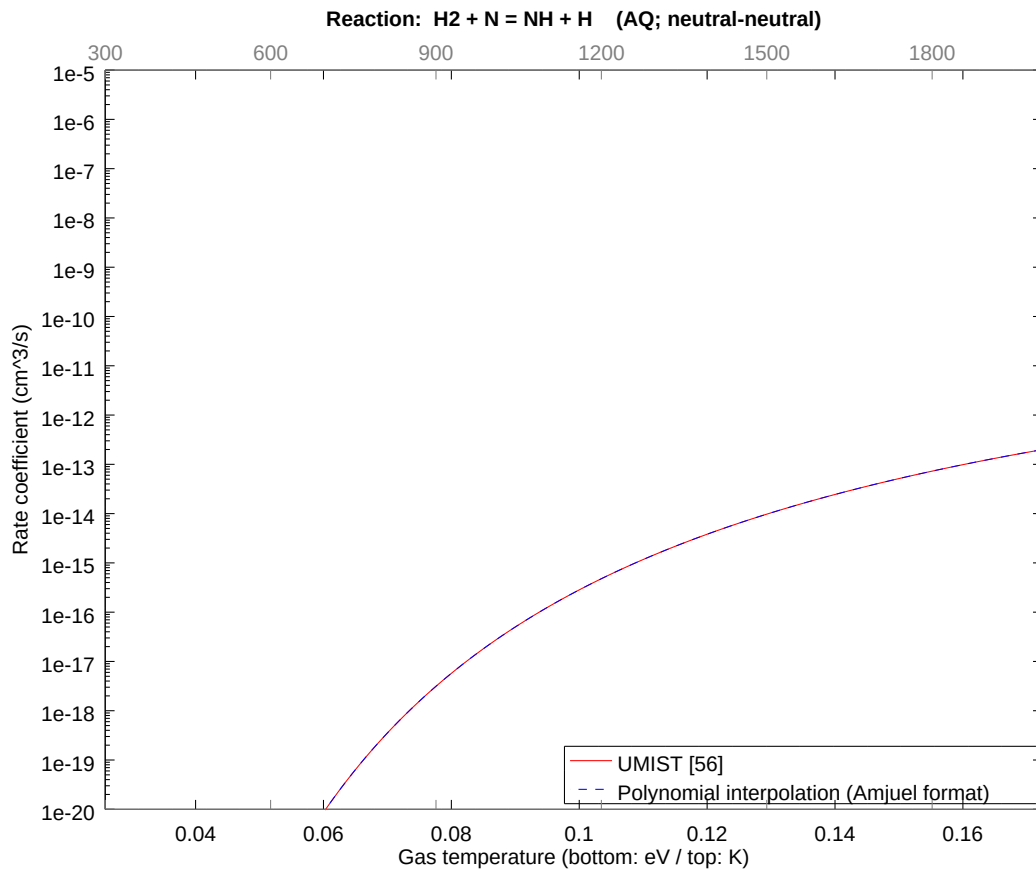
## 2.102 Reaction AP H2 + H = 3H

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.381915581932E+01 | b1 | 7.962151505816E-01  | b2 | -6.870440167331E+00 |
| b3 | -3.154677044982E+00 | b4 | -2.377259434294E+00 | b5 | -7.421744189427E-01 |
| b6 | -1.860864617664E-01 | b7 | -2.369415477215E-02 | b8 | -1.762459413704E-03 |



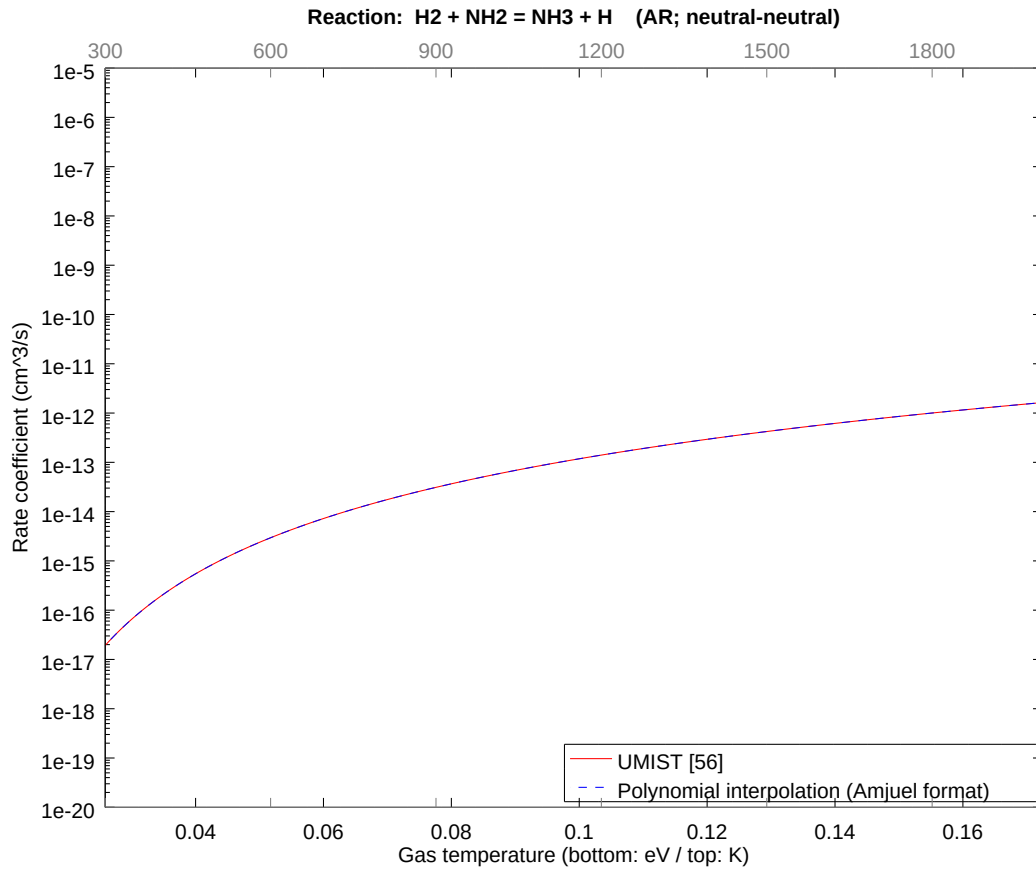
## 2.103 Reaction AQ $\text{H}_2 + \text{N} = \text{NH} + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.203677971440E+01 | b1 | 5.909547855171E-01  | b2 | -2.260374813605E+00 |
| b3 | -1.037888746589E+00 | b4 | -7.821183532573E-01 | b5 | -2.441753836278E-01 |
| b6 | -6.122244587988E-02 | b7 | -7.795376915334E-03 | b8 | -5.798491468765E-04 |



## 2.104 Reaction AR $\text{H}_2 + \text{NH}_2 = \text{NH}_3 + \text{H}$

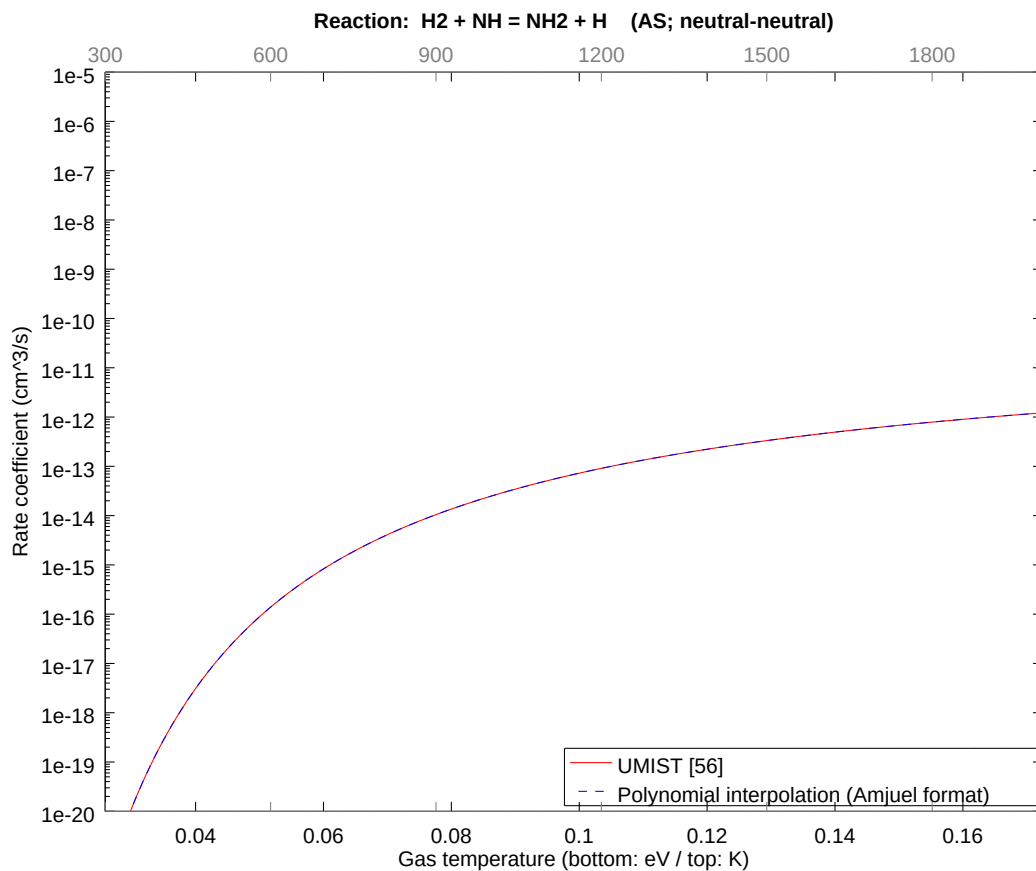
|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.974576937221E+01 | b1 | 3.935721838265E+00  | b2 | -1.748839337655E-01 |
| b3 | -8.030087161977E-02 | b4 | -6.051205881110E-02 | b5 | -1.889171257029E-02 |
| b6 | -4.736746304237E-03 | b7 | -6.031239388174E-04 | b8 | -4.486260315385E-05 |





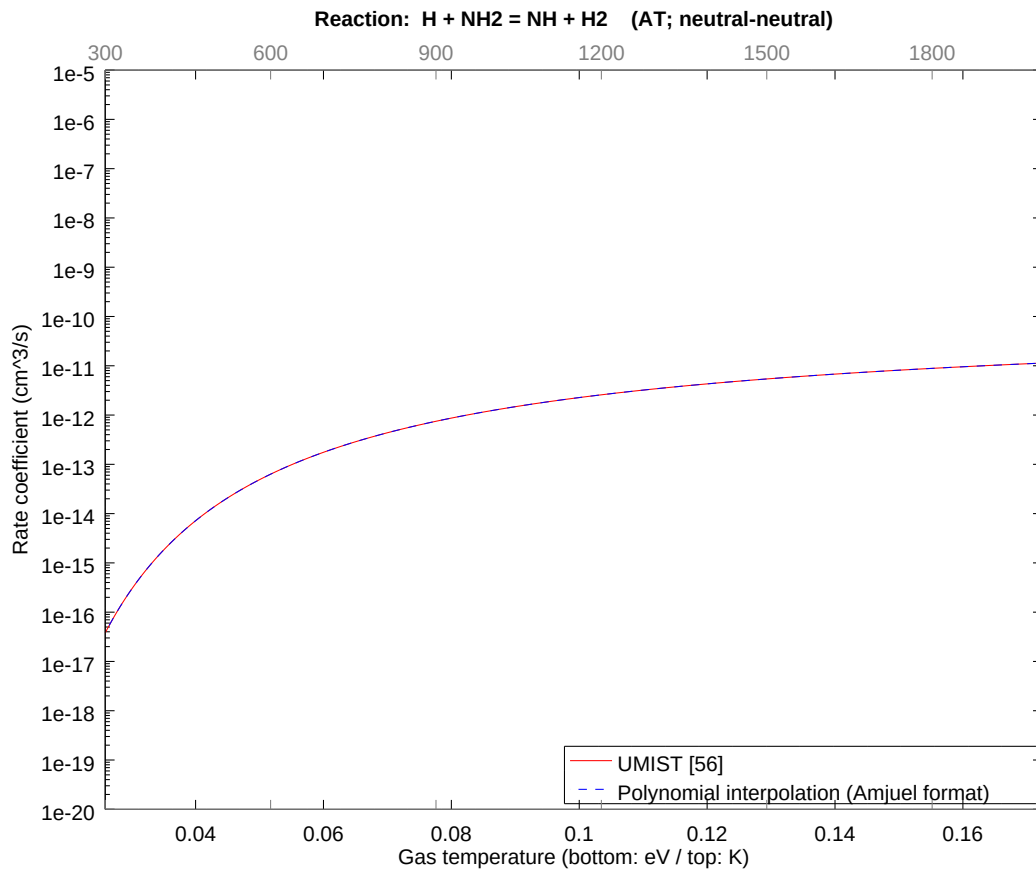
## 2.105 Reaction AS $\text{H}_2 + \text{NH} = \text{NH}_2 + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.433392660574E+01 | b1 | 2.541481071904E-01  | b2 | -9.721048352481E-01 |
| b3 | -4.463581308227E-01 | b4 | -3.363606022624E-01 | b5 | -1.050109343825E-01 |
| b6 | -2.632954285966E-02 | b7 | -3.352507521719E-03 | b8 | -2.493719857357E-04 |



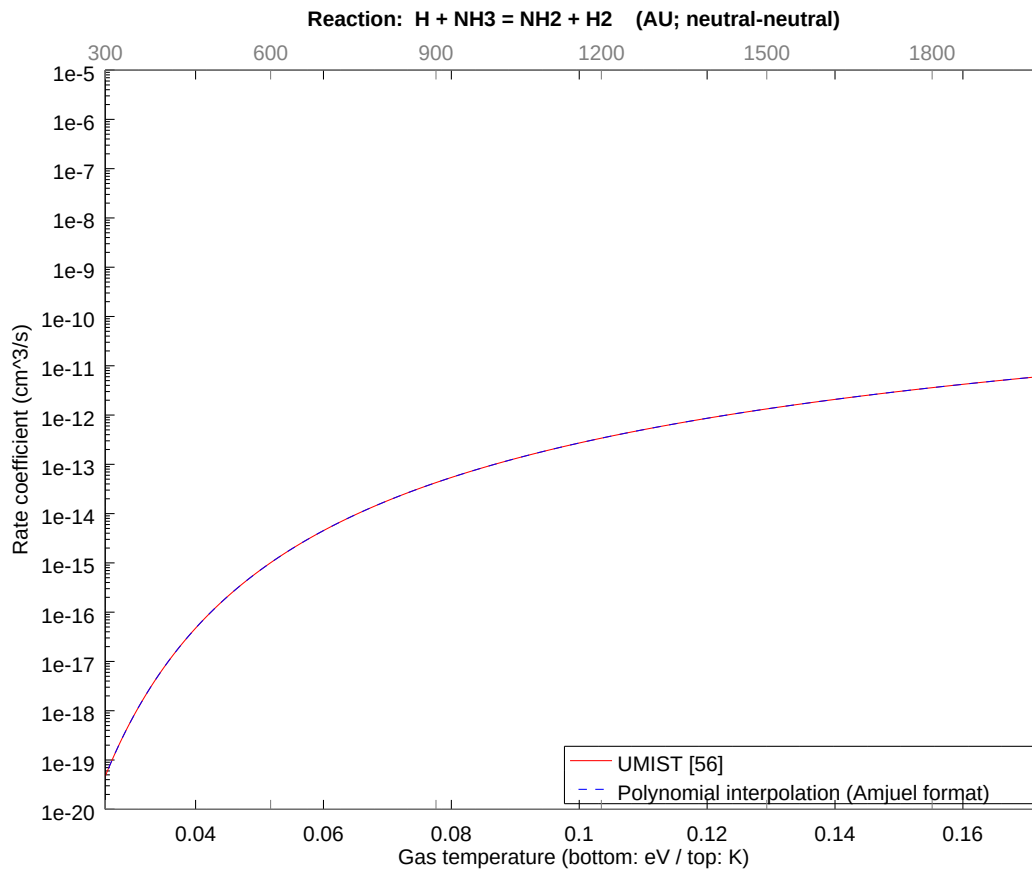
## 2.106 Reaction AT $\text{H} + \text{NH}_2 = \text{NH} + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.342912922359E+01 | b1 | 1.453301289331E-01  | b2 | -5.558810761840E-01 |
| b3 | -2.552420580269E-01 | b4 | -1.923419026233E-01 | b5 | -6.004865841419E-02 |
| b6 | -1.505608661547E-02 | b7 | -1.917072539614E-03 | b8 | -1.425989897052E-04 |



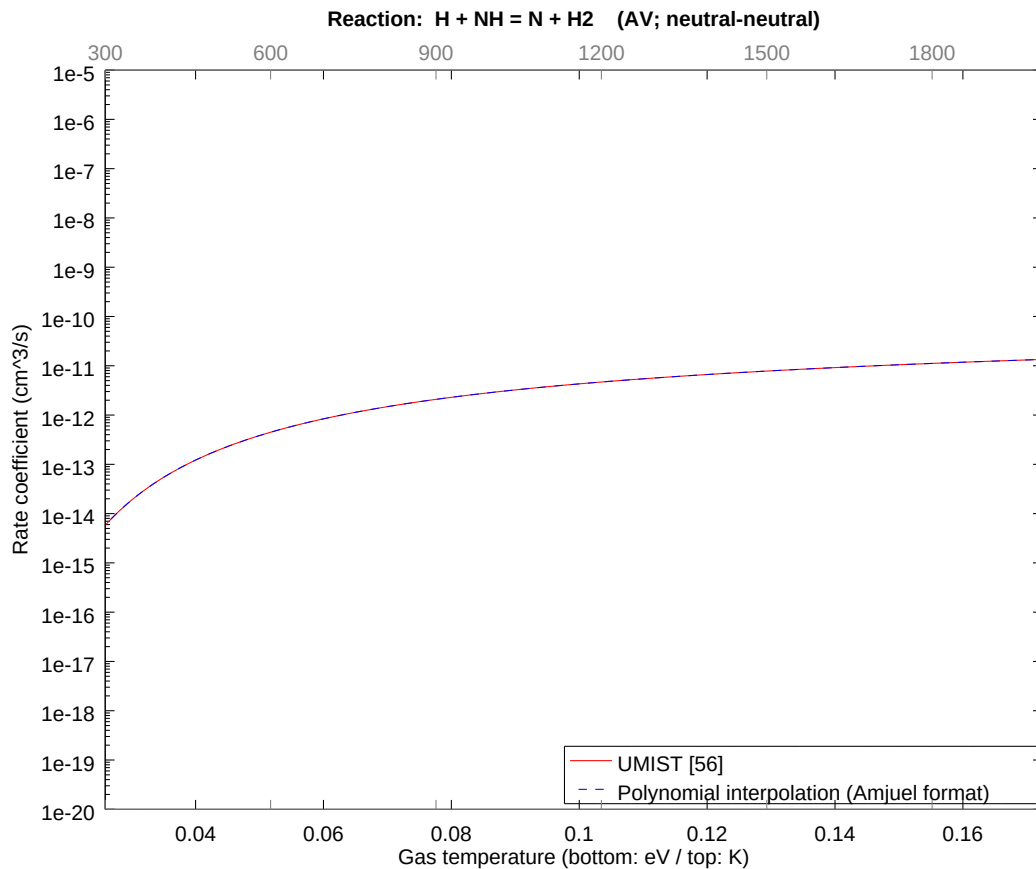
## 2.107 Reaction AU $\text{H} + \text{NH}_3 = \text{NH}_2 + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -1.961475355238E+01 | b1 | 2.562965700633E+00  | b2 | -6.233363002700E-01 |
| b3 | -2.862152455775E-01 | b4 | -2.156822654030E-01 | b5 | -6.733546099889E-02 |
| b6 | -1.688311717638E-02 | b7 | -2.149706042274E-03 | b8 | -1.599031358719E-04 |



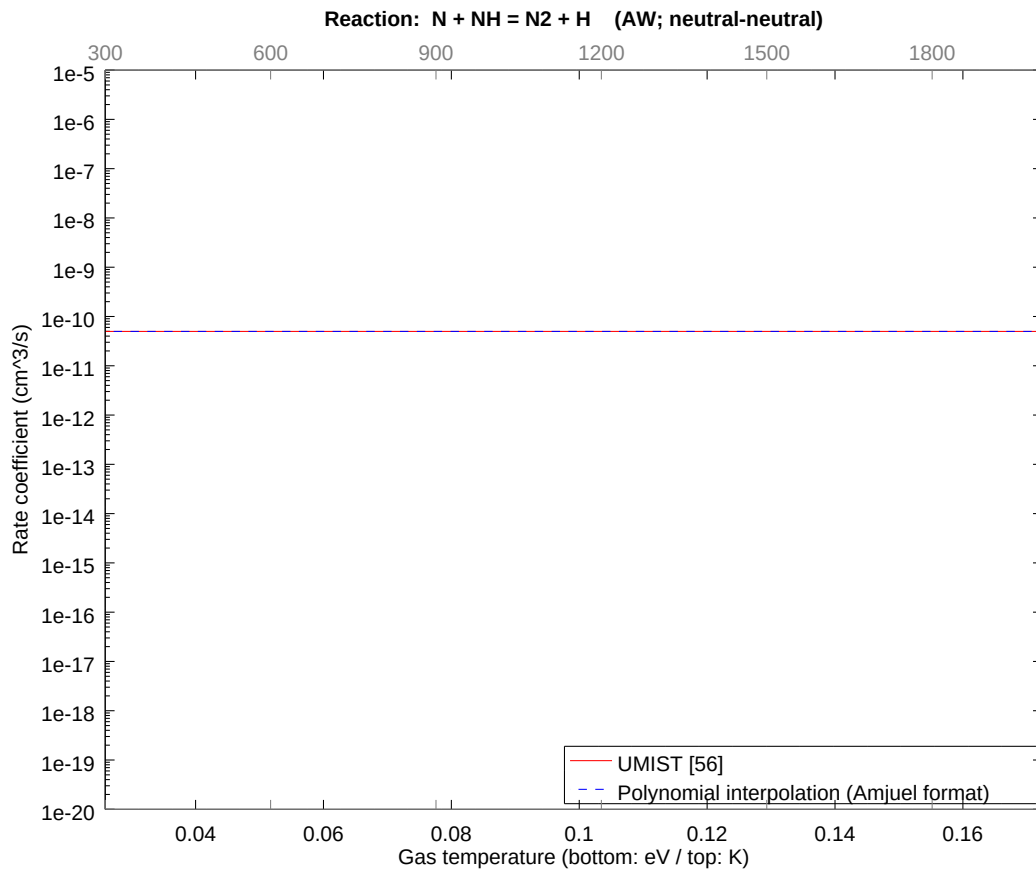
## 2.108 Reaction AV $\text{H} + \text{NH} = \text{N} + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.319669866637E+01 | b1 | 5.783802913881E-01  | b2 | -2.998010338650E-01 |
| b3 | -1.376586412086E-01 | b4 | -1.037349602663E-01 | b5 | -3.238579377985E-02 |
| b6 | -8.120136692671E-03 | b7 | -1.033926772738E-03 | b8 | -7.690732074648E-05 |



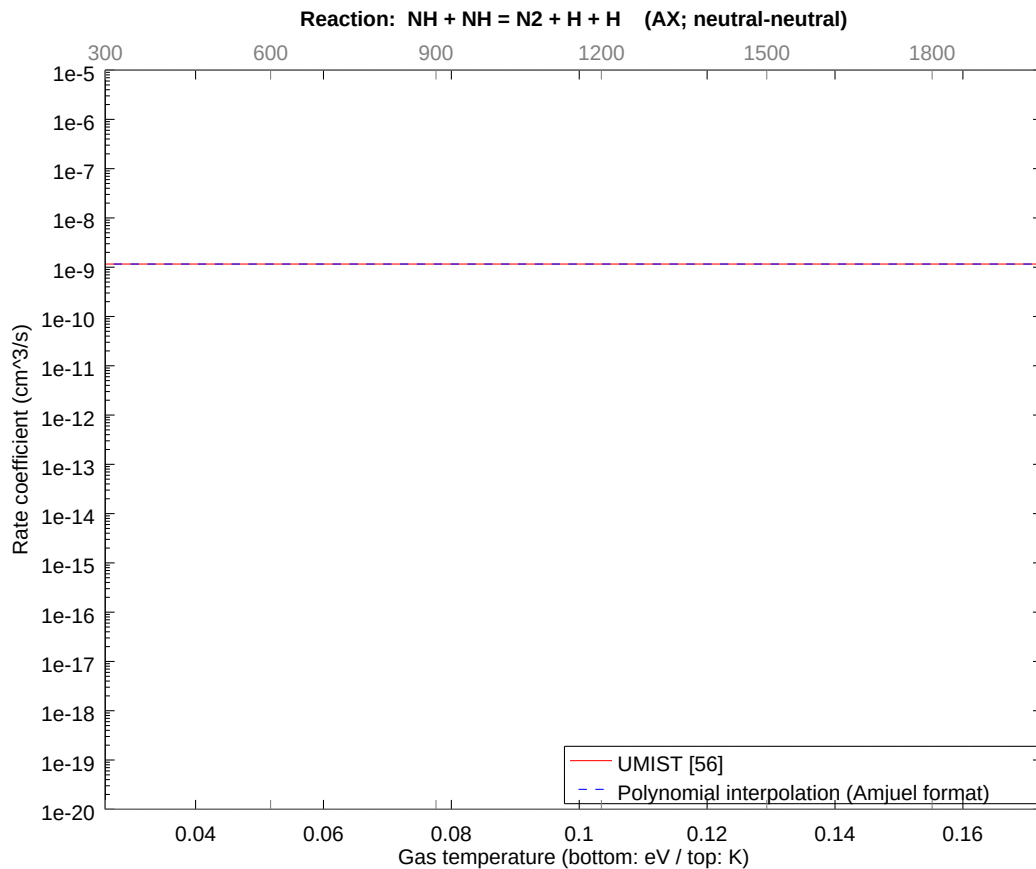
## 2.109 Reaction AW $\text{N} + \text{NH} = \text{N}_2 + \text{H}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.372300613190E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



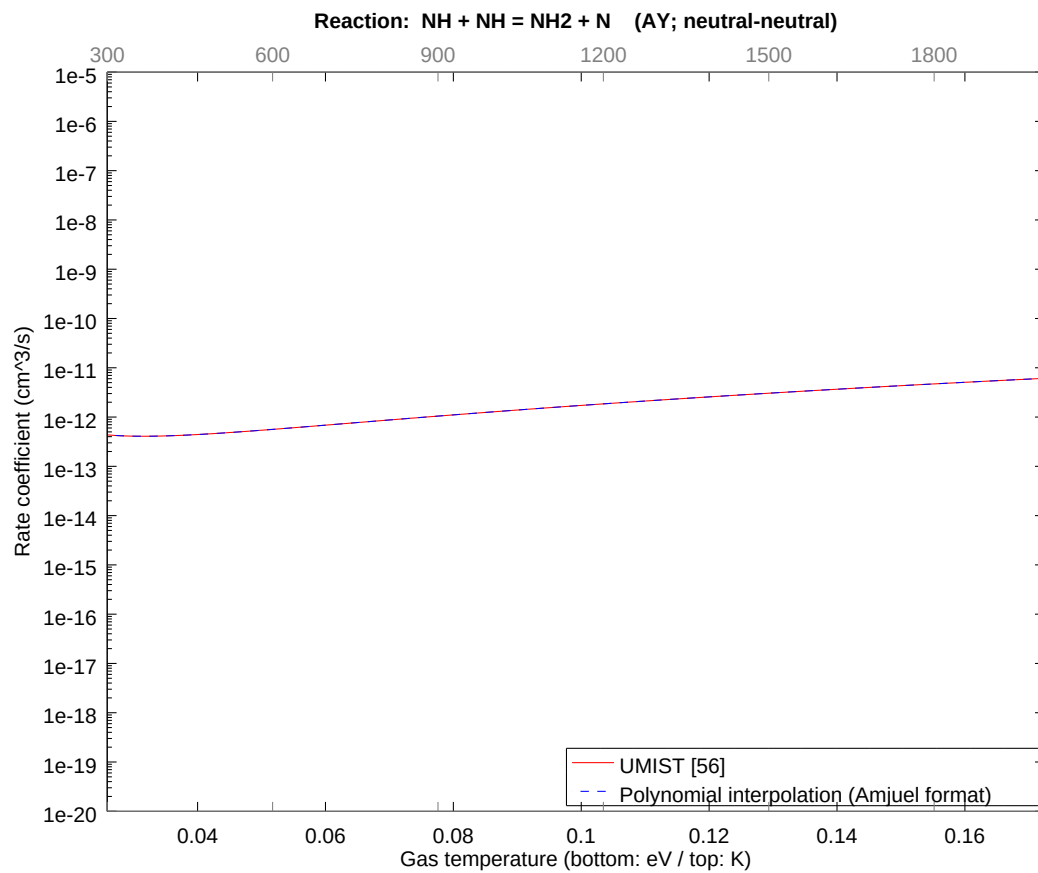
## 2.110 Reaction AX $\text{NH} + \text{NH} = \text{N}_2 + \text{H} + \text{H}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.057484583183E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



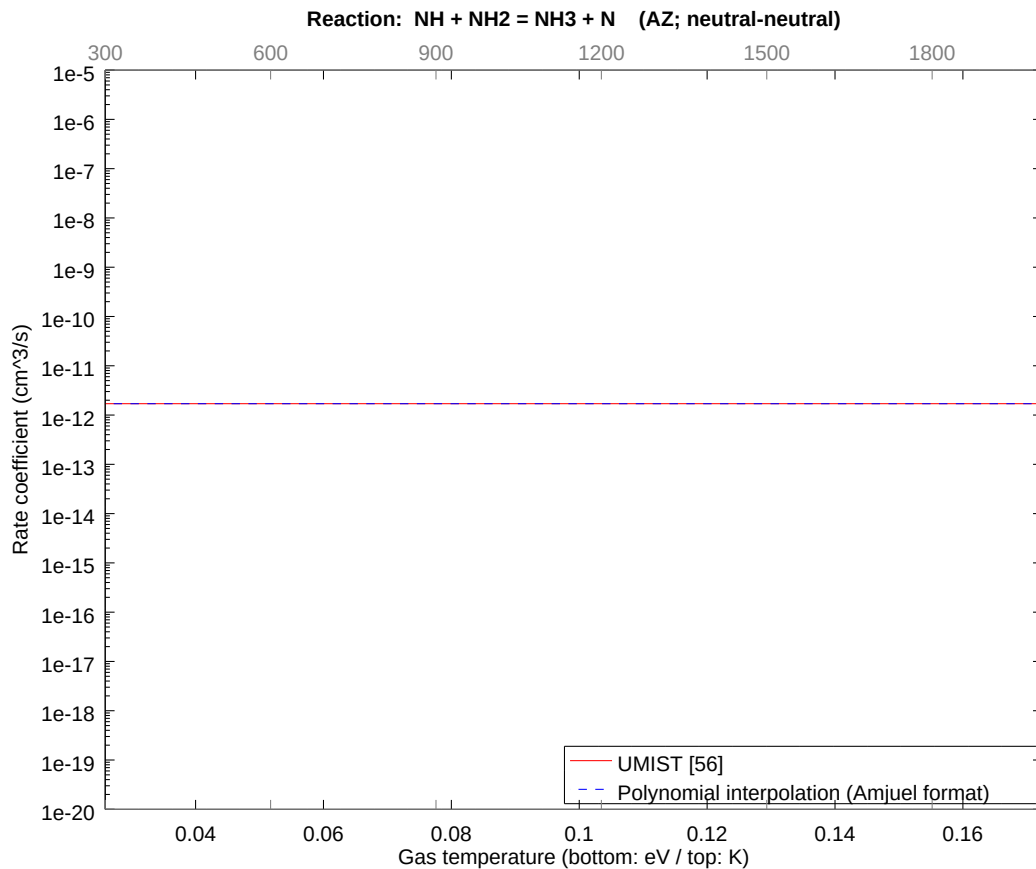
## 2.111 Reaction AY NH + NH = NH2 + N

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.087213986080E+01 | b1 | 3.033324552294E+00 | b2 | 1.402818964922E-01 |
| b3 | 6.441276966251E-02  | b4 | 4.853931554652E-02 | b5 | 1.515385231802E-02 |
| b6 | 3.799547234470E-03  | b7 | 4.837915635689E-04 | b8 | 3.598621695468E-05 |



## 2.112 Reaction AZ $\text{NH} + \text{NH}_2 = \text{NH}_3 + \text{N}$

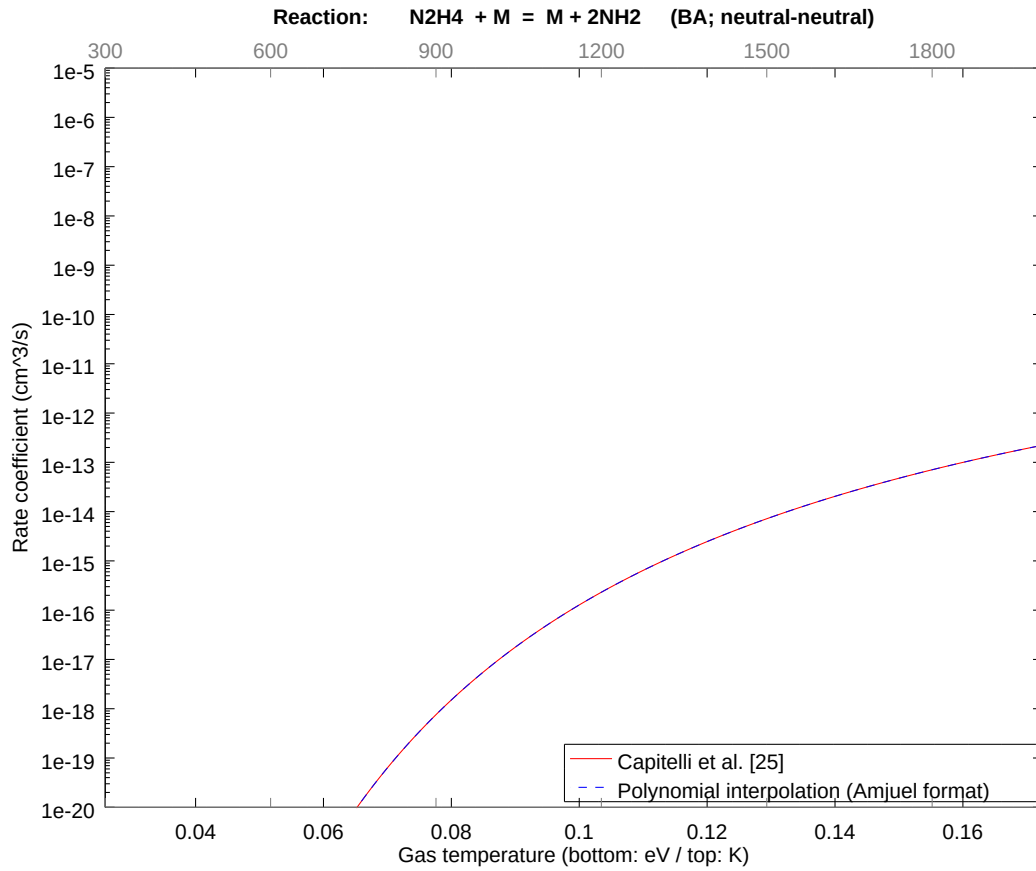
|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.710039286487E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |





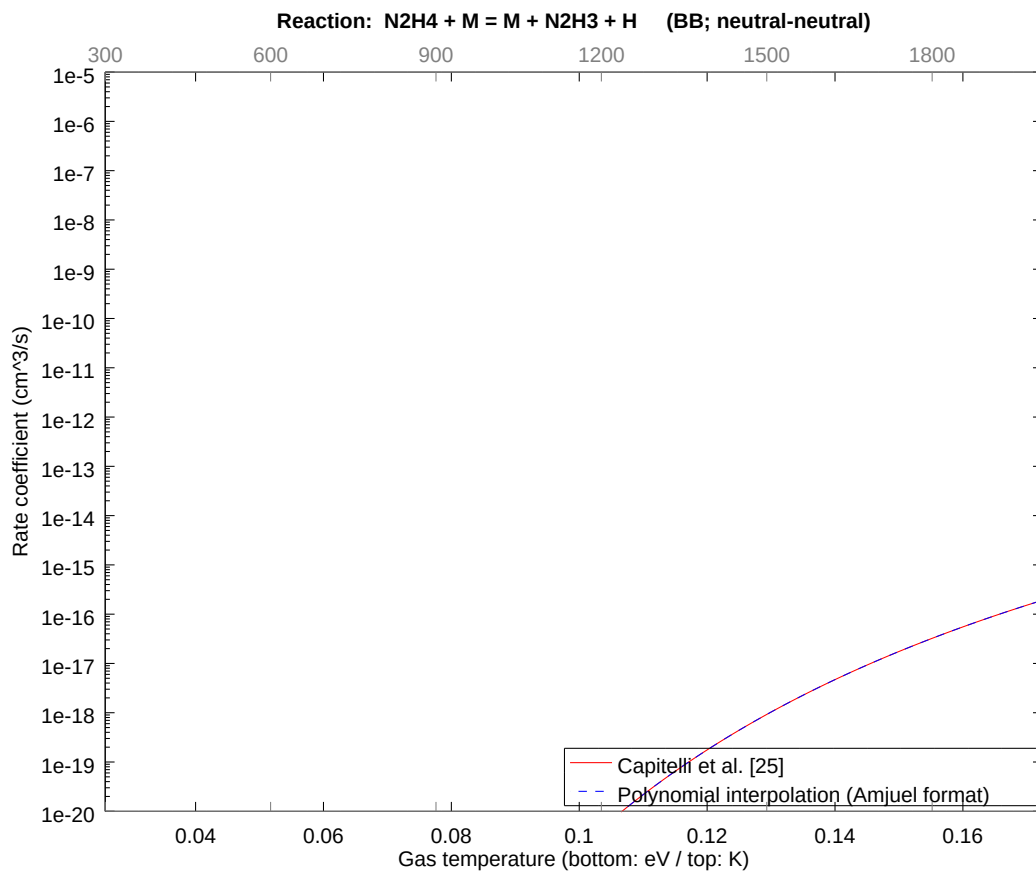
## 2.113 Reaction BA $\text{N}_2\text{H}_4 + \text{M} = \text{M} + 2\text{NH}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.092891758081E+01 | b1 | 6.727642154328E-01  | b2 | -2.573292141534E+00 |
| b3 | -1.181569952382E+00 | b4 | -8.903917175497E-01 | b5 | -2.779780557347E-01 |
| b6 | -6.969783852452E-02 | b7 | -8.874537981438E-03 | b8 | -6.601211627768E-04 |



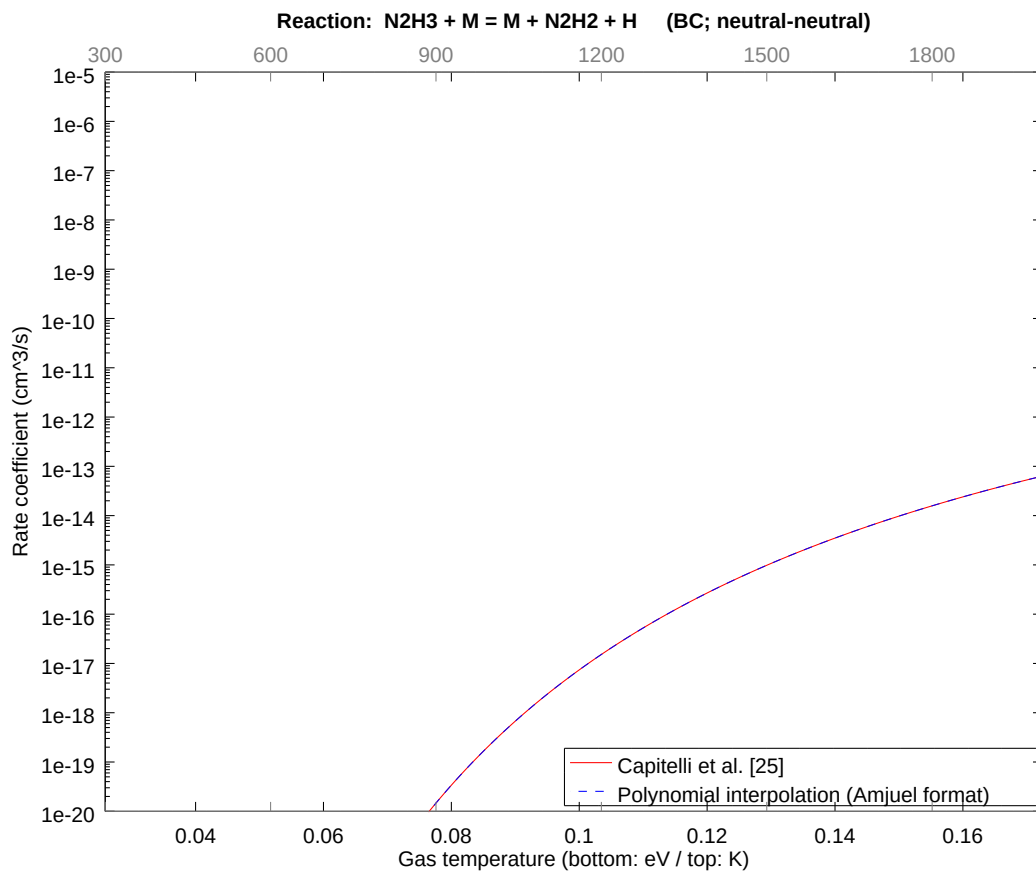
## 2.114 Reaction BB N2H4 + M = M + N2H3 + H

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.344346693000E+01 | b1 | 1.045070638427E+00  | b2 | -3.997346999314E+00 |
| b3 | -1.835448457155E+00 | b4 | -1.383132759255E+00 | b5 | -4.318105702736E-01 |
| b6 | -1.082684867080E-01 | b7 | -1.378569003483E-02 | b8 | -1.025430930953E-03 |



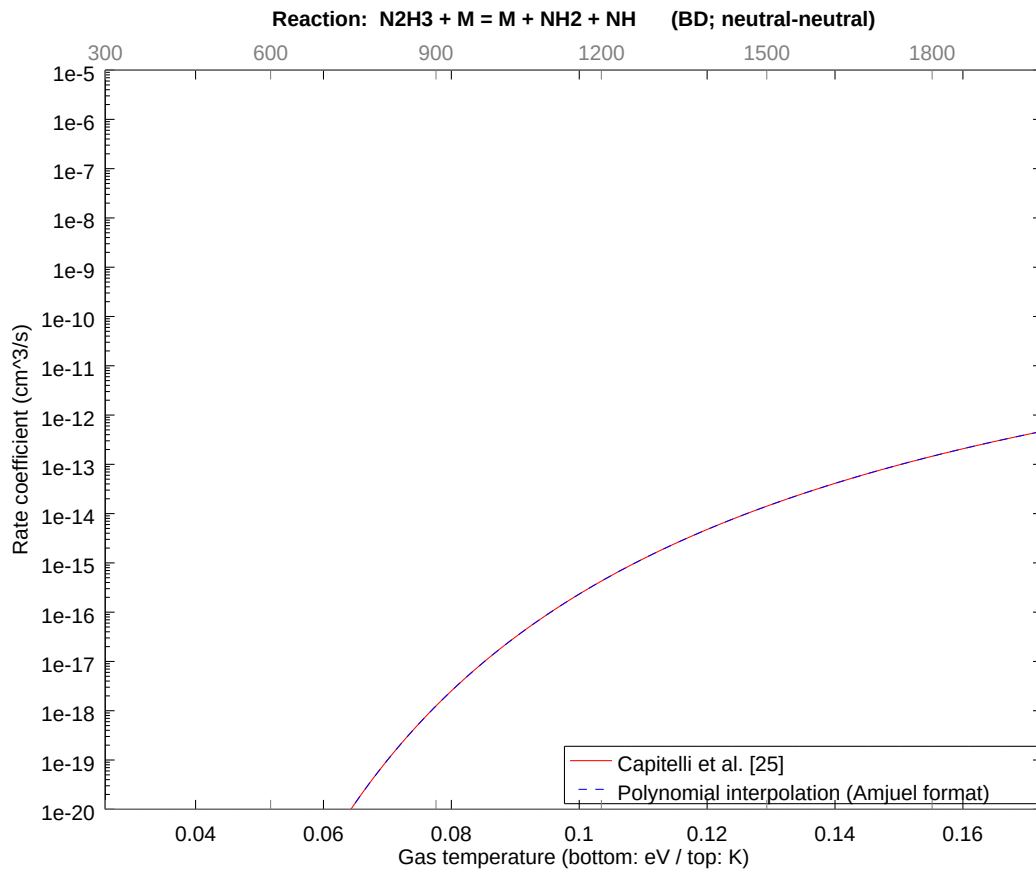
## 2.115 Reaction BC $\text{N}_2\text{H}_3 + \text{M} = \text{M} + \text{N}_2\text{H}_2 + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.042976292036E+01 | b1 | 8.164614287281E-01  | b2 | -3.122927353608E+00 |
| b3 | -1.433944115263E+00 | b4 | -1.080572472087E+00 | b5 | -3.373520092273E-01 |
| b6 | -8.458475546858E-02 | b7 | -1.077007036420E-02 | b8 | -8.011179159485E-04 |



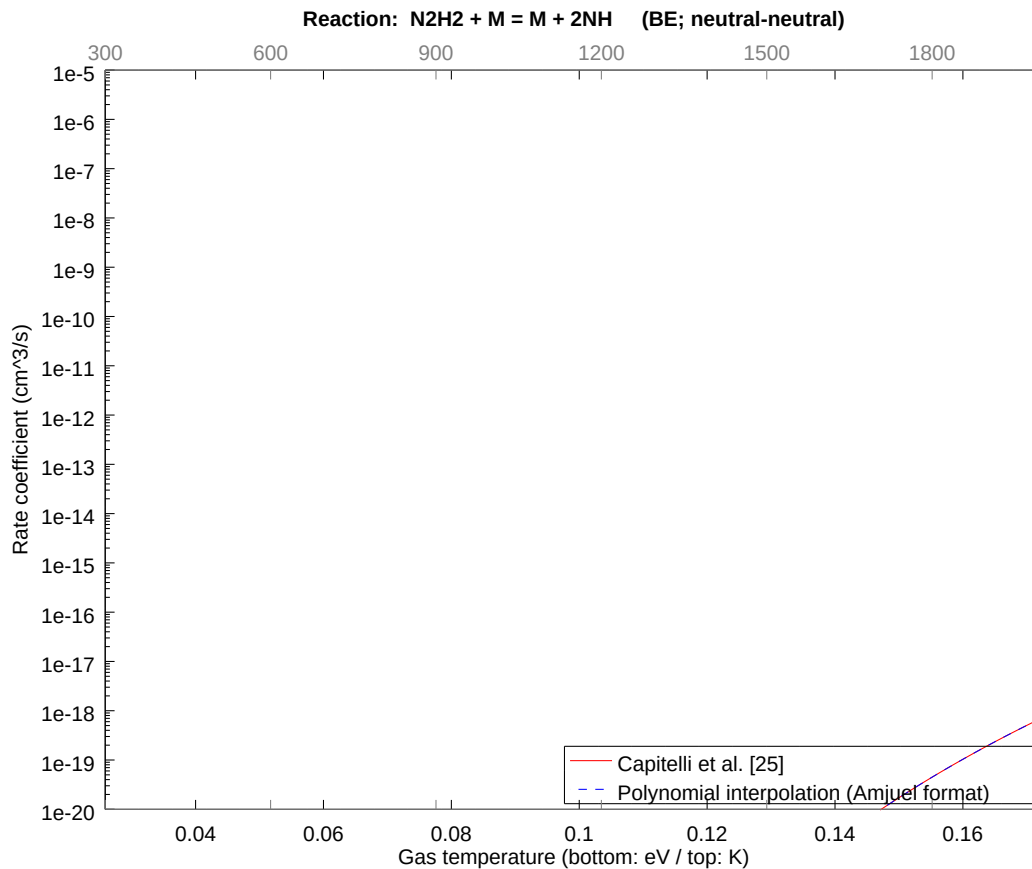
## 2.116 Reaction BD $\text{N}_2\text{H}_3 + \text{M} = \text{M} + \text{NH}_2 + \text{NH}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.002340925171E+01 | b1 | 6.858276010790E-01  | b2 | -2.623258975556E+00 |
| b3 | -1.204513055552E+00 | b4 | -9.076808758901E-01 | b5 | -2.833756875350E-01 |
| b6 | -7.105119455069E-02 | b7 | -9.046859101163E-03 | b8 | -6.729390491700E-04 |



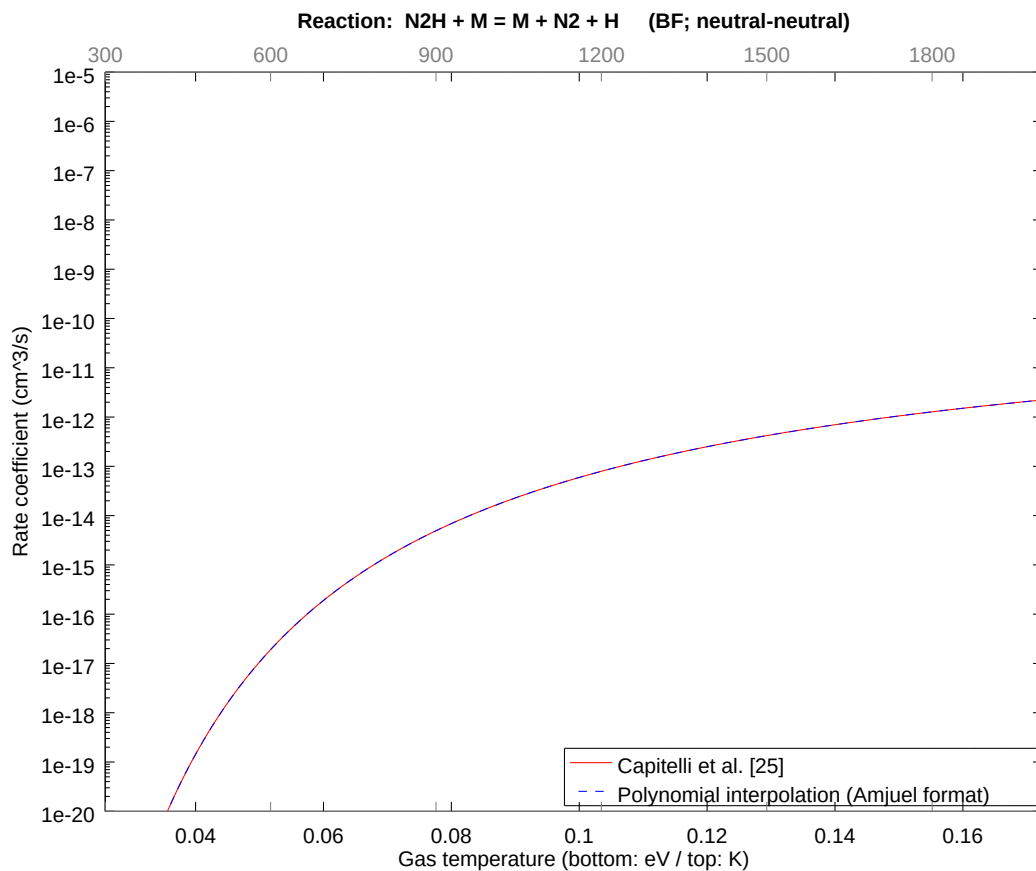
## 2.117 Reaction BE $\text{N}_2\text{H}_2 + \text{M} = \text{M} + 2\text{NH}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.185144297209E+01 | b1 | 1.632922861522E+00  | b2 | -6.245854701319E+00 |
| b3 | -2.867888225723E+00 | b4 | -2.161144941771E+00 | b5 | -6.747040176978E-01 |
| b6 | -1.691695107906E-01 | b7 | -2.154014071243E-02 | b8 | -1.602235831147E-03 |



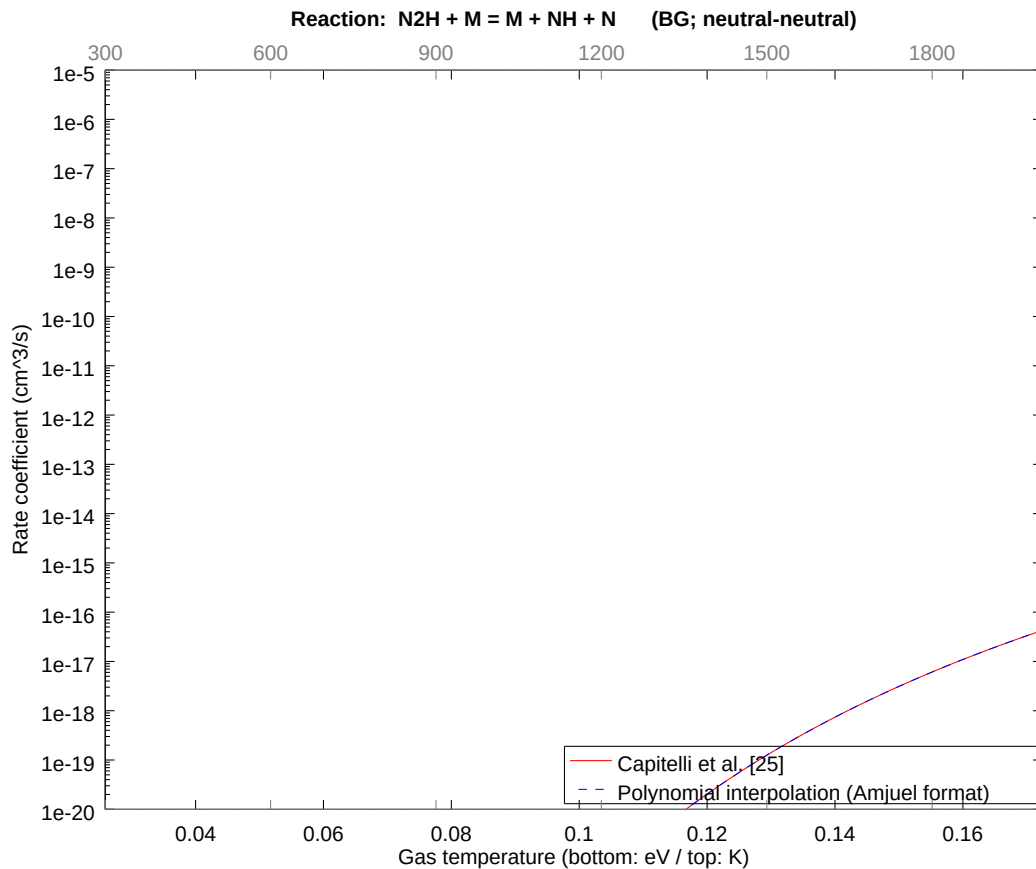
## 2.118 Reaction BF N2H + M = M + N2 + H

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.284781263330E+01 | b1 | 3.265845688574E-01  | b2 | -1.249170944978E+00 |
| b3 | -5.735776487824E-01 | b4 | -4.322289900869E-01 | b5 | -1.349408040612E-01 |
| b6 | -3.383390225510E-02 | b7 | -4.308028152669E-03 | b8 | -3.204471666919E-04 |



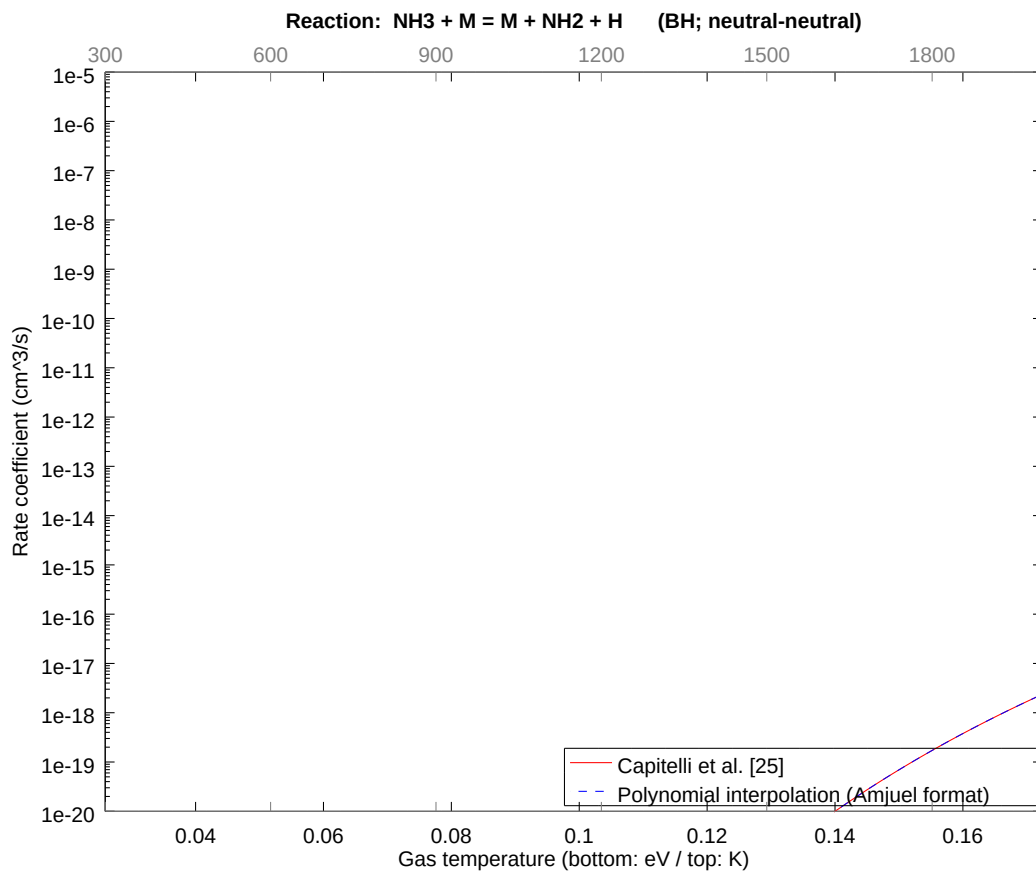
## 2.119 Reaction BG $\text{N}_2\text{H} + \text{M} = \text{M} + \text{NH} + \text{N}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.374823218580E+01 | b1 | 1.143045995553E+00  | b2 | -4.372098301507E+00 |
| b3 | -2.007521766427E+00 | b4 | -1.512801463379E+00 | b5 | -4.722928136757E-01 |
| b6 | -1.184186578008E-01 | b7 | -1.507809852559E-02 | b8 | -1.121565083068E-03 |



## 2.120 Reaction BH $\text{NH}_3 + \text{M} = \text{M} + \text{NH}_2 + \text{H}$

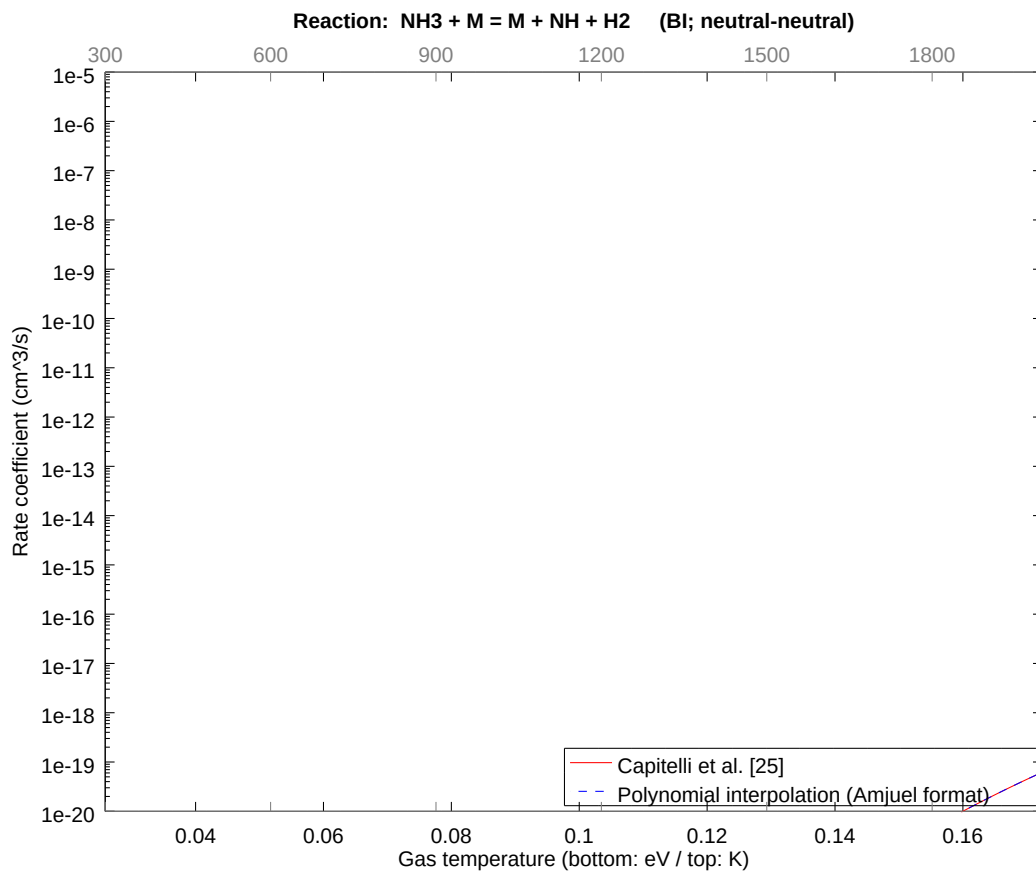
|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.178056950826E+01 | b1 | 1.541479169864E+00  | b2 | -5.896086853409E+00 |
| b3 | -2.707286496789E+00 | b4 | -2.040120830556E+00 | b5 | -6.369205943611E-01 |
| b6 | -1.596960184935E-01 | b7 | -2.033389286489E-02 | b8 | -1.512510626084E-03 |





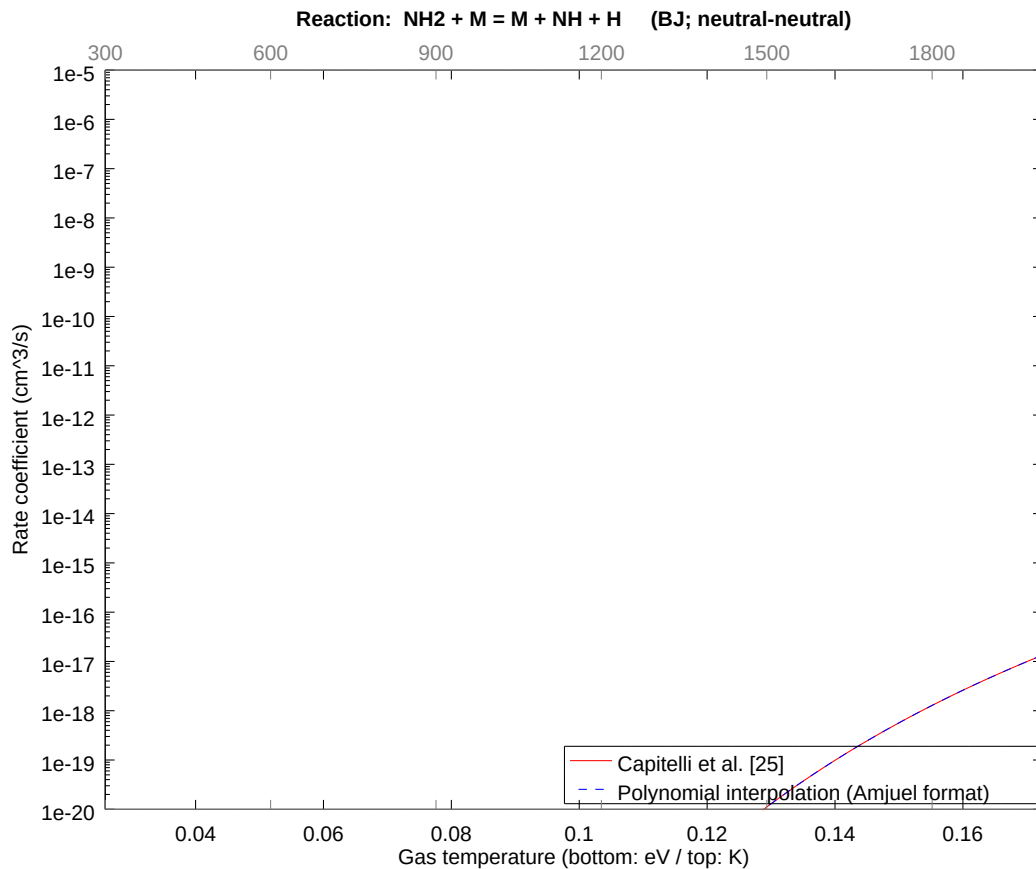
## 2.121 Reaction BI $\text{NH}_3 + \text{M} = \text{M} + \text{NH} + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.549792143959E+01 | b1 | 1.534947489548E+00  | b2 | -5.871103419390E+00 |
| b3 | -2.695814932102E+00 | b4 | -2.031476245129E+00 | b5 | -6.342217765633E-01 |
| b6 | -1.590193401233E-01 | b7 | -2.024773226688E-02 | b8 | -1.506101681115E-03 |



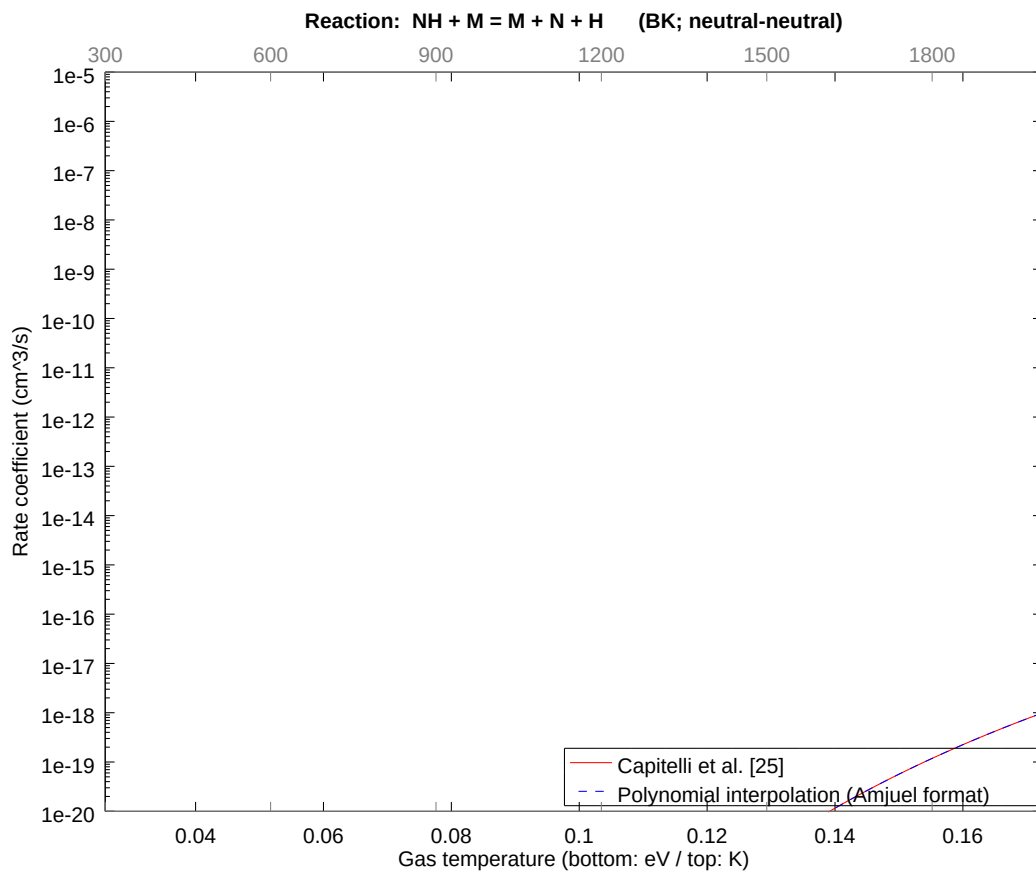
## 2.122 Reaction BJ $\text{NH}_2 + \text{M} = \text{M} + \text{NH} + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.404043242719E+01 | b1 | -4.977109602235E-01 | b2 | -5.746186315569E+00 |
| b3 | -2.638457160327E+00 | b4 | -1.988253342970E+00 | b5 | -6.207276952482E-01 |
| b6 | -1.556359497360E-01 | b7 | -1.981692943540E-02 | b8 | -1.474056963745E-03 |



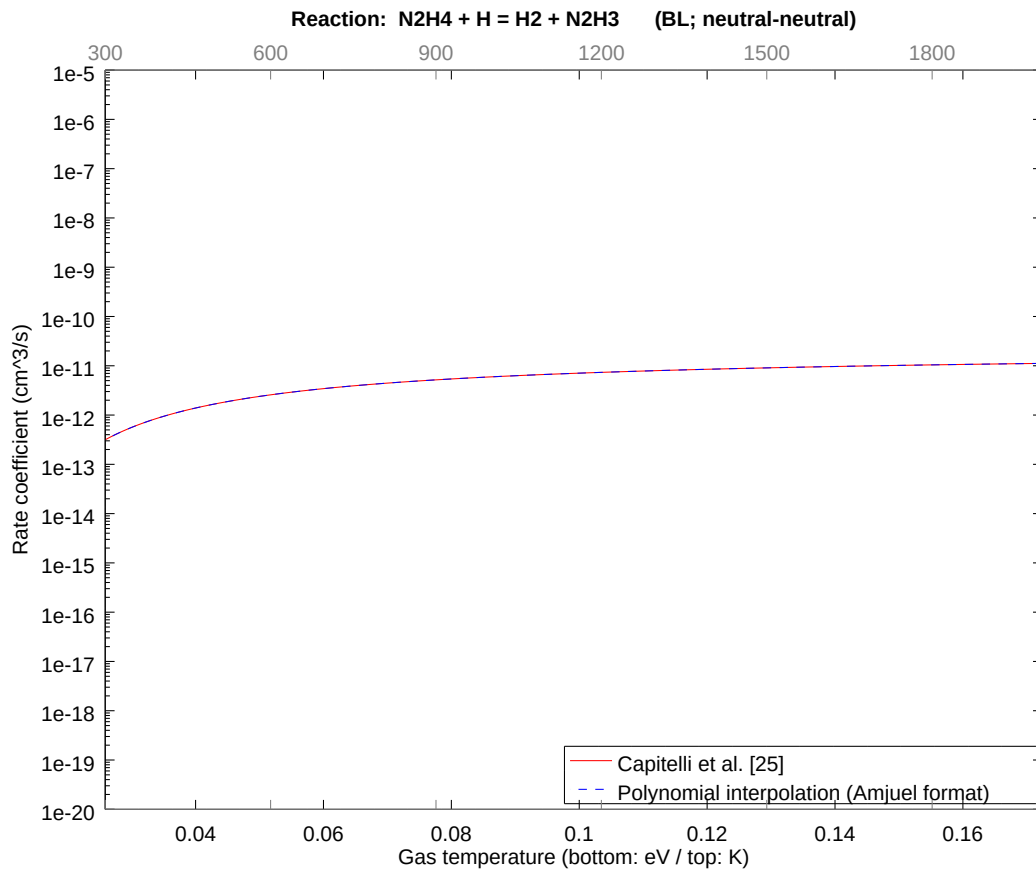
## 2.123 Reaction BK NH + M = M + N + H

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.823924894436E+01 | b1 | -6.283447873423E-01 | b2 | -5.246517936774E+00 |
| b3 | -2.409026100008E+00 | b4 | -1.815361746457E+00 | b5 | -5.667513734489E-01 |
| b6 | -1.421023887953E-01 | b7 | -1.809371816955E-02 | b8 | -1.345878096817E-03 |



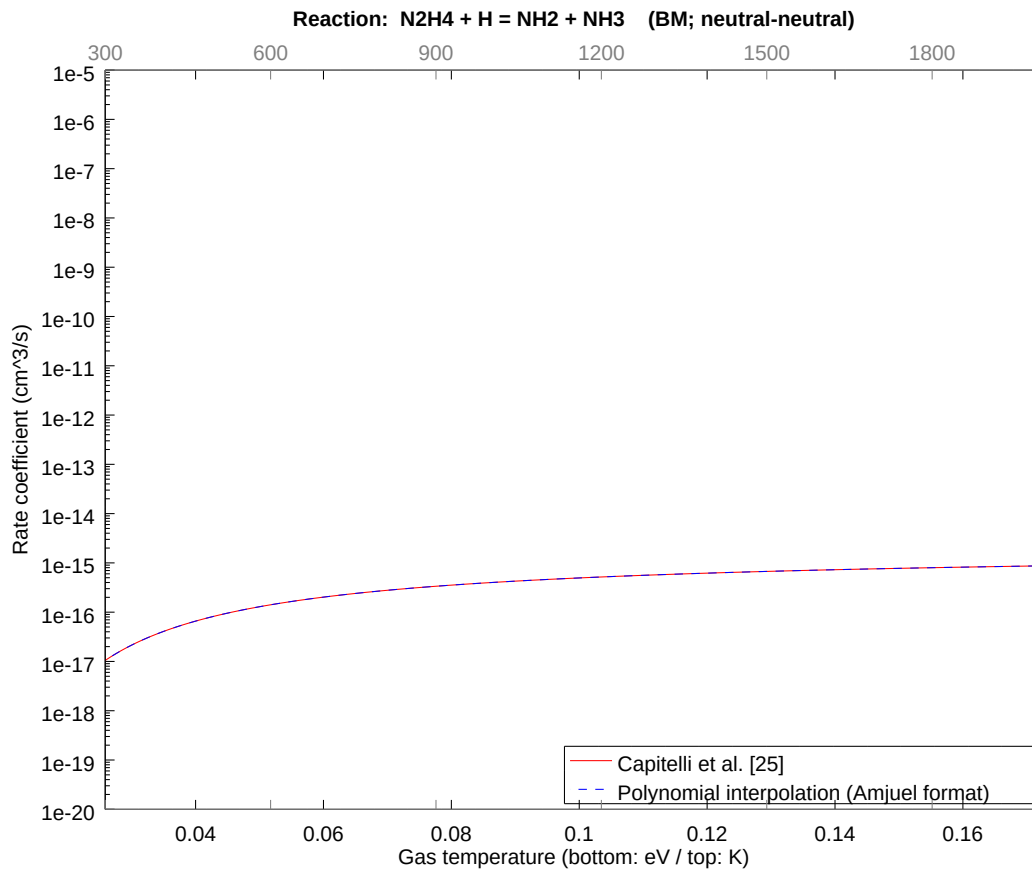
## 2.124 Reaction BL $\text{N}_2\text{H}_4 + \text{H} = \text{H}_2 + \text{N}_2\text{H}_3$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.471450008403E+01 | b1 | 4.114965527105E-02  | b2 | -1.573955393912E-01 |
| b3 | -7.227078380576E-02 | b4 | -5.446085268013E-02 | b5 | -1.700254125772E-02 |
| b6 | -4.263071667420E-03 | b7 | -5.428115447081E-04 | b8 | -4.037634284977E-05 |



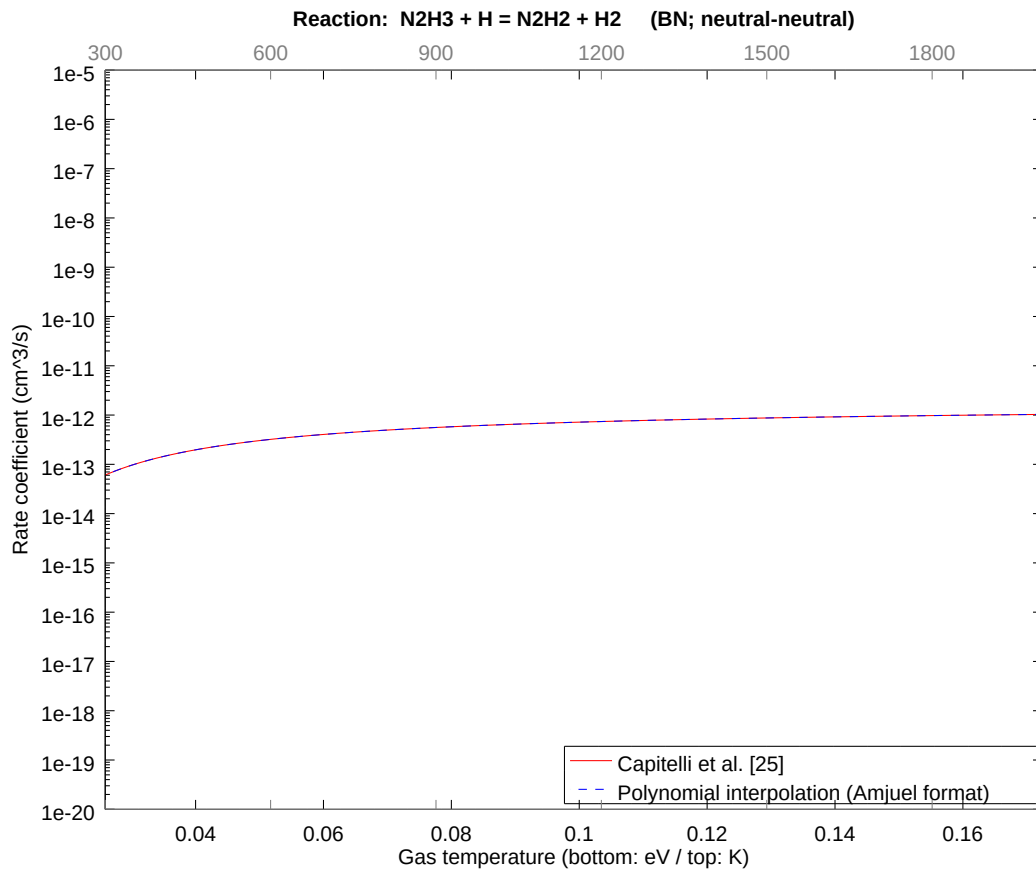
## 2.125 Reaction BM $\text{N}_2\text{H}_4 + \text{H} = \text{NH}_2 + \text{NH}_3$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.405540044124E+01 | b1 | 5.094718741288E-02  | b2 | -1.948706746692E-01 |
| b3 | -8.947811877476E-02 | b4 | -6.742772508512E-02 | b5 | -2.105076621879E-02 |
| b6 | -5.278088896147E-03 | b7 | -6.720524067615E-04 | b8 | -4.998975867108E-05 |



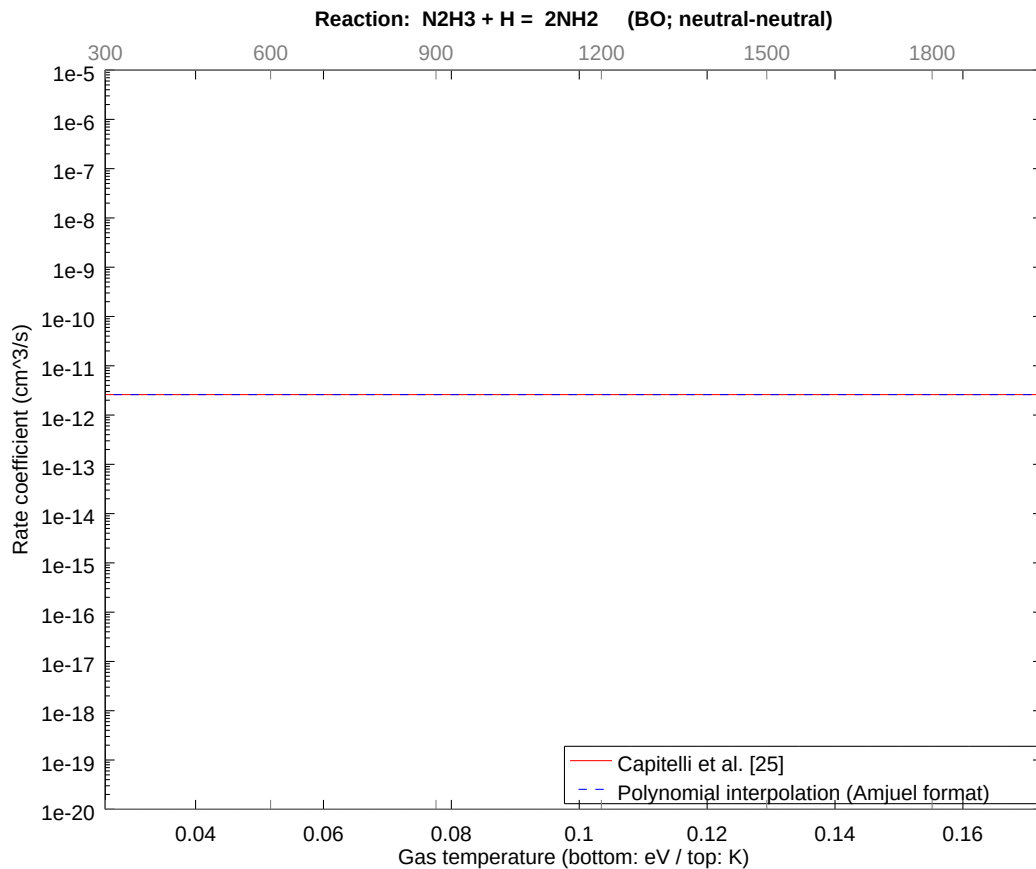
## 2.126 Reaction BN $\text{N}_2\text{H}_3 + \text{H} = \text{N}_2\text{H}_2 + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.720198128343E+01 | b1 | 3.265845275819E-02  | b2 | -1.249170998387E-01 |
| b3 | -5.735776876858E-02 | b4 | -4.322290075117E-02 | b5 | -1.349408089682E-02 |
| b6 | -3.383390310207E-03 | b7 | -4.308028234413E-04 | b8 | -3.204471700596E-05 |



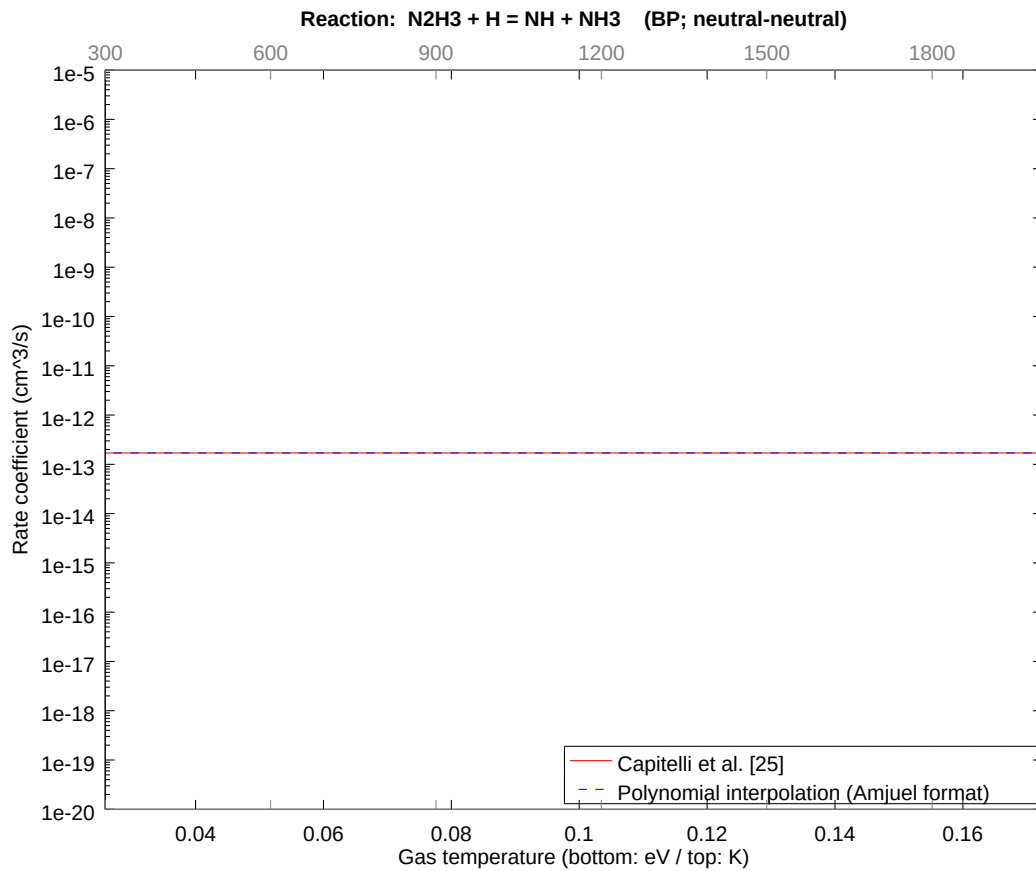
## 2.127 Reaction BO N2H3 + H = 2NH2

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.667550967090E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



## 2.128 Reaction BP N2H3 + H = NH + NH3

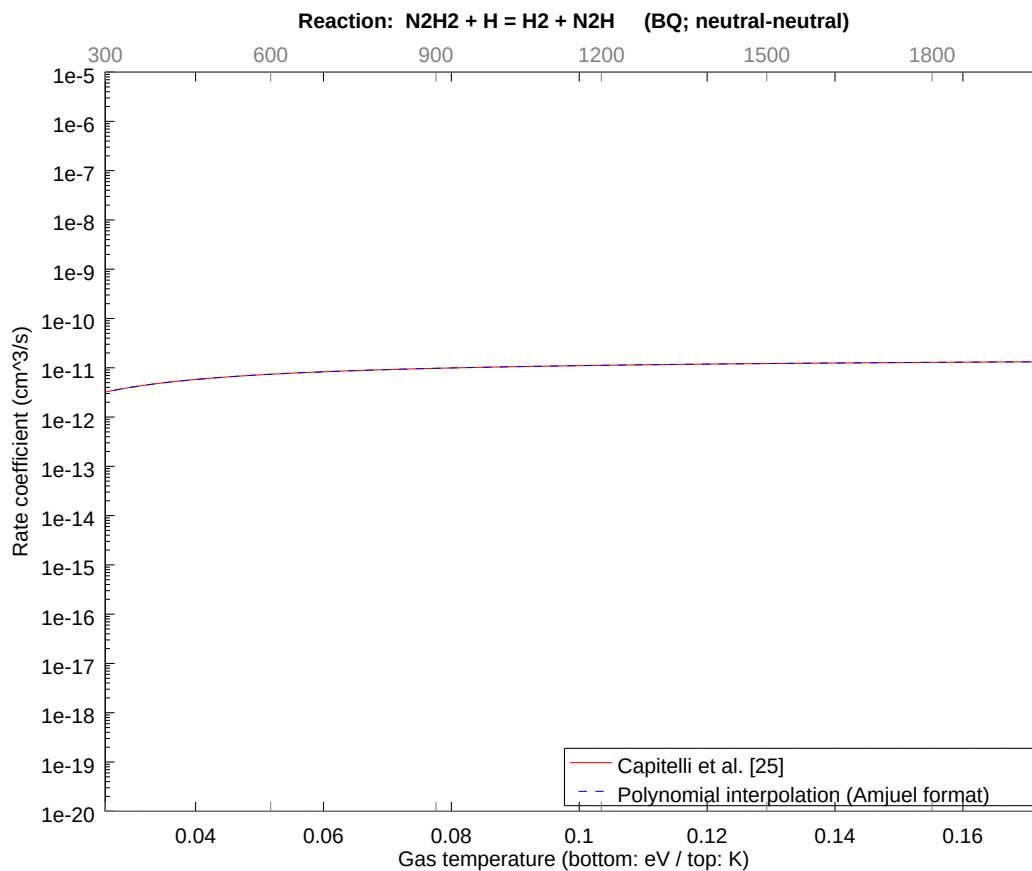
|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.940297795786E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |





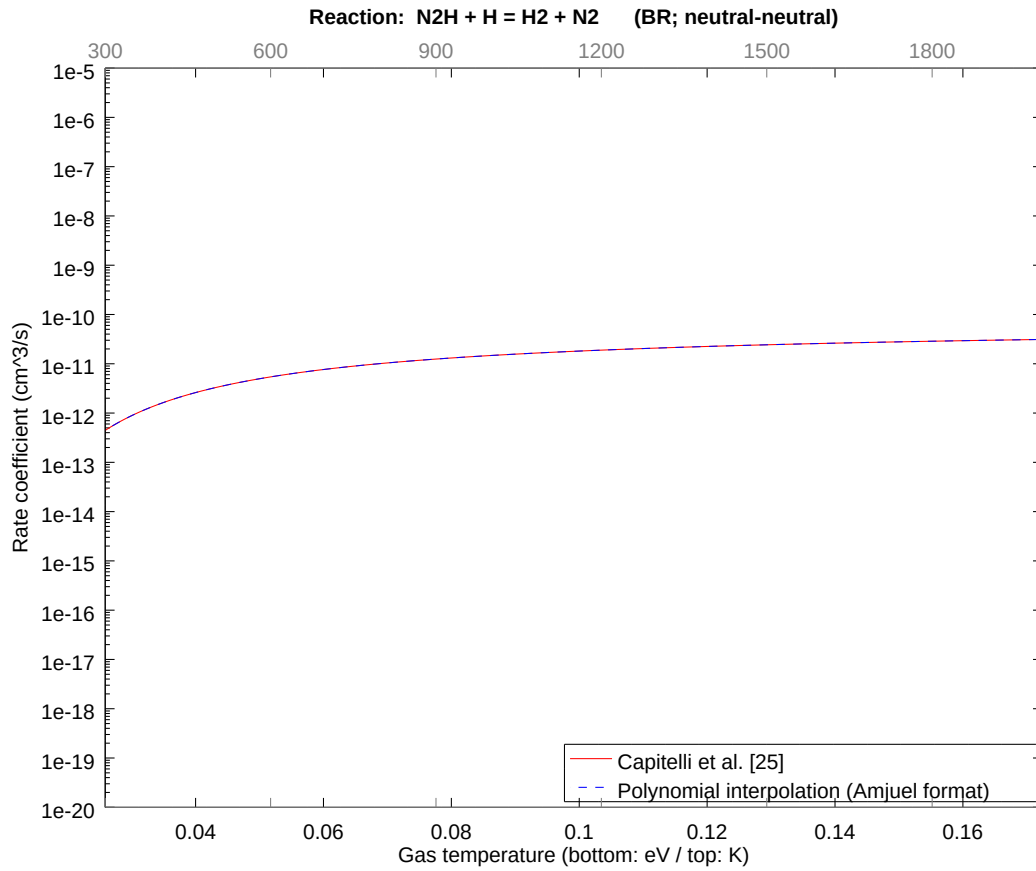
## 2.129 Reaction BQ N2H2 + H = H2 + N2H

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.484860198207E+01 | b1 | 1.632922359499E-02  | b2 | -6.245855355615E-02 |
| b3 | -2.867888706383E-02 | b4 | -2.161145159392E-02 | b5 | -6.747040798658E-03 |
| b6 | -1.691695217315E-03 | b7 | -2.154014179679E-04 | b8 | -1.602235877477E-05 |



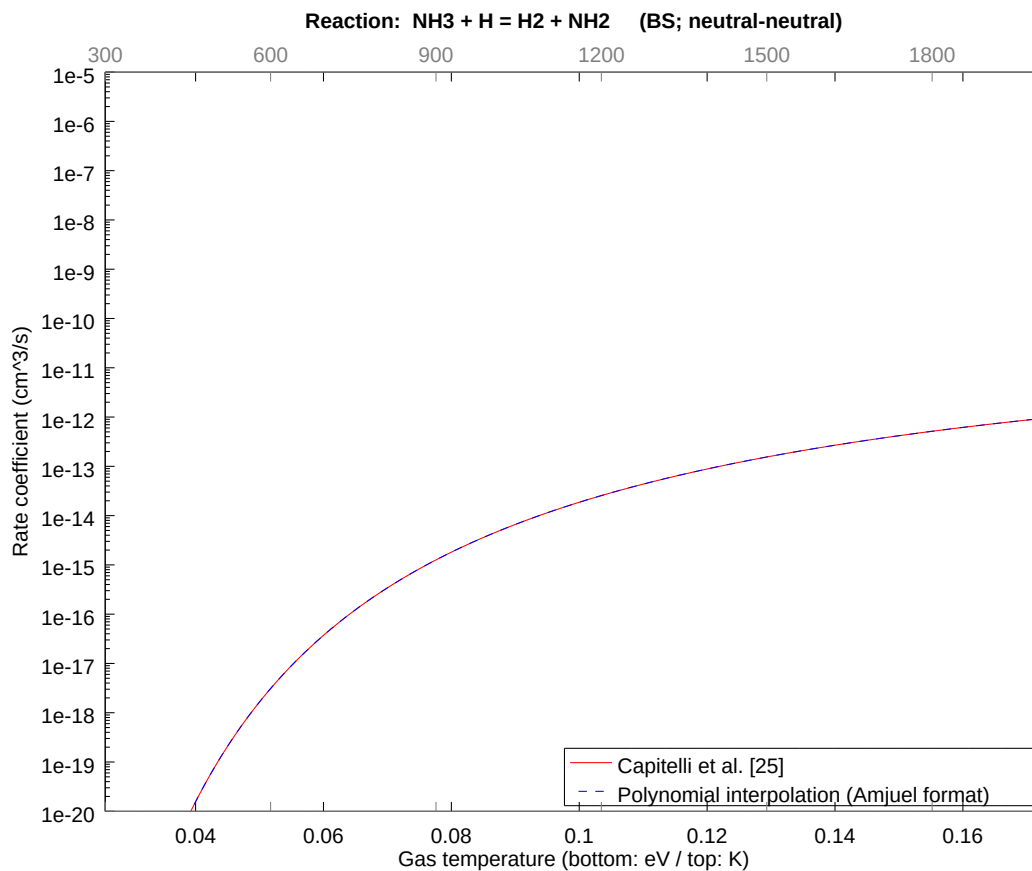
## 2.130 Reaction BR $\text{N}_2\text{H} + \text{H} = \text{H}_2 + \text{N}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.359374900180E+01 | b1 | 4.898767865117E-02  | b2 | -1.873756508272E-01 |
| b3 | -8.603665430488E-02 | b4 | -6.483435184066E-02 | b5 | -2.024112161349E-02 |
| b6 | -5.075085525879E-03 | b7 | -6.462042427391E-04 | b8 | -4.806707591320E-05 |



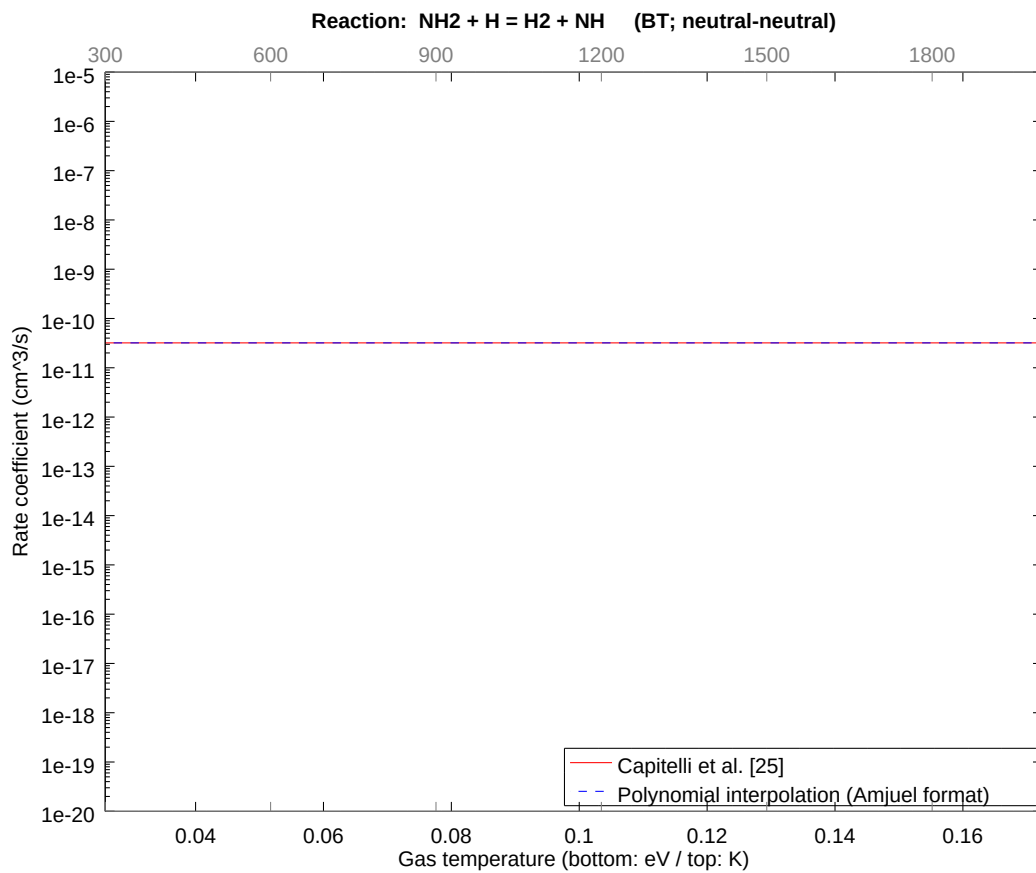
## 2.131 Reaction BS $\text{NH}_3 + \text{H} = \text{H}_2 + \text{NH}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.338310025915E+01 | b1 | 3.533645035060E-01  | b2 | -1.351602962390E+00 |
| b3 | -6.206110158630E-01 | b4 | -4.676717671848E-01 | b5 | -1.460059499565E-01 |
| b6 | -3.660828223069E-02 | b7 | -4.661286459931E-03 | b8 | -3.467238342890E-04 |



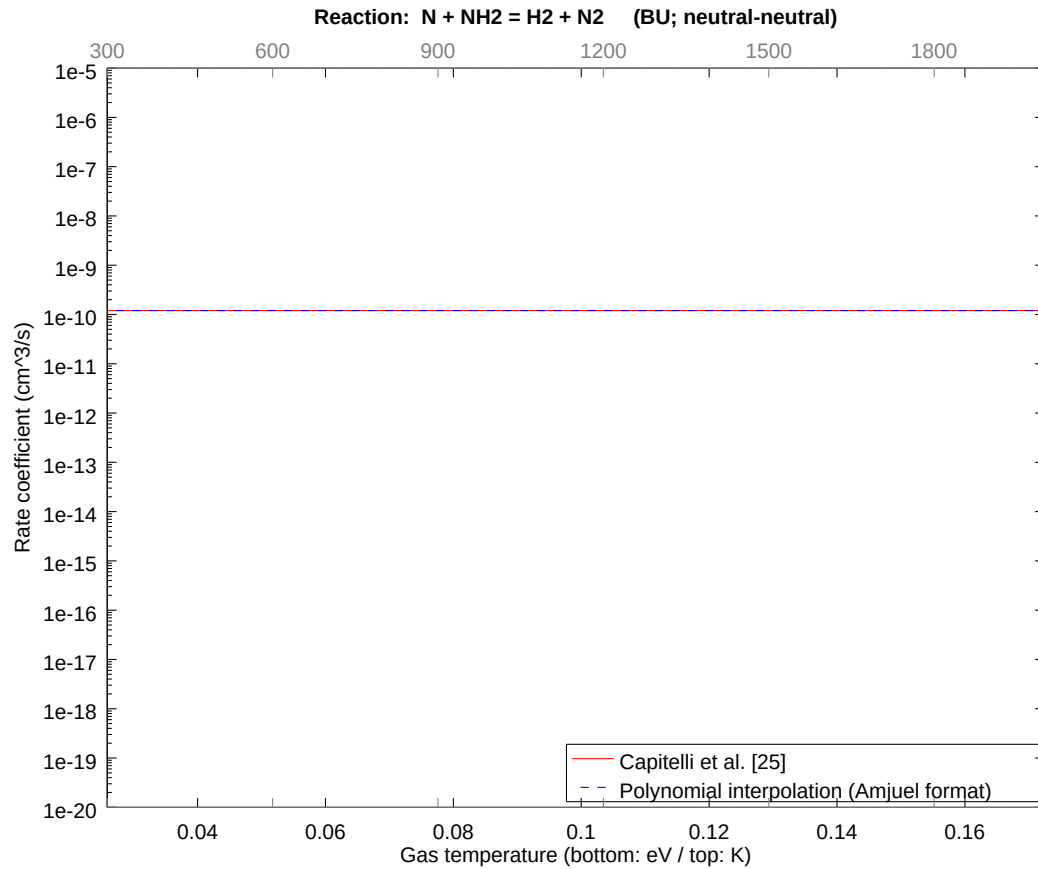
## 2.132 Reaction BT NH<sub>2</sub> + H = H<sub>2</sub> + NH

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.416528521313E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



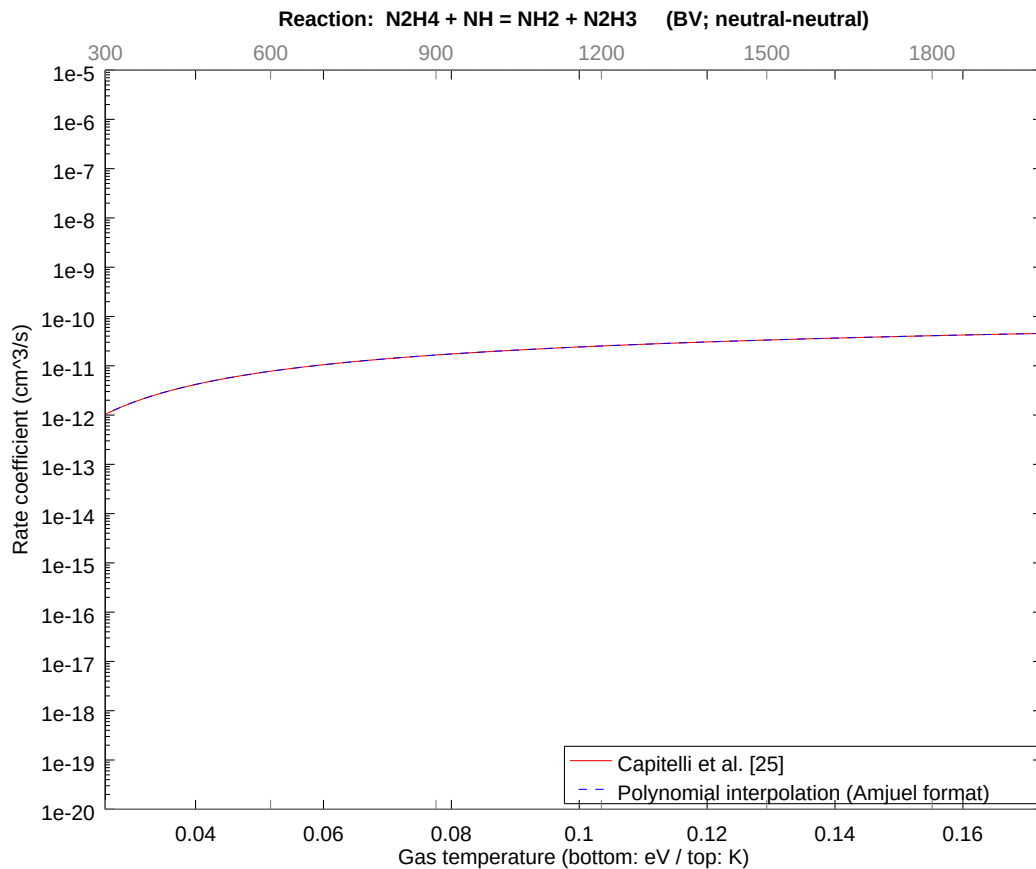
## 2.133 Reaction BU $\text{N} + \text{NH}_2 = \text{H}_2 + \text{N}_2$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.284352937315E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



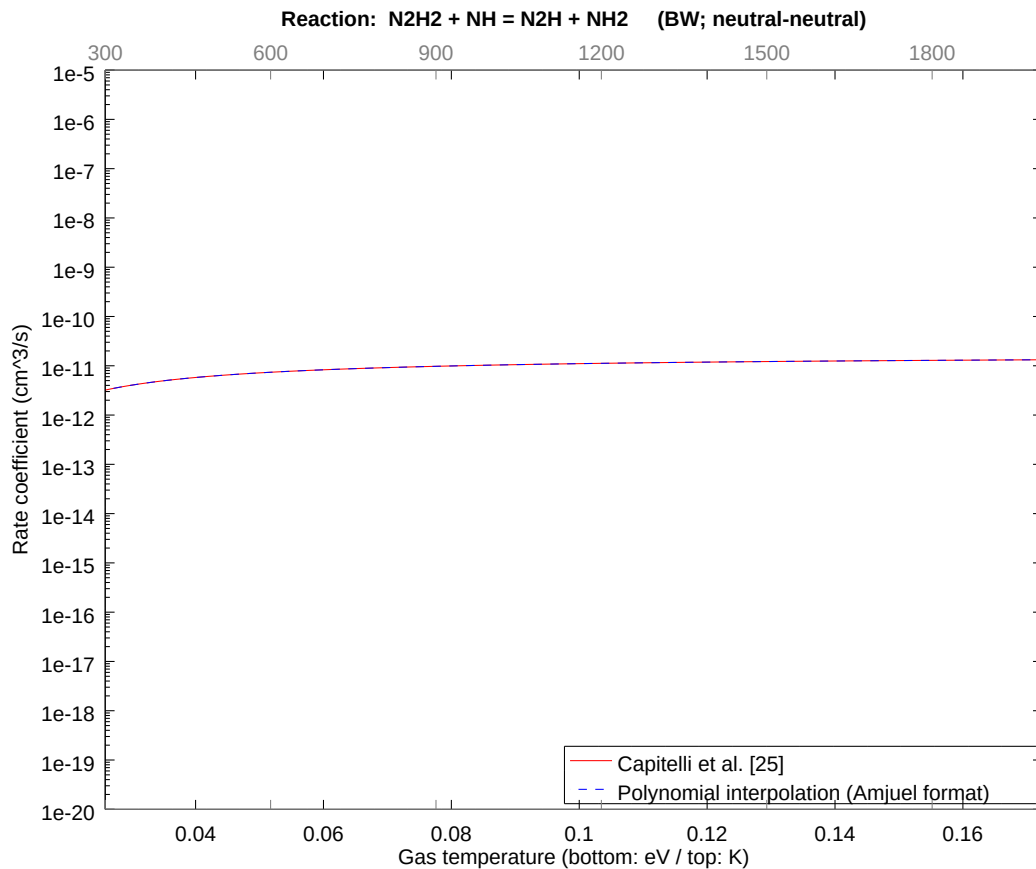
## 2.134 Reaction BV $\text{N}_2\text{H}_4 + \text{NH} = \text{NH}_2 + \text{N}_2\text{H}_3$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.253788555373E+01 | b1 | 5.326584523650E-01  | b2 | -1.249171003025E-01 |
| b3 | -5.735776906945E-02 | b4 | -4.322290086803E-02 | b5 | -1.349408092450E-02 |
| b6 | -3.383390314085E-03 | b7 | -4.308028237328E-04 | b8 | -3.204471701487E-05 |



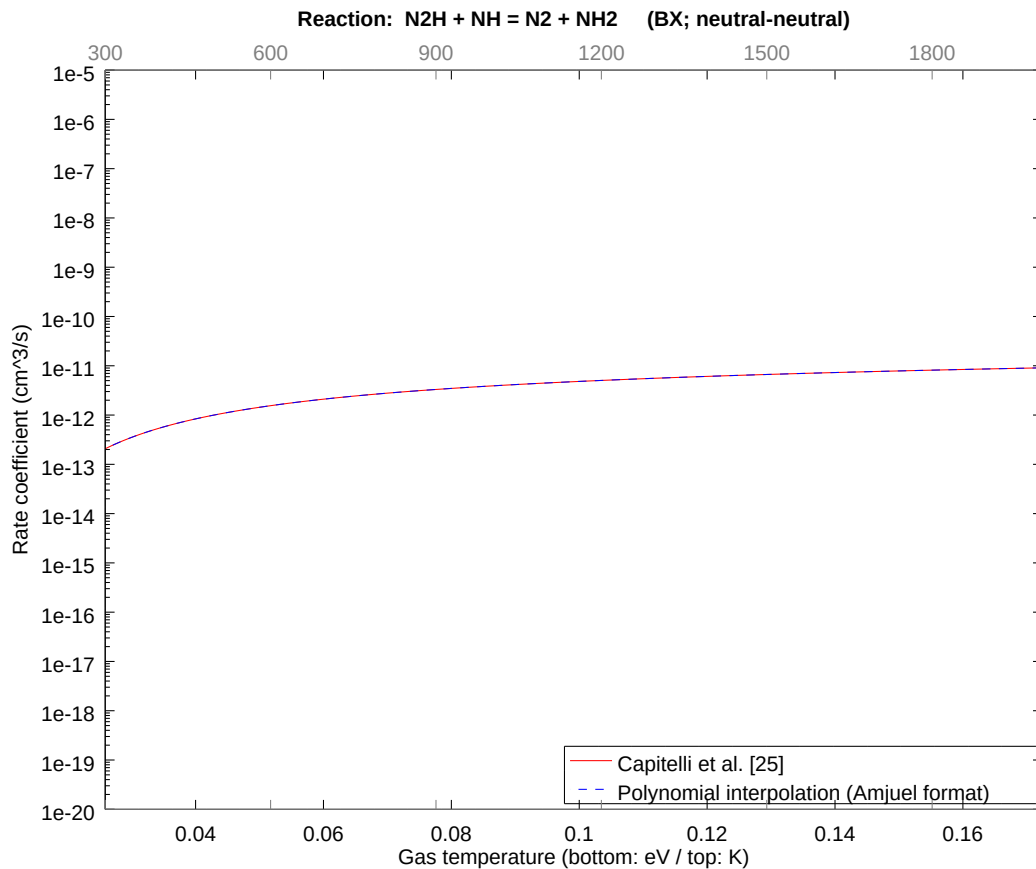
## 2.135 Reaction BW $\text{N}_2\text{H}_2 + \text{NH} = \text{N}_2\text{H} + \text{NH}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.484860198207E+01 | b1 | 1.632922359499E-02  | b2 | -6.245855355615E-02 |
| b3 | -2.867888706383E-02 | b4 | -2.161145159392E-02 | b5 | -6.747040798658E-03 |
| b6 | -1.691695217315E-03 | b7 | -2.154014179679E-04 | b8 | -1.602235877477E-05 |



## 2.136 Reaction BX $\text{N}_2\text{H} + \text{NH} = \text{N}_2 + \text{NH}_2$

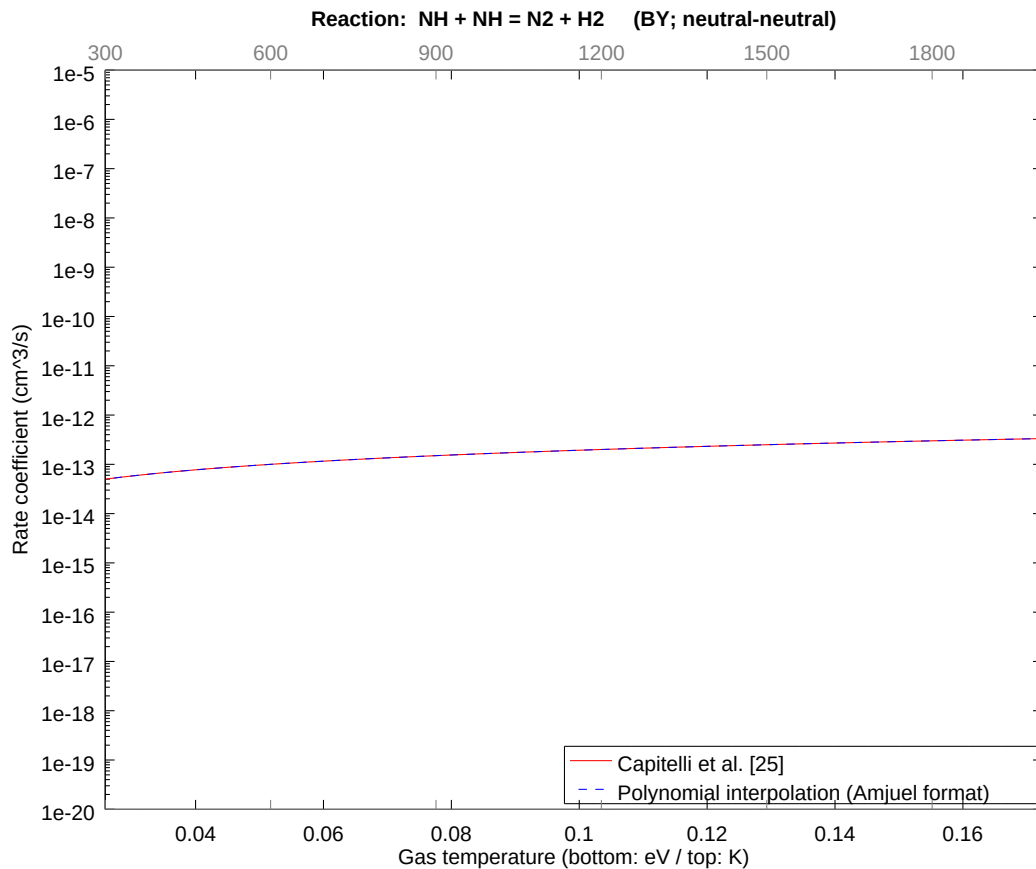
|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.414732346361E+01 | b1 | 5.326584602754E-01  | b2 | -1.249170896738E-01 |
| b3 | -5.735776098318E-02 | b4 | -4.322289705700E-02 | b5 | -1.349407978481E-02 |
| b6 | -3.383390102820E-03 | b7 | -4.308028015319E-04 | b8 | -3.204471600200E-05 |





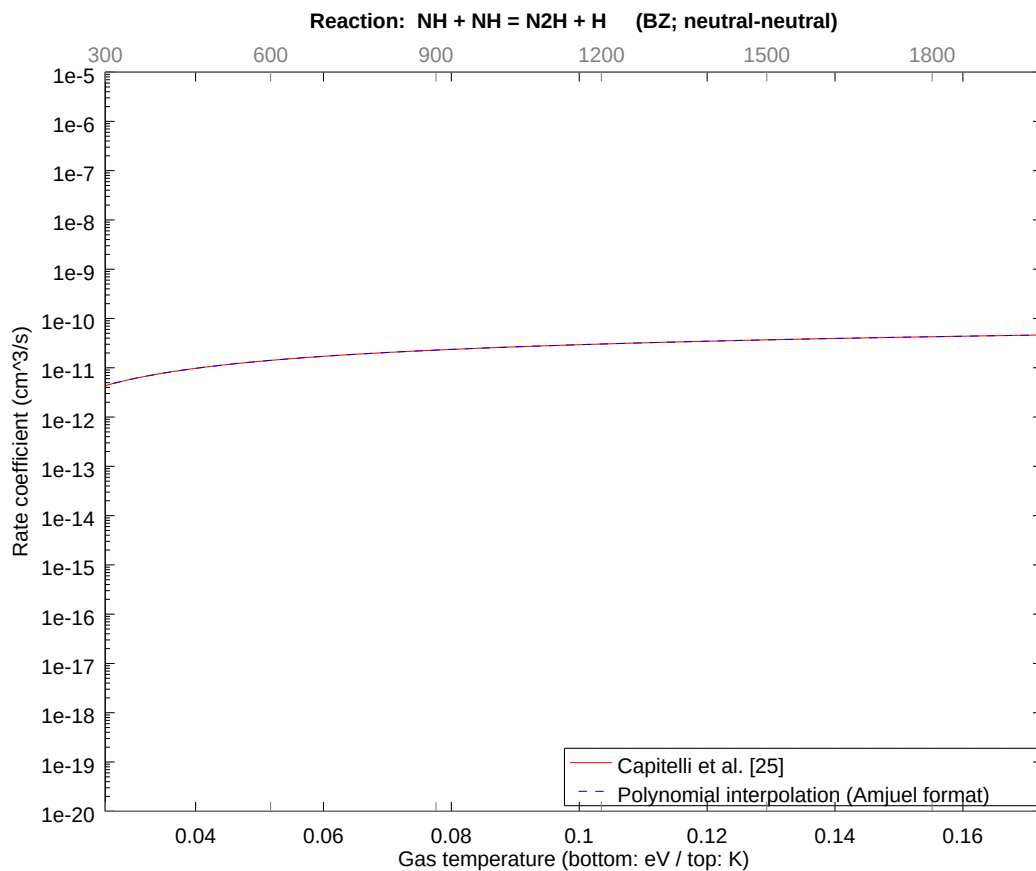
## 2.137 Reaction BY $\text{NH} + \text{NH} = \text{N}_2 + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.697189708924E+01 | b1 | 9.999999949711E-01  | b2 | -6.872542217226E-09 |
| b3 | -5.303904524598E-09 | b4 | -2.527570319855E-09 | b5 | -7.615316077845E-10 |
| b6 | -1.416645555477E-10 | b7 | -1.487898449876E-11 | b8 | -6.757021830936E-13 |



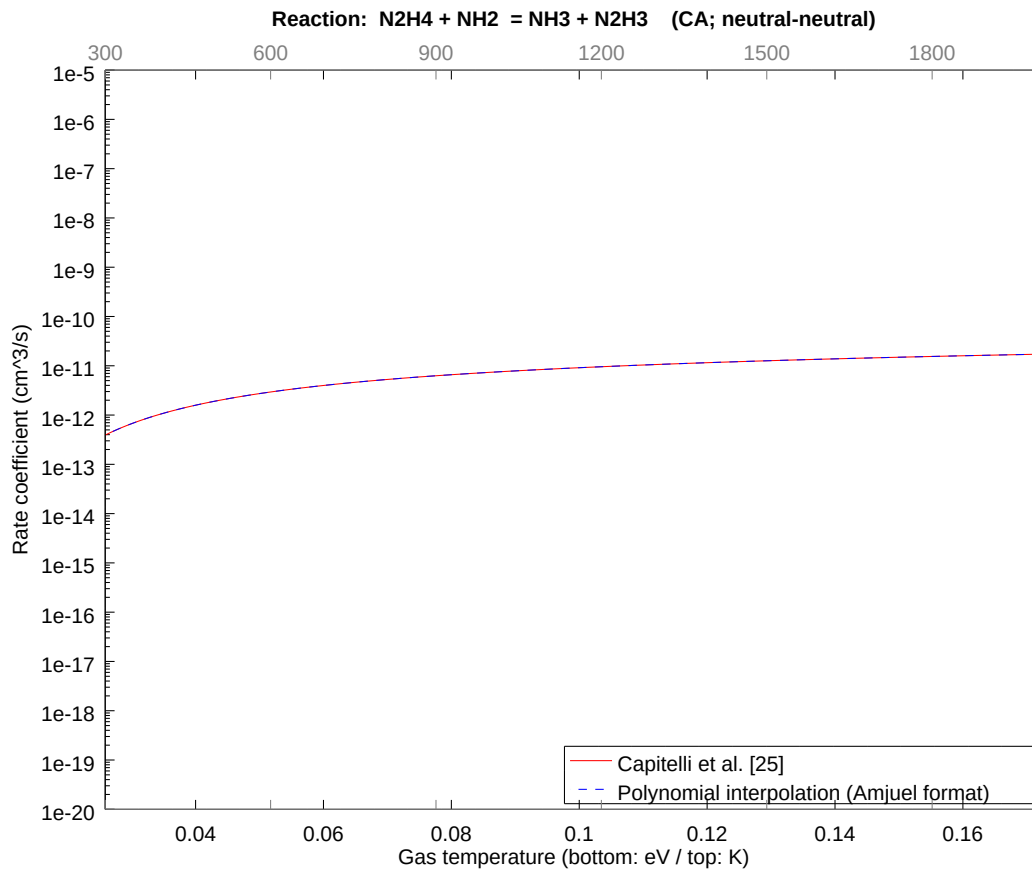
## 2.138 Reaction BZ $\text{NH} + \text{NH} = \text{N}_2\text{H} + \text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.271889295836E+01 | b1 | 5.163292262952E-01  | b2 | -6.245855016915E-02 |
| b3 | -2.867888467247E-02 | b4 | -2.161145055453E-02 | b5 | -6.747040513874E-03 |
| b6 | -1.691695169278E-03 | b7 | -2.154014134061E-04 | b8 | -1.602235858798E-05 |



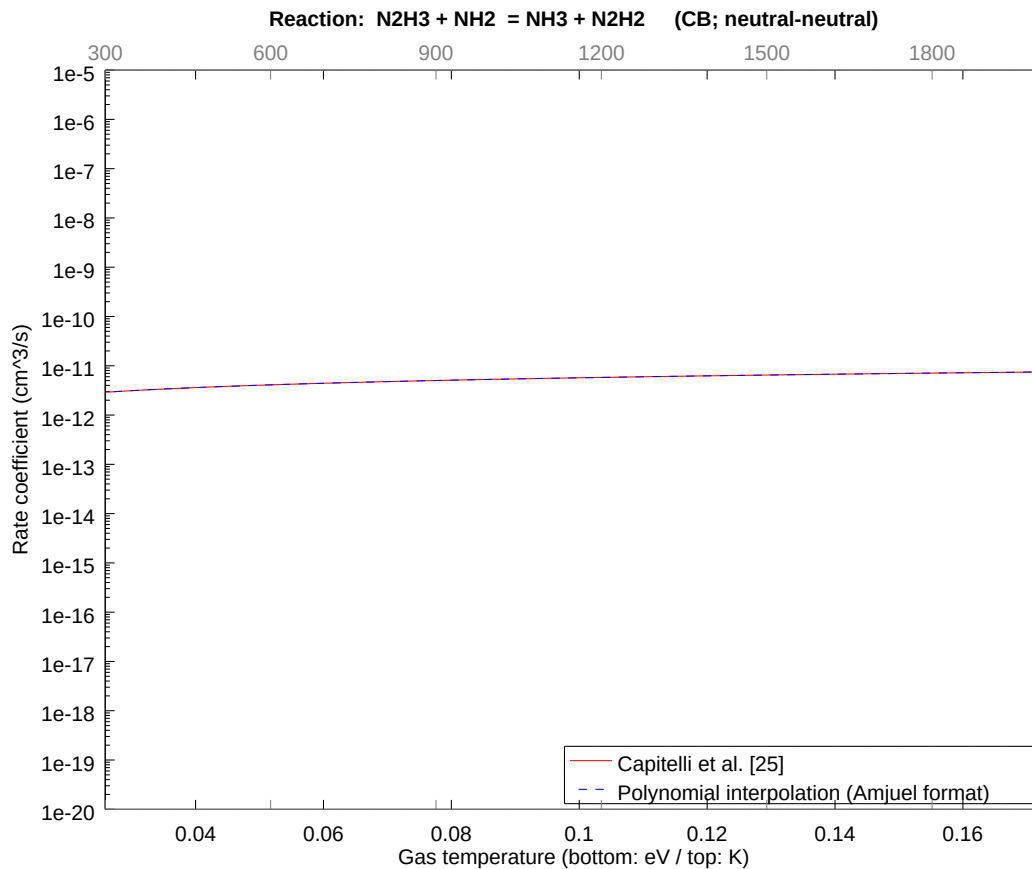
## 2.139 Reaction CA N2H4 + NH2 = NH3 + N2H3

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.350728610916E+01 | b1 | 5.326584576680E-01  | b2 | -1.249170933627E-01 |
| b3 | -5.735776393498E-02 | b4 | -4.322289851799E-02 | b5 | -1.349408024277E-02 |
| b6 | -3.383390191598E-03 | b7 | -4.308028112622E-04 | b8 | -3.204471646367E-05 |



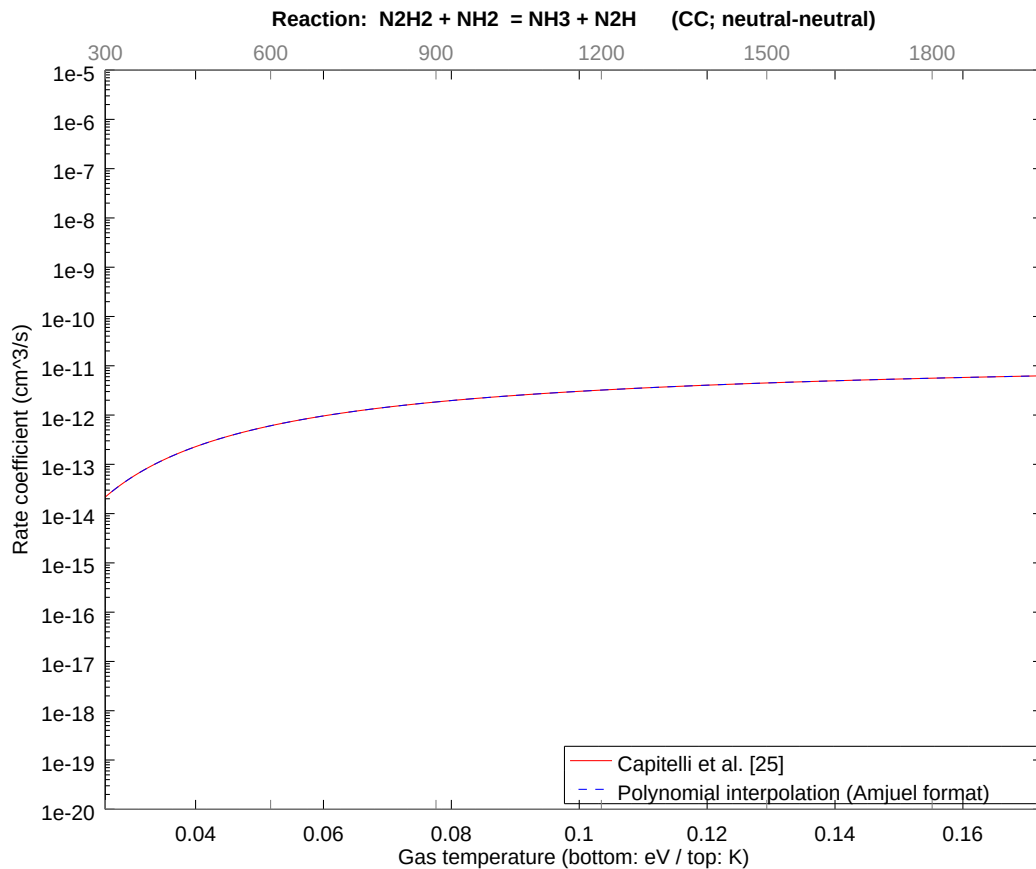
## 2.140 Reaction CB N2H3 + NH2 = NH3 + N2H2

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.473888223088E+01 | b1 | 4.999999909009E-01  | b2 | -1.252288180630E-08 |
| b3 | -9.740196382601E-09 | b4 | -4.682862879700E-09 | b5 | -1.425228340047E-09 |
| b6 | -2.682081542161E-10 | b7 | -2.854101745411E-11 | b8 | -1.315306849704E-12 |



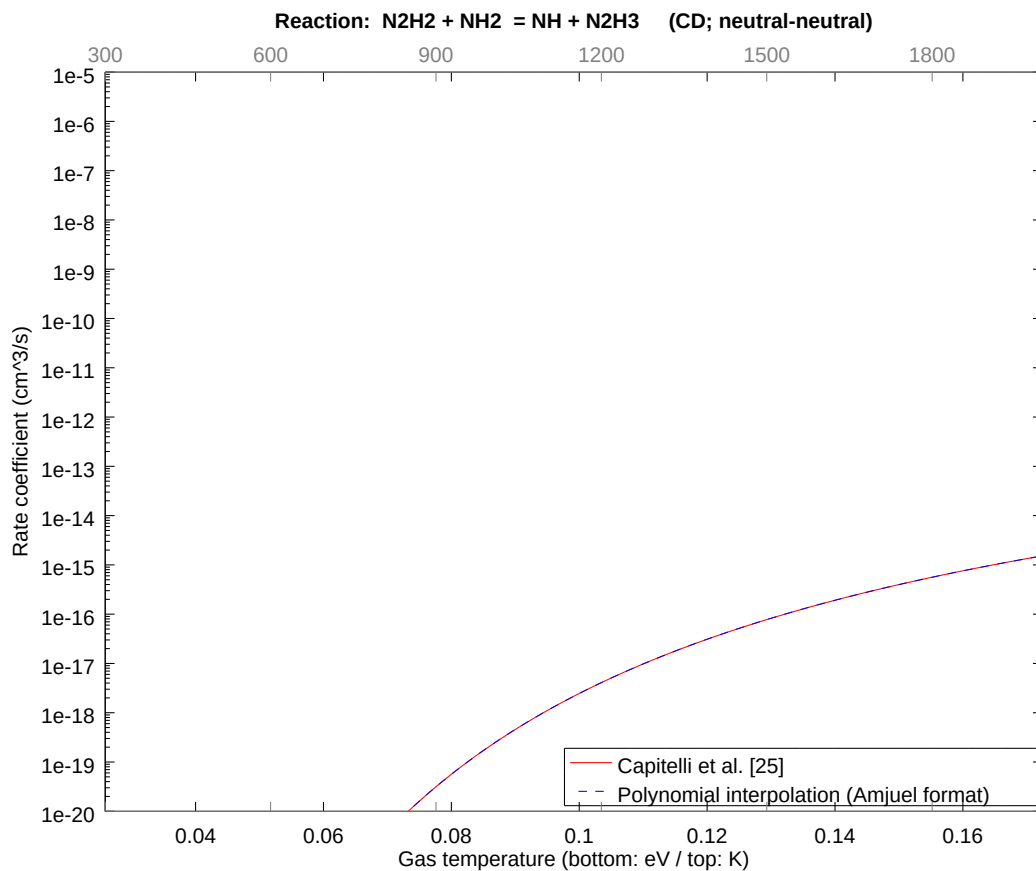
## 2.141 Reaction CC N2H2 + NH2 = NH3 + N2H

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.500098460761E+01 | b1 | 6.531690967044E-02  | b2 | -2.498341942865E-01 |
| b3 | -1.147155335862E-01 | b4 | -8.644579971549E-02 | b5 | -2.698816128360E-02 |
| b6 | -6.766780530776E-03 | b7 | -8.616056380250E-04 | b8 | -6.408943363558E-05 |



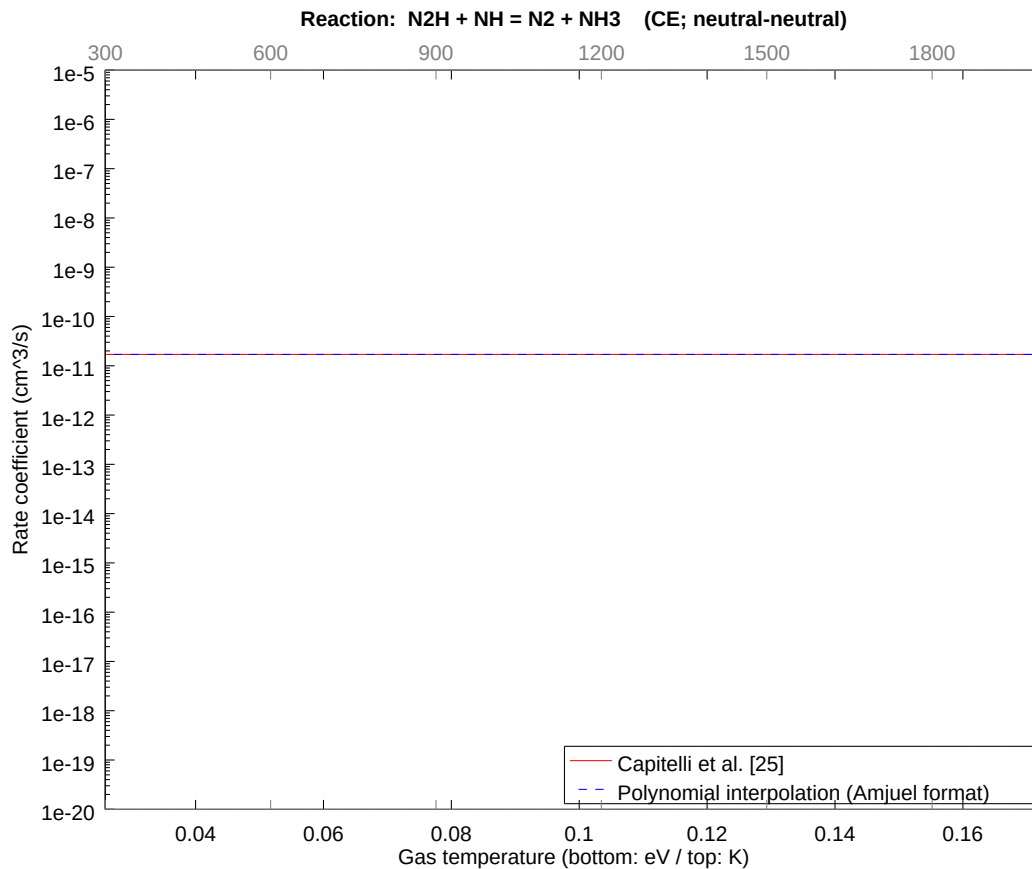
## 2.142 Reaction CD $\text{N}_2\text{H}_2 + \text{NH}_2 = \text{NH} + \text{N}_2\text{H}_3$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.646588531989E+01 | b1 | 1.055193767911E+00  | b2 | -2.123590605188E+00 |
| b3 | -9.750820018544E-01 | b4 | -7.347892825853E-01 | b5 | -2.293993667175E-01 |
| b6 | -5.751763379524E-02 | b7 | -7.323647855047E-03 | b8 | -5.447601831483E-04 |



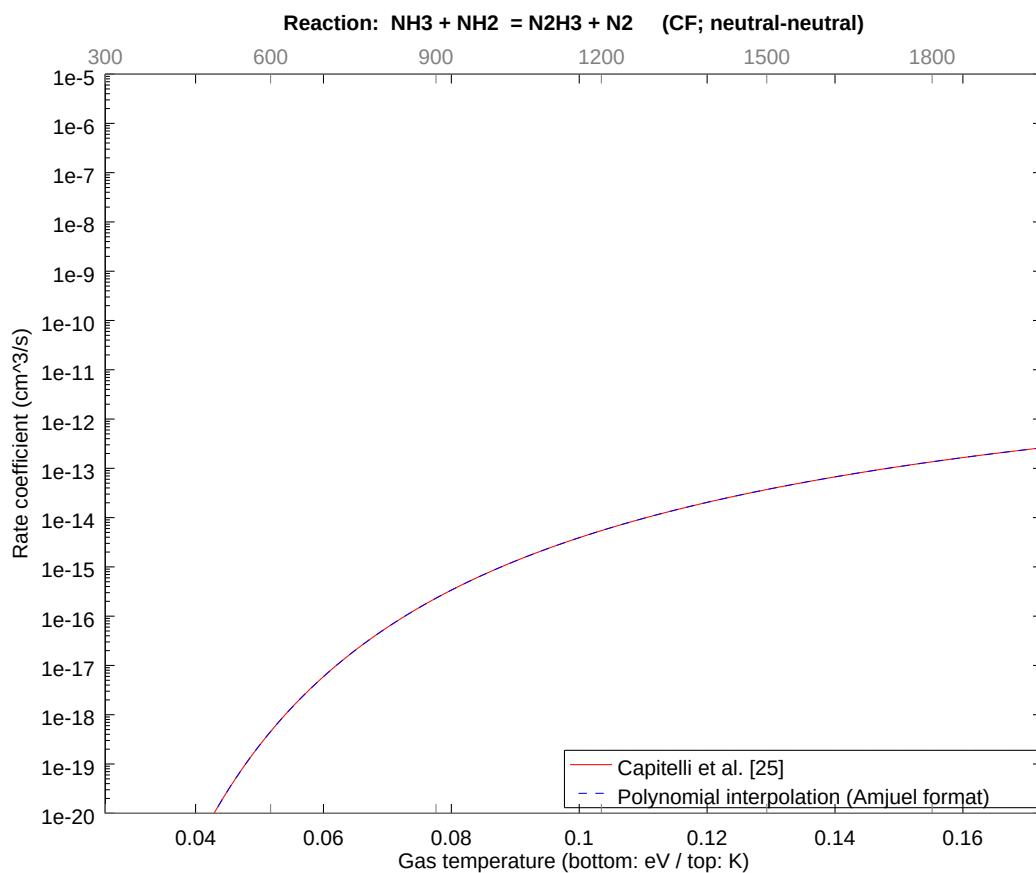
## 2.143 Reaction CE N2H + NH = N2 + NH3

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.479780777187E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



## 2.144 Reaction CF $\text{NH}_3 + \text{NH}_2 = \text{N}_2\text{H}_3 + \text{N}_2$

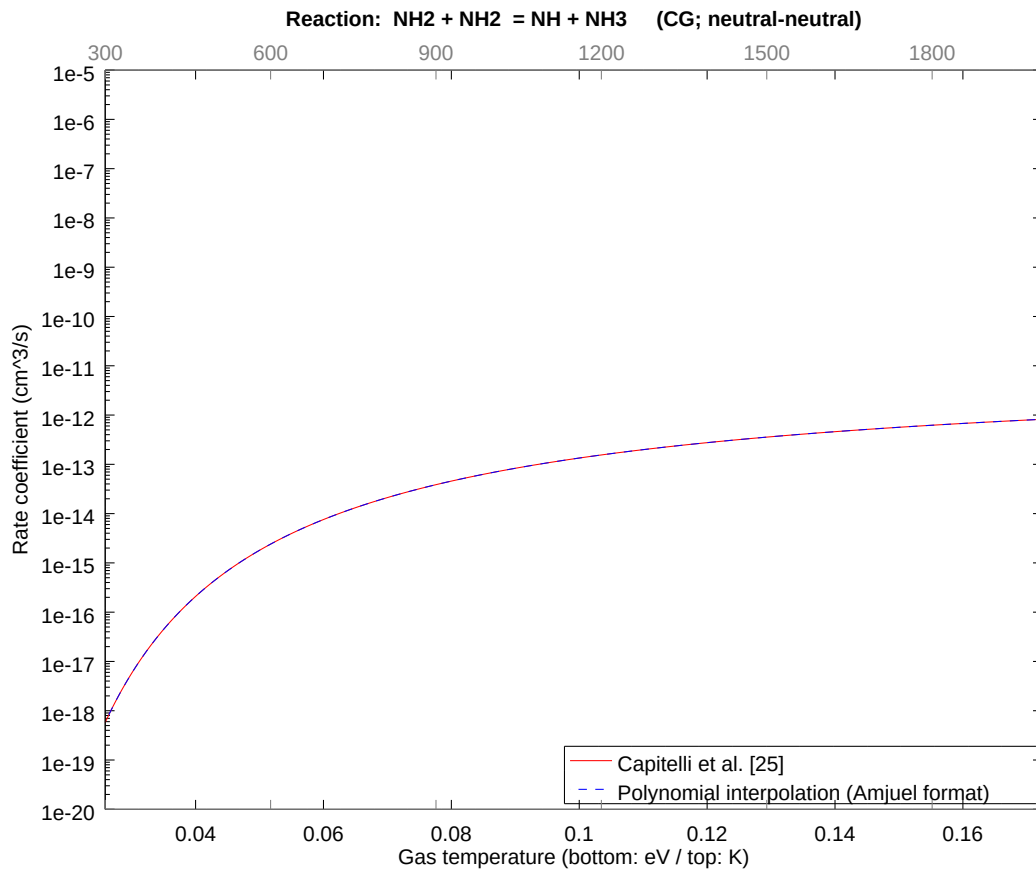
|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.377033307596E+01 | b1 | 8.543442558806E-01  | b2 | -1.355350477056E+00 |
| b3 | -6.223317502405E-01 | b4 | -4.689684548518E-01 | b5 | -1.464107725851E-01 |
| b6 | -3.670978397938E-02 | b7 | -4.674210549023E-03 | b8 | -3.476851760129E-04 |





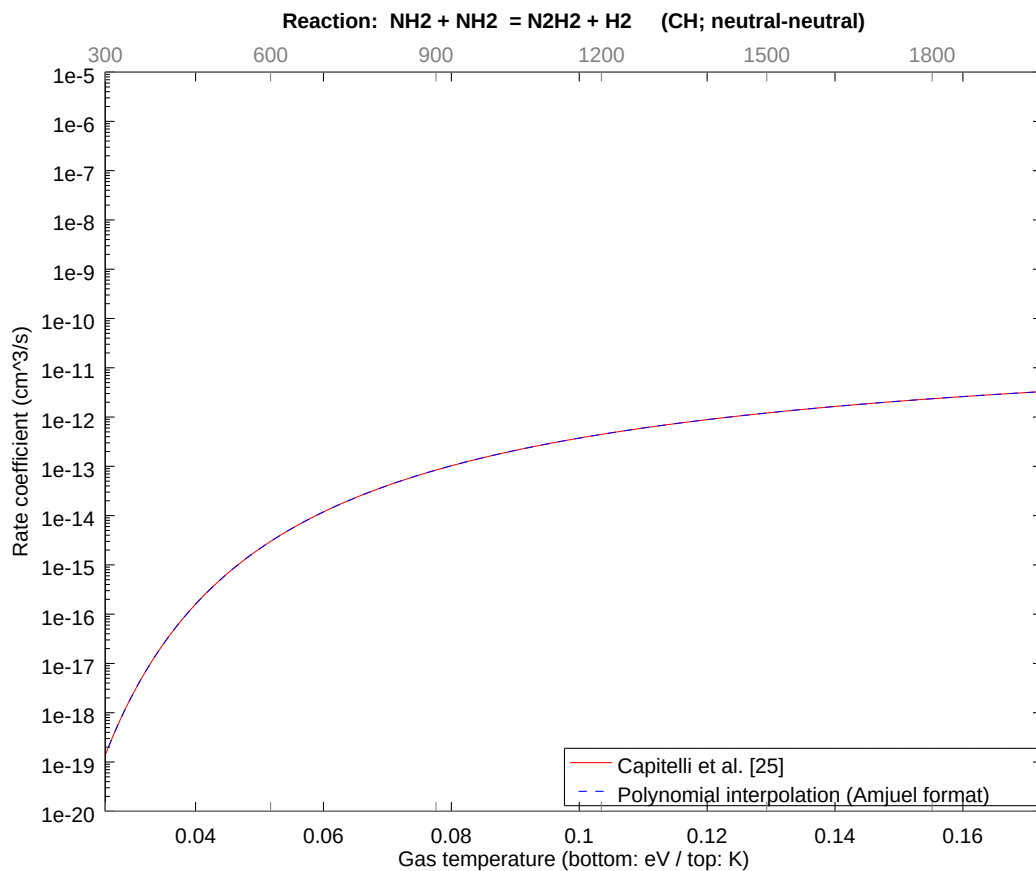
## 2.145 Reaction CG $\text{NH}_2 + \text{NH}_2 = \text{NH} + \text{NH}_3$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.583637811126E+01 | b1 | 1.632922770044E-01  | b2 | -6.245854823952E-01 |
| b3 | -2.867888318610E-01 | b4 | -2.161144985246E-01 | b5 | -6.747040305731E-02 |
| b6 | -1.691695131471E-02 | b7 | -2.154014095617E-03 | b8 | -1.602235842058E-04 |



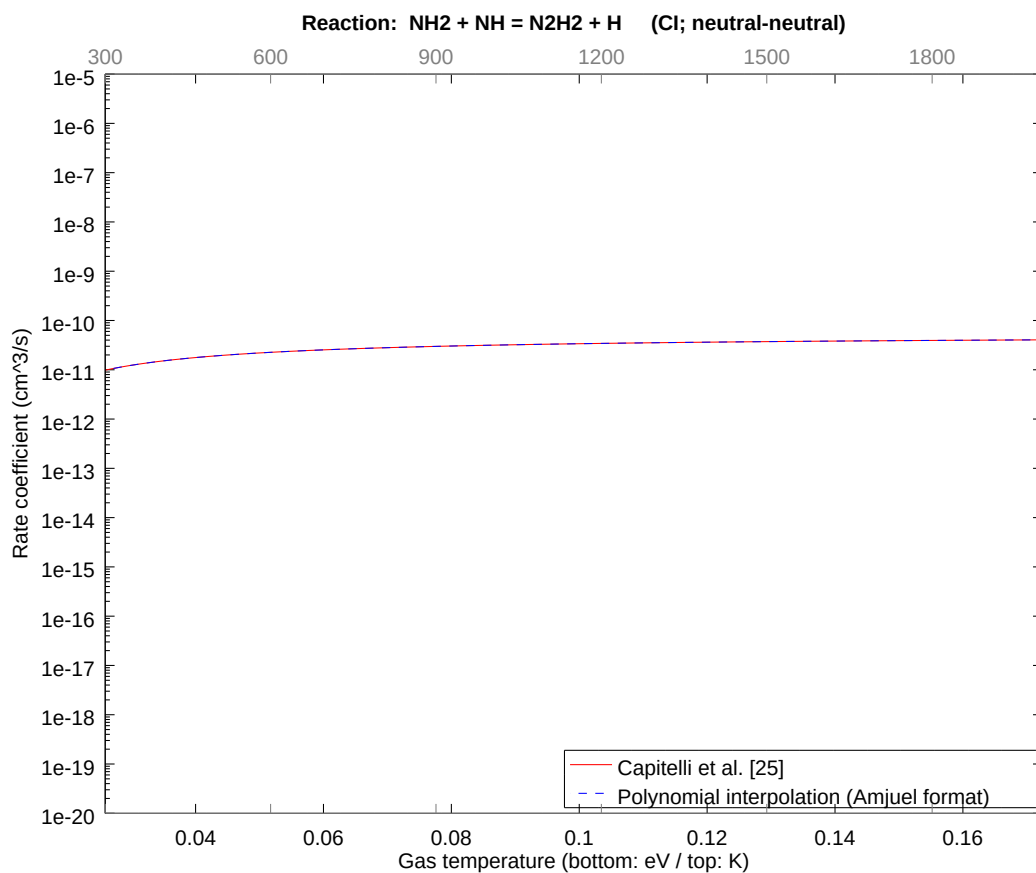
## 2.146 Reaction $\text{CH NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{H}_2$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.405089687724E+01 | b1 | 1.959507406339E-01  | b2 | -7.495025678272E-01 |
| b3 | -3.441465898399E-01 | b4 | -2.593373942811E-01 | b5 | -8.096448249087E-02 |
| b6 | -2.030034135994E-02 | b7 | -2.584816891937E-03 | b8 | -1.922683000102E-04 |



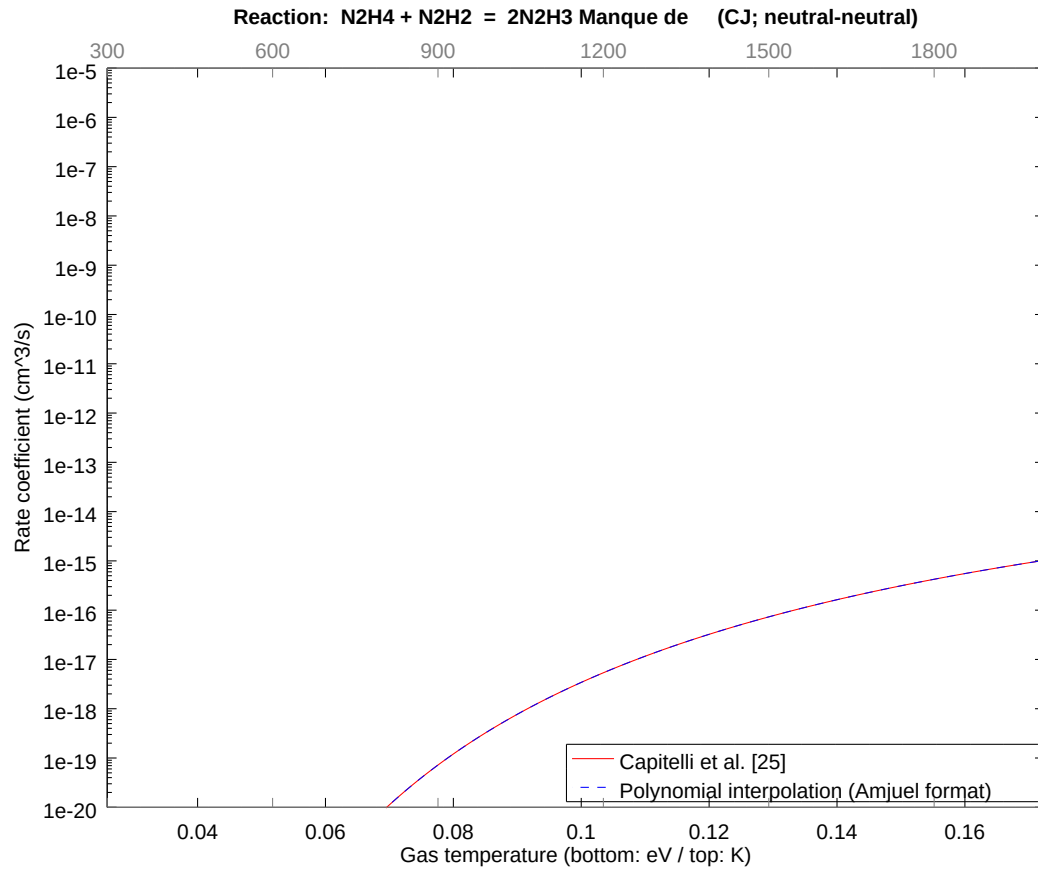
## 2.147 Reaction CI NH<sub>2</sub> + NH = N<sub>2</sub>H<sub>2</sub> + H

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.373057160757E+01 | b1 | 1.632922319726E-02  | b2 | -6.245855454798E-02 |
| b3 | -2.867888818227E-02 | b4 | -2.161145229905E-02 | b5 | -6.747041064367E-03 |
| b6 | -1.691695276958E-03 | b7 | -2.154014253426E-04 | b8 | -1.602235916207E-05 |



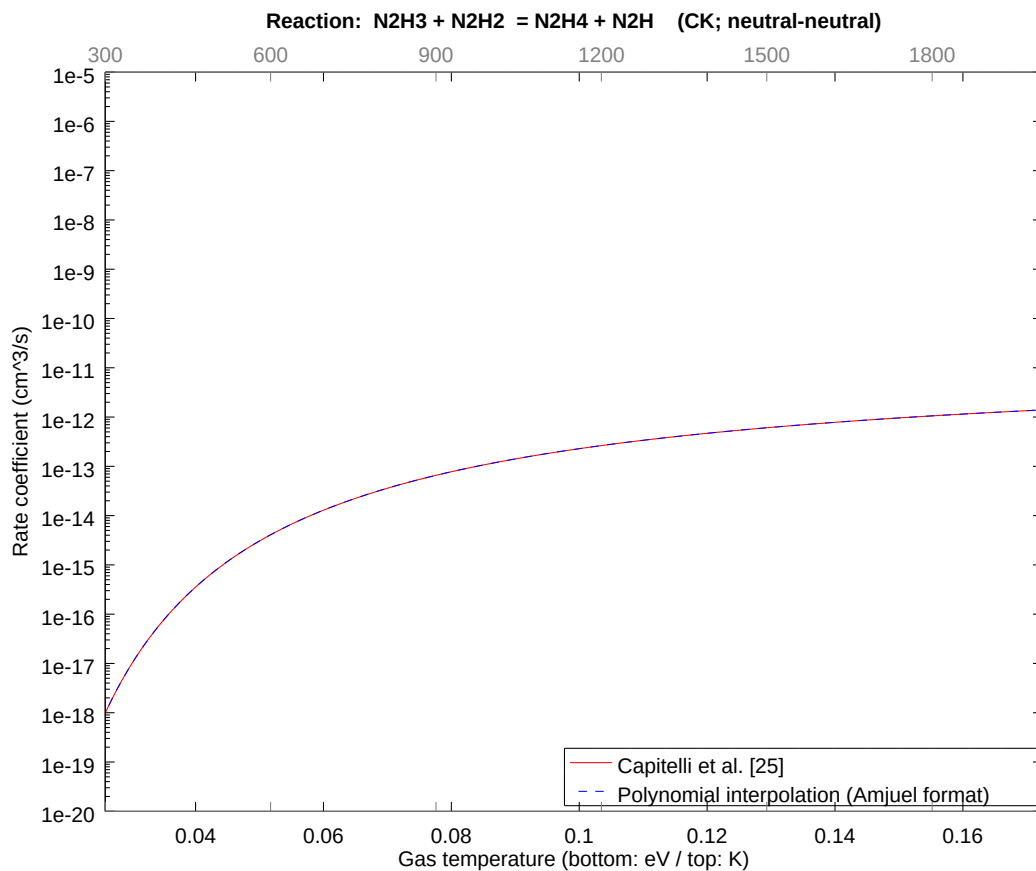
## 2.148 Reaction CJ $\text{N}_2\text{H}_4 + \text{N}_2\text{H}_2 = 2\text{N}_2\text{H}_3$ Manque de

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.765592328912E+01 | b1 | 9.898768553234E-01  | b2 | -1.873756414556E+00 |
| b3 | -8.603664708274E-01 | b4 | -6.483434839625E-01 | b5 | -2.024112057243E-01 |
| b6 | -5.075085331116E-02 | b7 | -6.462042221143E-03 | b8 | -4.806707496634E-04 |



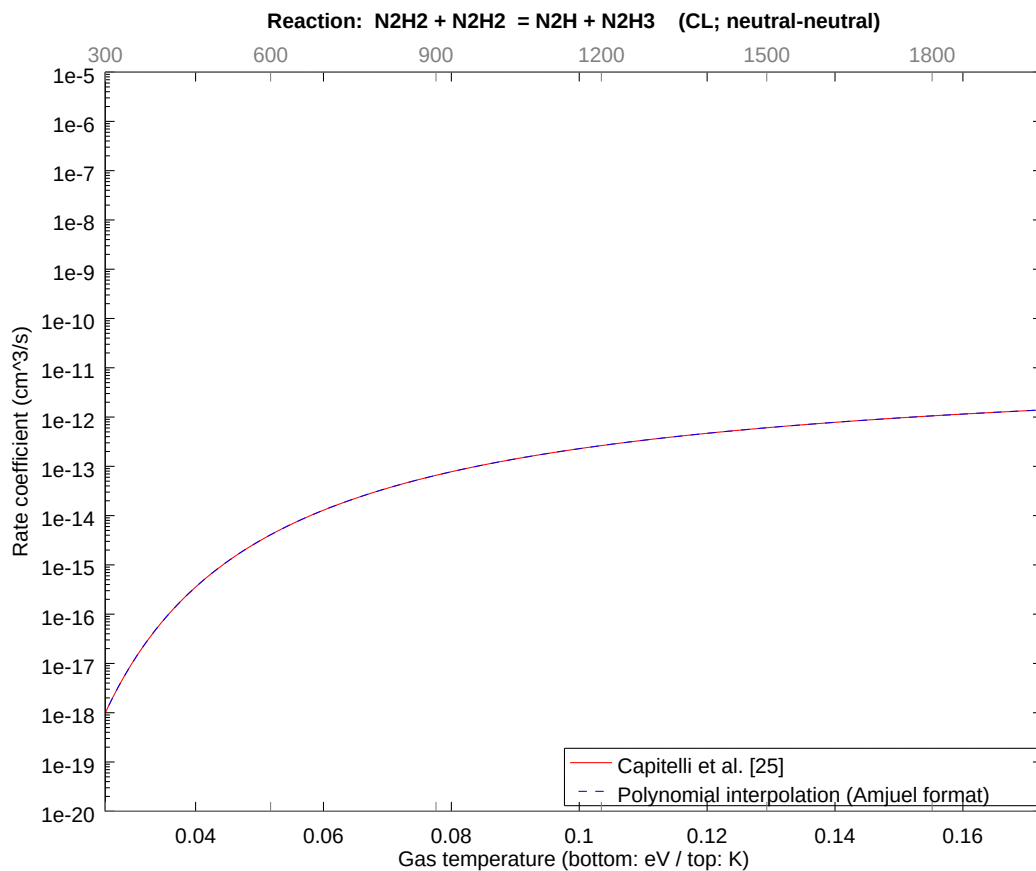
## 2.149 Reaction CK $\text{N}_2\text{H}_3 + \text{N}_2\text{H}_2 = \text{N}_2\text{H}_4 + \text{N}_2\text{H}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.530574985861E+01 | b1 | 1.632922819981E-01  | b2 | -6.245854756144E-01 |
| b3 | -2.867888266545E-01 | b4 | -2.161144960521E-01 | b5 | -6.747040231379E-02 |
| b6 | -1.691695117642E-02 | b7 | -2.154014081072E-03 | b8 | -1.602235835433E-04 |



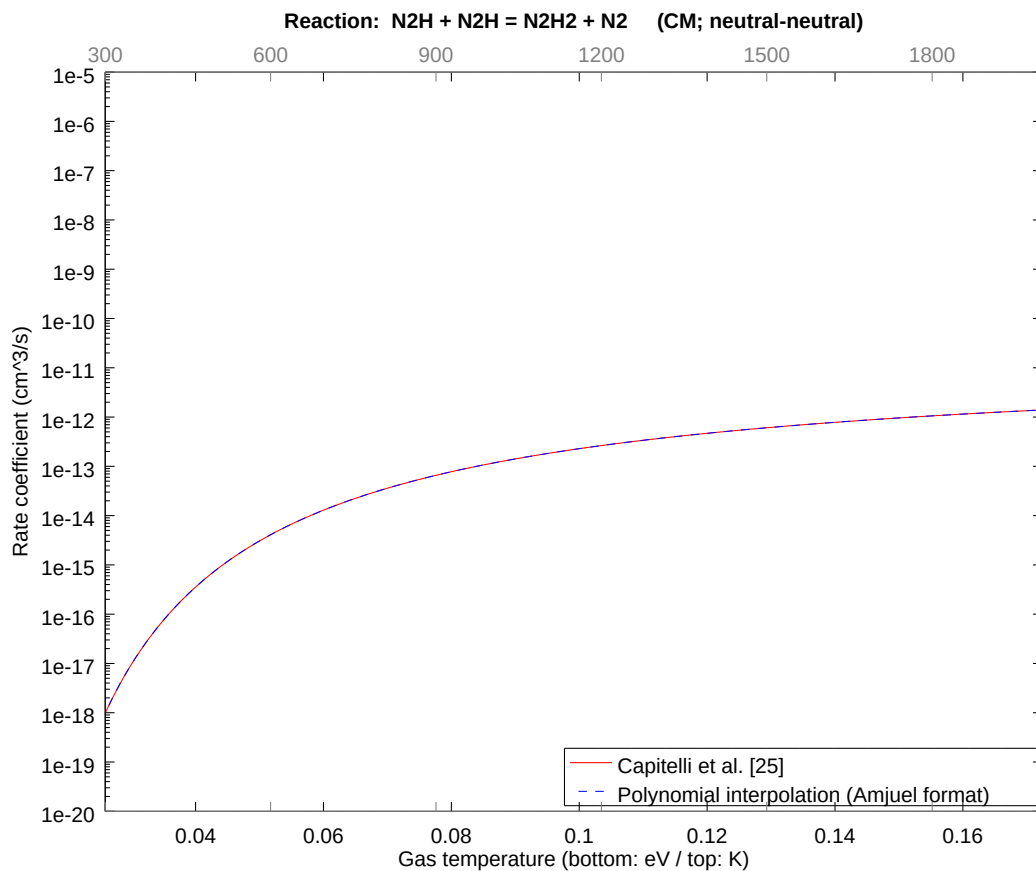
## 2.150 Reaction CL $\text{N}_2\text{H}_2 + \text{N}_2\text{H}_2 = \text{N}_2\text{H} + \text{N}_2\text{H}_3$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.530574985861E+01 | b1 | 1.632922819981E-01  | b2 | -6.245854756144E-01 |
| b3 | -2.867888266545E-01 | b4 | -2.161144960521E-01 | b5 | -6.747040231379E-02 |
| b6 | -1.691695117642E-02 | b7 | -2.154014081072E-03 | b8 | -1.602235835433E-04 |



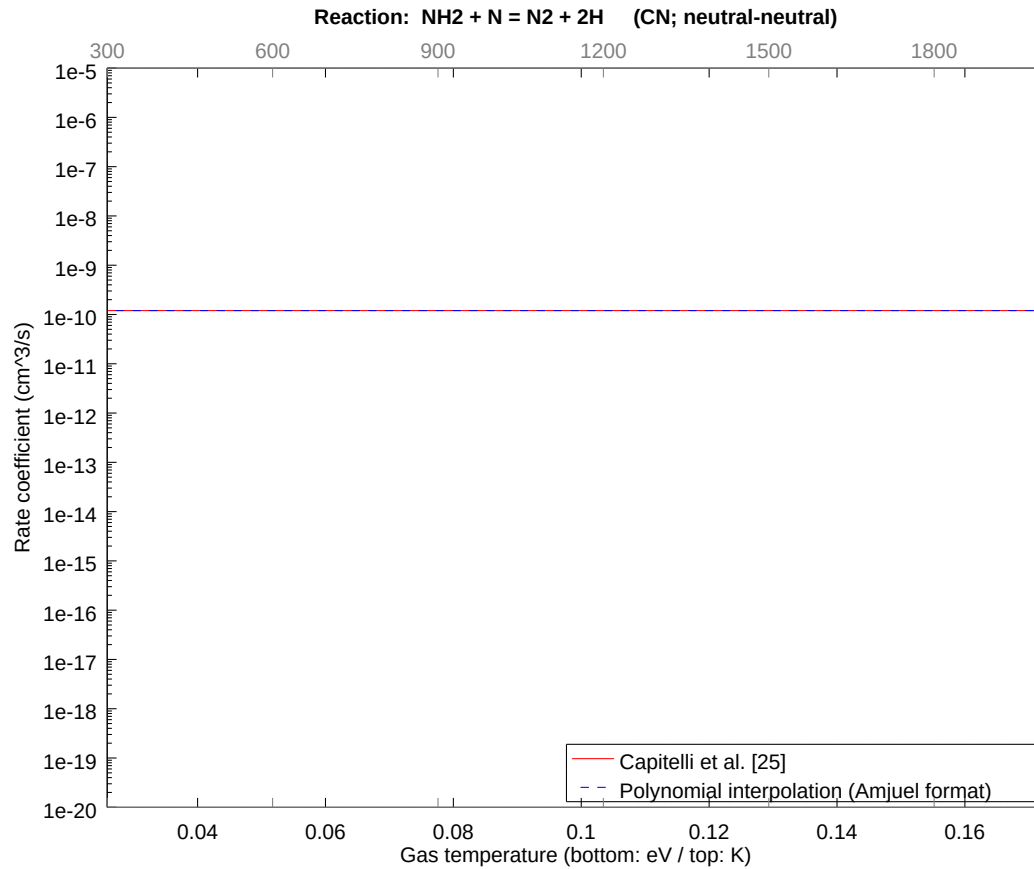
## 2.151 Reaction CM N2H + N2H = N2H2 + N2

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.530574985861E+01 | b1 | 1.632922819981E-01  | b2 | -6.245854756144E-01 |
| b3 | -2.867888266545E-01 | b4 | -2.161144960521E-01 | b5 | -6.747040231379E-02 |
| b6 | -1.691695117642E-02 | b7 | -2.154014081072E-03 | b8 | -1.602235835433E-04 |



## 2.152 Reaction CN NH2 + N = N2 + 2H

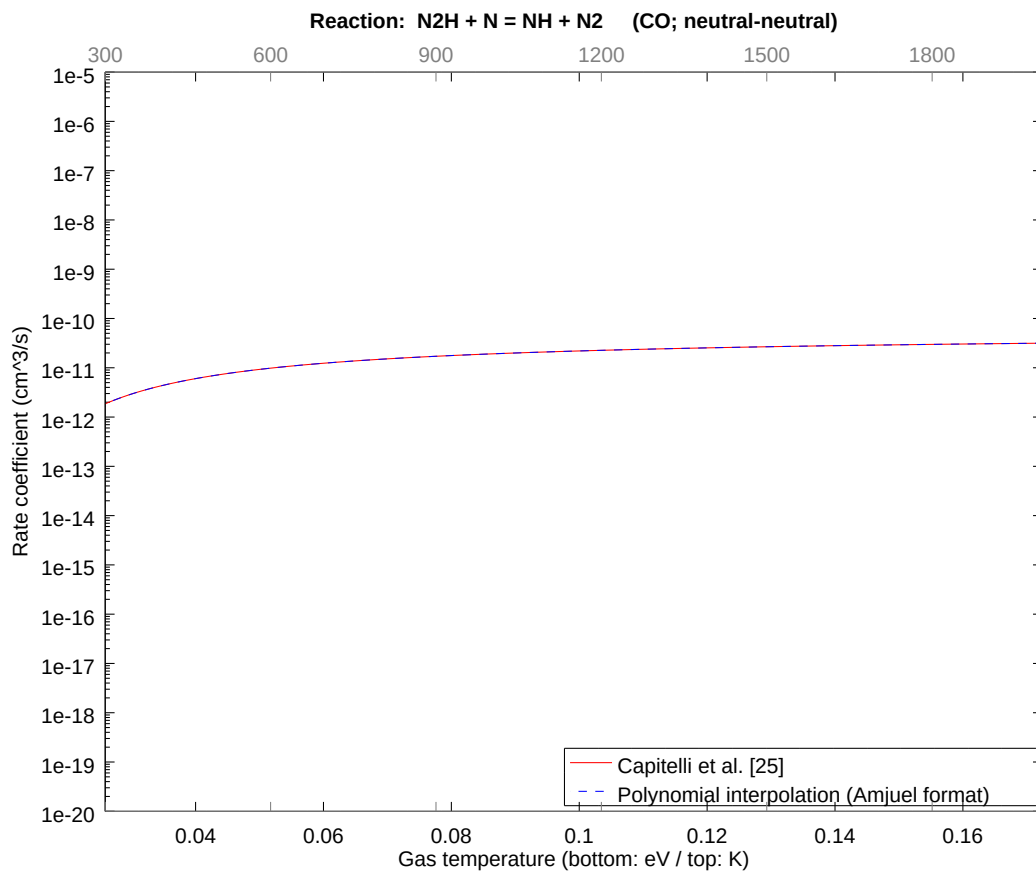
|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.284352937315E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |





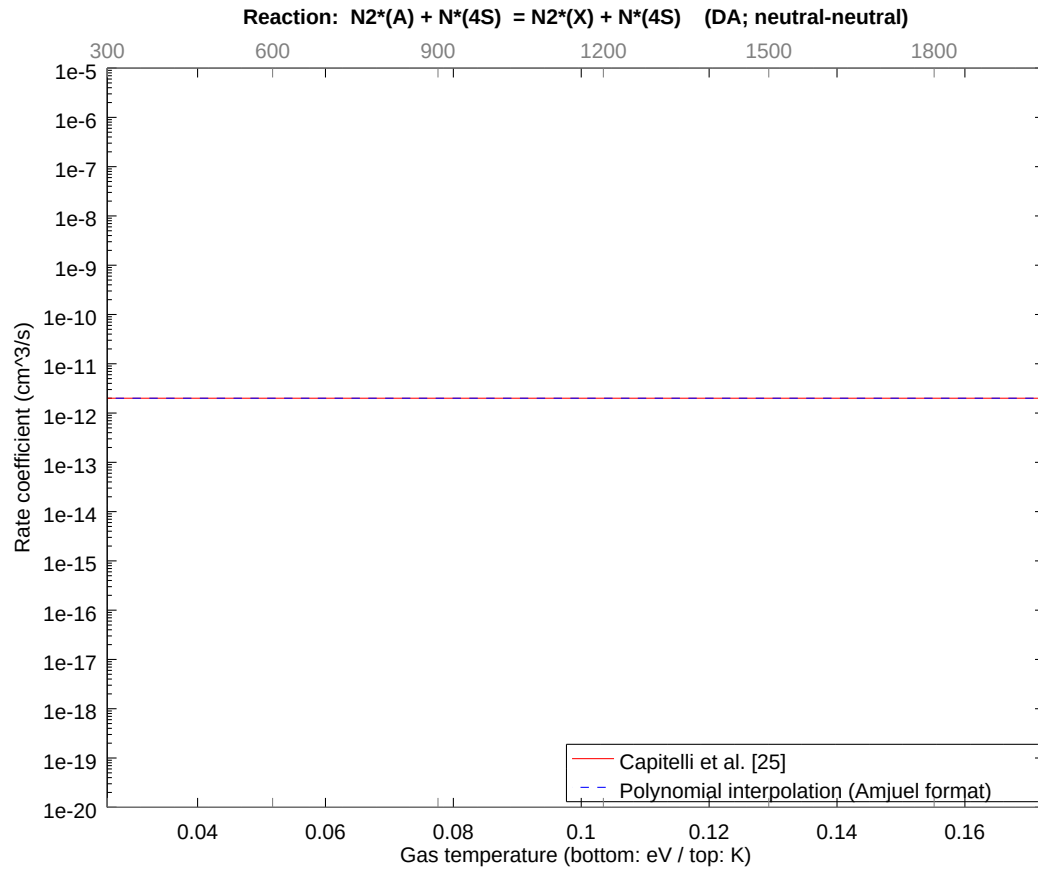
## 2.153 Reaction CO N2H + N = NH + N2

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.378136581554E+01 | b1 | 3.265845386299E-02  | b2 | -1.249170984012E-01 |
| b3 | -5.735776771641E-02 | b4 | -4.322290027883E-02 | b5 | -1.349408076421E-02 |
| b6 | -3.383390287591E-03 | b7 | -4.308028213161E-04 | b8 | -3.204471692274E-05 |



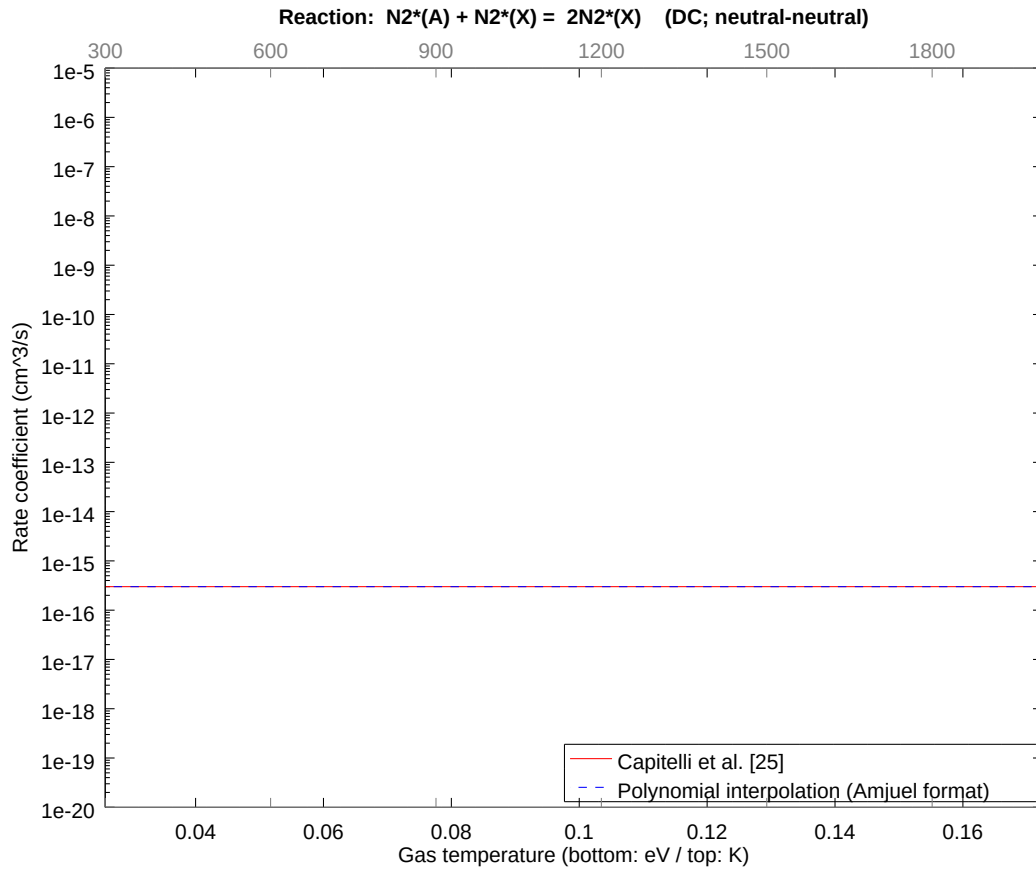
## 2.154 Reaction DA $\text{N}_2^*(\text{A}) + \text{N}^*(4\text{S}) = \text{N}_2^*(\text{X}) + \text{N}^*(4\text{S})$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.693787393537E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



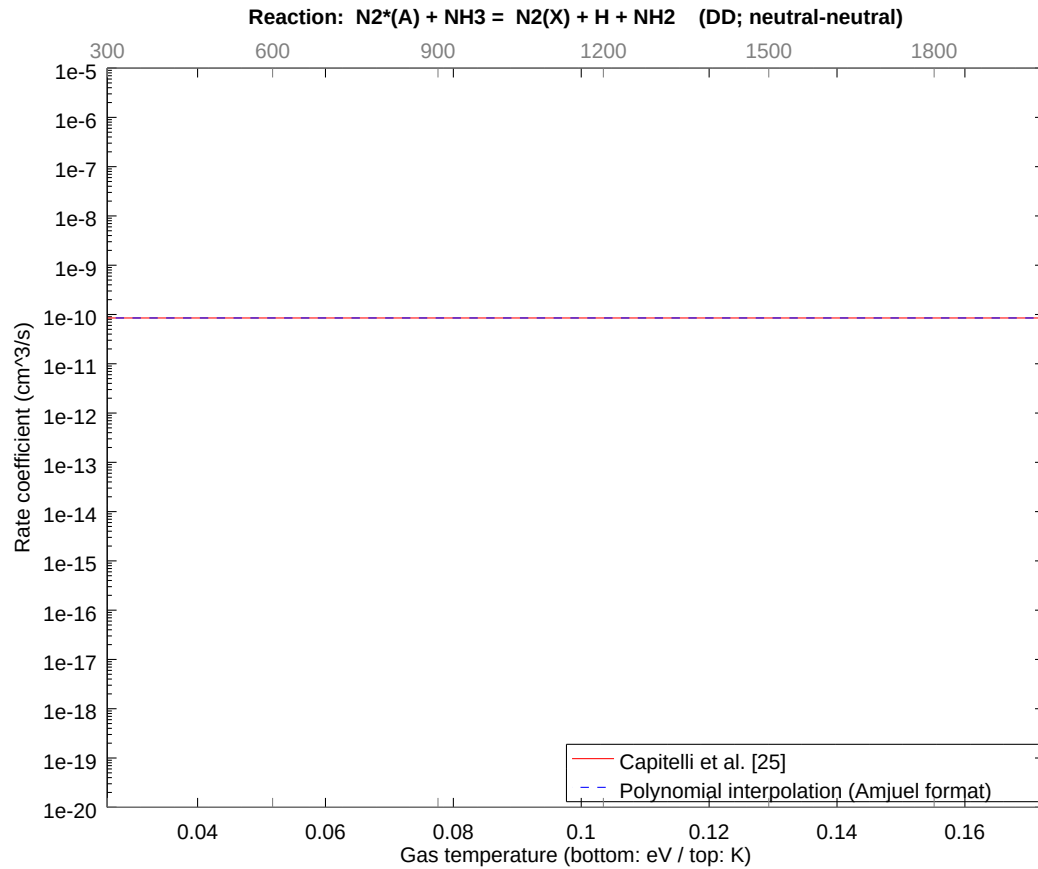
## 2.155 Reaction DC $\text{N2}^*(\text{A}) + \text{N2}^*(\text{X}) = 2\text{N2}^*(\text{X})$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -3.574274919924E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



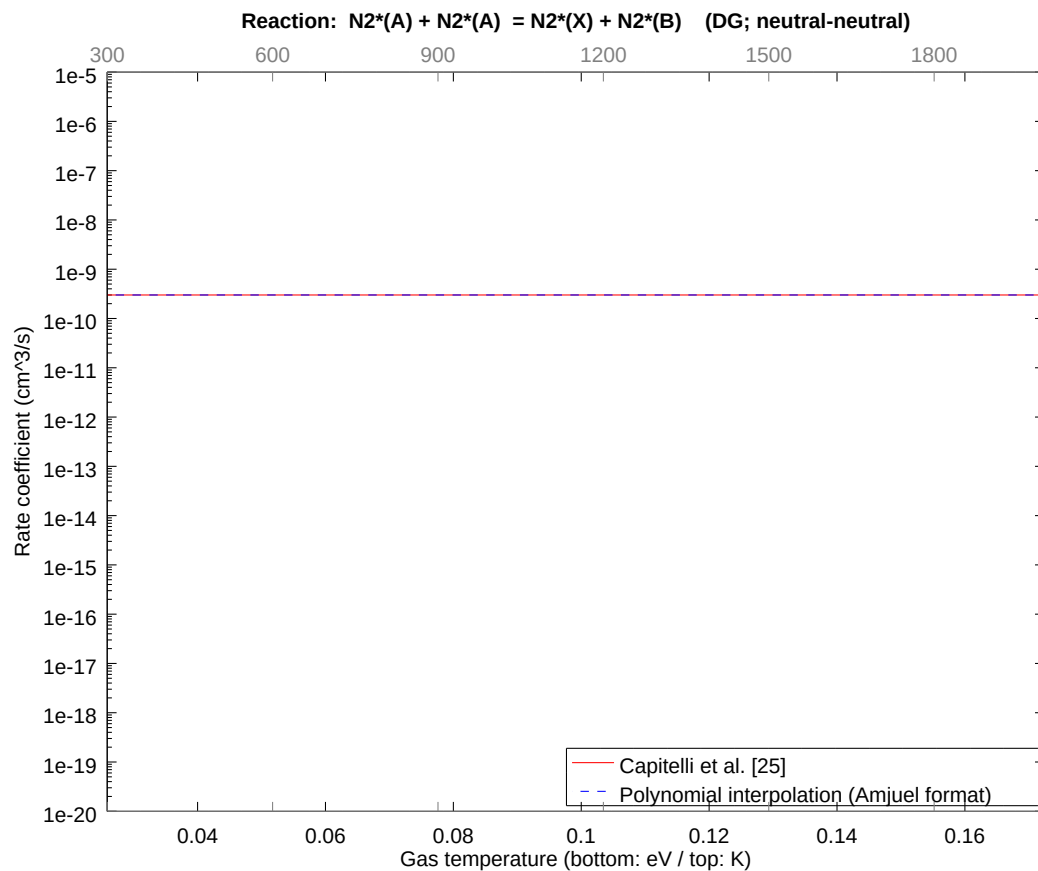
## 2.156 Reaction DD $\text{N}_2^*(\text{A}) + \text{NH}_3 = \text{N}_2(\text{X}) + \text{H} + \text{NH}_2$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.318836985944E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



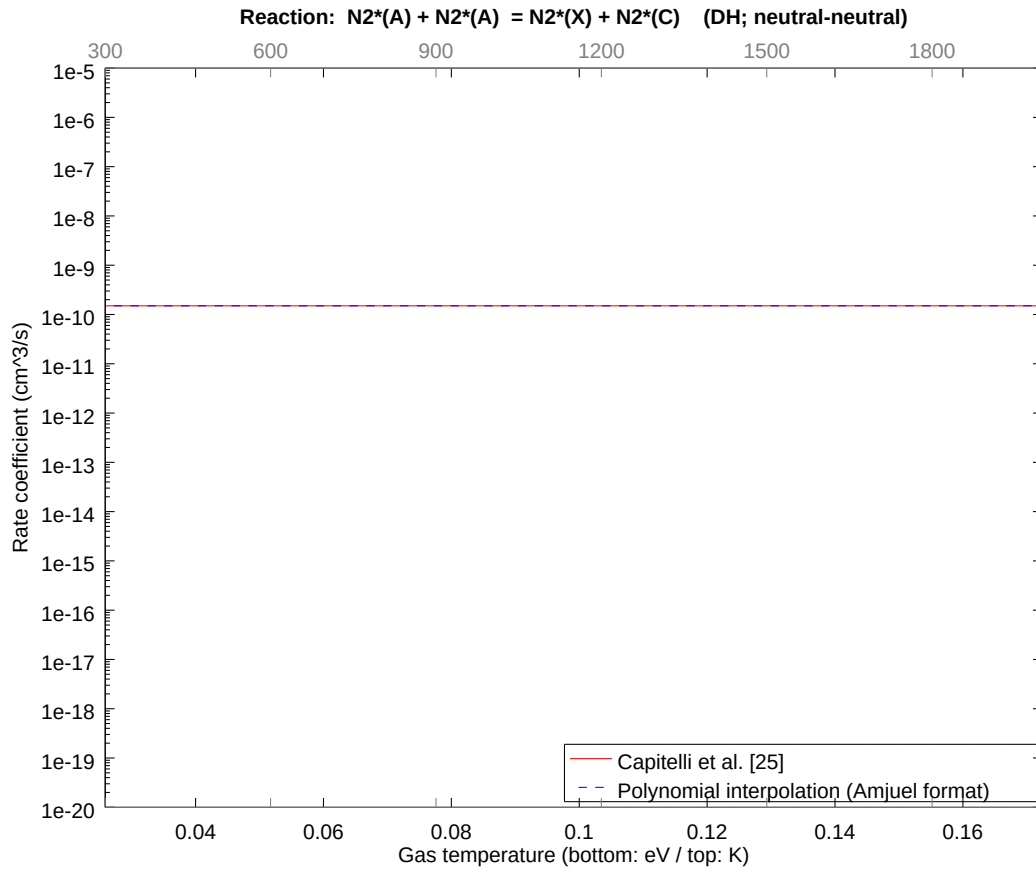
## 2.157 Reaction DG $\text{N2}^*(\text{A}) + \text{N2}^*(\text{A}) = \text{N2}^*(\text{X}) + \text{N2}^*(\text{B})$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.192723864127E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



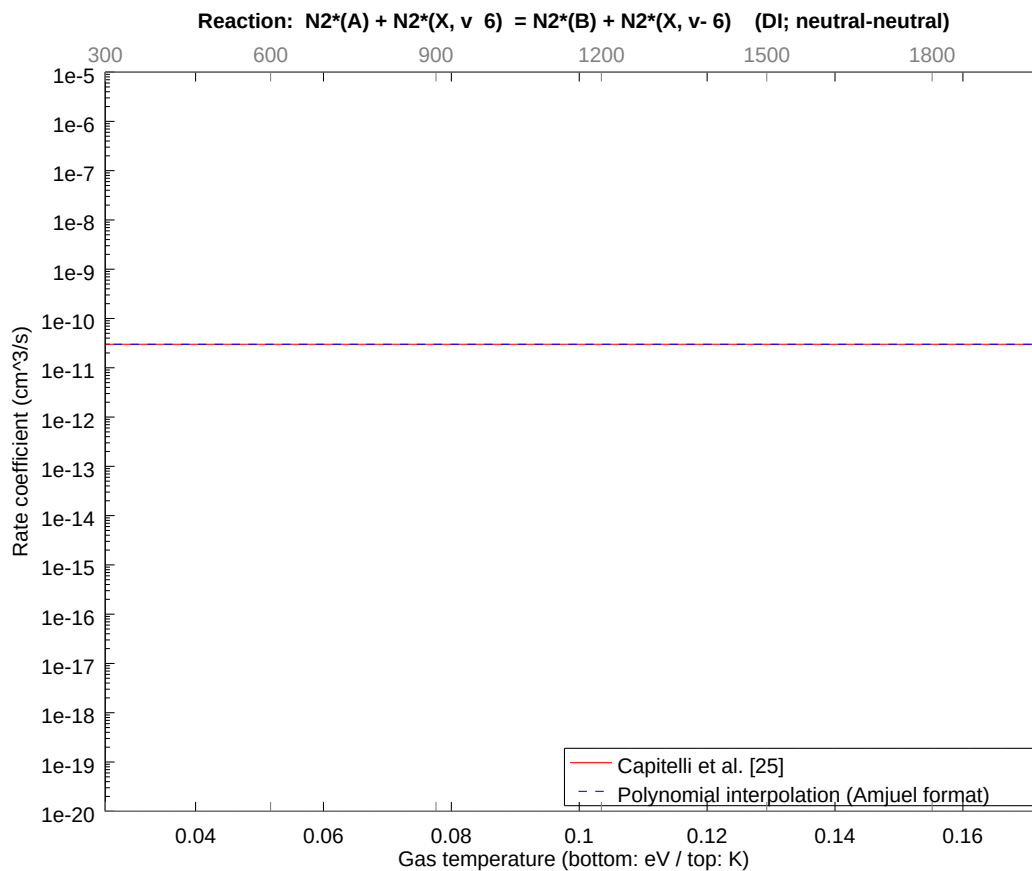
## 2.158 Reaction DH $\text{N}_2^*(\text{A}) + \text{N}_2^*(\text{A}) = \text{N}_2^*(\text{X}) + \text{N}_2^*(\text{C})$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.262038582183E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



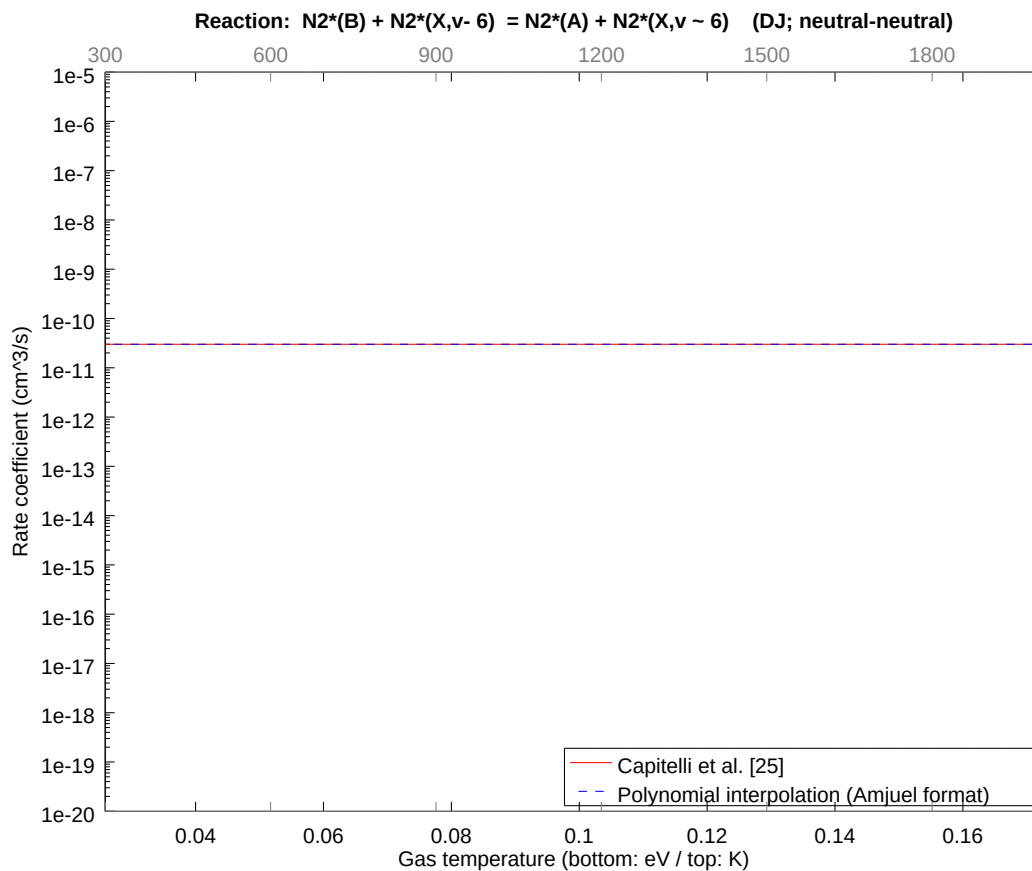
## 2.159 Reaction DI N2\*(A) + N2\*(X, v 6) = N2\*(B) + N2\*(X, v- 6)

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.422982373427E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



## 2.160 Reaction DJ $\text{N}_2^*(\text{B}) + \text{N}_2^*(\text{X}, v-6) = \text{N}_2^*(\text{A}) + \text{N}_2^*(\text{X}, v-6)$

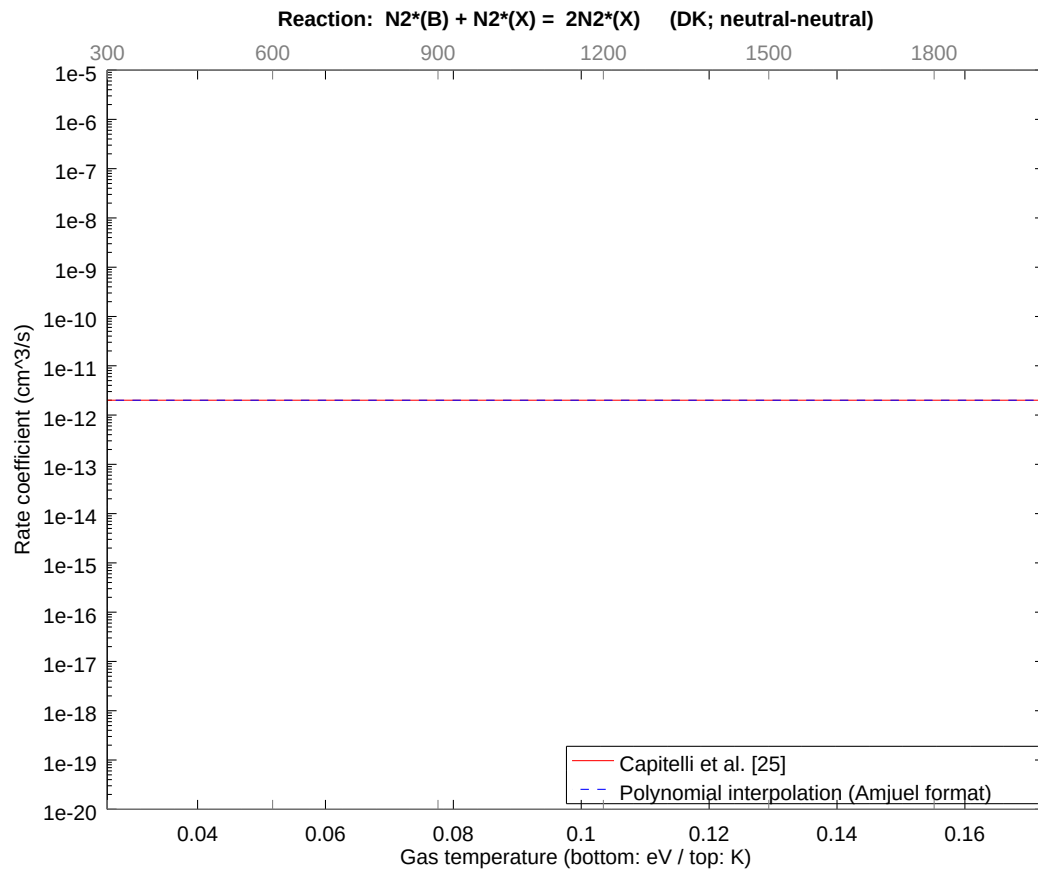
|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.422982373427E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |





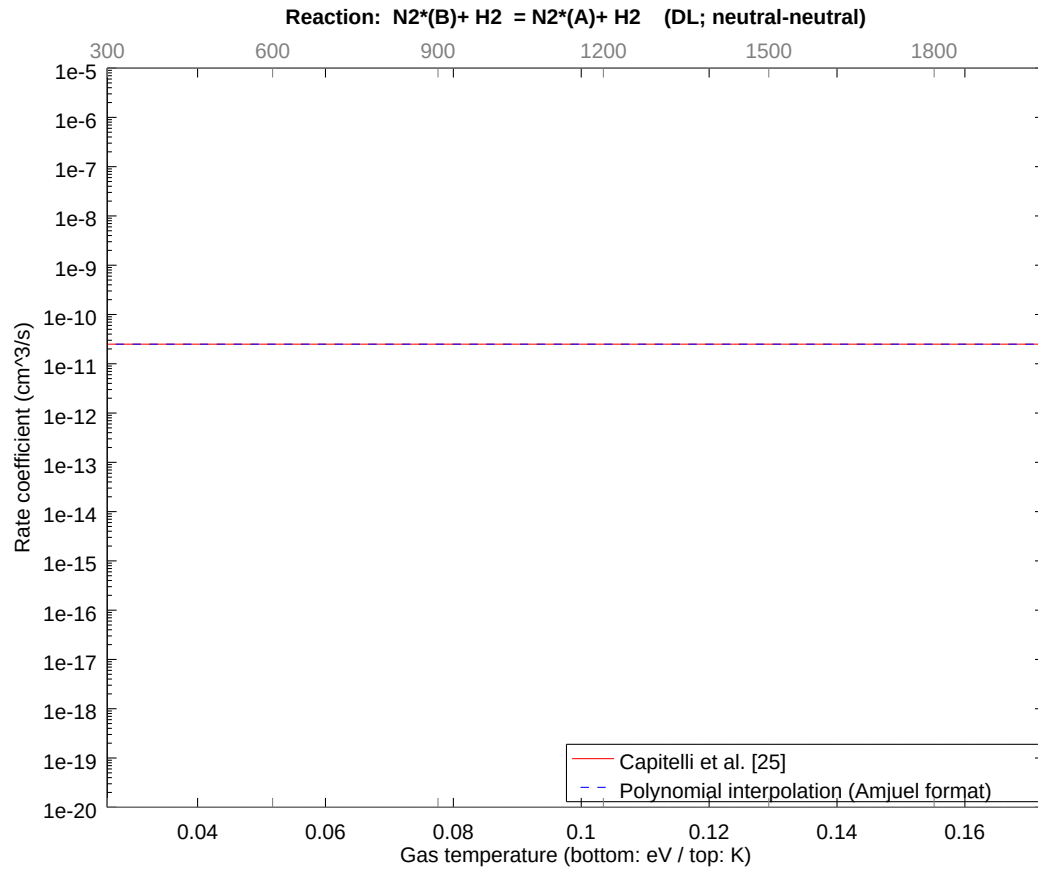
## 2.161 Reaction DK $\text{N}_2^*(\text{B}) + \text{N}_2^*(\text{X}) = 2\text{N}_2^*(\text{X})$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.693787393537E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



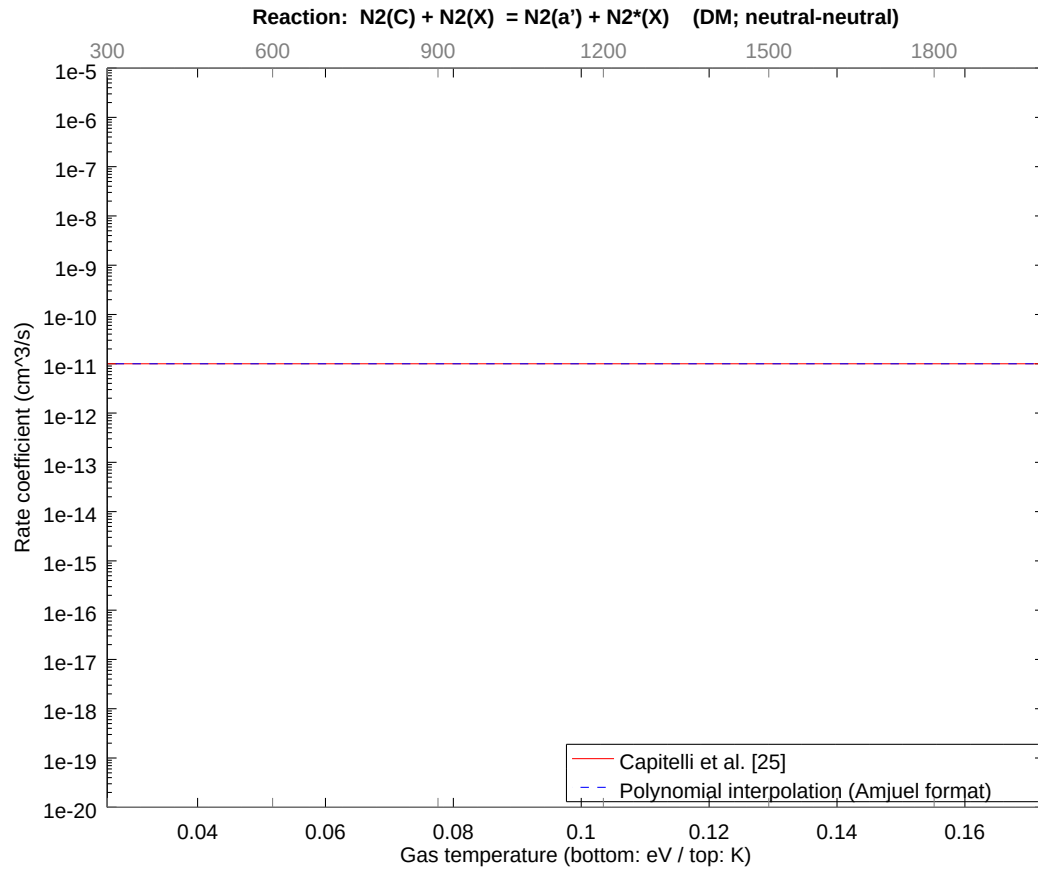
## 2.162 Reaction DL $\text{N}_2^*(\text{B}) + \text{H}_2 = \text{N}_2^*(\text{A}) + \text{H}_2$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.441214529106E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



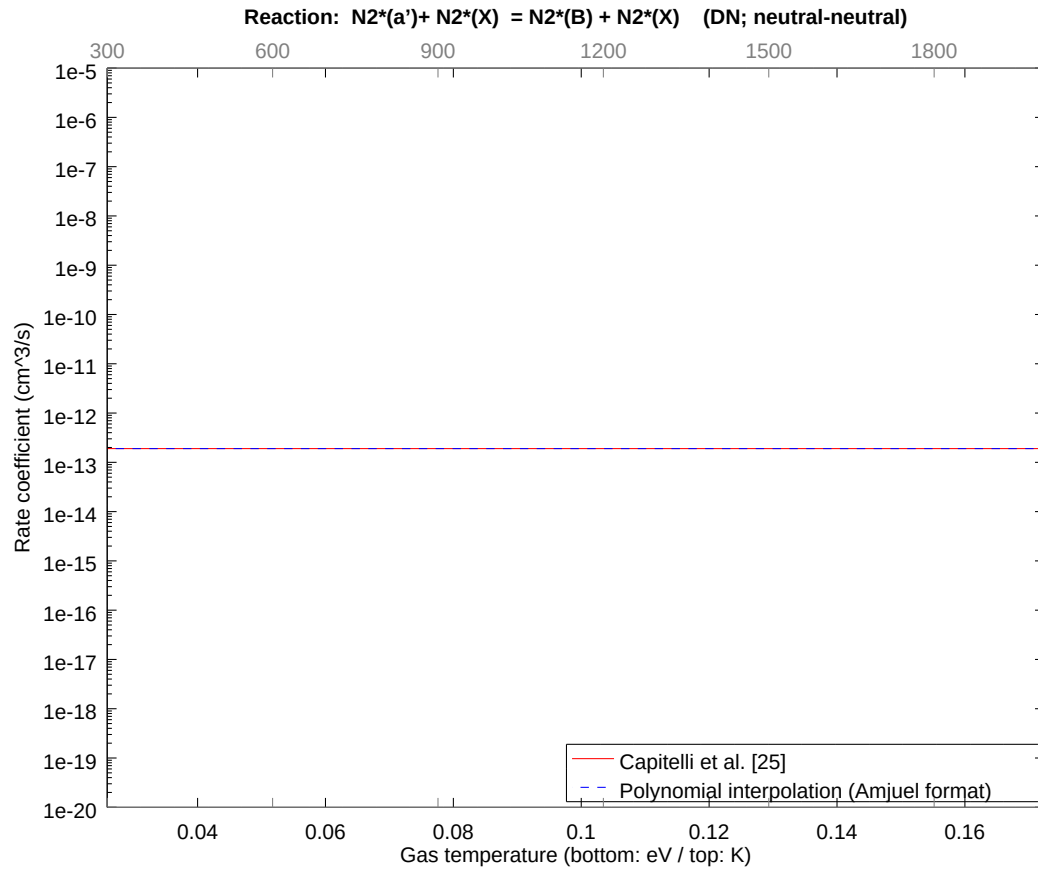
## 2.163 Reaction DM $\text{N}_2(\text{C}) + \text{N}_2(\text{X}) = \text{N}_2(\text{a}') + \text{N}_2^*(\text{X})$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.532843602293E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



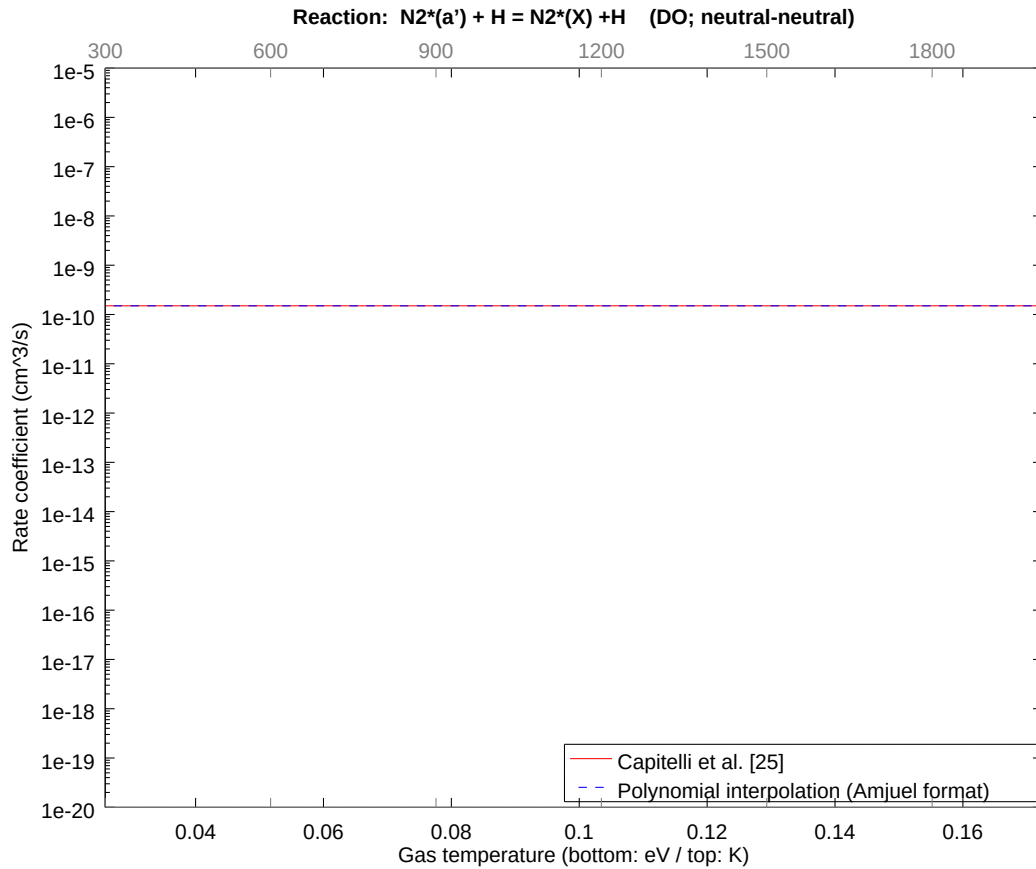
## 2.164 Reaction DN $\text{N}_2^*(a') + \text{N}_2^*(X) = \text{N}_2^*(B) + \text{N}_2^*(X)$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.929175232275E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



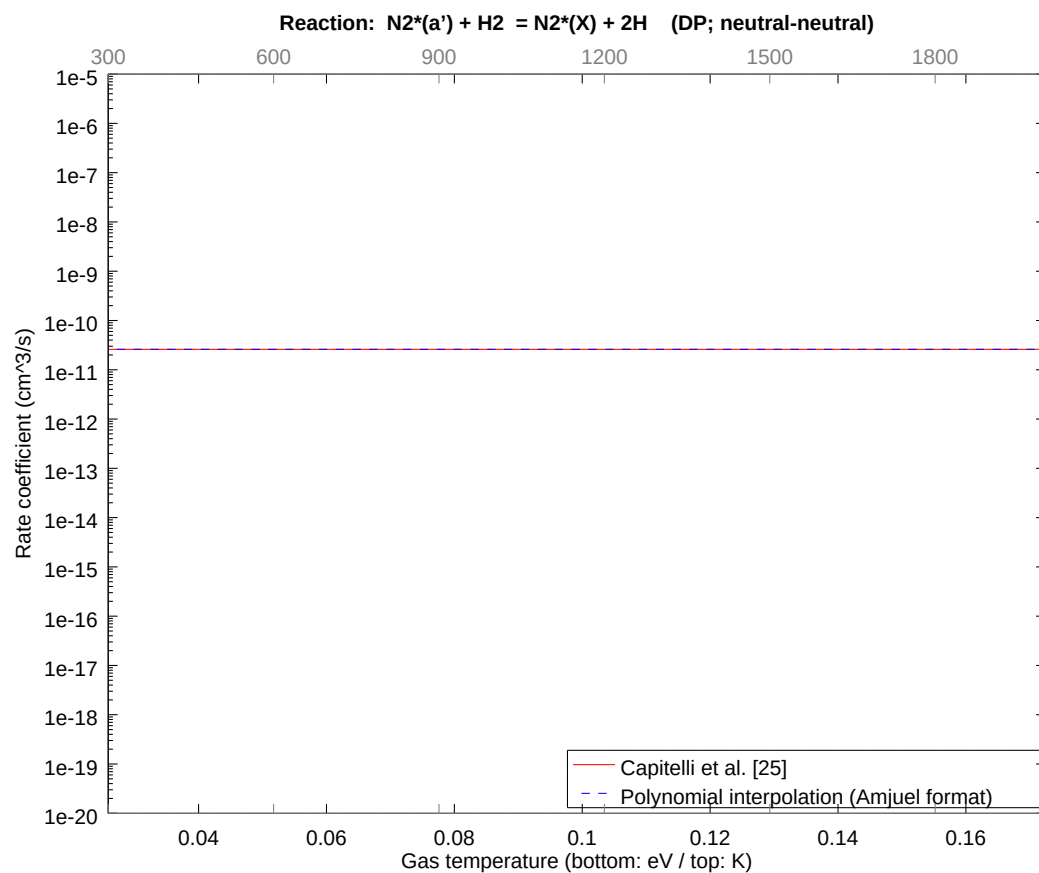
## 2.165 Reaction DO N2\*(a') + H = N2\*(X) +H

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.262038582183E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



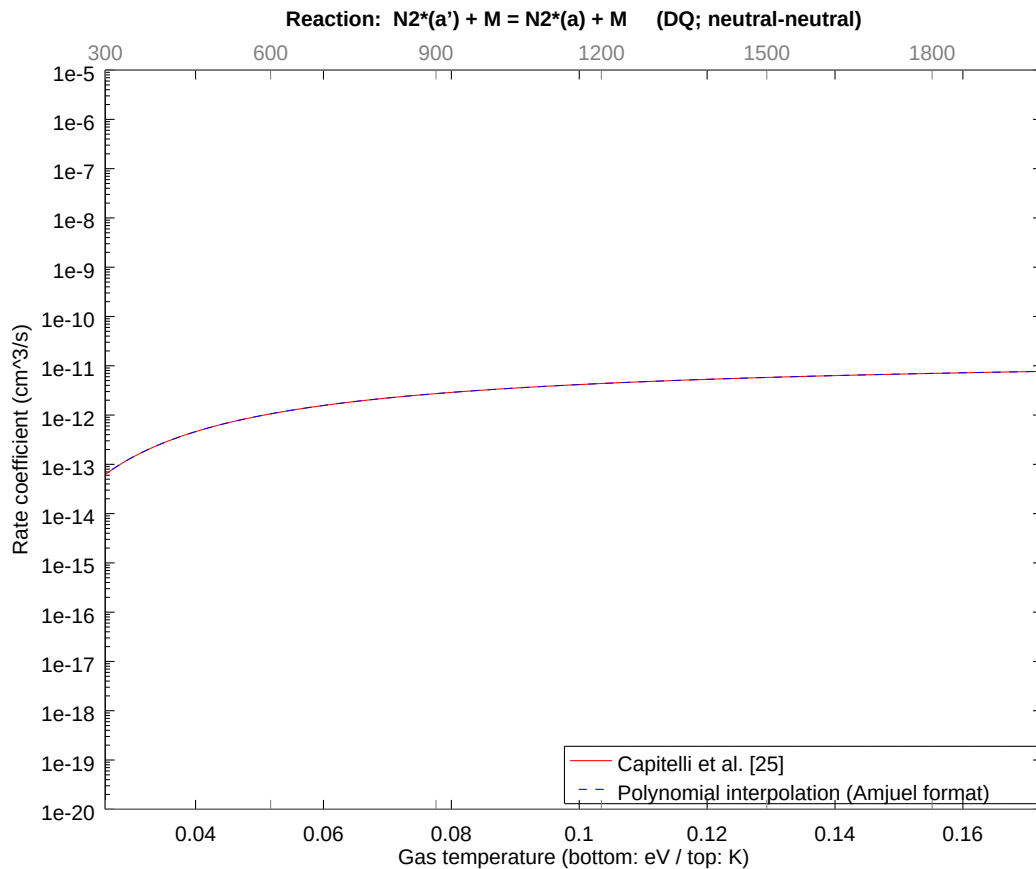
## 2.166 Reaction DP $\text{N}_2^*(a') + \text{H}_2 = \text{N}_2^*(X) + 2\text{H}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.437292457791E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



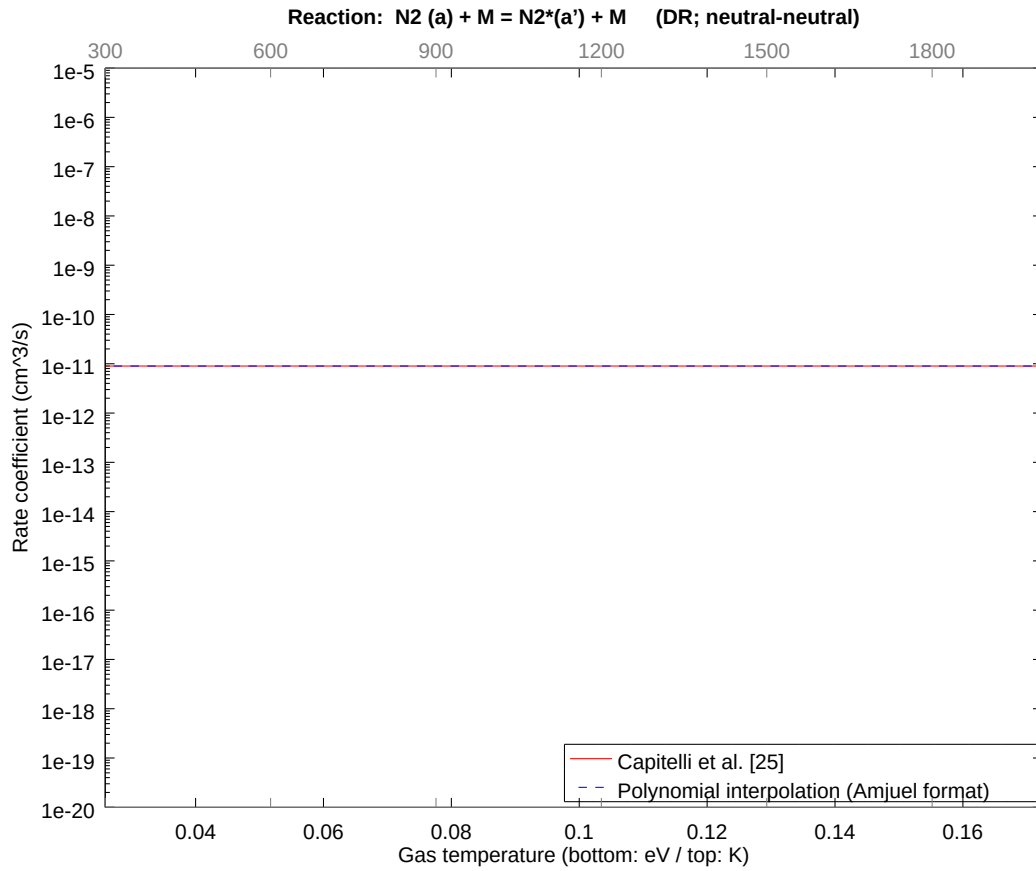
## 2.167 Reaction DQ $\text{N}_2^*(a') + \text{M} = \text{N}_2^*(a) + \text{M}$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -2.491334966794E+01 | b1 | 5.551937465612E-02  | b2 | -2.123590632336E-01 |
| b3 | -9.750820211534E-02 | b4 | -7.347892909455E-02 | b5 | -2.293993689667E-02 |
| b6 | -5.751763415969E-03 | b7 | -7.323647887183E-04 | b8 | -5.447601843037E-05 |



## 2.168 Reaction DR N2 (a) + M = N2\*(a') + M

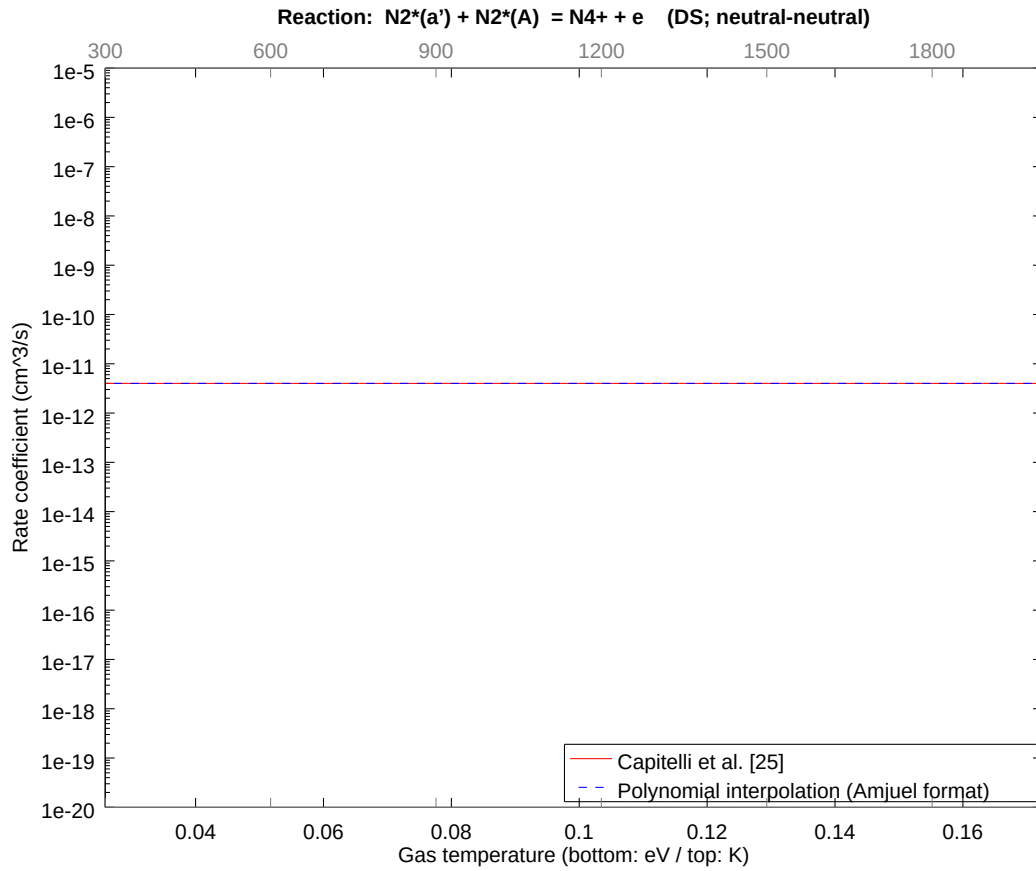
|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.543379653859E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |





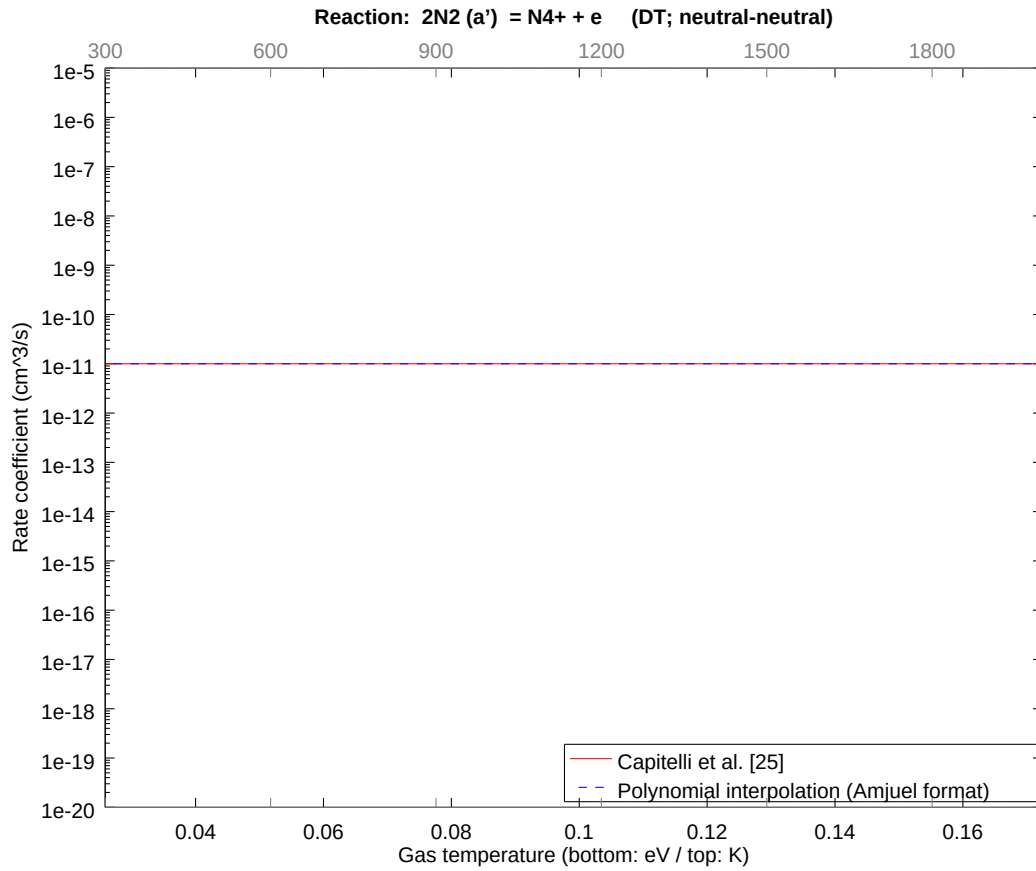
## 2.169 Reaction DS $\text{N2}^*(\text{a}') + \text{N2}^*(\text{A}) = \text{N4}^+ + \text{e}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.624472675481E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



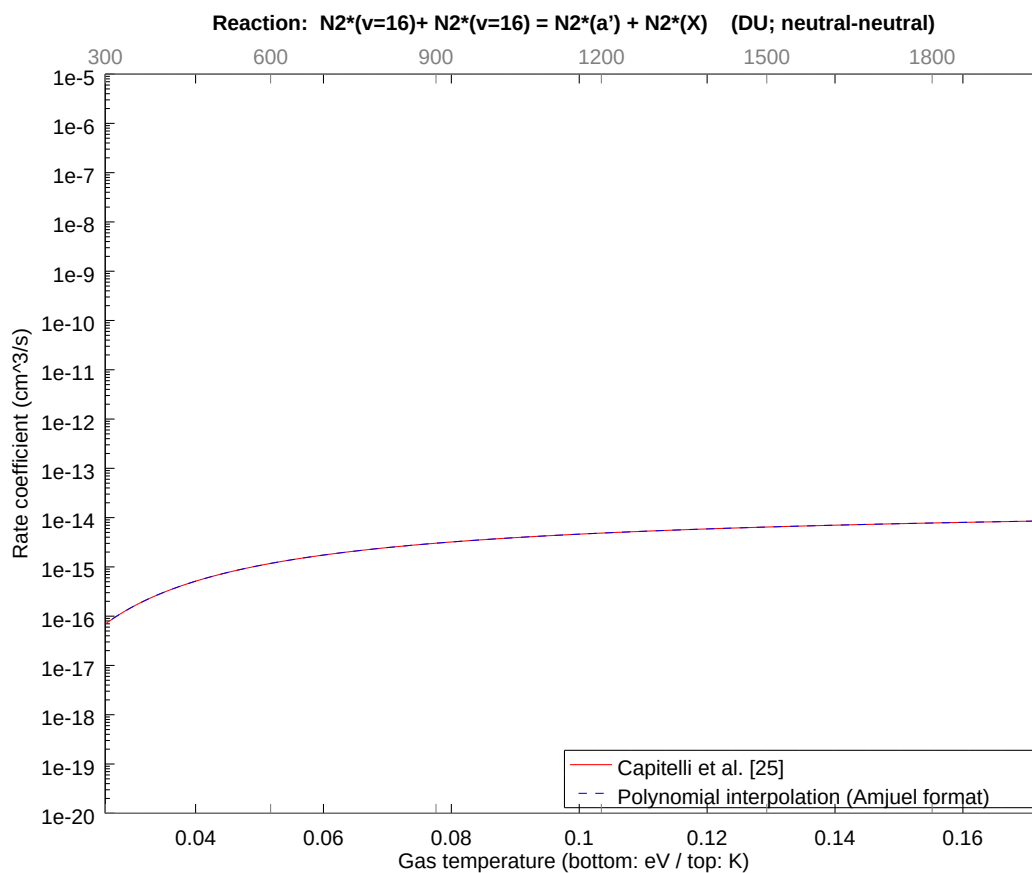
## 2.170 Reaction DT $2\text{N}_2(\text{a}') = \text{N}_4^+ + \text{e}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.532843602293E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



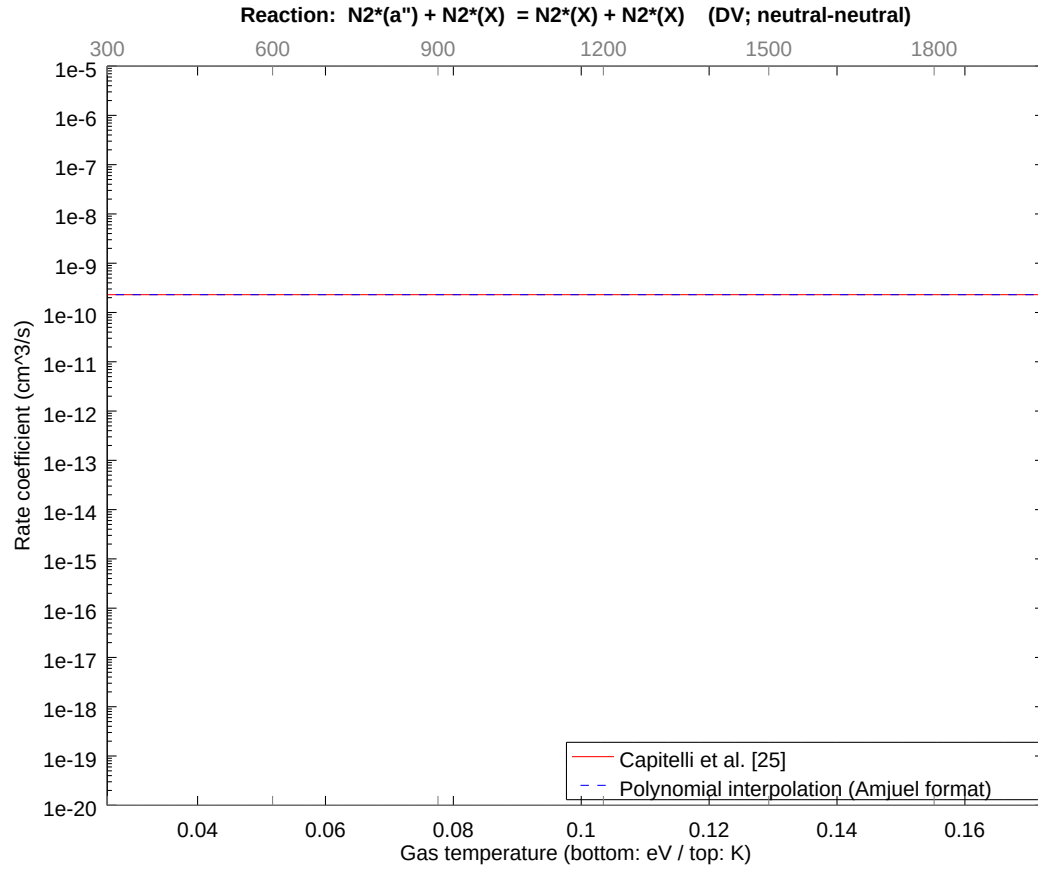
## 2.171 Reaction DU $\text{N}_2^*(v=16) + \text{N}_2^*(v=16) = \text{N}_2^*(a') + \text{N}_2^*(X)$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -3.171574443248E+01 | b1 | 5.551937076721E-02  | b2 | -2.123590686304E-01 |
| b3 | -9.750820635699E-02 | b4 | -7.347893115968E-02 | b5 | -2.293993753453E-02 |
| b6 | -5.751763538053E-03 | b7 | -7.323648019616E-04 | b8 | -5.447601905402E-05 |



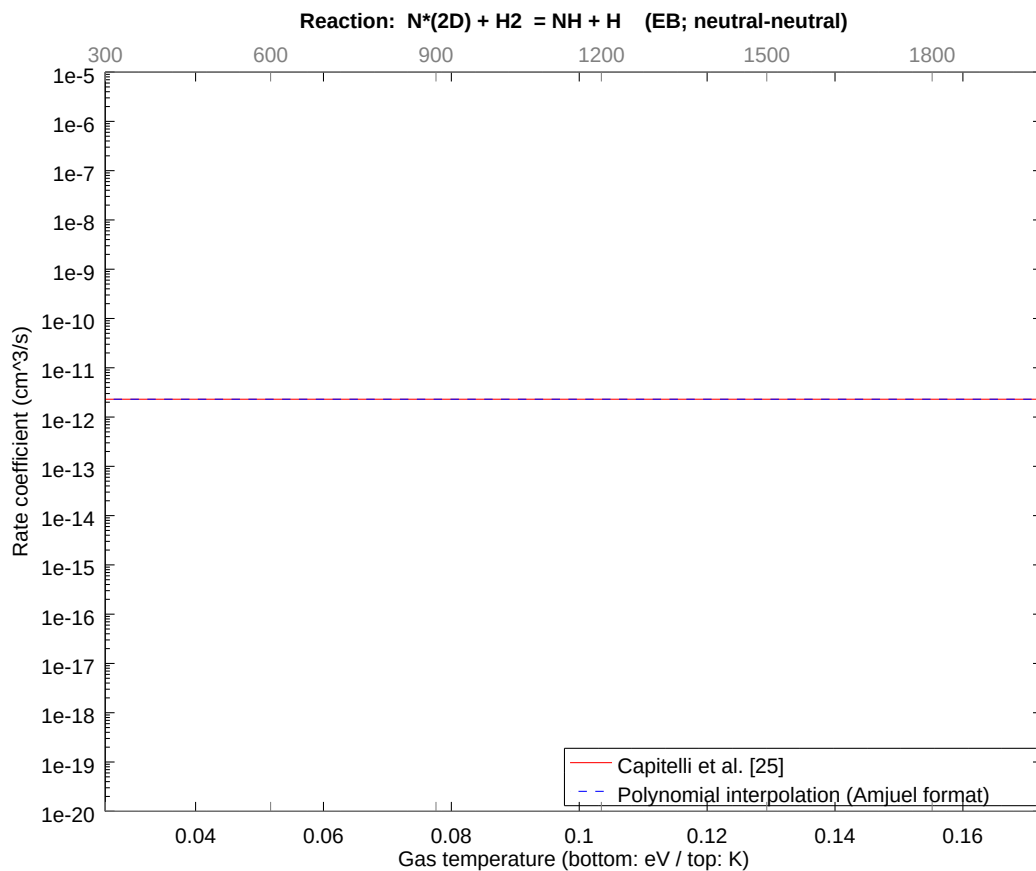
## 2.172 Reaction DV $\text{N2}^*(a'') + \text{N2}^*(X) = \text{N2}^*(X) + \text{N2}^*(X)$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.219294180701E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



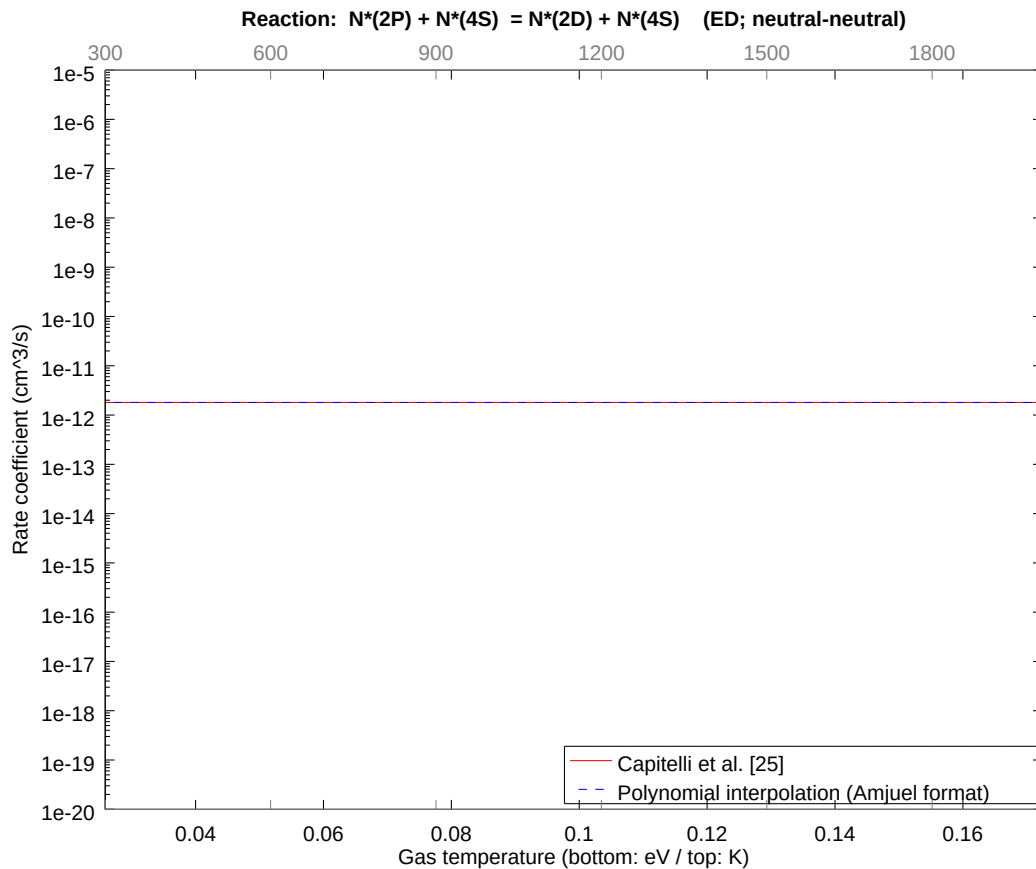
## 2.173 Reaction EB $\text{N}^*(2\text{D}) + \text{H}_2 = \text{NH} + \text{H}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.679811199299E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



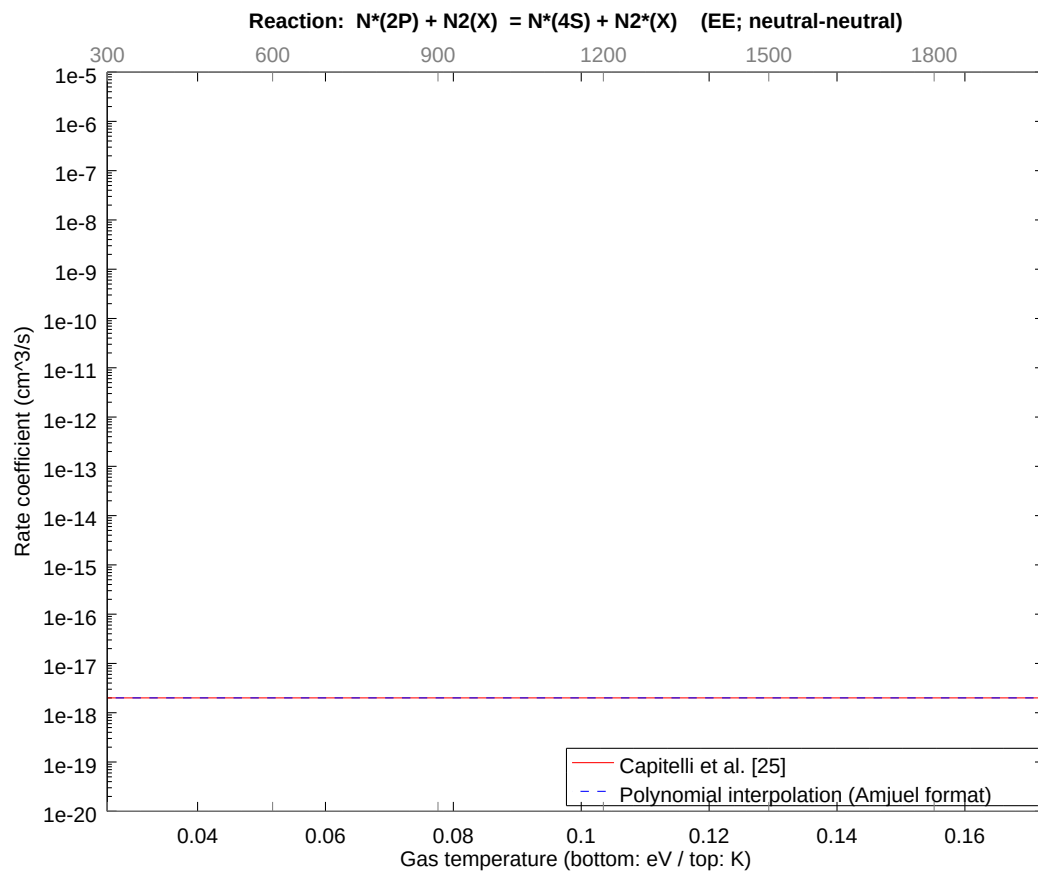
## 2.174 Reaction ED $\text{N}^*(2\text{P}) + \text{N}^*(4\text{S}) = \text{N}^*(2\text{D}) + \text{N}^*(4\text{S})$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -2.704323445103E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



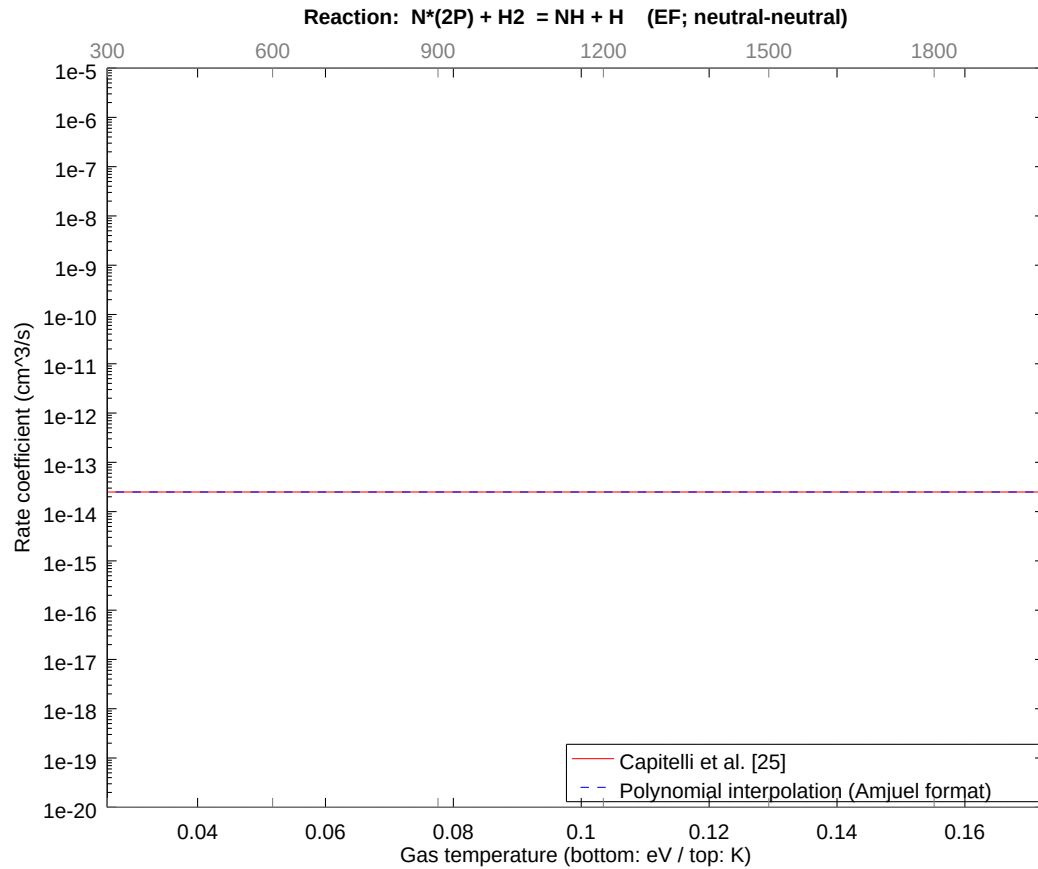
## 2.175 Reaction EE $\text{N}^*(2\text{P}) + \text{N}_2(\text{X}) = \text{N}^*(4\text{S}) + \text{N}_2^*(\text{X})$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -4.075338449333E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



## 2.176 Reaction EF $\text{N}^*(2\text{P}) + \text{H}_2 = \text{NH} + \text{H}$

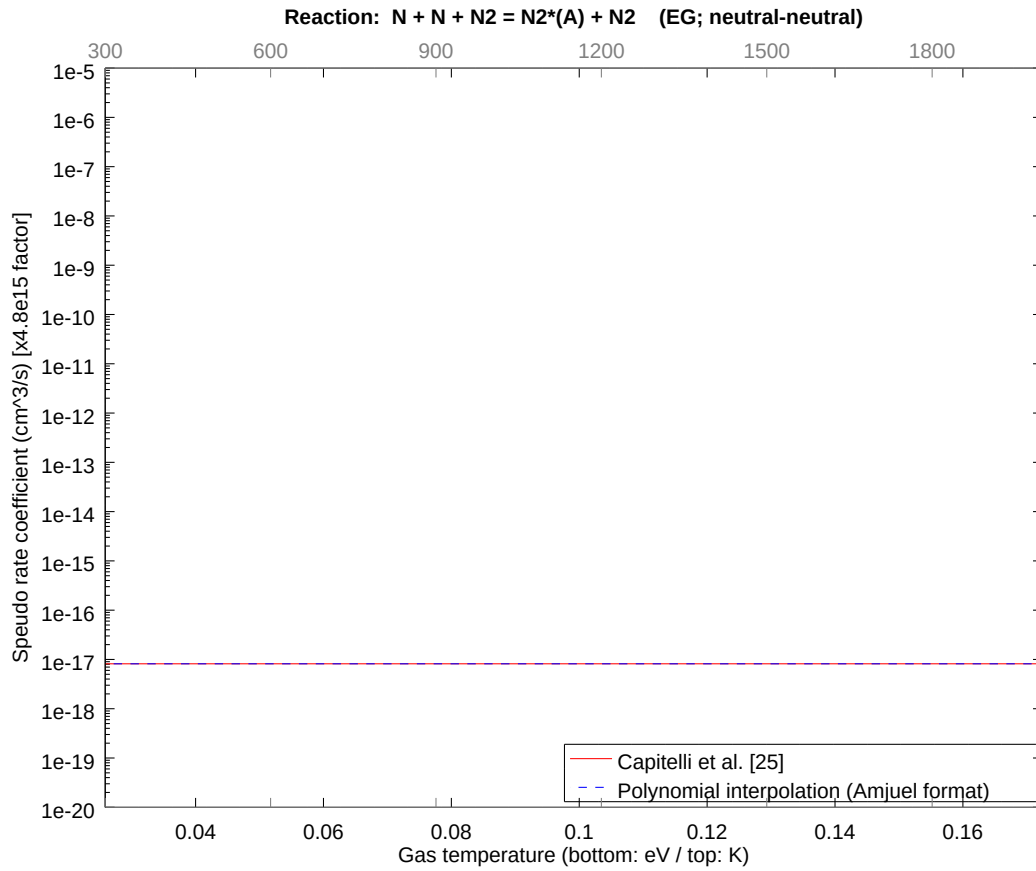
|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -3.131990057004E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |





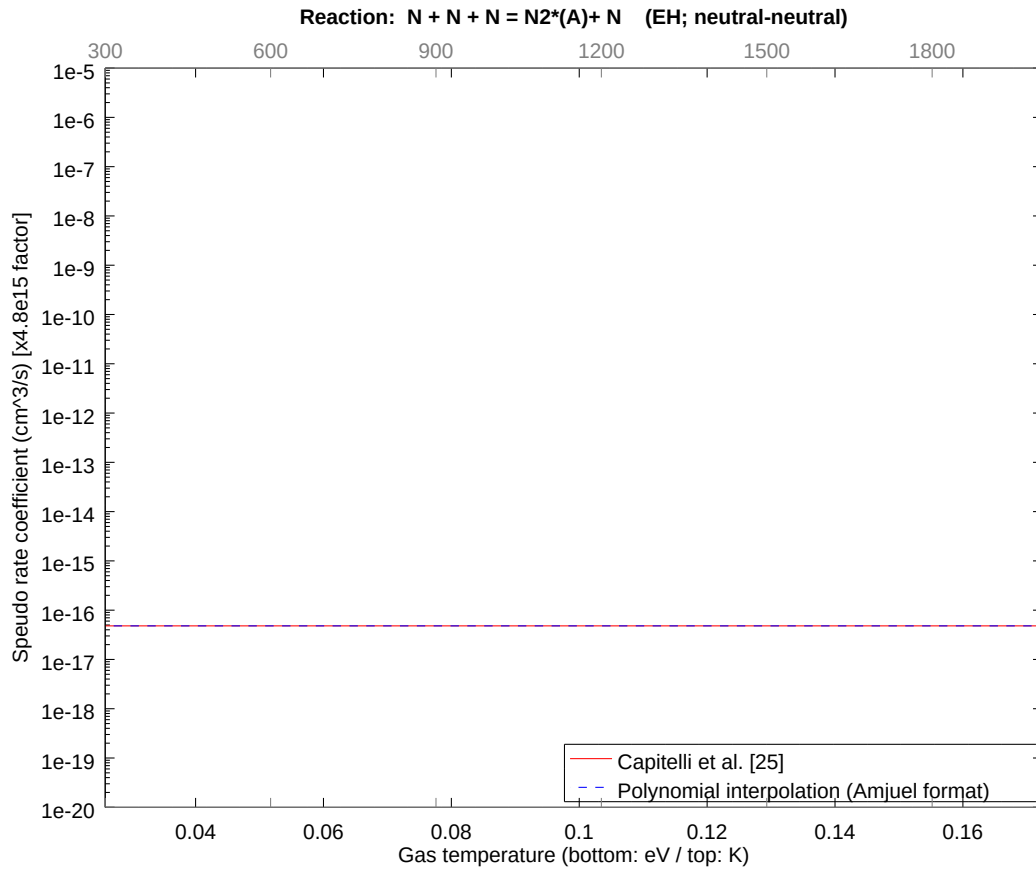
## 2.177 Reaction EG $\text{N} + \text{N} + \text{N}_2 = \text{N}_2^*(\text{A}) + \text{N}_2$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -3.934728750492E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



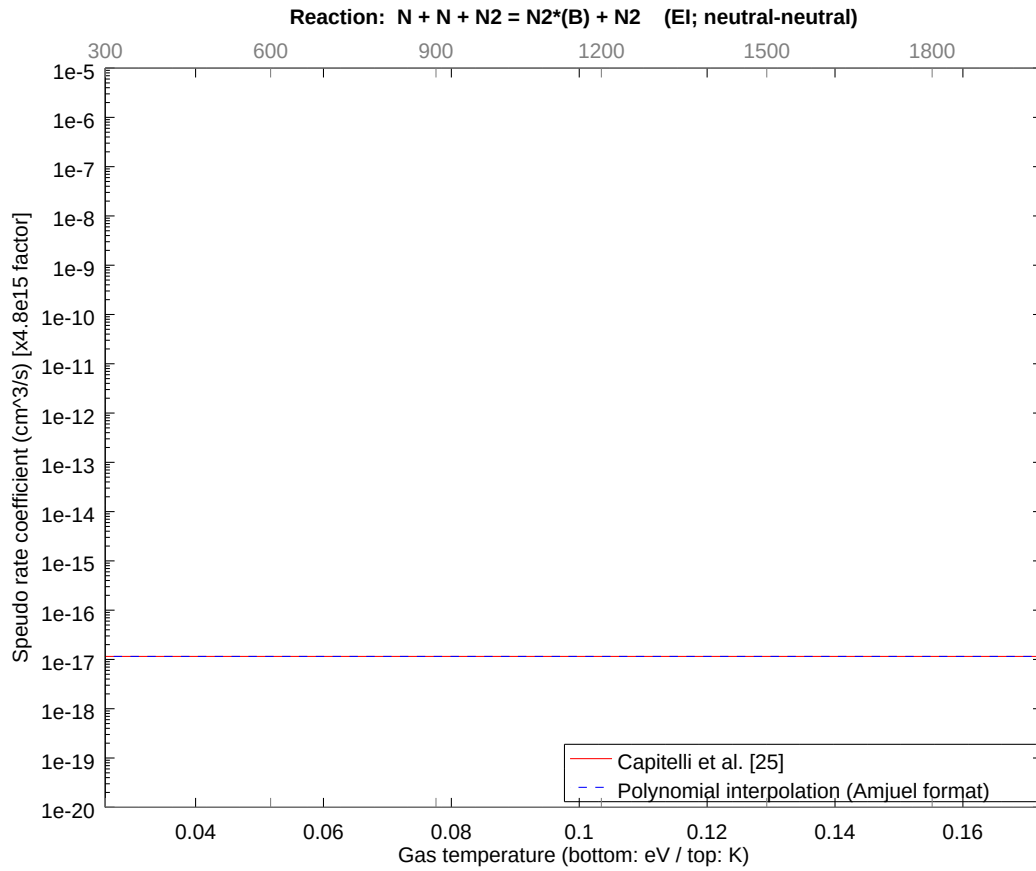
## 2.178 Reaction EH $N + N + N = N_2^*(A) + N$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -3.757533066298E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



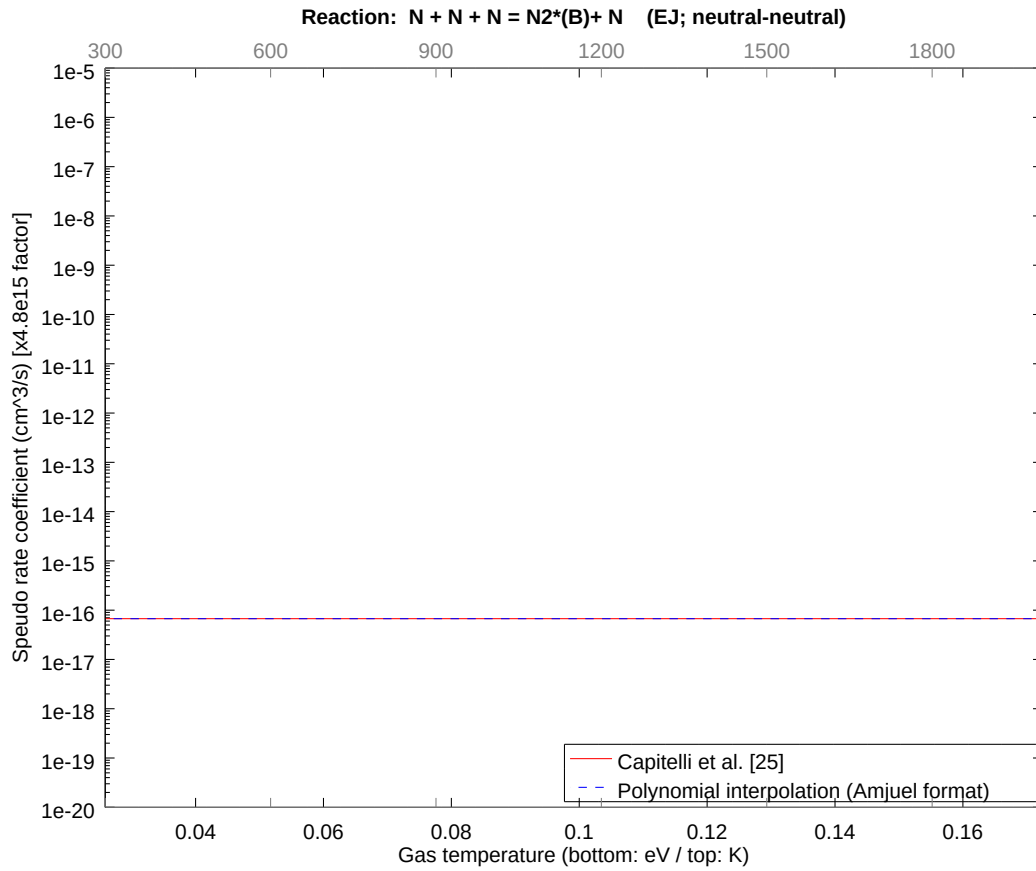
## 2.179 Reaction EI $\text{N} + \text{N} + \text{N}_2 = \text{N}_2^*(\text{B}) + \text{N}_2$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -3.900244701863E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



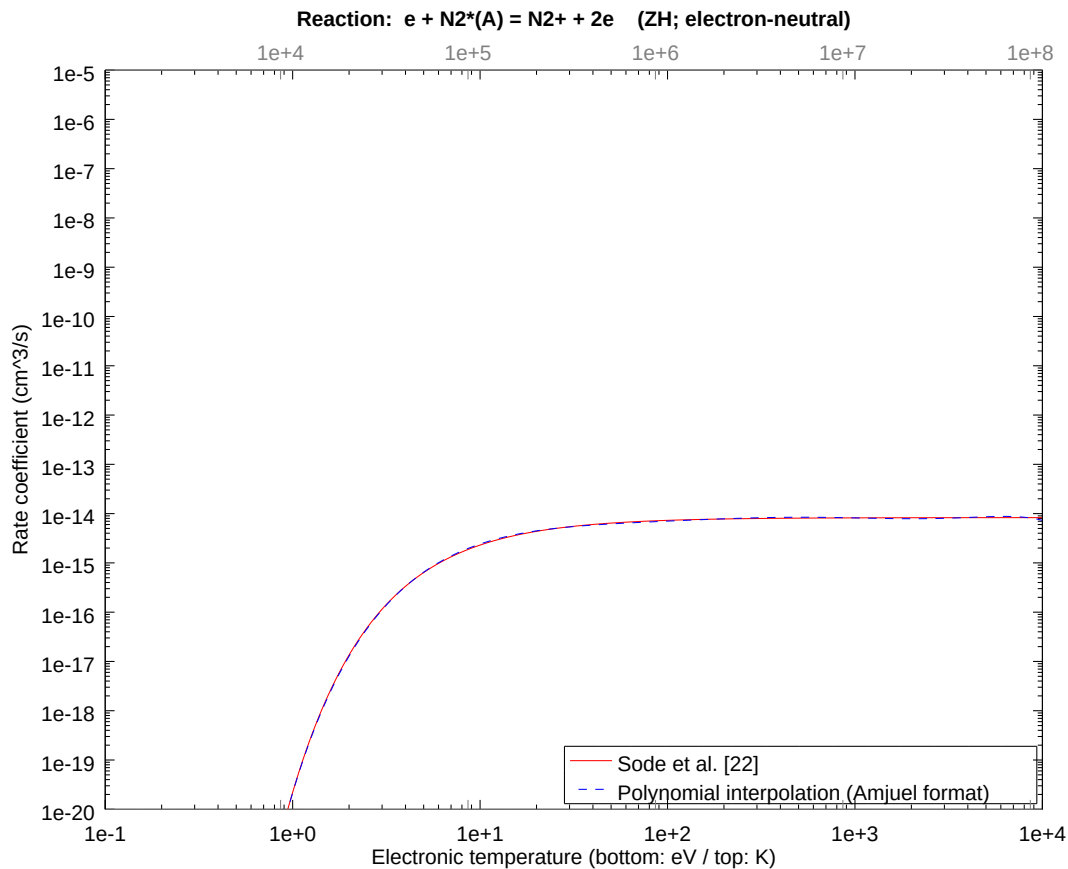
## 2.180 Reaction EJ $\text{N} + \text{N} + \text{N} = \text{N}_2^*(\text{B}) + \text{N}$

|    |                     |    |                    |    |                    |
|----|---------------------|----|--------------------|----|--------------------|
| b0 | -3.723885842636E+01 | b1 | 0.000000000000E+00 | b2 | 0.000000000000E+00 |
| b3 | 0.000000000000E+00  | b4 | 0.000000000000E+00 | b5 | 0.000000000000E+00 |
| b6 | 0.000000000000E+00  | b7 | 0.000000000000E+00 | b8 | 0.000000000000E+00 |



## 2.181 Reaction ZH $e + N_2^*(A) = N_2 + 2e$

|    |                     |    |                     |    |                     |
|----|---------------------|----|---------------------|----|---------------------|
| b0 | -4.526129534265E+01 | b1 | 1.282248936134E+01  | b2 | -6.360013862574E+00 |
| b3 | 2.051867418196E+00  | b4 | -4.656265922761E-01 | b5 | 7.421537485720E-02  |
| b6 | -7.867836650513E-03 | b7 | 4.934657549434E-04  | b8 | -1.373819547970E-05 |



### **3 Appendix**

## References

- [1] A.B.Ehrhardt, W.D.Langer, PPPL 2477 (1987)
- [2] R.K. Janev, B. Langer et al. Springer Series on Atoms+Plasmas, Vol 4, 1987